31) $r = 6$ on the interval $0 \le \theta \le \frac{\pi}{2}$

32) $r = e^{3\theta}$ on the interval $0 \le \theta \le 2$

Answer

$\frac{\sqrt{10}}{3}(e^6 - 1)$ units

33) $r = 6\cos\theta$ on the interval $0 \le \theta \le \frac{\pi}{2}$

34) $r = 8 + 8\cos\theta$ on the interval $0 \le \theta \le \pi$

Answer

32 units

35) $r = 1 - \sin\theta$ on the interval $0 \le \theta \le 2\pi$

In exercises 36 - 40, use the integration capabilities of a calculator to approximate the length of the curve.

36) [T] $r = 3\theta$ on the interval $0 \le \theta \le \frac{\pi}{2}$

Answer

6.238 units

37) [T] $r = \dfrac{2}{\theta}$ on the interval $\pi \le \theta \le 2\pi$

38) [T] $r = \sin^2\left(\frac{\theta}{2}\right)$ on the interval $0 \le \theta \le \pi$

Answer

2 units

39) [T] $r = 2\theta^2$ on the interval $0 \le \theta \le \pi$

40) [T] $r = \sin(3\cos\theta)$ on the interval $0 \le \theta \le \pi$

Answer

4.39 units

In exercises 41 - 43, use the familiar formula from geometry to find the area of the region described and then confirm by using the definite integral.

41) $r = 3\sin\theta$ on the interval $0 \le \theta \le \pi$

42) $r = \sin\theta + \cos\theta$ on the interval $0 \le \theta \le \pi$

Answer

$A = \pi\left(\frac{\sqrt{2}}{2}\right)^2 = \dfrac{\pi}{2}$ units^2 and $\dfrac{1}{2}\displaystyle\int_0^{\pi}(1 + 2\sin\theta\cos\theta)\,d\theta = \dfrac{\pi}{2}$ units^2

43) $r = 6\sin\theta + 8\cos\theta$ on the interval $0 \le \theta \le \pi$

In exercises 44 - 46, use the familiar formula from geometry to find the length of the curve and then confirm using the definite integral.

44) $r = 3\sin\theta$ on the interval $0 \le \theta \le \pi$

Answer

$C = 2\pi\left(\frac{3}{2}\right) = 3\pi$ units and $\displaystyle\int_0^{\pi} 3\,d\theta = 3\pi$ units

45) $r = \sin\theta + \cos\theta$ on the interval $0 \le \theta \le \pi$

46) $r = 6\sin\theta + 8\cos\theta$ on the interval $0 \le \theta \le \pi$

Answer

$C = 2\pi(5) = 10\pi$ units and $\int_0^{\pi} 10\,d\theta = 10\pi$ units

47) Verify that if $y = r\sin\theta = f(\theta)\sin\theta$ then $\dfrac{dy}{d\theta} = f'(\theta)\sin\theta + f(\theta)\cos\theta$.

In exercises 48 - 56, find the slope of a tangent line to a polar curve $r = f(\theta)$. Let $x = r\cos\theta = f(\theta)\cos\theta$ and $y = r\sin\theta = f(\theta)\sin\theta$, so the polar equation $r = f(\theta)$ is now written in parametric form.

48) Use the definition of the derivative $\dfrac{dy}{dx} = \dfrac{dy/d\theta}{dx/d\theta}$ and the product rule to derive the derivative of a polar equation.

Answer

$$\frac{dy}{dx} = \frac{f'(\theta)\sin\theta + f(\theta)\cos\theta}{f'(\theta)\cos\theta - f(\theta)\sin\theta}$$

49) $r = 1 - \sin\theta;\ \left(\frac{1}{2}, \frac{\pi}{6}\right)$

50) $r = 4\cos\theta;\ \left(2, \frac{\pi}{3}\right)$

Answer

The slope is $\frac{1}{\sqrt{3}}$.

51) $r = 8\sin\theta;\ \left(4, \frac{5\pi}{6}\right)$

52) $r = 4 + \sin\theta;\ \left(3, \frac{3\pi}{2}\right)$

Answer

The slope is 0.

53) $r = 6 + 3\cos\theta;\ (3, \pi)$

54) $r = 4\cos(2\theta)$; tips of the leaves

Answer

At $(4, 0)$, the slope is undefined. At $(-4, \frac{\pi}{2})$, the slope is 0.

55) $r = 2\sin(3\theta)$; tips of the leaves

56) $r = 2\theta;\ \left(\frac{\pi}{2}, \frac{\pi}{4}\right)$

Answer

The slope is undefined at $\theta = \frac{\pi}{4}$.

57) Find the points on the interval $-\pi \le \theta \le \pi$ at which the cardioid $r = 1 - \cos\theta$ has a vertical or horizontal tangent line.

58) For the cardioid $r = 1 + \sin\theta$, find the slope of the tangent line when $\theta = \frac{\pi}{3}$.

Answer

Slope = -1.

In exercises 59 - 62, find the slope of the tangent line to the given polar curve at the point given by the value of θ.

59) $r = 3\cos\theta,\ \theta = \frac{\pi}{3}$

60) $r = \theta,\ \theta = \frac{\pi}{2}$

Answer

Slope is $\frac{-2}{\pi}$.

61) $r = \ln\theta,\ \theta = e$

62) [T] Use technology: $r = 2 + 4\cos\theta$ at $\theta = \frac{\pi}{6}$

Answer

Calculator answer: -0.836.

In exercises 63 - 66, find the points at which the following polar curves have a horizontal or vertical tangent line.

63) $r = 4\cos\theta$

64) $r^2 = 4\cos(2\theta)$

Answer

Horizontal tangent at $\left(\pm\sqrt{2}, \frac{\pi}{6}\right),\ \left(\pm\sqrt{2}, -\frac{\pi}{6}\right)$.

65) $r = 2\sin(2\theta)$

66) The cardioid $r = 1 + \sin\theta$

Answer

Horizontal tangents at $\frac{\pi}{2},\ \frac{7\pi}{6},\ \frac{11\pi}{6}$.
Vertical tangents at $\frac{\pi}{6},\ \frac{5\pi}{6}$ and also at the pole $(0, 0)$.

67) Show that the curve $r = \sin\theta\tan\theta$ (called a **cissoid of Diocles**) has the line $x = 1$ as a vertical asymptote.

This page titled 11.4E: Exercises for Section 11.4 is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin "Jed" Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

11.5: Conic Sections

Learning Objectives

- Identify the equation of a parabola in standard form with given focus and directrix.
- Identify the equation of an ellipse in standard form with given foci.
- Identify the equation of a hyperbola in standard form with given foci.
- Recognize a parabola, ellipse, or hyperbola from its eccentricity value.
- Write the polar equation of a conic section with eccentricity e.
- Identify when a general equation of degree two is a parabola, ellipse, or hyperbola.

Conic sections have been studied since the time of the ancient Greeks, and were considered to be an important mathematical concept. As early as 320 BCE, such Greek mathematicians as Menaechmus, Appollonius, and Archimedes were fascinated by these curves. Appollonius wrote an entire eight-volume treatise on conic sections in which he was, for example, able to derive a specific method for identifying a conic section through the use of geometry. Since then, important applications of conic sections have arisen (for example, in astronomy), and the properties of conic sections are used in radio telescopes, satellite dish receivers, and even architecture. In this section we discuss the three basic conic sections, some of their properties, and their equations.

Conic sections get their name because they can be generated by intersecting a plane with a cone. A cone has two identically shaped parts called **nappes.** One nappe is what most people mean by "cone," having the shape of a party hat. A right circular cone can be generated by revolving a line passing through the origin around the y-axis as shown in Figure 11.5.1.

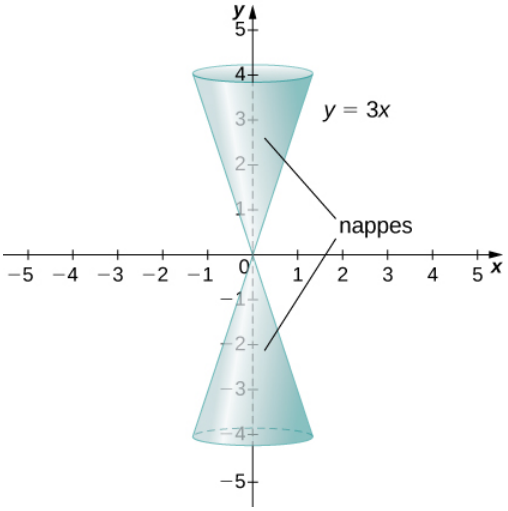


Figure 11.5.1: A cone generated by revolving the line $y = 3x$ around the y-axis.

Conic sections are generated by the intersection of a plane with a cone (Figure 11.5.2). If the plane is parallel to the axis of revolution (the y-axis), then the **conic section** is a hyperbola. If the plane is parallel to the generating line, the conic section is a parabola. If the plane is perpendicular to the axis of revolution, the conic section is a circle. If the plane intersects one nappe at an angle to the axis (other than **90°**), then the conic section is an ellipse.

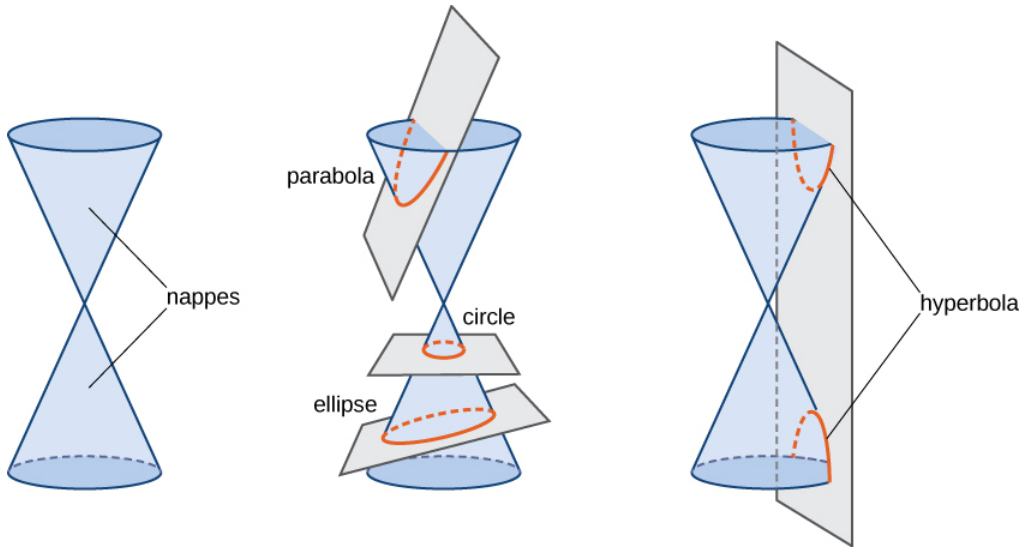


Figure 11.5.2: The four conic sections. Each conic is determined by the angle the plane makes with the axis of the cone.

Parabolas

A parabola is generated when a plane intersects a cone parallel to the generating line. In this case, the plane intersects only one of the nappes. A parabola can also be defined in terms of distances.

Definitions: The Focus, Directrix and Vertex

A parabola is the set of all points whose distance from a fixed point, called the *focus*, is equal to the distance from a fixed line, called the *directrix*. The point halfway between the focus and the directrix is called the *vertex* of the parabola.

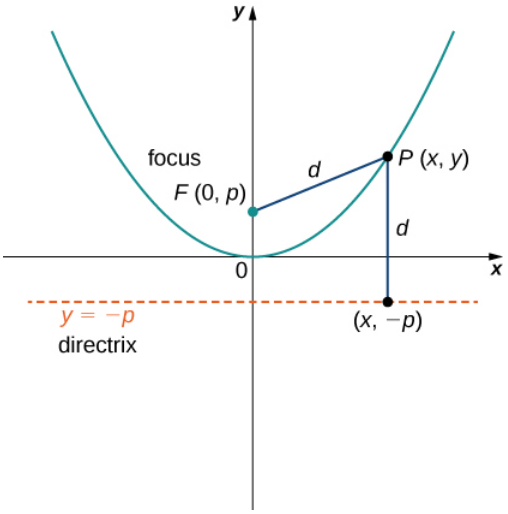


Figure 11.5.3: A typical parabola in which the distance from the focus to the vertex is represented by the variable p.

A graph of a typical parabola appears in Figure 11.5.3. Using this diagram in conjunction with the distance formula, we can derive an equation for a parabola. Recall the distance formula: Given point P with coordinates (x_1, y_1) and point Q with coordinates (x_2, y_2), the distance between them is given by the formula

$$d(P,Q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

Then from the definition of a parabola and Figure 11.5.3, we get

$$d(F,P) = d(P,Q)$$

$$\sqrt{(0-x)^2+(p-y)^2} = \sqrt{(x-x)^2+(-p-y)^2}.$$

Squaring both sides and simplifying yields

$$x^2+(p-y)^2 = 0^2+(-p-y)^2 \tag{11.5.1}$$
$$x^2+p^2-2py+y^2 = p^2+2py+y^2 \tag{11.5.2}$$
$$x^2-2py = 2py \tag{11.5.3}$$
$$x^2 = 4py. \tag{11.5.4}$$

Now suppose we want to relocate the vertex. We use the variables (h,k) to denote the coordinates of the vertex. Then if the focus is directly above the vertex, it has coordinates $(h,\ k+p)$ and the directrix has the equation $y = k-p$. Going through the same derivation yields the formula $(x-h)^2 = 4p(y-k)$. Solving this equation for y leads to the following theorem.

Equations for Parabolas: standard form

Given a parabola opening upward with vertex located at (h,k) and focus located at $(h,k+p)$, where p is a constant, the equation for the parabola is given by

$$y = \frac{1}{4p}(x-h)^2+k. \tag{11.5.5}$$

This is the *standard form* of a parabola.

We can also study the cases when the parabola opens down or to the left or the right. The equation for each of these cases can also be written in standard form as shown in the following graphs.

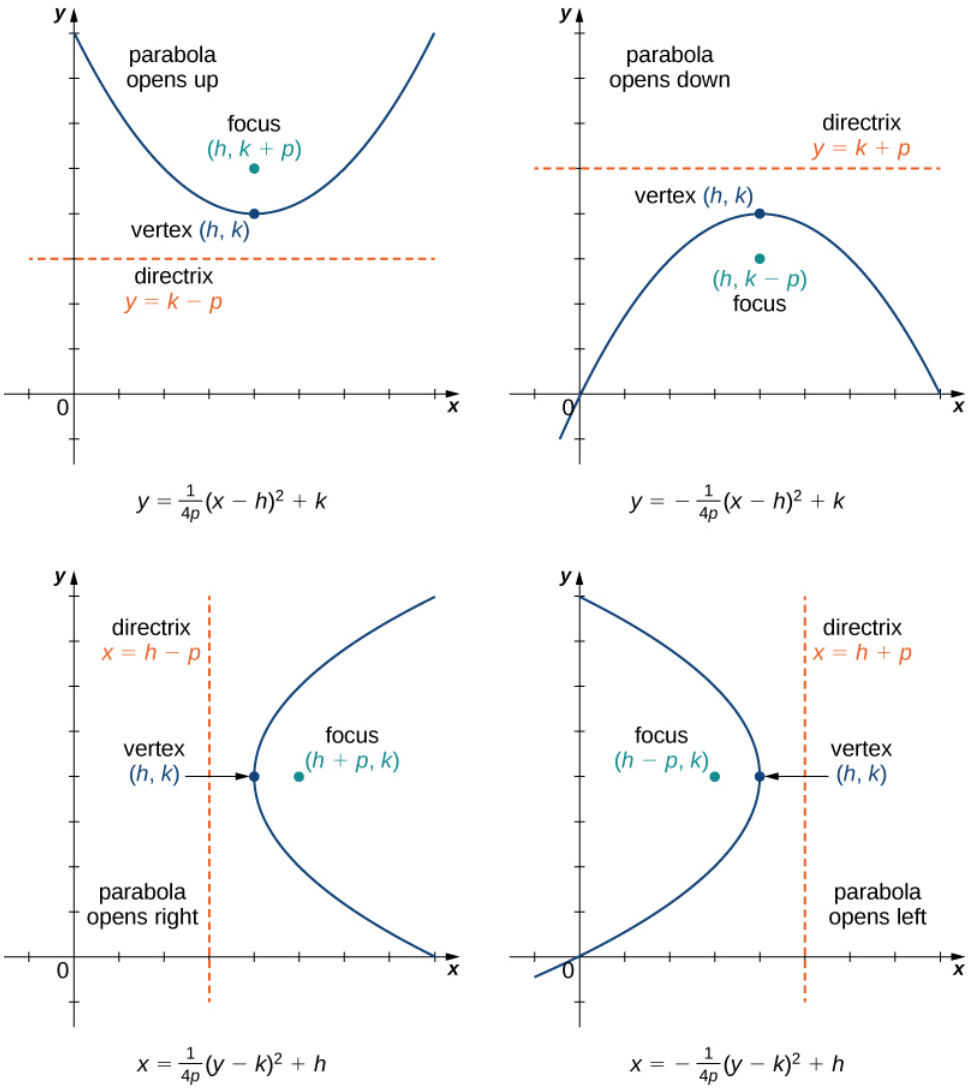


Figure 11.5.4: Four parabolas, opening in various directions, along with their equations in standard form.

In addition, the equation of a parabola can be written in the **general form**, though in this form the values of h, k, and p are not immediately recognizable. The general form of a parabola is written as

$$ax^2 + bx + cy + d = 0 \tag{11.5.6}$$

or

$$ay^2 + bx + cy + d = 0. \tag{11.5.7}$$

Equation 11.5.6 represents a parabola that opens either up or down. Equation 11.5.7 represents a parabola that opens either to the left or to the right. To put the equation into standard form, use the method of completing the square.

Example 11.5.1: Converting the Equation of a Parabola from General into Standard Form

Put the equation

$$x^2 - 4x - 8y + 12 = 0$$

into standard form and graph the resulting parabola.

Solution

Since y is not squared in this equation, we know that the parabola opens either upward or downward. Therefore we need to solve this equation for y, which will put the equation into standard form. To do that, first add $8y$ to both sides of the equation:

$$8y = x^2 - 4x + 12.$$

The next step is to complete the square on the right-hand side. Start by grouping the first two terms on the right-hand side using parentheses:

$$8y = (x^2 - 4x) + 12.$$

Next determine the constant that, when added inside the parentheses, makes the quantity inside the parentheses a perfect square trinomial. To do this, take half the coefficient of x and square it. This gives $\left(\frac{-4}{2}\right)^2 = 4$. Add 4 inside the parentheses and subtract 4 outside the parentheses, so the value of the equation is not changed:

$$8y = (x^2 - 4x + 4) + 12 - 4.$$

Now combine like terms and factor the quantity inside the parentheses:

$$8y = (x - 2)^2 + 8.$$

Finally, divide by 8:

$$y = \frac{1}{8}(x - 2)^2 + 1.$$

This equation is now in standard form. Comparing this to Equation 11.5.5 gives $h = 2$, $k = 1$, and $p = 2$. The parabola opens up, with vertex at $(2, 1)$, focus at $(2, 3)$, and directrix $y = -1$. The graph of this parabola appears as follows.

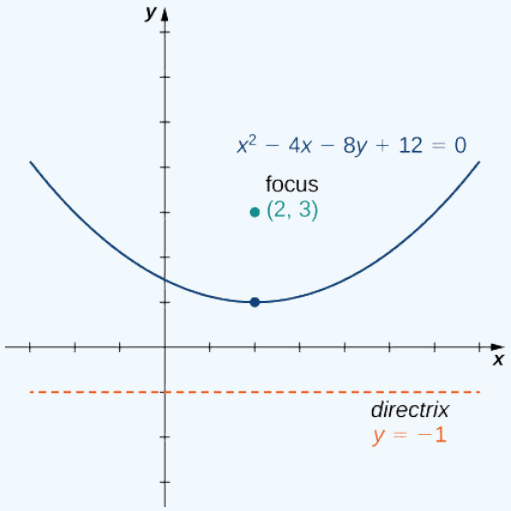


Figure 11.5.5: The parabola in Example 11.5.1.

? Exercise 11.5.1

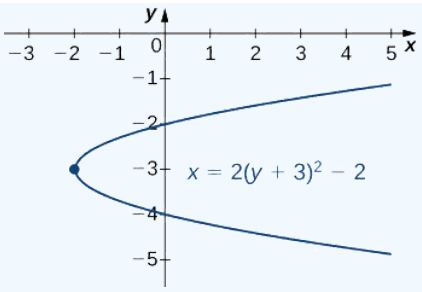
Put the equation $2y^2 - x + 12y + 16 = 0$ into standard form and graph the resulting parabola.

Hint

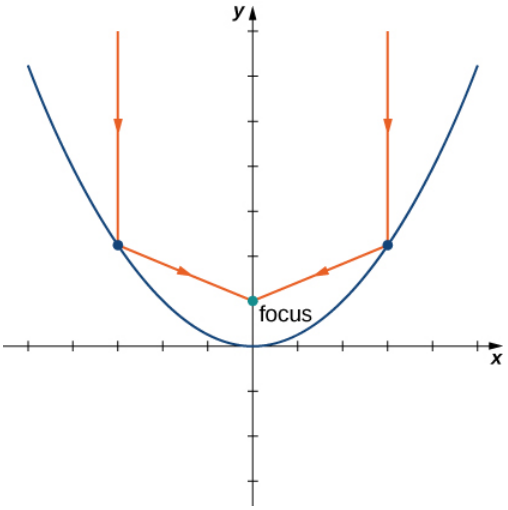
Solve for x. Check which direction the parabola opens.

Answer

$$x = 2(y + 3)^2 - 2$$



The axis of symmetry of a vertical (opening up or down) parabola is a vertical line passing through the vertex. The parabola has an interesting reflective property. Suppose we have a satellite dish with a parabolic cross section. If a beam of electromagnetic waves, such as light or radio waves, comes into the dish in a straight line from a satellite (parallel to the axis of symmetry), then the waves reflect off the dish and collect at the focus of the parabola as shown.



Consider a parabolic dish designed to collect signals from a satellite in space. The dish is aimed directly at the satellite, and a receiver is located at the focus of the parabola. Radio waves coming in from the satellite are reflected off the surface of the parabola to the receiver, which collects and decodes the digital signals. This allows a small receiver to gather signals from a wide angle of sky. Flashlights and headlights in a car work on the same principle, but in reverse: the source of the light (that is, the light bulb) is located at the focus and the reflecting surface on the parabolic mirror focuses the beam straight ahead. This allows a small light bulb to illuminate a wide angle of space in front of the flashlight or car.

Ellipses

An ellipse can also be defined in terms of distances. In the case of an ellipse, there are two foci (plural of focus), and two directrices (plural of directrix). We look at the directrices in more detail later in this section.

Definition: Ellipse

An ellipse is the set of all points for which the sum of their distances from two fixed points (the foci) is constant.

A graph of a typical ellipse is shown in Figure 11.5.6. In this figure the foci are labeled as F and F'. Both are the same fixed distance from the origin, and this distance is represented by the variable c. Therefore the coordinates of F are $(c, 0)$ and the coordinates of F' are $(-c, 0)$. The points P and P' are located at the ends of the **major axis** of the ellipse, and have coordinates $(a, 0)$ and $(-a, 0)$, respectively. The major axis is always the longest distance across the ellipse, and can be horizontal or vertical. Thus, the length of the major axis in this ellipse is $2a$. Furthermore, P and P' are called the vertices of the ellipse. The points Q and Q' are located at the ends of the **minor axis** of the ellipse, and have coordinates $(0, b)$ and $(0, -b)$, respectively. The minor axis is the shortest distance across the ellipse. The minor axis is perpendicular to the major axis.

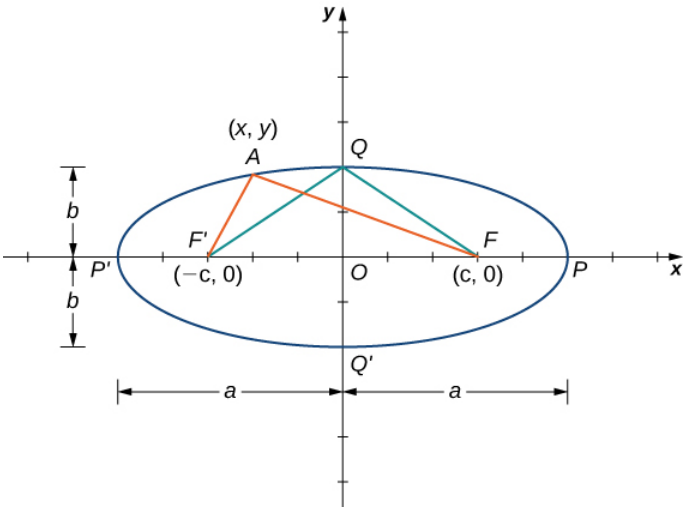


Figure 11.5.6: A typical ellipse in which the sum of the distances from any point on the ellipse to the foci is constant.

According to the definition of the ellipse, we can choose any point on the ellipse and the sum of the distances from this point to the two foci is constant. Suppose we choose the point P. Since the coordinates of point P are $(a, 0)$, the sum of the distances is

$$d(P, F) + d(P, F') = (a - c) + (a + c) = 2a.$$

Therefore the sum of the distances from an arbitrary point A with coordinates (x, y) is also equal to $2a$. Using the distance formula, we get

$$d(A, F) + d(A, F') = 2a.$$

$$\sqrt{(x - c)^2 + y^2} + \sqrt{(x + c)^2 + y^2} = 2a$$

Subtract the second radical from both sides and square both sides:

$$\sqrt{(x - c)^2 + y^2} = 2a - \sqrt{(x + c)^2 + y^2}$$

$$(x - c)^2 + y^2 = 4a^2 - 4a\sqrt{(x + c)^2 + y^2} + (x + c)^2 + y^2$$

$$x^2 - 2cx + c^2 + y^2 = 4a^2 - 4a\sqrt{(x + c)^2 + y^2} + x^2 + 2cx + c^2 + y^2$$

$$-2cx = 4a^2 - 4a\sqrt{(x + c)^2 + y^2} + 2cx.$$

Now isolate the radical on the right-hand side and square again:

$$-2cx = 4a^2 - 4a\sqrt{(x + c)^2 + y^2} + 2cx$$

$$4a\sqrt{(x + c)^2 + y^2} = 4a^2 + 4cx$$

$$\sqrt{(x + c)^2 + y^2} = a + \frac{cx}{a}$$

$$(x + c)^2 + y^2 = a^2 + 2cx + \frac{c^2x^2}{a^2}$$

$$x^2 + 2cx + c^2 + y^2 = a^2 + 2cx + \frac{c^2x^2}{a^2}$$

$$x^2 + c^2 + y^2 = a^2 + \frac{c^2x^2}{a^2}.$$

Isolate the variables on the left-hand side of the equation and the constants on the right-hand side:

$$x^2 - \frac{c^2x^2}{a^2} + y^2 = a^2 - c^2$$

$$\frac{(a^2 - c^2)x^2}{a^2} + y^2 = a^2 - c^2.$$

Divide both sides by $a^2 - c^2$. This gives the equation

$$\frac{x^2}{a^2} + \frac{y^2}{a^2 - c^2} = 1.$$

If we refer back to Figure 11.5.6, then the length of each of the two green line segments is equal to a. This is true because the sum of the distances from the point Q to the foci F and F' is equal to $2a$, and the lengths of these two line segments are equal. This line segment forms a right triangle with hypotenuse length a and leg lengths b and c. From the Pythagorean theorem, $b^2 + c^2 = a^2$ and $b^2 = a^2 - c^2$. Therefore the equation of the ellipse becomes

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

Finally, if the center of the ellipse is moved from the origin to a point (h, k), we have the following standard form of an ellipse.

Equation of an Ellipse in Standard Form

Consider the ellipse with center (h, k), a horizontal major axis with length $2a$, and a vertical minor axis with length $2b$. Then the equation of this ellipse in standard form is

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1 \tag{11.5.8}$$

and the foci are located at $(h \pm c, k)$, where $c^2 = a^2 - b^2$. The equations of the directrices are $x = h \pm \frac{a^2}{c}$.

If the major axis is vertical, then the equation of the ellipse becomes

$$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1 \tag{11.5.9}$$

and the foci are located at $(h, k \pm c)$, where $c^2 = a^2 - b^2$. The equations of the directrices in this case are $y = k \pm \frac{a^2}{c}$.

If the major axis is horizontal, then the ellipse is called horizontal, and if the major axis is vertical, then the ellipse is called vertical. The equation of an ellipse is in general form if it is in the form

$$Ax^2 + By^2 + Cx + Dy + E = 0,$$

where *A* and *B* are either both positive or both negative. To convert the equation from general to standard form, use the method of **completing the square**.

Example 11.5.2: Finding the Standard Form of an Ellipse

Put the equation

$$9x^2 + 4y^2 - 36x + 24y + 36 = 0$$

into standard form and graph the resulting ellipse.

Solution

First subtract 36 from both sides of the equation:

$$9x^2 + 4y^2 - 36x + 24y = -36.$$

Next group the x terms together and the y terms together, and factor out the common factor:

$$(9x^2 - 36x) + (4y^2 + 24y) = -36$$

$$9(x^2-4x)+4(y^2+6y)=-36.$$

We need to determine the constant that, when added inside each set of parentheses, results in a perfect square. In the first set of parentheses, take half the coefficient of x and square it. This gives $\left(\frac{-4}{2}\right)^2=4$. In the second set of parentheses, take half the coefficient of y and square it. This gives $\left(\frac{6}{2}\right)^2=9$. Add these inside each pair of parentheses. Since the first set of parentheses has a 9 in front, we are actually adding 36 to the left-hand side. Similarly, we are adding 36 to the second set as well. Therefore the equation becomes

$$9(x^2-4x+4)+4(y^2+6y+9)=-36+36+36$$

$$9(x^2-4x+4)+4(y^2+6y+9)=36.$$

Now factor both sets of parentheses and divide by 36:

$$9(x-2)^2+4(y+3)^2=36$$

$$\frac{9(x-2)^2}{36}+\frac{4(y+3)^2}{36}=1$$

$$\frac{(x-2)^2}{4}+\frac{(y+3)^2}{9}=1.$$

The equation is now in standard form. Comparing this to Equation 11.5.9 gives $h=2, k=-3, a=3$, and $b=2$. This is a vertical ellipse with center at $(2,-3)$, major axis 6, and minor axis 4. The graph of this ellipse appears as follows.

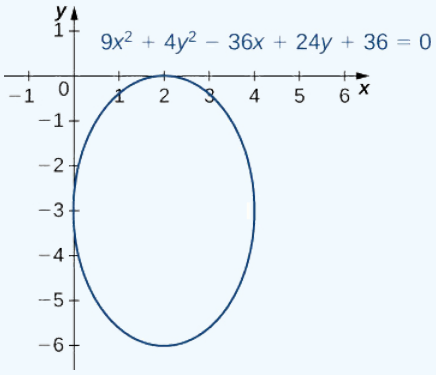


Figure 11.5.7: The ellipse in Example 11.5.2.

? Exercise 11.5.2

Put the equation

$$9x^2+16y^2+18x-64y-71=0$$

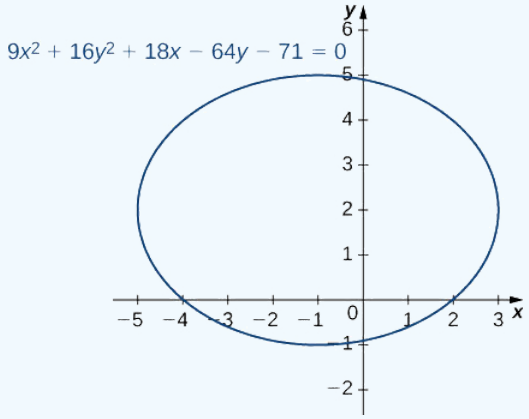
into standard form and graph the resulting ellipse.

Hint

Move the constant over and complete the square.

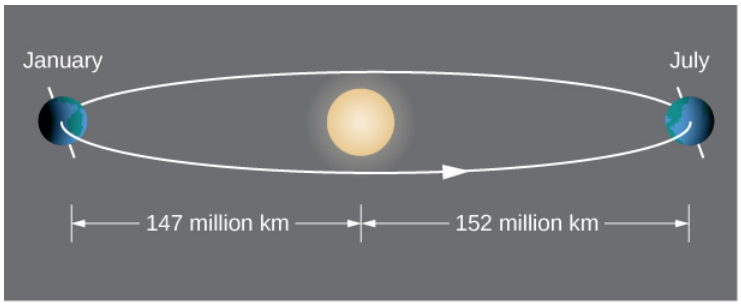
Answer

$$\frac{(x+1)^2}{16}+\frac{(y-2)^2}{9}=1$$



According to Kepler's first law of planetary motion, the orbit of a planet around the Sun is an ellipse with the Sun at one of the foci as shown in Figure 11.5.8*A*. Because Earth's orbit is an ellipse, the distance from the Sun varies throughout the year. A commonly held misconception is that Earth is closer to the Sun in the summer. In fact, in summer for the northern hemisphere, Earth is farther from the Sun than during winter. The difference in season is caused by the tilt of Earth's axis in the orbital plane. Comets that orbit the Sun, such as Halley's Comet, also have elliptical orbits, as do moons orbiting the planets and satellites orbiting Earth.

Ellipses also have interesting reflective properties: A light ray emanating from one focus passes through the other focus after mirror reflection in the ellipse. The same thing occurs with a sound wave as well. The National Statuary Hall in the U.S. Capitol in Washington, DC, is a famous room in an elliptical shape as shown in Figure 11.5.8*B*. This hall served as the meeting place for the U.S. House of Representatives for almost fifty years. The location of the two foci of this semi-elliptical room are clearly identified by marks on the floor, and even if the room is full of visitors, when two people stand on these spots and speak to each other, they can hear each other much more clearly than they can hear someone standing close by. Legend has it that John Quincy Adams had his desk located on one of the foci and was able to eavesdrop on everyone else in the House without ever needing to stand. Although this makes a good story, it is unlikely to be true, because the original ceiling produced so many echoes that the entire room had to be hung with carpets to dampen the noise. The ceiling was rebuilt in 1902 and only then did the now-famous whispering effect emerge. Another famous whispering gallery—the site of many marriage proposals—is in Grand Central Station in New York City.



(a)



(b)

Figure 11.5.8: (a) Earth's orbit around the Sun is an ellipse with the Sun at one focus. (b) Statuary Hall in the U.S. Capitol is a whispering gallery with an elliptical cross section.

Hyperbolas

A hyperbola can also be defined in terms of distances. In the case of a hyperbola, there are two foci and two directrices. Hyperbolas also have two asymptotes.

Definition: hyperbola

A hyperbola is the set of all points where the difference between their distances from two fixed points (the foci) is constant.

A graph of a typical hyperbola appears as follows.

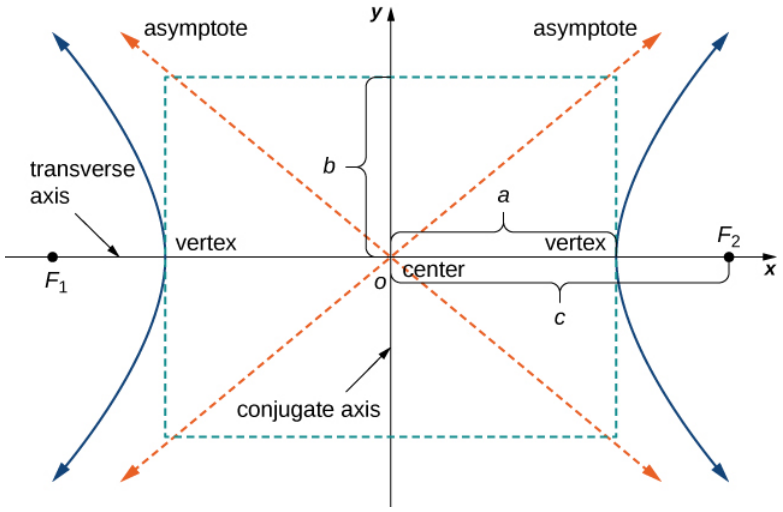


Figure 11.5.9: A typical hyperbola in which the difference of the distances from any point on the hyperbola to the foci is constant. The transverse axis is also called the major axis, and the conjugate axis is also called the minor axis.

The derivation of the equation of a hyperbola in standard form is virtually identical to that of an ellipse. One slight hitch lies in the definition: The difference between two numbers is always positive. Let P be a point on the hyperbola with coordinates (x, y). Then the definition of the hyperbola gives $|d(P, F_1) - d(P, F_2)| = constant$. To simplify the derivation, assume that P is on the right branch of the hyperbola, so the absolute value bars drop. If it is on the left branch, then the subtraction is reversed. The vertex of the right branch has coordinates $(a, 0)$, so

$$d(P, F_1) - d(P, F_2) = (c + a) - (c - a) = 2a.$$

This equation is therefore true for any point on the hyperbola. Returning to the coordinates (x, y) for P:

$$d(P, F_1) - d(P, F_2) = 2a$$

$$\sqrt{(x + c)^2 + y^2} - \sqrt{(x - c)^2 + y^2} = 2a.$$

Isolate the second radical and square both sides:

$$\sqrt{(x - c)^2 + y^2} = -2a + \sqrt{(x + c)^2 + y^2}$$

$$(x - c)^2 + y^2 = 4a^2 - 4a\sqrt{(x + c)^2 + y^2} + (x + c)^2 + y^2$$

$$x^2 - 2cx + c^2 + y^2 = 4a^2 - 4a\sqrt{(x + c)^2 + y^2} + x^2 + 2cx + c^2 + y^2$$

$$-2cx = 4a^2 - 4a\sqrt{(x + c)^2 + y^2} + 2cx.$$

Now isolate the radical on the right-hand side and square again:

$$-2cx = 4a^2 - 4a\sqrt{(x + c)^2 + y^2} + 2cx$$

$$-4a\sqrt{(x + c)^2 + y^2} = -4a^2 - 4cx$$

$$-\sqrt{(x + c)^2 + y^2} = -a - \frac{cx}{a}$$

$$(x + c)^2 + y^2 = a^2 + 2cx + \frac{c^2x^2}{a^2}$$

$$x^2 + 2cx + c^2 + y^2 = a^2 + 2cx + \frac{c^2x^2}{a^2}$$

$$x^2 + c^2 + y^2 = a^2 + \frac{c^2x^2}{a^2}.$$

Isolate the variables on the left-hand side of the equation and the constants on the right-hand side:

$$x^2 - \frac{c^2x^2}{a^2} + y^2 = a^2 - c^2$$

$$\frac{(a^2 - c^2)x^2}{a^2} + y^2 = a^2 - c^2.$$

Finally, divide both sides by $a^2 - c^2$. This gives the equation

$$\frac{x^2}{a^2} + \frac{y^2}{a^2 - c^2} = 1.$$

We now define b so that $b^2 = c^2 - a^2$. This is possible because $c > a$. Therefore the equation of the hyperbola becomes

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

Finally, if the center of the hyperbola is moved from the origin to the point (h, k), we have the following standard form of a hyperbola.

Equation of a Hyperbola in Standard Form

Consider the hyperbola with center (h, k), a horizontal major axis, and a vertical minor axis. Then the equation of this hyperbola is

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1 \tag{11.5.10}$$

and the foci are located at $(h \pm c, k)$, where $c^2 = a^2 + b^2$. The equations of the asymptotes are given by $y = k \pm \frac{b}{a}(x - h)$. The equations of the directrices are

$$x = h \pm \frac{a^2}{\sqrt{a^2 + b^2}} = h \pm \frac{a^2}{c}$$

If the major axis is vertical, then the equation of the hyperbola becomes

$$\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$$

and the foci are located at $(h, k \pm c)$, where $c^2 = a^2 + b^2$. The equations of the asymptotes are given by $y = k \pm \frac{a}{b}(x - h)$. The equations of the directrices are

$$y = k \pm \frac{a^2}{\sqrt{a^2 + b^2}} = k \pm \frac{a^2}{c}.$$

If the major axis (transverse axis) is horizontal, then the hyperbola is called horizontal, and if the major axis is vertical then the hyperbola is called vertical. The equation of a hyperbola is in general form if it is in the form

$$Ax^2 + By^2 + Cx + Dy + E = 0,$$

where A and B have opposite signs. In order to convert the equation from general to standard form, use the method of completing the square.

Example 11.5.3: Finding the Standard Form of a Hyperbola

Put the equation $9x^2 - 16y^2 + 36x + 32y - 124 = 0$ into standard form and graph the resulting hyperbola. What are the equations of the asymptotes?

Solution

First add 124 to both sides of the equation:

$$9x^2 - 16y^2 + 36x + 32y = 124.$$

Next group the x terms together and the y terms together, then factor out the common factors:

$$(9x^2+36x)-(16y^2-32y)=124$$

$$9(x^2+4x)-16(y^2-2y)=124.$$

We need to determine the constant that, when added inside each set of parentheses, results in a perfect square. In the first set of parentheses, take half the coefficient of x and square it. This gives $\left(\frac{4}{2}\right)^2=4$. In the second set of parentheses, take half the coefficient of y and square it. This gives $\left(\frac{-2}{2}\right)^2=1$. Add these inside each pair of parentheses. Since the first set of parentheses has a 9 in front, we are actually adding 36 to the left-hand side. Similarly, we are subtracting 16 from the second set of parentheses. Therefore the equation becomes

$$9(x^2+4x+4)-16(y^2-2y+1)=124+36-16$$

$$9(x^2+4x+4)-16(y^2-2y+1)=144.$$

Next factor both sets of parentheses and divide by 144:

$$9(x+2)^2-16(y-1)^2=144$$

$$\frac{9(x+2)^2}{144}-\frac{16(y-1)^2}{144}=1$$

$$\frac{(x+2)^2}{16}-\frac{(y-1)^2}{9}=1.$$

The equation is now in standard form. Comparing this to Equation 11.5.10 gives $h=-2$, $k=1$, $a=4$, and $b=3$. This is a horizontal hyperbola with center at $(-2,1)$ and asymptotes given by the equations $y=1\pm\frac{3}{4}(x+2)$. The graph of this hyperbola appears in Figure 11.5.10.

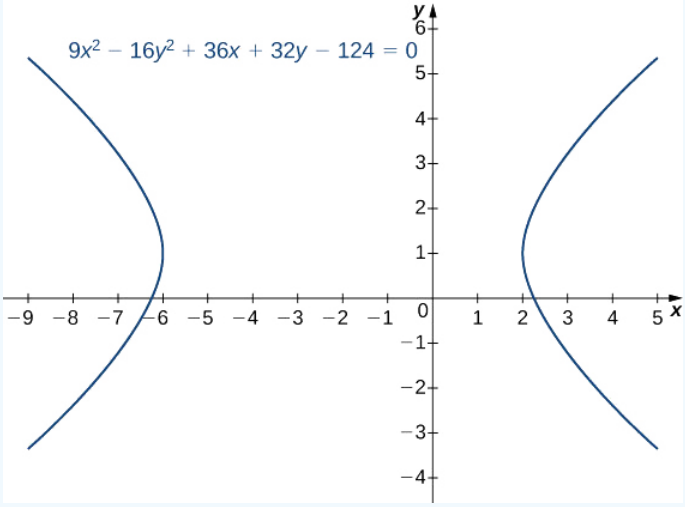


Figure 11.5.10: Graph of the hyperbola in Example 11.5.3.

Exercise 11.5.3

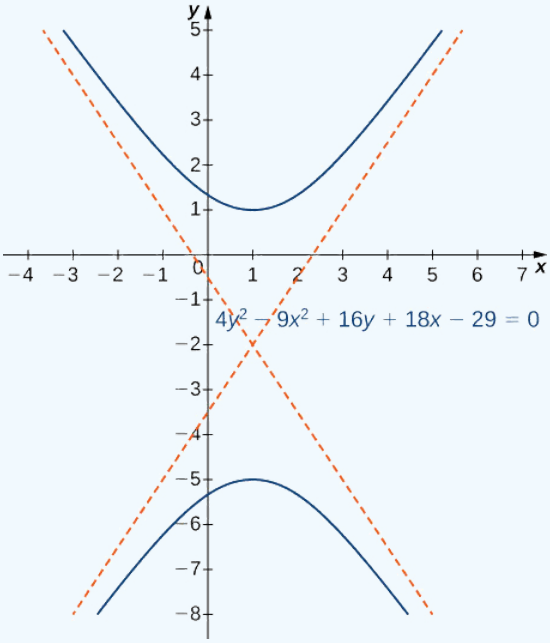
Put the equation $4y^2-9x^2+16y+18x-29=0$ into standard form and graph the resulting hyperbola. What are the equations of the asymptotes?

Hint

Move the constant over and complete the square. Check which direction the hyperbola opens

Answer

$\frac{(y+2)^2}{9}-\frac{(x-1)^2}{4}=1$. This is a vertical hyperbola. Asymptotes $y=-2\pm\frac{3}{2}(x-1)$.



Hyperbolas also have interesting reflective properties. A ray directed toward one focus of a hyperbola is reflected by a hyperbolic mirror toward the other focus. This concept is illustrated in Figure 11.5.11.

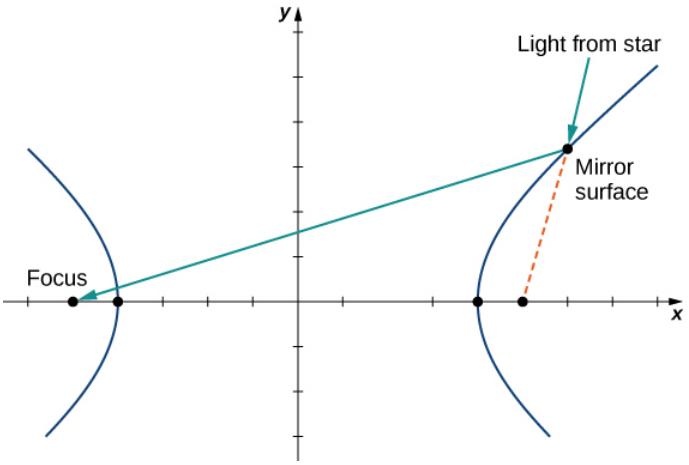


Figure 11.5.11: A hyperbolic mirror used to collect light from distant stars.

This property of the hyperbola has important applications. It is used in radio direction finding (since the difference in signals from two towers is constant along hyperbolas), and in the construction of mirrors inside telescopes (to reflect light coming from the parabolic mirror to the eyepiece). Another interesting fact about hyperbolas is that for a comet entering the solar system, if the speed is great enough to escape the Sun's gravitational pull, then the path that the comet takes as it passes through the solar system is hyperbolic.

Eccentricity and Directrix

An alternative way to describe a conic section involves the directrices, the foci, and a new property called eccentricity. We will see that the value of the eccentricity of a conic section can uniquely define that conic.

Definition: Eccentricity and Directrices

The *eccentricity* e of a conic section is defined to be the distance from any point on the conic section to its focus, divided by the perpendicular distance from that point to the nearest directrix. This value is constant for any conic section, and can define the conic section as well:

1. If $e = 1$, the conic is a parabola.
2. If $e < 1$, it is an ellipse.
3. If $e > 1$, it is a hyperbola.

The eccentricity of a circle is zero. The *directrix* of a conic section is the line that, together with the point known as the focus, serves to define a conic section. Hyperbolas and noncircular ellipses have two foci and two associated directrices. Parabolas have one focus and one directrix.

The three conic sections with their directrices appear in Figure 11.5.12.

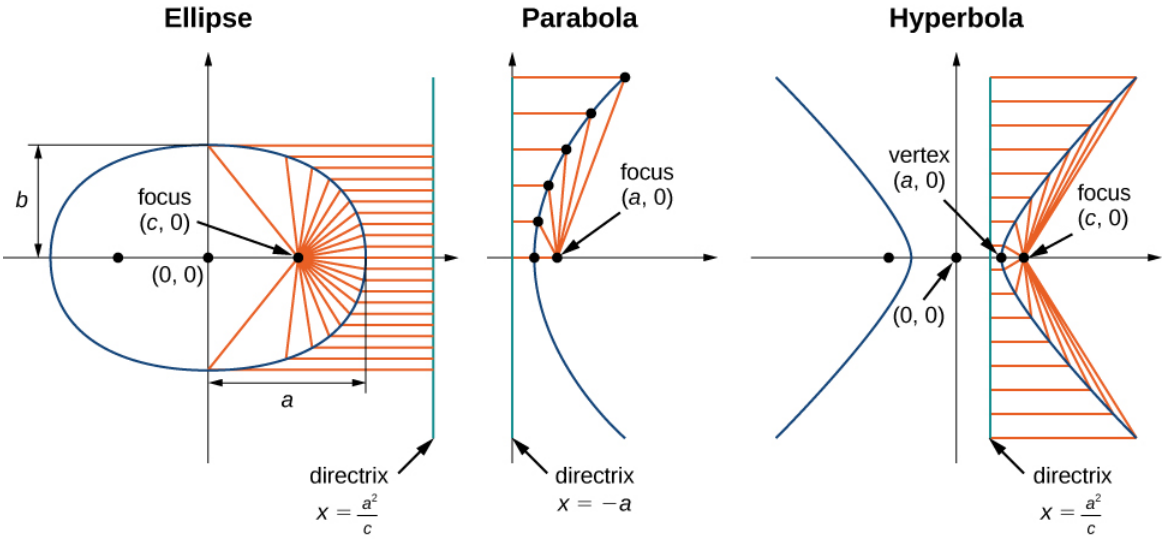


Figure 11.5.12: The three conic sections with their foci and directrices.

Recall from the definition of a parabola that the distance from any point on the parabola to the focus is equal to the distance from that same point to the directrix. Therefore, by definition, the eccentricity of a parabola must be 1. The equations of the directrices of a horizontal ellipse are $x = \pm\frac{a^2}{c}$. The right vertex of the ellipse is located at $(a, 0)$ and the right focus is $(c, 0)$. Therefore the distance from the vertex to the focus is $a - c$ and the distance from the vertex to the right directrix is $\frac{a^2}{c} - c$. This gives the eccentricity as

$$e = \frac{a-c}{\frac{a^2}{c} - a} = \frac{c(a-c)}{a^2 - ac} = \frac{c(a-c)}{a(a-c)} = \frac{c}{a}.$$

Since $c < a$, this step proves that the eccentricity of an ellipse is less than 1. The directrices of a horizontal hyperbola are also located at $x = \pm\frac{a^2}{c}$, and a similar calculation shows that the eccentricity of a hyperbola is also $e = \frac{c}{a}$. However in this case we have $c > a$, so the eccentricity of a hyperbola is greater than 1.

Example 11.5.4: Determining Eccentricity of a Conic Section

Determine the eccentricity of the ellipse described by the equation

$$\frac{(x-3)^2}{16} + \frac{(y+2)^2}{25} = 1.$$

Solution

From the equation we see that $a = 5$ and $b = 4$. The value of c can be calculated using the equation $a^2 = b^2 + c^2$ for an ellipse. Substituting the values of a and b and solving for c gives $c = 3$. Therefore the eccentricity of the ellipse is $e = \frac{c}{a} = \frac{3}{5} = 0.6$.

Exercise 11.5.4

Determine the eccentricity of the hyperbola described by the equation

$$\frac{(y-3)^2}{49} - \frac{(x+2)^2}{25} = 1.$$

Hint

First find the values of a and b, then determine c using the equation $c^2 = a^2 + b^2$.

Answer

$e = \frac{c}{a} = \frac{\sqrt{74}}{7} \approx 1.229$

Polar Equations of Conic Sections

Sometimes it is useful to write or identify the equation of a conic section in polar form. To do this, we need the concept of the focal parameter. The **focal parameter** of a conic section p is defined as the distance from a focus to the nearest directrix. The following table gives the focal parameters for the different types of conics, where a is the length of the semi-major axis (i.e., half the length of the major axis), c is the distance from the origin to the focus, and e is the eccentricity. In the case of a parabola, a represents the distance from the vertex to the focus.

Table 11.5.1: Eccentricities and Focal Parameters of the Conic Sections

Conic	e	p
Ellipse	$0 < e < 1$	$\frac{a^2 - c^2}{c} = \frac{a(1-e^2)}{c}$
Parabola	$e = 1$	$2a$
Hyperbola	$e > 1$	$\frac{c^2 - a^2}{c} = \frac{a(e^2-1)}{c}$

Using the definitions of the focal parameter and eccentricity of the conic section, we can derive an equation for any conic section in polar coordinates. In particular, we assume that one of the foci of a given conic section lies at the pole. Then using the definition of the various conic sections in terms of distances, it is possible to prove the following theorem.

Polar Equation of Conic Sections

The polar equation of a conic section with focal parameter p is given by

$$r = \frac{ep}{1 \pm e\cos\theta} \text{ or } r = \frac{ep}{1 \pm e\sin\theta}.$$

In the equation on the left, the major axis of the conic section is horizontal, and in the equation on the right, the major axis is vertical. To work with a conic section written in polar form, first make the constant term in the denominator equal to 1. This can be done by dividing both the numerator and the denominator of the fraction by the constant that appears in front of the plus or minus in the denominator. Then the coefficient of the sine or cosine in the denominator is the eccentricity. This value identifies the conic. If cosine appears in the denominator, then the conic is horizontal. If sine appears, then the conic is vertical. If both appear then the axes are rotated. The center of the conic is not necessarily at the origin. The center is at the origin only if the conic is a circle (i.e., $e = 0$).

Example 11.5.5: Graphing a Conic Section in Polar Coordinates

Identify and create a graph of the conic section described by the equation

$$r = \frac{3}{1 + 2\cos\theta}.$$

Solution

The constant term in the denominator is 1, so the eccentricity of the conic is 2. This is a hyperbola. The focal parameter p can be calculated by using the equation $ep = 3$. Since $e = 2$, this gives $p = \frac{3}{2}$. The cosine function appears in the denominator, so the hyperbola is horizontal. Pick a few values for θ and create a table of values. Then we can graph the hyperbola (Figure 11.5.13).

θ	r	θ	r
0	1	π	-3
$\frac{\pi}{4}$	$\frac{3}{1+\sqrt{2}} \approx 1.2426$	$\frac{5\pi}{4}$	$\frac{3}{1-\sqrt{2}} \approx -7.2426$
$\frac{\pi}{2}$	3	$\frac{3\pi}{2}$	3
$\frac{3\pi}{4}$	$\frac{3}{1-\sqrt{2}} \approx -7.2426$	$\frac{7\pi}{4}$	$\frac{3}{1+\sqrt{2}} \approx 1.2426$

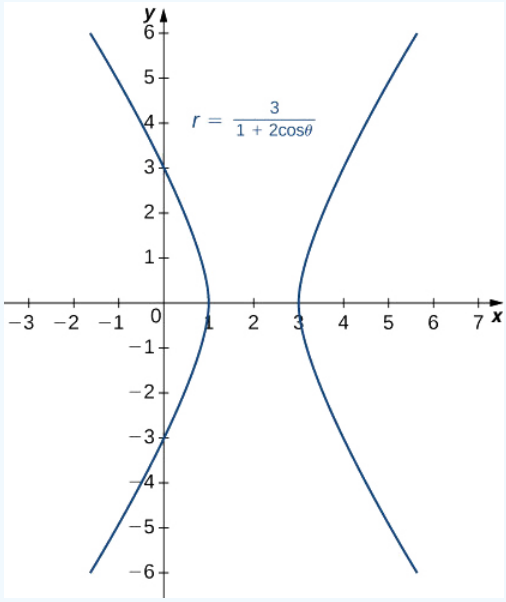


Figure 11.5.13: Graph of the hyperbola described in Example 11.5.5.

? Exercise 11.5.5

Identify and create a graph of the conic section described by the equation

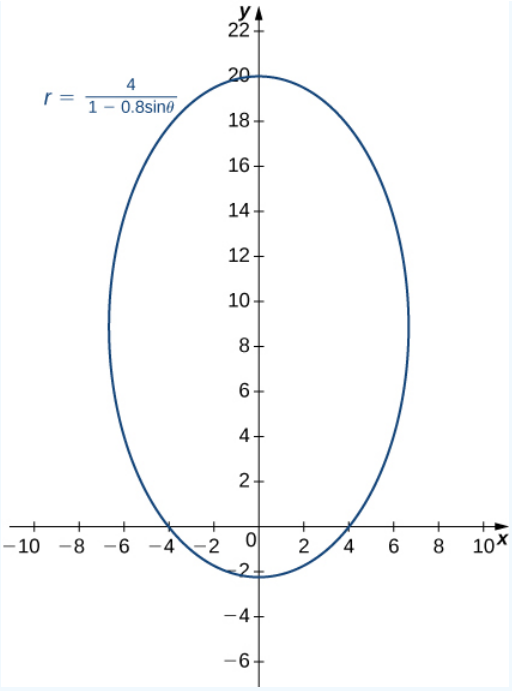
$$r = \frac{4}{1 - 0.8\sin\theta}.$$

Hint

First find the values of e and p, and then create a table of values.

Answer

Here $e = 0.8$ and $p = 5$. This conic section is an ellipse.



General Equations of Degree Two

A general equation of degree two can be written in the form

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0.$$

The graph of an equation of this form is a conic section. If $B \neq 0$ then the coordinate axes are rotated. To identify the conic section, we use the discriminant of the conic section $4AC - B^2$.

Identifying the Conic Section

One of the following cases must be true:

1. $4AC - B^2 > 0$. If so, the graph is an ellipse.
2. $4AC - B^2 = 0$. If so, the graph is a parabola.
3. $4AC - B^2 < 0$. If so, the graph is a hyperbola.

The simplest example of a second-degree equation involving a cross term is $xy = 1$. This equation can be solved for y to obtain $y = \frac{1}{x}$. The graph of this function is called a rectangular hyperbola as shown.

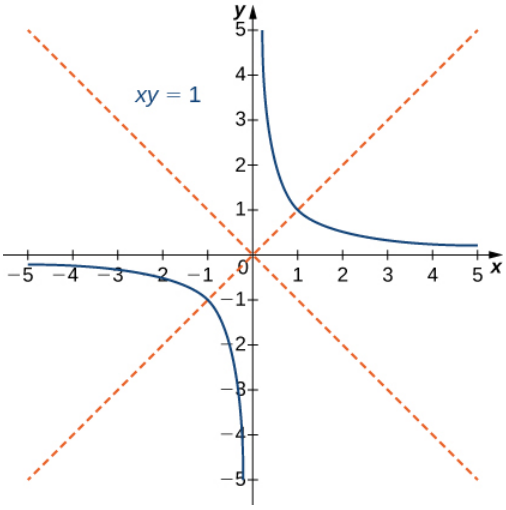


Figure 11.5.14: Graph of the equation $xy = 1$; The red lines indicate the rotated axes.

The asymptotes of this hyperbola are the x and y coordinate axes. To determine the angle θ of rotation of the conic section, we use the formula $\cot 2\theta = \frac{A-C}{B}$. In this case $A = C = 0$ and $B = 1$, so $\cot 2\theta = (0-0)/1 = 0$ and $\theta = 45°$. The method for graphing a conic section with rotated axes involves determining the coefficients of the conic in the rotated coordinate system. The new coefficients are labeled A', B', C', D', E', and F', and are given by the formulas

$$\begin{aligned} A' &= A\cos^2\theta + B\cos\theta\sin\theta + C\sin^2\theta \\ B' &= 0 \\ C' &= A\sin^2\theta - B\sin\theta\cos\theta + C\cos^2\theta \\ D' &= D\cos\theta + E\sin\theta \\ E' &= -D\sin\theta + E\cos\theta \\ F' &= F. \end{aligned}$$

Procedure: graphing a rotated conic

The procedure for graphing a rotated conic is the following:

1. Identify the conic section using the discriminant $4AC - B^2$.
2. Determine θ using the formula

$$\cot 2\theta = \frac{A-C}{B}. \tag{11.5.11}$$

3. Calculate A', B', C', D', E', and F'.
4. Rewrite the original equation using A', B', C', D', E', and F'.
5. Draw a graph using the rotated equation.

Example 11.5.6: Identifying a Rotated Conic

Identify the conic and calculate the angle of rotation of axes for the curve described by the equation

$$13x^2 - 6\sqrt{3}xy + 7y^2 - 256 = 0.$$

Solution

In this equation, $A = 13,\ B = -6\sqrt{3},\ C = 7,\ D = 0,\ E = 0$, and $F = -256$. The discriminant of this equation is

$$4AC - B^2 = 4(13)(7) - (-6\sqrt{3})^2 = 364 - 108 = 256.$$

Therefore this conic is an ellipse.

To calculate the angle of rotation of the axes, use Equation 11.5.11

$$\cot 2\theta = \frac{A-C}{B}.$$

This gives

$$\cot 2\theta = \frac{A-C}{B} = \frac{13-7}{-6\sqrt{3}} = -\frac{\sqrt{3}}{3}.$$

Therefore $2\theta = 120^o$ and $\theta = 60^o$, which is the angle of the rotation of the axes.

To determine the rotated coefficients, use the formulas given above:

$$\begin{aligned}
A' &= A\cos^2\theta + B\cos\theta\sin\theta + C\sin^2\theta \\
&= 13\cos^2 60 + (-6\sqrt{3})\cos 60\sin 60 + 7\sin^2 60 \\
&= 13\left(\frac{1}{2}\right)^2 - 6\sqrt{3}\left(\frac{1}{2}\right)\left(\frac{\sqrt{3}}{2}\right) + 7\left(\frac{\sqrt{3}}{2}\right)^2 \\
&= 4, \\
B' &= 0 \\
C' &= A\sin^2\theta - B\sin\theta\cos\theta + C\cos^2\theta \\
&= 13\sin^2 60 + (6\sqrt{3})\sin 60\cos 60 + 7\cos^2 60 \\
&= 13\left(\frac{\sqrt{3}}{2}\right)^2 + 6\sqrt{3}\left(\frac{\sqrt{3}}{2}\right)\left(\frac{1}{2}\right) + 7\left(\frac{1}{2}\right)^2 \\
&= 16, \\
D' &= D\cos\theta + E\sin\theta \\
&= (0)\cos 60 + (0)\sin 60 \\
&= 0, \\
E' &= -D\sin\theta + E\cos\theta \\
&= -(0)\sin 60 + (0)\cos 60 \\
&= 0 \\
F' &= F \\
&= -256.
\end{aligned}$$

The equation of the conic in the rotated coordinate system becomes

$$\begin{aligned}
4(x')^2 + 16(y')^2 &= 256 \\
\frac{(x')^2}{64} + \frac{(y')^2}{16} &= 1.
\end{aligned}$$

A graph of this conic section appears as follows.

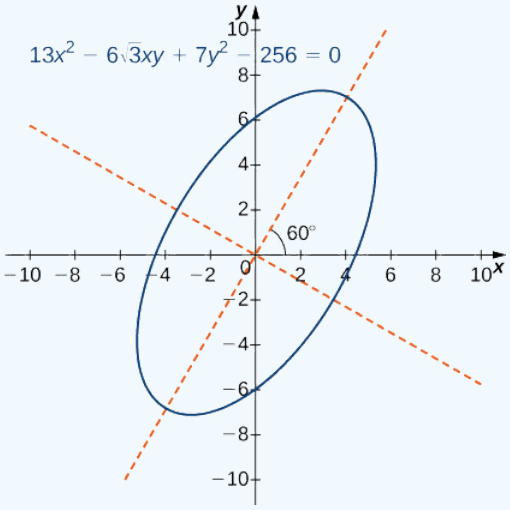


Figure 11.5.15: Graph of the ellipse described by the equation $13x^2 - 6\sqrt{3}xy + 7y^2 - 256 = 0$. The axes are rotated 60°. The red dashed lines indicate the rotated axes.

? Exercise 11.5.6

Identify the conic and calculate the angle of rotation of axes for the curve described by the equation

$$3x^2 + 5xy - 2y^2 - 125 = 0.$$

Hint

Follow steps 1 and 2 of the five-step method outlined above

Answer

The conic is a hyperbola and the angle of rotation of the axes is $\theta = 22.5°$.

Key Concepts

- The equation of a vertical parabola in standard form with given focus and directrix is $y = \frac{1}{4p}(x-h)^2 + k$ where p is the distance from the vertex to the focus and (h, k) are the coordinates of the vertex.
- The equation of a horizontal ellipse in standard form is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$ where the center has coordinates (h, k), the major axis has length $2a$, the minor axis has length $2b$, and the coordinates of the foci are $(h \pm c, k)$, where $c^2 = a^2 - b^2$.
- The equation of a horizontal hyperbola in standard form is $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$ where the center has coordinates (h, k), the vertices are located at $(h \pm a, k)$, and the coordinates of the foci are $(h \pm c, k)$, where $c^2 = a^2 + b^2$.
- The eccentricity of an ellipse is less than 1, the eccentricity of a parabola is equal to 1, and the eccentricity of a hyperbola is greater than 1. The eccentricity of a circle is 0.
- The polar equation of a conic section with eccentricity e is $r = \frac{ep}{1 \pm e\cos\theta}$ or $r = \frac{ep}{1 \pm e\sin\theta}$, where p represents the focal parameter.
- To identify a conic generated by the equation $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$,first calculate the discriminant $D = 4AC - B^2$. If $D > 0$ then the conic is an ellipse, if $D = 0$ then the conic is a parabola, and if $D < 0$ then the conic is a hyperbola.

Glossary

conic section

a conic section is any curve formed by the intersection of a plane with a cone of two nappes

directrix

a directrix (plural: directrices) is a line used to construct and define a conic section; a parabola has one directrix; ellipses and hyperbolas have two

discriminant

the value $4AC - B^2$, which is used to identify a conic when the equation contains a term involving xy, is called a discriminant

focus

a focus (plural: foci) is a point used to construct and define a conic section; a parabola has one focus; an ellipse and a hyperbola have two

eccentricity

the eccentricity is defined as the distance from any point on the conic section to its focus divided by the perpendicular distance from that point to the nearest directrix

focal parameter

the focal parameter is the distance from a focus of a conic section to the nearest directrix

general form

an equation of a conic section written as a general second-degree equation

major axis

the major axis of a conic section passes through the vertex in the case of a parabola or through the two vertices in the case of an ellipse or hyperbola; it is also an axis of symmetry of the conic; also called the transverse axis

minor axis

the minor axis is perpendicular to the major axis and intersects the major axis at the center of the conic, or at the vertex in the case of the parabola; also called the conjugate axis

nappe

a nappe is one half of a double cone

standard form

an equation of a conic section showing its properties, such as location of the vertex or lengths of major and minor axes

vertex

a vertex is an extreme point on a conic section; a parabola has one vertex at its turning point. An ellipse has two vertices, one at each end of the major axis; a hyperbola has two vertices, one at the turning point of each branch

---

This page titled 11.5: Conic Sections is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

11.5E: Exercises for Section 11.5

In exercises 1 - 8, determine the equation of the parabola using the information given.

1) Focus $(4,0)$ and directrix $x=-4$

Answer

$y^2=16x$

2) Focus $(0,-3)$ and directrix $y=3$

3) Focus $(0,0.5)$ and directrix $y=-0.5$

Answer

$x^2=2y$

4) Focus $(2,3)$ and directrix $x=-2$

5) Focus $(0,2)$ and directrix $y=4$

Answer

$x^2=-4(y-3)$

6) Focus $(-1,4)$ and directrix $x=5$

7) Focus $(-3,5)$ and directrix $y=1$

Answer

$(x+3)^2=8(y-3)$

8) Focus $\left(\frac{5}{2},-4\right)$ and directrix $x=\frac{7}{2}$

In exercises 9 - 16, determine the equation of the ellipse using the information given.

9) Endpoints of major axis at $(4,0),\ (-4,0)$ and foci located at $(2,0),\ (-2,0)$

Answer

$\frac{x^2}{16}+\frac{y^2}{12}=1$

10) Endpoints of major axis at $(0,5),\ (0,-5)$ and foci located at $(0,3),\ (0,-3)$

11) Endpoints of major axis at $(0,2),\ (0,-2)$ and foci located at $(3,0),\ (-3,0)$

Answer

$\frac{x^2}{13}+\frac{y^2}{4}=1$

12) Endpoints of major axis at $(-3,3),\ (7,3)$ and foci located at $(-2,3),\ (6,3)$

13) Endpoints of major axis at $(-3,5),\ (-3,-3)$ and foci located at $(-3,3),\ (-3,-1)$

Answer

$\frac{(y-1)^2}{16}+\frac{(x+3)^2}{12}=1$

14) Endpoints of major axis at $(0,0),\ (0,4)$ and foci located at $(5,2),\ (-5,2)$

15) Foci located at $(2,0),\ (-2,0)$ and eccentricity of $\frac{1}{2}$

Answer

$$\frac{x^2}{16}+\frac{y^2}{12}=1$$

16) Foci located at $(0,-3),\ (0,3)$ and eccentricity of $\frac{3}{4}$

In exercises 17 - 24, determine the equation of the hyperbola using the information given.

17) Vertices located at $(5,0),\ (-5,0)$ and foci located at $(6,0),\ (-6,0)$

Answer

$\frac{x^2}{25}-\frac{y^2}{11}=1$

18) Vertices located at $(0,2),\ (0,-2)$ and foci located at $(0,3),\ (0,-3)$

19) Endpoints of the conjugate axis located at $(0,3),\ (0,-3)$ and foci located $(4,0),\ (-4,0)$

Answer

$$\frac{x^2}{7}-\frac{y^2}{9}=1$$

20) Vertices located at $(0,1),\ (6,1)$ and focus located at $(8,1)$

21) Vertices located at $(-2,0),\ (-2,-4)$ and focus located at $(-2,-8)$

Answer

$$\frac{(y+2)^2}{4}-\frac{(x+2)^2}{32}=1$$

22) Endpoints of the conjugate axis located at $(3,2),\ (3,4)$ and focus located at $(3,7)$

23) Foci located at $(6,-0),\ (6,0)$ and eccentricity of 3

Answer

$$\frac{x^2}{4}-\frac{y^2}{32}=1$$

24) $(0,10),\ (0,-10)$ and eccentricity of 2.5

In exercises 25 - 30, consider the following polar equations of conics. Determine the eccentricity and identify the conic.

25) $r=\dfrac{-1}{1+\cos\theta}$

Answer

$e=1$, parabola

26) $r=\dfrac{8}{2-\sin\theta}$

27) $r=\dfrac{5}{2+\sin\theta}$

Answer

$e=\frac{1}{2}$, ellipse

28) $r=\dfrac{5}{-1+2\sin\theta}$

29) $r=\dfrac{3}{2-6\sin\theta}$

Answer

$e=3$, hyperbola

30) $r=\dfrac{3}{-4+3\sin\theta}$

In exercises 31 - 34, find a polar equation of the conic with focus at the origin and eccentricity and directrix as given.

31) Directrix: $x = 4;\ e = \frac{1}{5}$

Answer

$$r = \frac{4}{5 + \cos\theta}$$

32) Directrix: $x = -4;\ e = 5$

33) Directrix: $y = 2;\ e = 2$

Answer

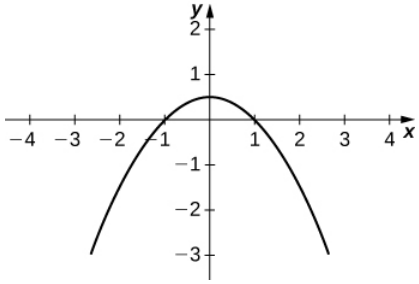
$$r = \frac{4}{1 + 2\sin\theta}$$

34) Directrix: $y = -2;\ e = \frac{1}{2}$

In exercises 35 - 51, sketch the graph of each conic.

35) $r = \dfrac{1}{1 + \sin\theta}$

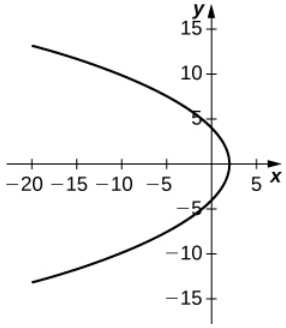
Answer



36) $r = \dfrac{1}{1 - \cos\theta}$

37) $r = \dfrac{4}{1 + \cos\theta}$

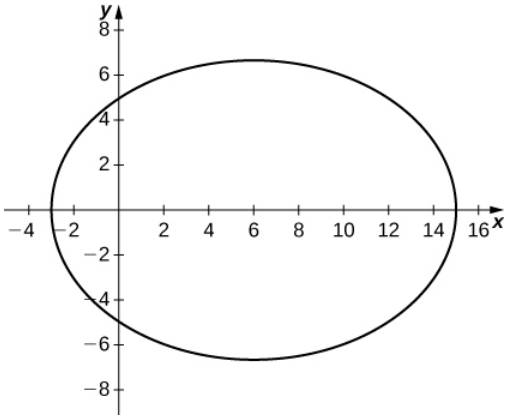
Answer



38) $r = \dfrac{10}{5 + 4\sin\theta}$

39) $r = \dfrac{15}{3 - 2\cos\theta}$

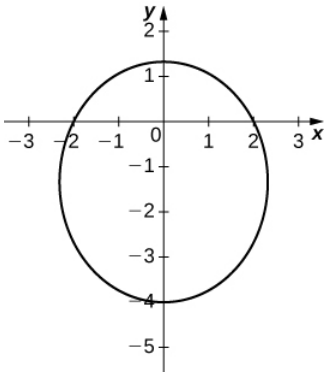
Answer



40) $r = \dfrac{32}{3 + 5\sin\theta}$

41) $r(2 + \sin\theta) = 4$

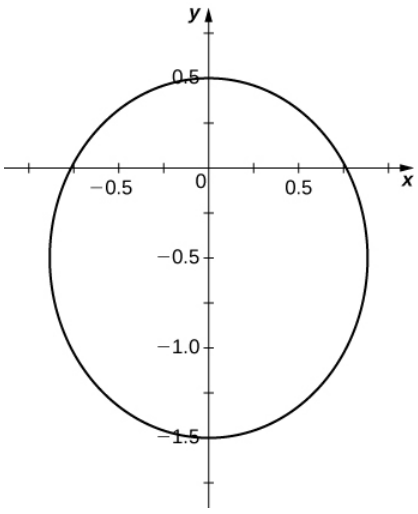
Answer



42) $r = \dfrac{3}{2 + 6\sin\theta}$

43) $r = \dfrac{3}{-4 + 2\sin\theta}$

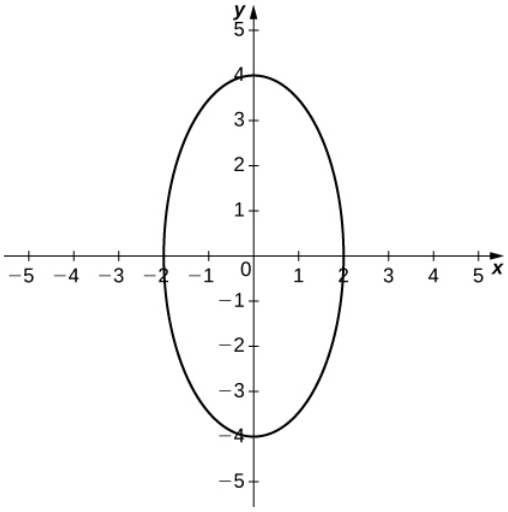
Answer



44) $\dfrac{x^2}{9} + \dfrac{y^2}{4} = 1$

45) $\frac{x^2}{4} + \frac{y^2}{16} = 1$

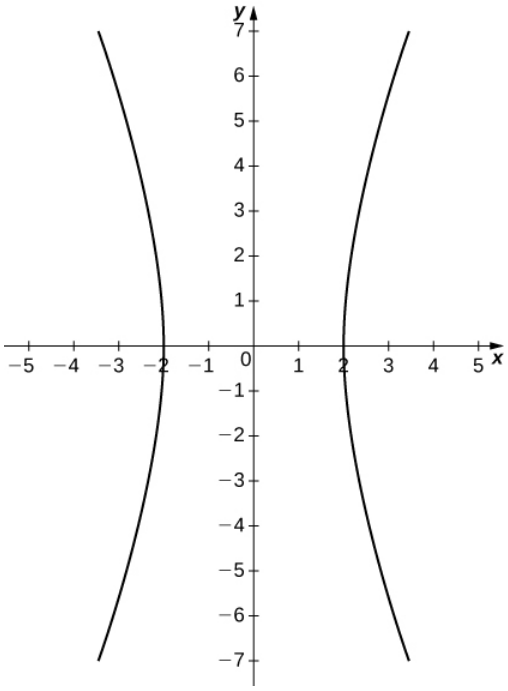
Answer



46) $4x^2 + 9y^2 = 36$

47) $25x^2 - 4y^2 = 100$

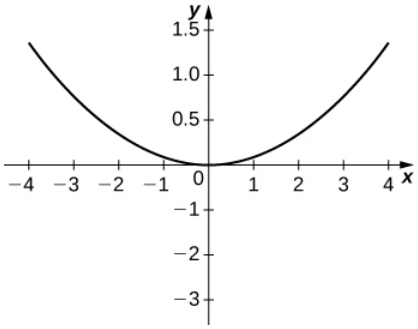
Answer



48) $\frac{x^2}{16} - \frac{y^2}{9} = 1$

49) $x^2 = 12y$

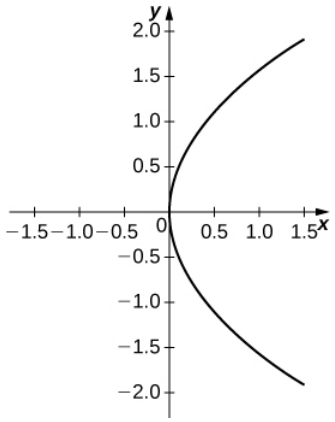
Answer



50) $y^2 = 20x$

51) $12x = 5y^2$

Answer



For the equations in exercises 52 - 57, determine which of the conic sections is described.

52) $xy = 4$

53) $x^2 + 4xy - 2y^2 - 6 = 0$

Answer

Hyperbola

54) $x^2 + 2\sqrt{3}xy + 3y^2 - 6 = 0$

55) $x^2 - xy + y^2 - 2 = 0$

Answer

Ellipse

56) $34x^2 - 24xy + 41y^2 - 25 = 0$

57) $52x^2 - 72xy + 73y^2 + 40x + 30y - 75 = 0$

Answer

Ellipse

58) The mirror in an automobile headlight has a parabolic cross section, with the lightbulb at the focus. On a schematic, the equation of the parabola is given as $x^2 = 4y$. At what coordinates should you place the lightbulb?

59) A satellite dish is shaped like a paraboloid of revolution. The receiver is to be located at the focus. If the dish is 12 feet across at its opening and 4 feet deep at its center, where should the receiver be placed?

Answer

At the point 2.25 feet above the vertex.

60) Consider the satellite dish of the preceding problem. If the dish is 8 feet across at the opening and 2 feet deep, where should we place the receiver?

61) A searchlight is shaped like a paraboloid of revolution. A light source is located 1 foot from the base along the axis of symmetry. If the opening of the searchlight is 3 feet across, find the depth.

Answer

0.5625 feet

62) Whispering galleries are rooms designed with elliptical ceilings. A person standing at one focus can whisper and be heard by a person standing at the other focus because all the sound waves that reach the ceiling are reflected to the other person. If a whispering gallery has a length of 120 feet and the foci are located 30 feet from the center, find the height of the ceiling at the center.

63) A person is standing 8 feet from the nearest wall in a whispering gallery. If that person is at one focus and the other focus is 80 feet away, what is the length and the height at the center of the gallery?

Answer

Length is 96 feet and height is approximately 26.53 feet.

In exercises 64 - 67, determine the polar equation form of the orbit given the length of the major axis and eccentricity for the orbits of the comets or planets. Distance is given in astronomical units (AU).

64) Halley's Comet: length of major axis = 35.88, eccentricity = 0.967

65) Hale-Bopp Comet: length of major axis = 525.91, eccentricity = 0.995

Answer

$$r = \frac{2.616}{1 + 0.995 \cos \theta}$$

66) Mars: length of major axis = 3.049, eccentricity = 0.0934

67) Jupiter: length of major axis = 10.408, eccentricity = 0.0484

Answer

$$r = \frac{5.192}{1 + 0.0484 \cos \theta}$$

This page titled 11.5E: Exercises for Section 11.5 is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin "Jed" Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

11.R: Chapter 11 Review Exercises

***True or False?* Justify your answer with a proof or a counterexample.**

1) The rectangular coordinates of the point $\left(4, \frac{5\pi}{6}\right)$ are $\left(2\sqrt{3}, -2\right)$.

2) The equations $x = \cosh(3t),\ y = 2\sinh(3t)$ represent a hyperbola.

Answer

True

3) The arc length of the spiral given by $r = \dfrac{\theta}{2}$ for $0 \le \theta \le 3\pi$ is $\frac{9}{4}\pi^3$ units.

4) Given $x = f(t)$ and $y = g(t)$, if $\dfrac{dx}{dy} = \dfrac{dy}{dx}$, then $f(t) = g(t) + C$, where C is a constant.

Answer

False. Imagine $y = t + 1,\ x = -t + 1$.

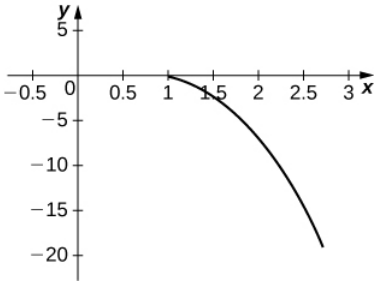
In exercises 5 -8, sketch the parametric curve and eliminate the parameter to find the Cartesian equation of the curve.

5) $x = 1 + t,\ y = t^2 - 1, \quad -1 \le t \le 1$

6) $x = e^t,\ y = 1 - e^{3t}, \quad 0 \le t \le 1$

Answer

$y = 1 - x^3$

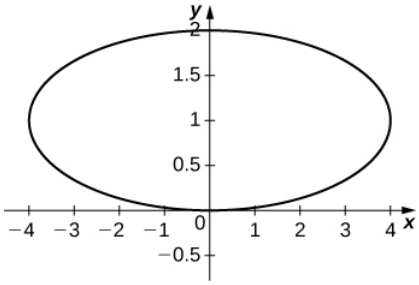


7) $x = \sin\theta,\ y = 1 - \csc\theta, \quad 0 \le \theta \le 2\pi$

8) $x = 4\cos\phi,\ y = 1 - \sin\phi, \quad 0 \le \phi \le 2\pi$

Answer

$\dfrac{x^2}{16} + (y-1)^2 = 1$



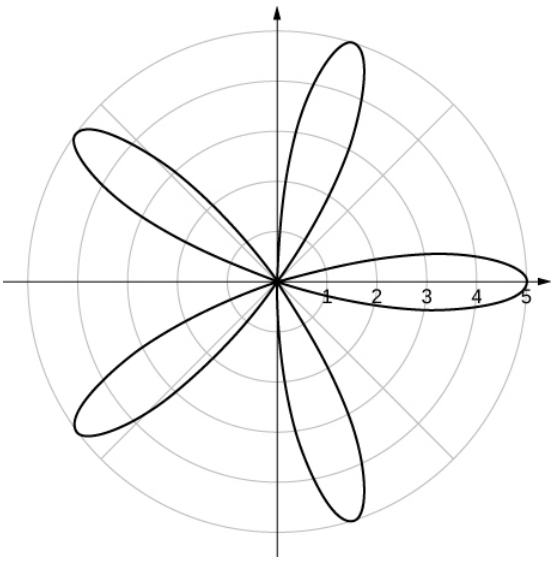
In exercises 9 - 10, sketch the polar curve and determine what type of symmetry exists, if any.

9) $r = 4\sin\left(\frac{\theta}{3}\right)$

10) $r = 5\cos(5\theta)$

Answer

Symmetric about polar axis



In exercises 11 - 12, find the polar equation for the curve given as a Cartesian equation.

11) $x + y = 5$

12) $y^2 = 4 + x^2$

Answer

$$r^2 = \frac{4}{\sin^2\theta - \cos^2\theta}$$

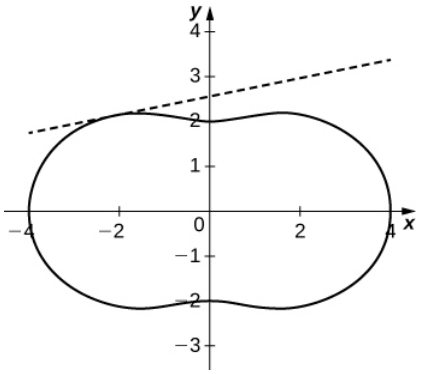
In exercises 13 - 14, find the equation of the tangent line to the given curve. Graph both the function and its tangent line.

13) $x = \ln(t),\ y = t^2 - 1,\ t = 1$

14) $r = 3 + \cos(2\theta),\ \theta = \frac{3\pi}{4}$

Answer

$y = \frac{3\sqrt{2}}{2} + \frac{1}{5}\left(x + \frac{3\sqrt{2}}{2}\right)$



15) Find $\frac{dy}{dx}$, $\frac{dx}{dy}$, and $\frac{d^2x}{dy^2}$ of $y = (2 + e^{-t}),\ x = 1 - \sin t$

In exercises 16 -17, find the area of the region.

16) $x = t^2,\ y = \ln(t), \quad 0 \le t \le e$

Answer

$\frac{e^2}{2}$ units^2

17) $r = 1 - \sin\theta$ in the first quadrant

In exercises 18 - 19, find the arc length of the curve over the given interval.

18) $x = 3t + 4,\ y = 9t - 2, \quad 0 \le t \le 3$

Answer

$9\sqrt{10}$ units

19) $r = 6\cos\theta, \quad 0 \le \theta \le 2\pi$. Check your answer by geometry.

In exercises 20 - 22, find the Cartesian equation describing the given shapes.

20) A parabola with focus $(2, -5)$ and directrix $x = 6$

Answer

$(y+5)^2 = -8x + 32$

21) An ellipse with a major axis length of 10 and foci at $(-7, 2)$ and $(1, 2)$

22) A hyperbola with vertices at $(3, -2)$ and $(-5, -2)$ and foci at $(-2, -6)$ and $(-2, 4)$

Answer

$\frac{(y+1)^2}{16} - \frac{(x+2)^2}{9} = 1$

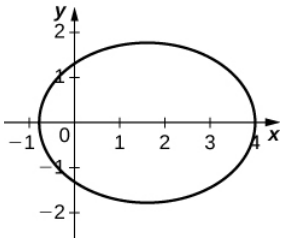
In exercises 23 - 25, determine the eccentricity and identify the conic. Sketch the conic.

23) $r = \frac{6}{1 + 3\cos\theta}$

24) $r = \frac{4}{3 - 2\cos\theta}$

Answer

$e = \frac{2}{3}$, ellipse



25) $r = \frac{7}{5 - 5\cos\theta}$

26) Determine the Cartesian equation describing the orbit of Pluto, the most eccentric orbit around the Sun. The length of the major axis is 39.26 AU and minor axis is 38.07 AU. What is the eccentricity?

Answer

$\frac{y^2}{19.03^2} + \frac{x^2}{19.63^2} = 1, \quad e = 0.2447$

27) The C/1980 E1 comet was observed in 1980. Given an eccentricity of 1.057 and a perihelion (point of closest approach to the Sun) of 3.364 AU, find the Cartesian equations describing the comet's trajectory. Are we guaranteed to see this comet again? (Hint: Consider the Sun at point $(0, 0)$.)

This page titled 11.R: Chapter 11 Review Exercises is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin "Jed" Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

CHAPTER OVERVIEW

12: Vectors in Space

A quantity that has magnitude and direction is called a **vector**. Vectors have many real-life applications, including situations involving force or velocity. For example, consider the forces acting on a boat crossing a river. The boat's motor generates a force in one direction, and the current of the river generates a force in another direction. Both forces are vectors. We must take both the magnitude and direction of each force into account if we want to know where the boat will go.

12.0: Prelude to Vectors in Space
12.1: Vectors in the Plane
12.1E: Exercises for Section 12.1
12.2: Vectors in Three Dimensions
12.2E: Exercises for Section 12.2
12.3: The Dot Product
12.3E: Exercises for Section 12.3
12.4: The Cross Product
12.4E: Exercises for Section 12.4
12.5: Equations of Lines and Planes in Space
12.5E: Exercises for Section 12.5
12.6: Quadric Surfaces
12.6E: Exercises for Section 12.6
12.7: Cylindrical and Spherical Coordinates
12.7E: Exercises for Section 12.7
12.R: Chapter 12 Review Exercises

This page titled 12: Vectors in Space is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin "Jed" Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

12.0: Prelude to Vectors in Space

Modern astronomical observatories often consist of a large number of parabolic reflectors, connected by computers, used to analyze radio waves. Each dish focuses the incoming parallel beams of radio waves to a precise focal point, where they can be synchronized by computer. If the surface of one of the parabolic reflectors is described by the equation $\frac{x^2}{100} + \frac{y^2}{100} = \frac{z}{4}$, where is the focal point of the reflector?



Figure 12.0.1: The Karl G. Jansky Very Large Array, located in Socorro, New Mexico, consists of a large number of radio telescopes that can collect radio waves and collate them as if they were gathering waves over a huge area with no gaps in coverage. (credit: modification of work by CGP Grey, Wikimedia Commons)

We are now about to begin a new part of the calculus course, when we study functions of two or three independent variables in multidimensional space. Many of the computations are similar to those in the study of single-variable functions, but there are also a lot of differences. In this first chapter, we examine coordinate systems for working in three-dimensional space, along with vectors, which are a key mathematical tool for dealing with quantities in more than one dimension. Let's start here with the basic ideas and work our way up to the more general and powerful tools of mathematics in later chapters.

This page titled 12.0: Prelude to Vectors in Space is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin "Jed" Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

12.1: Vectors in the Plane

Learning Objectives

- Describe a plane vector, using correct notation.
- Perform basic vector operations (scalar multiplication, addition, subtraction).
- Express a vector in component form.
- Explain the formula for the magnitude of a vector.
- Express a vector in terms of unit vectors.
- Give two examples of vector quantities.

When describing the movement of an airplane in flight, it is important to communicate two pieces of information: the direction in which the plane is traveling and the plane's speed. Together these form the plane's velocity. When measuring a force, such as the thrust of the plane's engines, it is important to describe not only the strength of that force, but also the direction in which it is applied. Some quantities, such as velocity and force, are defined in terms of both size (also called magnitude) and direction. A quantity that has magnitude and direction is called a **vector**. In textbooks, vectors are often denoted by boldface letters, such as $\mathbf{v}$. Since it is difficult to write in clear boldface when we write vectors by hand, we will also include an arrow or harpoon above the letter representing the vector. To make the vectors clearer in this textbook (and to reinforce the way you should write them by hand), we will generally use arrows or harpoons above boldface (or italic) letters to represent vectors, giving us $\vec{v}$ or $\vec{\mathbf{v}}$. Note that some figures will still only use boldface letters to denote vectors.

Definition: Vector

A vector is a quantity that has both magnitude and direction.

Vector Representation

A vector in a plane is represented by a directed line segment (an arrow). The endpoints of the segment are called the **initial point** and the **terminal point** of the vector. An arrow from the initial point to the terminal point indicates the direction of the vector. The length of the line segment represents its **magnitude**. We use the notation $\|\vec{\mathbf{v}}\|$ to denote the magnitude of the vector $\vec{\mathbf{v}}$. A vector with an initial point and terminal point that are the same is called the **zero vector**, denoted $\vec{\mathbf{0}}$. The zero vector is the only vector without a direction, and by convention can be considered to have any direction convenient to the problem at hand.

Vectors with the same magnitude and direction are called equivalent vectors. We treat equivalent vectors as equal, even if they have different initial points. Thus, if $\vec{\mathbf{v}}$ and $\vec{\mathbf{w}}$ are equivalent, we write

$$\vec{\mathbf{v}} = \vec{\mathbf{w}}.$$

Definition: Equivalent Vectors

Vectors are said to be *equivalent vectors* if they have the same magnitude and direction.

The arrows in Figure 12.1.1 are equivalent. Each arrow has the same length and direction. A closely related concept is the idea of parallel vectors. Two vectors are said to be parallel if they have the same or opposite directions. We explore this idea in more detail later in the chapter. A vector is defined by its magnitude and direction, regardless of where its initial point is located.

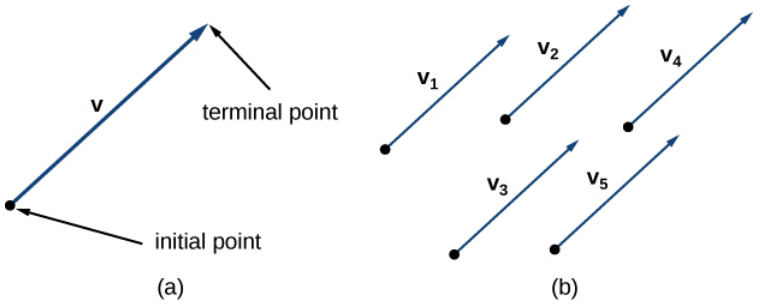


Figure 12.1.1: (a) A vector is represented by a directed line segment from its initial point to its terminal point. (b) Vectors $\vec{\mathbf{v}}_1$ through $\vec{\mathbf{v}}_5$ are equivalent.

The use of boldface, lowercase letters to name vectors is a common representation in print, but there are alternative notations. When writing the name of a vector by hand, for example, it is easier to sketch an arrow over the variable than to show it is a vector: $\vec{v}$. When a vector has initial point P and terminal point Q, the notation $\overrightarrow{PQ}$ is useful because it indicates the direction and location of the vector.

Example 12.1.1: Sketching Vectors

Sketch a vector in the plane from initial point $P(1,1)$ to terminal point $Q(8,5)$.

Solution

See Figure 12.1.2. Because the vector goes from point P to point Q, we name it $\overrightarrow{PQ}$.

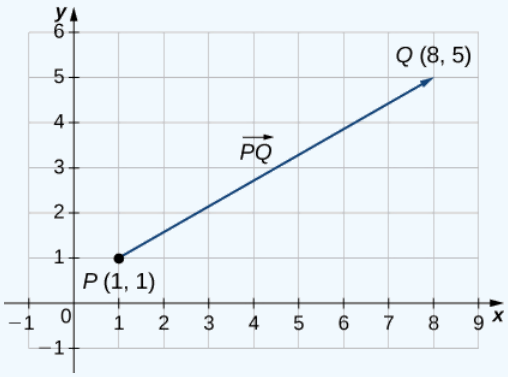


Figure 12.1.2: The vector with initial point $(1,1)$ and terminal point $(8,5)$ is named $\overrightarrow{PQ}$.

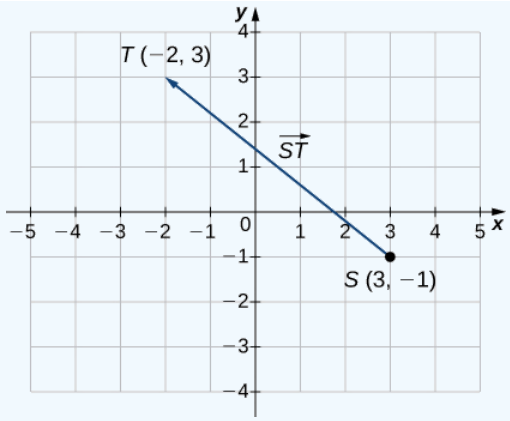
Exercise 12.1.1

Sketch the vector $\overrightarrow{ST}$ where S is point $(3,-1)$ and T is point $(-2,3)$.

Hint

The first point listed in the name of the vector is the initial point of the vector.

Answer



Combining Vectors

Vectors have many real-life applications, including situations involving force or velocity. For example, consider the forces acting on a boat crossing a river. The boat's motor generates a force in one direction, and the current of the river generates a force in another direction. Both forces are vectors. We must take both the magnitude and direction of each force into account if we want to know where the boat will go.

A second example that involves vectors is a quarterback throwing a football. The quarterback does not throw the ball parallel to the ground; instead, he aims up into the air. The velocity of his throw can be represented by a vector. If we know how hard he throws the ball (magnitude—in this case, speed), and the angle (direction), we can tell how far the ball will travel down the field.

A real number is often called a **scalar** in mathematics and physics. Unlike vectors, scalars are generally considered to have a magnitude only, but no direction. Multiplying a vector by a scalar changes the vector's magnitude. This is called scalar multiplication. Note that changing the magnitude of a vector does not indicate a change in its direction. For example, wind blowing from north to south might increase or decrease in speed while maintaining its direction from north to south.

Definition: Scalar Multiplication

The product $k\vec{\mathrm{v}}$ of a vector $\vec{\mathrm{v}}$ and a scalar k is a vector with a magnitude that is $|k|$ times the magnitude of $\vec{\mathrm{v}}$, and with a direction that is the same as the direction of $\vec{\mathrm{v}}$ if $k > 0$, and opposite the direction of $\vec{\mathrm{v}}$ if $k < 0$. This is called *scalar multiplication*. If $k = 0$ or $\vec{\mathrm{v}} = \vec{0}$, then $k\vec{\mathrm{v}} = \vec{0}$.

As you might expect, if $k = -1$, we denote the product $k\vec{\mathrm{v}}$ as

$$k\vec{\mathrm{v}} = (-1)\vec{\mathrm{v}} = -\vec{\mathrm{v}}.$$

Note that $-\vec{\mathrm{v}}$ has the same magnitude as $\vec{\mathrm{v}}$, but has the opposite direction (Figure 12.1.3).

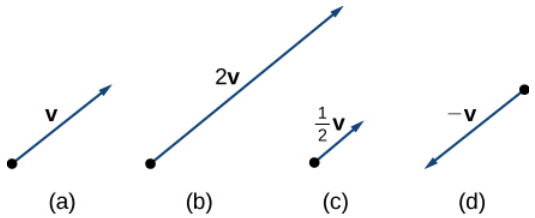


Figure 12.1.3: (a) The original vector $\vec{\mathrm{v}}$ has length n units. (b) The length of $2\vec{\mathrm{v}}$ equals $2n$ units. (c) The length of $\vec{\mathrm{v}}/2$ is $n/2$ units. (d) The vectors $\vec{\mathrm{v}}$ and $-\vec{\mathrm{v}}$ have the same length but opposite directions.

Another operation we can perform on vectors is to add them together in vector addition, but because each vector may have its own direction, the process is different from adding two numbers. The most common graphical method for adding two vectors is to place the initial point of the second vector at the terminal point of the first, as in Figure 12.1.4. To see why this makes sense, suppose, for example, that both vectors represent displacement. If an object moves first from the initial point to the terminal point of vector $\vec{\mathrm{v}}$, then from the initial point to the terminal point of vector $\vec{\mathrm{w}}$, the overall displacement is the same as if the object had made just one

movement from the initial point to the terminal point of the vector $\vec{v} + \vec{w}$. For obvious reasons, this approach is called the **triangle method**. Notice that if we had switched the order, so that $\vec{w}$ was our first vector and $\vec{v}$ was our second vector, we would have ended up in the same place. (Again, see Figure 12.1.4.) Thus,

$$\vec{v} + \vec{w} = \vec{w} + \vec{v}.$$

A second method for adding vectors is called the **parallelogram method**. With this method, we place the two vectors so they have the same initial point, and then we draw a parallelogram with the vectors as two adjacent sides, as in Figure 12.1.4. The length of the diagonal of the parallelogram is the sum. Comparing Figure 12.1.4 and Figure 12.1.4, we can see that we get the same answer using either method. The vector $\vec{v} + \vec{w}$ is called the **vector sum.**

Definition: Vector Addition

The sum of two vectors $\vec{v}$ and $\vec{w}$ can be constructed graphically by placing the initial point of $\vec{w}$ at the terminal point of $\vec{v}$. Then, the vector sum, $\vec{v} + \vec{w}$, is the vector with an initial point that coincides with the initial point of $\vec{v}$ and has a terminal point that coincides with the terminal point of $\vec{w}$. This operation is known as *vector addition***.**

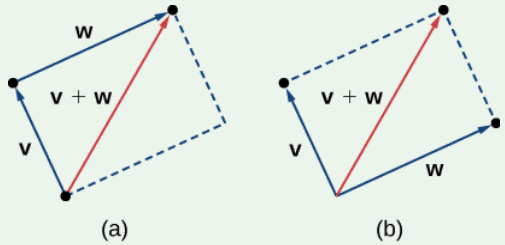


Figure 12.1.4: (a) When adding vectors by the triangle method, the initial point of $\vec{w}$ is the terminal point of $\vec{v}$. (b) When adding vectors by the parallelogram method, the vectors $\vec{v}$ and $\vec{w}$ have the same initial point.

It is also appropriate here to discuss vector subtraction. We define $\vec{v} - \vec{w}$ as $\vec{v} + (-\vec{w}) = \vec{v} + (-1)\vec{w}$. The vector $\vec{v} - \vec{w}$ is called the **vector difference**. Graphically, the vector $\vec{v} - \vec{w}$ is depicted by drawing a vector from the terminal point of $\vec{w}$ to the terminal point of $\vec{v}$ (Figure 12.1.5).

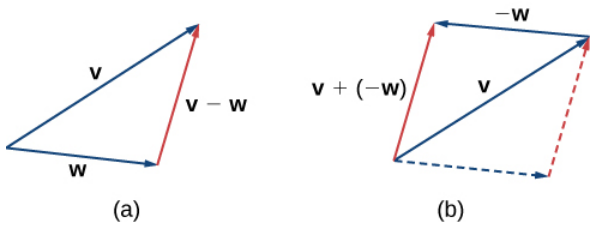


Figure 12.1.5: (a) The vector difference $\vec{v} - \vec{w}$ is depicted by drawing a vector from the terminal point of $\vec{w}$ to the terminal point of $\vec{v}$. (b) The vector$\vec{v} - \vec{w}$ is equivalent to the vector $\vec{v} + (-\vec{w})$.

In Figure 12.1.4, the initial point of $\vec{v} + \vec{w}$ is the initial point of $\vec{v}$. The terminal point of $\vec{v} + \vec{w}$ is the terminal point of $\vec{w}$. These three vectors form the sides of a triangle. It follows that the length of any one side is less than the sum of the lengths of the remaining sides. So we have

$$\|\vec{v} + \vec{w}\| \leq \|\vec{v}\| + \|\vec{w}\|.$$

This is known more generally as the **triangle inequality**. There is one case, however, when the resultant vector $\vec{u} + \vec{v}$ has the same magnitude as the sum of the magnitudes of $\vec{u}$ and $\vec{v}$. This happens only when $\vec{u}$ and $\vec{v}$ have the same direction.

Example 12.1.2: Combining Vectors

Given the vectors $\vec{v}$ and $\vec{w}$ shown in Figure 12.1.6, sketch the vectors

a. $3\vec{w}$
b. $\vec{v} + \vec{w}$
c. $2\vec{v} - \vec{w}$

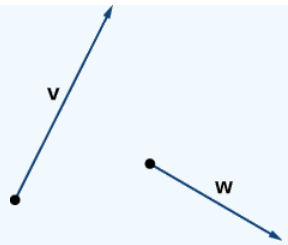
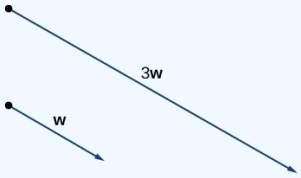


Figure 12.1.6: Vectors $\vec{v}$ and $\vec{w}$ lie in the same plane.

Solution

a. The vector $3\vec{w}$ has the same direction as $\vec{w}$; it is three times as long as $\vec{w}$.



Vector $3\vec{w}$ has the same direction as $\vec{w}$ and is three times as long.

b. Use either addition method to find $\vec{v} + \vec{w}$.

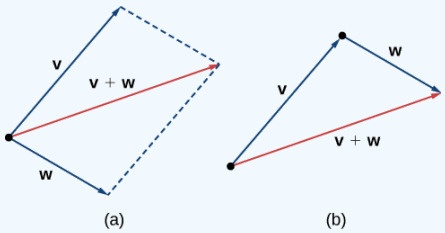


Figure 12.1.7: To find $\vec{v} + \vec{w}$, align the vectors at their initial points or place the initial point of one vector at the terminal point of the other. (a) The vector$\vec{v} + \vec{w}$ is the diagonal of the parallelogram with sides $\vec{v}$ and $\vec{w}$. (b) The vector $\vec{v} + \vec{w}$ is the third side of a triangle formed with $\vec{w}$ placed at the terminal point of $\vec{v}$.

c. To find $2\vec{v} - \vec{w}$, we can first rewrite the expression as $2\vec{v} + (-\vec{w})$. Then we can draw the vector $-\vec{w}$, then add it to the vector $2\vec{v}$.

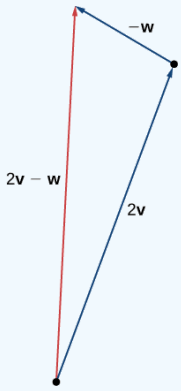


Figure 12.1.8: To find $2\vec{v} - \vec{w}$, simply add $2\vec{v} + (-\vec{w})$.

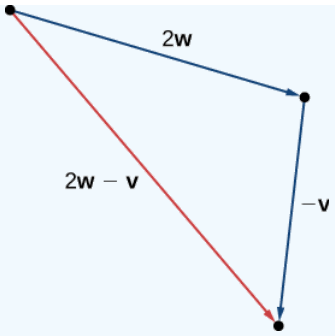
? Exercise 12.1.2

Using vectors $\vec{v}$ and $\vec{w}$ from Example 12.1.2, sketch the vector $2\vec{w} - \vec{v}$.

Hint

First sketch vectors $2\vec{w}$ and $-\vec{v}$.

Answer



Vector Components

Working with vectors in a plane is easier when we are working in a coordinate system. When the initial points and terminal points of vectors are given in Cartesian coordinates, computations become straightforward.

Example 12.1.3: Comparing Vectors

Are $\vec{v}$ and $\vec{w}$ equivalent vectors?

a.

- $\vec{v}$ has initial point $(3, 2)$ and terminal point $(7, 2)$
- $\vec{w}$ has initial point $(1, -4)$ and terminal point $(1, 0)$

b.

- $\vec{v}$ has initial point $(0, 0)$ and terminal point $(1, 1)$
- $\vec{w}$ has initial point $(-2, 2)$ and terminal point $(-1, 3)$

Solution

a. The vectors are each 4 units long, but they are oriented in different directions. So $\vec{v}$ and $\vec{w}$ are not equivalent (Figure 12.1.9).

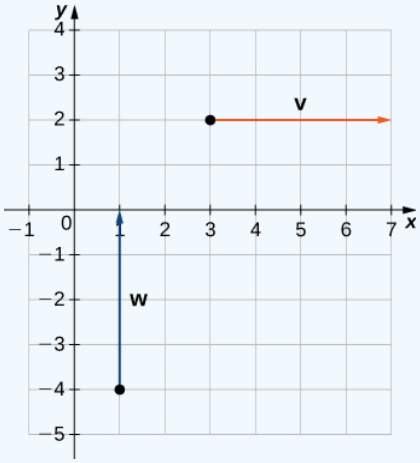


Figure 12.1.9: These vectors are not equivalent.

b. Based on Figure 12.1.10, and using a bit of geometry, it is clear these vectors have the same length and the same direction, so $\vec{v}$ and $\vec{w}$ are equivalent.

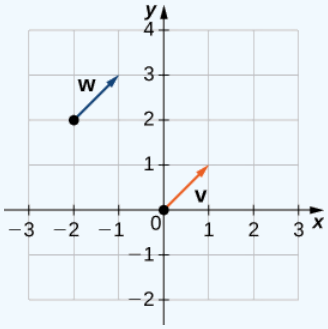
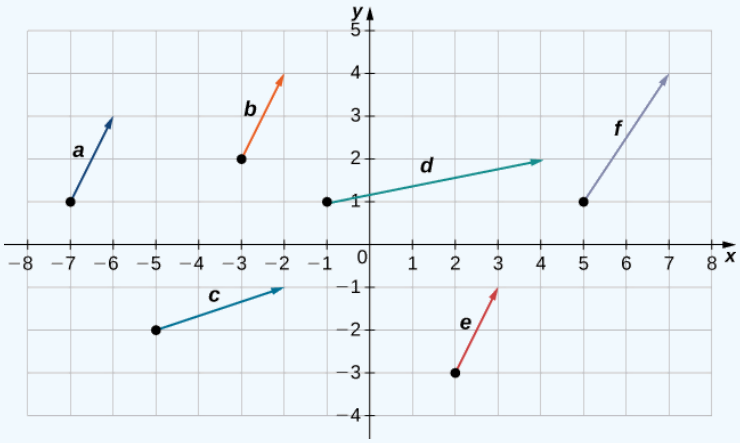


Figure 12.1.10: These vectors are equivalent.

Exercise 12.1.3

Which of the following vectors are equivalent?



Hint

Equivalent vectors have both the same magnitude and the same direction.

Answer

Vectors $\vec{a}$, $\vec{b}$, and $\vec{e}$ are equivalent.

We have seen how to plot a vector when we are given an initial point and a terminal point. However, because a vector can be placed anywhere in a plane, it may be easier to perform calculations with a vector when its initial point coincides with the origin. We call a vector with its initial point at the origin a **standard-position vector.** Because the initial point of any vector in standard position is known to be $(0, 0)$, we can describe the vector by looking at the coordinates of its terminal point. Thus, if vector $\vec{v}$ has its initial point at the origin and its terminal point at (x, y), we write the vector in component form as

$$\vec{v} = \langle x, y \rangle.$$

When a vector is written in component form like this, the scalars x and y are called the **components** of $\vec{v}$.

Definition: Vector components

The vector with initial point $(0, 0)$ and terminal point (x, y) can be written in component form as

$$\vec{v} = \langle x, y \rangle.$$

The scalars x and y are called the components of $\vec{v}$.

Recall that vectors are named with lowercase letters in bold type or by drawing an arrow over their name. We have also learned that we can name a vector by its component form, with the coordinates of its terminal point in angle brackets. However, when writing the component form of a vector, it is important to distinguish between $\langle x, y \rangle$ and (x, y). The first ordered pair uses angle brackets to describe a vector, whereas the second uses parentheses to describe a point in a plane. The initial point of $\langle x, y \rangle$ is $(0, 0)$; the terminal point of $\langle x, y \rangle$ is (x, y).

When we have a vector not already in standard position, we can determine its component form in one of two ways. We can use a geometric approach, in which we sketch the vector in the coordinate plane, and then sketch an equivalent standard-position vector. Alternatively, we can find it algebraically, using the coordinates of the initial point and the terminal point. To find it algebraically, we subtract the x-coordinate of the initial point from the x-coordinate of the terminal point to get the x-component, and we subtract the y-coordinate of the initial point from the y-coordinate of the terminal point to get the y-component.

Rule: Component Form of a Vector

Let $\vec{\mathbf{v}}$ be a vector with initial point (x_i, y_i) and terminal point (x_t, y_t). Then we can express $\vec{\mathbf{v}}$ in component form as $\vec{\mathbf{v}} = \langle x_t - x_i, y_t - y_i \rangle$.

Example 12.1.4: Expressing Vectors in Component Form

Express vector $\vec{\mathbf{v}}$ with initial point $(-3, 4)$ and terminal point $(1, 2)$ in component form.

Solution:

a. Geometric

1. Sketch the vector in the coordinate plane (Figure 12.1.11).
2. The terminal point is 4 units to the right and 2 units down from the initial point.
3. Find the point that is 4 units to the right and 2 units down from the origin.
4. In standard position, this vector has initial point $(0, 0)$ and terminal point $(4, -2)$:

$$\vec{\mathbf{v}} = \langle 4, -2 \rangle.$$

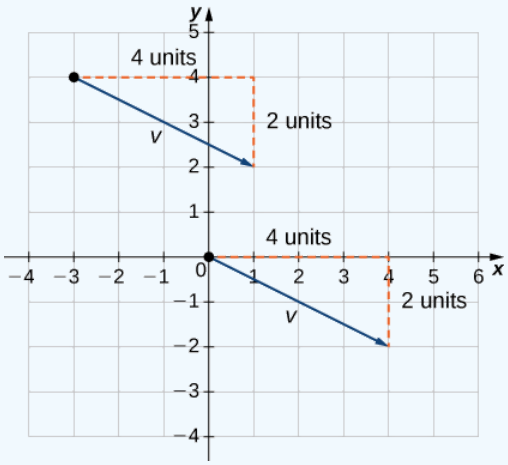


Figure 12.1.11: These vectors are equivalent.

b. Algebraic

In the first solution, we used a sketch of the vector to see that the terminal point lies 4 units to the right. We can accomplish this algebraically by finding the difference of the x-coordinates:

$$x_t - x_i = 1 - (-3) = 4.$$

Similarly, the difference of the y-coordinates shows the vertical length of the vector.

$$y_t - y_i = 2 - 4 = -2.$$

So, in component form,

$$\vec{v} = \langle x_t - x_i, y_t - y_i \rangle = \langle 1 - (-3), 2 - 4 \rangle = \langle 4, -2 \rangle.$$

Exercise 12.1.4

Vector $\vec{w}$ has initial point $(-4, -5)$ and terminal point $(-1, 2)$. Express $\vec{w}$ in component form.

Hint

You may use either geometric or algebraic method.

Answer

$\langle 3, 7 \rangle$

To find the magnitude of a vector, we calculate the distance between its initial point and its terminal point. The **magnitude** of vector $\vec{v} = \langle x, y \rangle$ is denoted $\|\vec{v}\|$, or $|\vec{v}|$, and can be computed using the formula

$$\|\vec{v}\| = \sqrt{x^2 + y^2}.$$

Note that because this vector is written in component form, it is equivalent to a vector in standard position, with its initial point at the origin and terminal point (x, y). Thus, it suffices to calculate the magnitude of the vector in standard position. Using the distance formula to calculate the distance between initial point $(0, 0)$ and terminal point (x, y), we have

$$\|\vec{v}\| = \sqrt{(x-0)^2 + (y-0)^2} = \sqrt{x^2 + y^2}.$$

Based on this formula, it is clear that for any vector $\vec{v}$, $\|\vec{v}\| \geq 0$, and $\|\vec{v}\| = 0$ if and only if $\vec{v} = \vec{0}$.

The magnitude of a vector can also be derived using the Pythagorean theorem, as in the following figure.

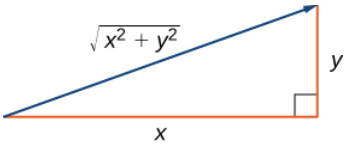


Figure 12.1.12: If you use the components of a vector to define a right triangle, the magnitude of the vector is the length of the triangle's hypotenuse.

We have defined scalar multiplication and vector addition geometrically. Expressing vectors in component form allows us to perform these same operations algebraically.

Definition: Scalar multiplication and Vector addition

Let $\vec{v} = \langle x_1, y_1 \rangle$ and $\vec{w} = \langle x_2, y_2 \rangle$ be vectors, and let k be a scalar.

- *Scalar multiplication:*

$$k\vec{v} = \langle kx_1, ky_1 \rangle$$

- *Vector addition:*

$$\vec{v} + \vec{w} = \langle x_1, y_1 \rangle + \langle x_2, y_2 \rangle = \langle x_1 + x_2, y_1 + y_2 \rangle$$

Example 12.1.5: Performing Operations in Component Form

Let $\vec{v}$ be the vector with initial point $(2, 5)$ and terminal point $(8, 13)$, and let $\vec{w} = \langle -2, 4 \rangle$.

a. Express $\vec{v}$ in component form and find $\|\vec{v}\|$. Then, using algebra, find
b. $\vec{v} + \vec{w}$,
c. $3\vec{v}$, and

d. $\vec{v} - 2\vec{w}$.

Solution

a. To place the initial point of $\vec{v}$ at the origin, we must translate the vector **2** units to the left and **5** units down (Figure 12.1.13). Using the algebraic method, we can express $\vec{v}$ as $\vec{v} = \langle 8-2, 13-5\rangle = \langle 6, 8\rangle$:

$$\|\vec{v}\| = \sqrt{6^2+8^2} = \sqrt{36+64} = \sqrt{100} = 10.$$

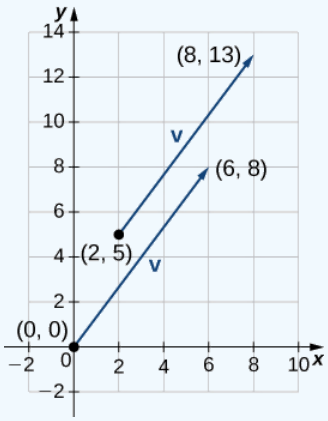


Figure 12.1.13: In component form, $\vec{v} = \langle 6, 8\rangle$.

b. To find $\vec{v} + \vec{w}$, add the x-components and the y-components separately:

$$\vec{v} + \vec{w} = \langle 6, 8\rangle + \langle -2, 4\rangle = \langle 4, 12\rangle.$$

c. To find $3\vec{v}$, multiply $\vec{v}$ by the scalar $k = 3$:

$$3\vec{v} = 3 \cdot \langle 6, 8\rangle = \langle 3 \cdot 6, 3 \cdot 8\rangle = \langle 18, 24\rangle.$$

d. To find $\vec{v} - 2\vec{w}$, find $-2\vec{w}$ and add it to $\vec{v}$:

$$\vec{v} - 2\vec{w} = \langle 6, 8\rangle - 2 \cdot \langle -2, 4\rangle = \langle 6, 8\rangle + \langle 4, -8\rangle = \langle 10, 0\rangle.$$

? Exercise 12.1.5A

Let $\vec{a} = \langle 7, 1\rangle$ and let $\vec{b}$ be the vector with initial point $(3, 2)$ and terminal point $(-1, -1)$.

a. Find $\|\vec{a}\|$.
b. Express $\vec{b}$ in component form.
c. Find $3\vec{a} - 4\vec{b}$.

Hint

Use the Pythagorean Theorem to find $\|\vec{a}\|$. To find $3\vec{a} - 4\vec{b}$, start by finding the scalar multiples $3\vec{a}$ and $-4\vec{b}$.

Answer a

$\|\vec{a}\| = 5\sqrt{2}$,

Answer b

$\vec{b} = \langle -4, -3\rangle$,

Answer c

$3\vec{a} - 4\vec{b} = \langle 37, 15\rangle$

Now that we have established the basic rules of vector arithmetic, we can state the properties of vector operations. We will prove two of these properties. The others can be proved in a similar manner.

Properties of Vector Operations

Let $\vec{\mathbf{u}}$, $\vec{\mathbf{v}}$, and $\vec{\mathbf{w}}$ be vectors in a plane. Let r and s be scalars.

i. *Commutative property*

$$\vec{\mathbf{u}} + \vec{\mathbf{v}} = \vec{\mathbf{v}} + \vec{\mathbf{u}} \tag{12.1.1}$$

ii. *Associative property*

$$(\vec{\mathbf{u}} + \vec{\mathbf{v}}) + \vec{\mathbf{w}} = \vec{\mathbf{u}} + (\vec{\mathbf{v}} + \vec{\mathbf{w}})$$

iii. *Additive identity property*

$$\vec{\mathbf{u}} + \vec{\mathbf{0}} = \vec{\mathbf{u}}$$

iv. *Additive inverse property*

$$\vec{\mathbf{u}} + (-\vec{\mathbf{u}}) = \vec{\mathbf{0}}$$

v. *Associativity of scalar multiplication*

$$r(s\vec{\mathbf{u}}) = (rs)\vec{\mathbf{u}}$$

vi. *Distributive property*

$$(r+s)\vec{\mathbf{u}} = r\vec{\mathbf{u}} + s\vec{\mathbf{u}} \tag{12.1.2}$$

vii. *Distributive property*

$$r(\vec{\mathbf{u}} + \vec{\mathbf{v}}) = r\vec{\mathbf{u}} + r\vec{\mathbf{v}}$$

viii. *Identity and zero properties*

$$1\vec{\mathbf{u}} = \vec{\mathbf{u}},\ 0\vec{\mathbf{u}} = \vec{\mathbf{0}}$$

Proof of Commutative Property

Let $\vec{\mathbf{u}} = \langle x_1, y_1 \rangle$ and $\vec{\mathbf{v}} = \langle x_2, y_2 \rangle$. Apply the commutative property for real numbers:

$$\begin{aligned} \vec{\mathbf{u}} + \vec{\mathbf{v}} &= \langle x_1 + x_2, y_1 + y_2 \rangle \\ &= \langle x_2 + x_1, y_2 + y_1 \rangle \\ &= \vec{\mathbf{v}} + \vec{\mathbf{u}}. \end{aligned}$$

□

Proof of Distributive Property

Apply the distributive property for real numbers:

$$\begin{aligned} r(\vec{\mathbf{u}} + \vec{\mathbf{v}}) &= r \cdot \langle x_1 + x_2, y_1 + y_2 \rangle \\ &= \langle r(x_1 + x_2), r(y_1 + y_2) \rangle \\ &= \langle rx_1 + rx_2, ry_1 + ry_2 \rangle \\ &= \langle rx_1, ry_1 \rangle + \langle rx_2, ry_2 \rangle \\ &= r\vec{\mathbf{u}} + r\vec{\mathbf{v}}. \end{aligned}$$

□

Exercise 12.1.5B

Prove the additive inverse property.

Hint

Use the component form of the vectors.

We have found the components of a vector given its initial and terminal points. In some cases, we may only have the magnitude and direction of a vector, not the points. For these vectors, we can identify the horizontal and vertical components using trigonometry (Figure 12.1.14).

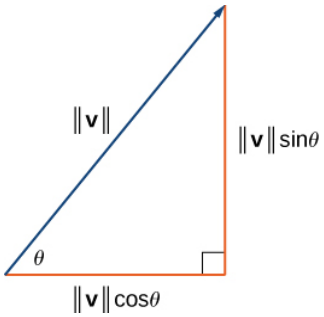


Figure 12.1.14: The components of a vector form the legs of a right triangle, with the vector as the hypotenuse.

Consider the angle θ formed by the vector $\vec{v}$ and the positive x-axis. We can see from the triangle that the components of vector $\vec{v}$ are $\langle \|\vec{v}\| \cos\theta, \|\vec{v}\| \sin\theta \rangle$. Therefore, given an angle and the magnitude of a vector, we can use the cosine and sine of the angle to find the components of the vector.

Example 12.1.6: Finding the Component Form of a Vector Using Trigonometry

Find the component form of a vector with magnitude 4 that forms an angle of $-45°$ with the x-axis.

Solution

Let x and y represent the components of the vector (Figure 12.1.15). Then $x = 4\cos(-45°) = 2\sqrt{2}$ and $y = 4\sin(-45°) = -2\sqrt{2}$. The component form of the vector is $\langle 2\sqrt{2}, -2\sqrt{2} \rangle$.

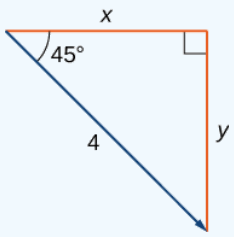


Figure 12.1.15: Use trigonometric ratios, $x = \|\vec{v}\| \cos\theta$ and $y = \|\vec{v}\| \sin\theta$, to identify the components of the vector.

Exercise 12.1.6

Find the component form of vector $\vec{v}$ with magnitude 10 that forms an angle of $120°$ with the positive x-axis.

Hint

$x = \|\vec{v}\| \cos\theta$ and $y = \|\vec{v}\| \sin\theta$

Answer

$\vec{v} = \langle -5, 5\sqrt{3} \rangle$

Unit Vectors

A **unit vector** is a vector with magnitude 1. For any nonzero vector $\vec{v}$, we can use scalar multiplication to find a unit vector $\vec{u}$ that has the same direction as $\vec{v}$. To do this, we multiply the vector by the reciprocal of its magnitude:

$$\vec{u} = \frac{1}{\|\vec{v}\|}\vec{v}.$$

Recall that when we defined scalar multiplication, we noted that $\|k\vec{v}\| = |k| \cdot \|\vec{v}\|$. For $\vec{u} = \frac{1}{\|\vec{v}\|}\vec{v}$, it follows that $\|\vec{u}\| = \frac{1}{\|\vec{v}\|}(\|\vec{v}\|) = 1$. We say that $\vec{u}$ is the *unit vector in the direction of* $\vec{v}$ (Figure 12.1.16). The process of using scalar multiplication to find a unit vector with a given direction is called **normalization.**



Figure 12.1.16: The vector $\vec{v}$ and associated unit vector $\vec{u} = \frac{1}{\|\vec{v}\|}\vec{v}$. In this case, $\|\vec{v}\| > 1$.

Example 12.1.7: Finding a Unit Vector

Let $\vec{v} = \langle 1, 2\rangle$.

a. Find a unit vector with the same direction as $\vec{v}$.
b. Find a vector $\vec{w}$ with the same direction as $\vec{v}$ such that $\|\vec{w}\| = 7$.

Solution:

a. First, find the magnitude of $\vec{v}$, then divide the components of $\vec{v}$ by the magnitude:

$$\|\vec{v}\| = \sqrt{1^2 + 2^2} = \sqrt{1+4} = \sqrt{5}$$

$$\vec{u} = \frac{1}{\|\vec{v}\|}\vec{v} = \frac{1}{\sqrt{5}}\langle 1, 2\rangle = \langle \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\rangle.$$

b. The vector $\vec{u}$ is in the same direction as $\vec{v}$ and $\|\vec{u}\| = 1$. Use scalar multiplication to increase the length of $\vec{u}$ without changing direction:

$$\vec{w} = 7\vec{u} = 7\langle \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\rangle = \langle \frac{7}{\sqrt{5}}, \frac{14}{\sqrt{5}}\rangle.$$

Exercise 12.1.7

Let $\vec{v} = \langle 9, 2\rangle$. Find a vector with magnitude 5 in the opposite direction as $\vec{v}$.

Hint

First, find a unit vector in the same direction as $\vec{v}$.

Answer

$$\langle -\frac{45}{\sqrt{85}}, -\frac{10}{\sqrt{85}}\rangle$$

We have seen how convenient it can be to write a vector in component form. Sometimes, though, it is more convenient to write a vector as a sum of a horizontal vector and a vertical vector. To make this easier, let's look at standard unit vectors. The **standard unit vectors** are the vectors $\hat{i} = \langle 1, 0\rangle$ and $\hat{j} = \langle 0, 1\rangle$ (Figure 12.1.17).

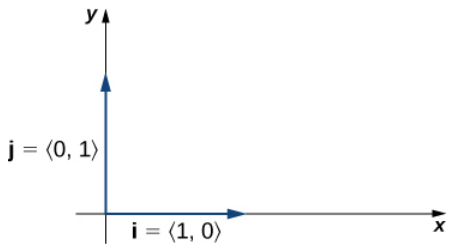


Figure 12.1.17: The standard unit vectors $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$.

By applying the properties of vectors, it is possible to express any vector in terms of $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ in what we call a *linear combination:*

$$\vec{\mathbf{v}} = \langle x, y\rangle = \langle x, 0\rangle + \langle 0, y\rangle = x\langle 1, 0\rangle + y\langle 0, 1\rangle = x\hat{\mathbf{i}} + y\hat{\mathbf{j}}.$$

Thus, $\vec{\mathbf{v}}$ is the sum of a horizontal vector with magnitude x, and a vertical vector with magnitude y, as in Figure 12.1.18.

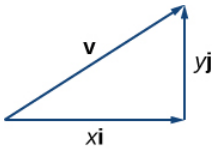


Figure 12.1.18: The vector $\vec{\mathbf{v}}$ is the sum of $x\hat{\mathbf{i}}$ and $y\hat{\mathbf{j}}$.

Example 12.1.8: Using Standard Unit Vectors

a. Express the vector $\vec{\mathbf{w}} = \langle 3, -4\rangle$ in terms of standard unit vectors.

b. Vector $\vec{\mathbf{u}}$ is a unit vector that forms an angle of $60°$ with the positive x-axis. Use standard unit vectors to describe $\vec{\mathbf{u}}$.

Solution:

a. Resolve vector $\vec{\mathbf{w}}$ into a vector with a zero y-component and a vector with a zero x-component:

$$\vec{\mathbf{w}} = \langle 3, -4\rangle = 3\hat{\mathbf{i}} - 4\hat{\mathbf{j}}.$$

b. Because $\vec{\mathbf{u}}$ is a unit vector, the terminal point lies on the unit circle when the vector is placed in standard position (Figure 12.1.19).

$$\begin{aligned}\vec{\mathbf{u}} &= \langle \cos 60°, \sin 60°\rangle \\ &= \langle \frac{1}{2}, \frac{\sqrt{3}}{2}\rangle \\ &= \frac{1}{2}\hat{\mathbf{i}} + \frac{\sqrt{3}}{2}\hat{\mathbf{j}}.\end{aligned}$$

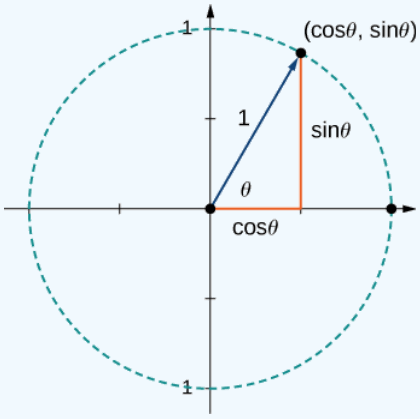


Figure 12.1.19: The terminal point of $\vec{\mathbf{u}}$ lies on the unit circle $(\cos\theta, \sin\theta)$.

Exercise 12.1.8

Let $\vec{\mathbf{a}} = \langle 16, -11 \rangle$ and let $\vec{\mathbf{b}}$ be a unit vector that forms an angle of $225°$ with the positive x-axis. Express $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ in terms of the standard unit vectors.

Hint

Use sine and cosine to find the components of $\vec{\mathbf{b}}$.

Answer

$$\vec{\mathbf{a}} = 16\hat{\mathbf{i}} - 11\hat{\mathbf{j}}, \quad \vec{\mathbf{b}} = -\frac{\sqrt{2}}{2}\hat{\mathbf{i}} - \frac{\sqrt{2}}{2}\hat{\mathbf{j}}$$

Applications of Vectors

Because vectors have both direction and magnitude, they are valuable tools for solving problems involving such applications as motion and **force.** Recall the boat example and the quarterback example we described earlier. Here we look at two other examples in detail.

Example 12.1.9A: Finding Resultant Force

Jane's car is stuck in the mud. Lisa and Jed come along in a truck to help pull her out. They attach one end of a tow strap to the front of the car and the other end to the truck's trailer hitch, and the truck starts to pull. Meanwhile, Jane and Jed get behind the car and push. The truck generates a horizontal force of **300** lb on the car. Jane and Jed are pushing at a slight upward angle and generate a force of **150** lb on the car. These forces can be represented by vectors, as shown in Figure 12.1.20. The angle between these vectors is **15°**. Find the resultant force (the vector sum) and give its magnitude to the nearest tenth of a pound and its direction angle from the positive x-axis.



Figure 12.1.20: Two forces acting on a car in different directions.

Solution

To find the effect of combining the two forces, add their representative vectors. First, express each vector in component form or in terms of the standard unit vectors. For this purpose, it is easiest if we align one of the vectors with the positive x-axis. The horizontal vector, then, has initial point $(0, 0)$ and terminal point $(300, 0)$. It can be expressed as $\langle 300, 0 \rangle$ or $300\hat{\mathbf{i}}$.

The second vector has magnitude 150 and makes an angle of $15°$ with the first, so we can express it as $\langle 150\cos(15°), 150\sin(15°) \rangle$, or $150\cos(15°)\hat{\mathbf{i}} + 150\sin(15°)\hat{\mathbf{j}}$. Then, the sum of the vectors, or resultant vector, is $\vec{\mathbf{r}} = \langle 300, 0 \rangle + \langle 150\cos(15°), 150\sin(15°) \rangle$, and we have

$$\|\vec{\mathbf{r}}\| = \sqrt{(300 + 150\cos(15°))^2 + (150\sin(15°))^2} \approx 446.6.$$

The angle θ made by $\vec{\mathbf{r}}$ and the positive x-axis has $\tan\theta = \dfrac{150\sin 15°}{(300 + 150\cos 15°)} \approx 0.09$, so $\theta \approx \tan^{-1}(0.09) \approx 5°$, which means the resultant force $\vec{\mathbf{r}}$ has an angle of $5°$ above the horizontal axis.

Example 12.1.9B: Finding Resultant Velocity

An airplane flies due west at an airspeed of 425 mph. The wind is blowing from the northeast at 40 mph. What is the ground speed of the airplane? What is the bearing of the airplane?

Solution

Let's start by sketching the situation described (Figure 12.1.21).

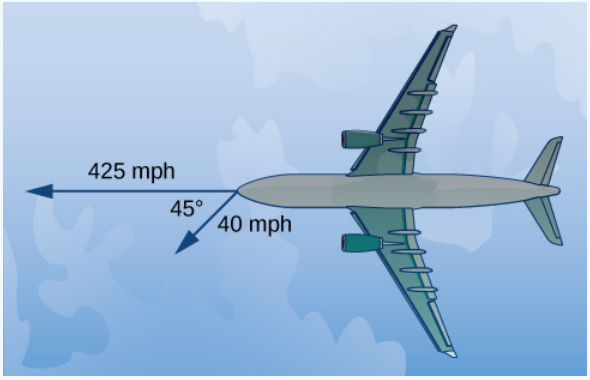


Figure 12.1.21: Initially, the plane travels due west. The wind is from the northeast, so it is blowing to the southwest. The angle between the plane's course and the wind is 45°. (Figure not drawn to scale.)

Set up a sketch so that the initial points of the vectors lie at the origin. Then, the plane's velocity vector is $\vec{\mathbf{p}} = -425\hat{\mathbf{i}}$. The vector describing the wind makes an angle of 225° with the positive x-axis:

$$\vec{\mathbf{w}} = \langle 40\cos(225°), 40\sin(225°)\rangle = \langle -\frac{40}{\sqrt{2}}, -\frac{40}{\sqrt{2}}\rangle = -\frac{40}{\sqrt{2}}\hat{\mathbf{i}} - \frac{40}{\sqrt{2}}\hat{\mathbf{j}}.$$

When the airspeed and the wind act together on the plane, we can add their vectors to find the resultant force:

$$\vec{\mathbf{p}} + \vec{\mathbf{w}} = -425\hat{\mathbf{i}} + (-\frac{40}{\sqrt{2}}\hat{\mathbf{i}} - \frac{40}{\sqrt{2}}\hat{\mathbf{j}}) = (-425 - \frac{40}{\sqrt{2}})\hat{\mathbf{i}} - \frac{40}{\sqrt{2}}\hat{\mathbf{j}}.$$

The magnitude of the resultant vector shows the effect of the wind on the ground speed of the airplane:

$$\|\vec{\mathbf{p}} + \vec{\mathbf{w}}\| = \sqrt{(-425 - \frac{40}{\sqrt{2}})^2 + (-\frac{40}{\sqrt{2}})^2} \approx 454.17 \text{ mph}$$

As a result of the wind, the plane is traveling at approximately 454 mph relative to the ground.

To determine the bearing of the airplane, we want to find the direction of the vector $\vec{\mathbf{p}} + \vec{\mathbf{w}}$:

$$\tan\theta = \frac{-\frac{40}{\sqrt{2}}}{(-425 - \frac{40}{\sqrt{2}})} \approx 0.06$$

$$\theta \approx 3.57°.$$

The overall direction of the plane is 3.57° south of west.

? Exercise 12.1.9

An airplane flies due north at an airspeed of 550 mph. The wind is blowing from the northwest at 50 mph. What is the ground speed of the airplane?

Hint

Sketch the vectors with the same initial point and find their sum.

Answer

Approximately 516 mph

Key Concepts

- Vectors are used to represent quantities that have both magnitude and direction.
- We can add vectors by using the parallelogram method or the triangle method to find the sum. We can multiply a vector by a scalar to change its length or give it the opposite direction.
- Subtraction of vectors is defined in terms of adding the negative of the vector.
- A vector is written in component form as $\vec{\mathbf{v}} = \langle x, y \rangle$.
- The magnitude of a vector is a scalar: $\|\vec{\mathbf{v}}\| = \sqrt{x^2 + y^2}$.
- A unit vector $\vec{\mathbf{u}}$ has magnitude 1 and can be found by dividing a vector by its magnitude: $\vec{\mathbf{u}} = \dfrac{1}{\|\vec{\mathbf{v}}\|}\vec{\mathbf{v}}$. The standard unit vectors are $\hat{\mathbf{i}} = \langle 1, 0 \rangle$ and $\hat{\mathbf{j}} = \langle 0, 1 \rangle$. A vector $\vec{\mathbf{v}} = \langle x, y \rangle$ can be expressed in terms of the standard unit vectors as $\vec{\mathbf{v}} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}}$.
- Vectors are often used in physics and engineering to represent forces and velocities, among other quantities.

Glossary

component

a scalar that describes either the vertical or horizontal direction of a vector

equivalent vectors

vectors that have the same magnitude and the same direction

initial point

the starting point of a vector

magnitude

the length of a vector

normalization

using scalar multiplication to find a unit vector with a given direction

parallelogram method

a method for finding the sum of two vectors; position the vectors so they share the same initial point; the vectors then form two adjacent sides of a parallelogram; the sum of the vectors is the diagonal of that parallelogram

scalar

a real number

scalar multiplication

a vector operation that defines the product of a scalar and a vector

standard-position vector

a vector with initial point $(0, 0)$

standard unit vectors

unit vectors along the coordinate axes: $\hat{\mathbf{i}} = \langle 1, 0 \rangle$, $\hat{\mathbf{j}} = \langle 0, 1 \rangle$

terminal point

the endpoint of a vector

triangle inequality

the length of any side of a triangle is less than the sum of the lengths of the other two sides

triangle method

a method for finding the sum of two vectors; position the vectors so the terminal point of one vector is the initial point of the other; these vectors then form two sides of a triangle; the sum of the vectors is the vector that forms the third side; the initial point of the sum is the initial point of the first vector; the terminal point of the sum is the terminal point of the second vector

unit vector

a vector with magnitude 1

vector

a mathematical object that has both magnitude and direction

vector addition

a vector operation that defines the sum of two vectors

vector difference

the vector difference $\vec{v} - \vec{w}$ is defined as $\vec{v} + (-\vec{w}) = \vec{v} + (-1)\vec{w}$

vector sum

the sum of two vectors, $\vec{v}$ and $\vec{w}$, can be constructed graphically by placing the initial point of $\vec{w}$ at the terminal point of $\vec{v}$; then the vector sum $\vec{v} + \vec{w}$ is the vector with an initial point that coincides with the initial point of $\vec{v}$, and with a terminal point that coincides with the terminal point of $\vec{w}$

zero vector

the vector with both initial point and terminal point $(0, 0)$

This page titled 12.1: Vectors in the Plane is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin "Jed" Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

12.1E: Exercises for Section 12.1

For exercises 1 - 10, consider points $P(-1,3)$, $Q(1,5)$, and $R(-3,7)$. Determine the requested vectors and express each of them

a. in component form and

b. by using standard unit vectors.

1) $\overrightarrow{PQ}$

Answer

a. $\overrightarrow{PQ} = \langle 2,2 \rangle$
b. $\overrightarrow{PQ} = 2\hat{\mathbf{i}} + 2\hat{\mathbf{j}}$

2) $\overrightarrow{PR}$

3) $\overrightarrow{QP}$

Answer

a. $\overrightarrow{QP} = \langle -2,-2 \rangle$
b. $\overrightarrow{QP} = -2\hat{\mathbf{i}} - 2\hat{\mathbf{j}}$

4) $\overrightarrow{RP}$

5) $\overrightarrow{PQ} + \overrightarrow{PR}$

Answer

a. $\overrightarrow{PQ} + \overrightarrow{PR} = \langle 0,6 \rangle$
b. $\overrightarrow{PQ} + \overrightarrow{PR} = 6\hat{\mathbf{j}}$

6) $\overrightarrow{PQ} - \overrightarrow{PR}$

7) $2\overrightarrow{PQ} - 2\overrightarrow{PR}$

Answer

a. $2\overrightarrow{PQ} \rightarrow -2\overrightarrow{PR} = \langle 8,-4 \rangle$
b. $2\overrightarrow{PQ} - 2\overrightarrow{PR} = 8\hat{\mathbf{i}} - 4\hat{\mathbf{j}}$

8) $2\overrightarrow{PQ} + \frac{1}{2}\overrightarrow{PR}$

9) The unit vector in the direction of $\overrightarrow{PQ}$

Answer

a. $\left\langle \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right\rangle$
b. $\frac{\sqrt{2}}{2}\hat{\mathbf{i}} + \frac{\sqrt{2}}{2}\hat{\mathbf{j}}$

10) The unit vector in the direction of $\overrightarrow{PR}$

11) A vector $\vec{\mathbf{v}}$ has initial point $(-1,-3)$ and terminal point $(2,1)$. Find the unit vector in the direction of $\vec{\mathbf{v}}$. Express the answer in component form.

Answer

$\langle \frac{3}{5}, \frac{4}{5} \rangle$

12) A vector $\vec{\mathbf{v}}$ has initial point $(-2,5)$ and terminal point $(3,-1)$. Find the unit vector in the direction of $\vec{\mathbf{v}}$. Express the answer in component form.

13) The vector $\vec{v}$ has initial point $P(1,0)$ and terminal point Q that is on the y-axis and above the initial point. Find the coordinates of terminal point Q such that the magnitude of the vector $\vec{v}$ is $\sqrt{5}$.

Answer

$Q(0,2)$

14) The vector $\vec{v}$ has initial point $P(1,1)$ and terminal point Q that is on the x-axis and left of the initial point. Find the coordinates of terminal point Q such that the magnitude of the vector $\vec{v}$ is $\sqrt{10}$.

For exercises 15 and 16, use the given vectors $\vec{a}$ and $\vec{b}$.

a. Determine the vector sum $\vec{a}+\vec{b}$ and express it in both the component form and by using the standard unit vectors.

b. Find the vector difference $\vec{a}-\vec{b}$ and express it in both the component form and by using the standard unit vectors.

c. Verify that the vectors $\vec{a}$, $\vec{b}$, and $\vec{a}+\vec{b}$, and, respectively, $\vec{a}$, $\vec{b}$, and $\vec{a}-\vec{b}$ satisfy the triangle inequality.

d. Determine the vectors $2\vec{a}$, $-\vec{b}$, and $2\vec{a}-\vec{b}$. Express the vectors in both the component form and by using standard unit vectors.

15) $\vec{a}=2\hat{i}+\hat{j}, \quad \vec{b}=\hat{i}+3\hat{j}$

Answer

a. $\vec{a}+\vec{b}=\langle 3,4\rangle, \quad \vec{a}+\vec{b}=3\hat{i}+4\hat{j}$
b. $\vec{a}-\vec{b}=\langle 1,-2\rangle, \quad \vec{a}-\vec{b}=\hat{i}-2\hat{j}$
c. Answers will vary
d. $2\vec{a}=\langle 4,2\rangle, \quad 2\vec{a}=4\hat{i}+2\hat{j}, \quad -\vec{b}=\langle -1,-3\rangle, \quad -\vec{b}=-\hat{i}-3\hat{j}, \quad 2\vec{a}-\vec{b}=\langle 3,-1\rangle, \quad 2\vec{a}-\vec{b}=3\hat{i}-\hat{j}$

16) $\vec{a}=2\hat{i}, \quad \vec{b}=-2\hat{i}+2\hat{j}$

17) Let $\vec{a}$ be a standard-position vector with terminal point $(-2,-4)$. Let $\vec{b}$ be a vector with initial point $(1,2)$ and terminal point $(-1,4)$. Find the magnitude of vector $-3\vec{a}+\vec{b}-4\hat{i}+\hat{j}$.

Answer

15

18) Let $\vec{a}$ be a standard-position vector with terminal point at $(2,5)$. Let $\vec{b}$ be a vector with initial point $(-1,3)$ and terminal point $(1,0)$. Find the magnitude of vector $\vec{a}-3\vec{b}+14\hat{i}-14\hat{j}$.

19) Let $\vec{u}$ and $\vec{v}$ be two nonzero vectors that are nonequivalent. Consider the vectors $\vec{a}=4\vec{u}+5\vec{v}$ and $\vec{b}=\vec{u}+2\vec{v}$ defined in terms of $\vec{u}$ and $\vec{v}$. Find the scalar λ such that vectors $\vec{a}+\lambda\vec{b}$ and $\vec{u}-\vec{v}$ are equivalent.

Answer

$\lambda=-3$

20) Let $\vec{u}$ and $\vec{v}$ be two nonzero vectors that are nonequivalent. Consider the vectors $\vec{a}=2\vec{u}-4\vec{v}$ and $\vec{b}=3\vec{u}-7\vec{v}$ defined in terms of $\vec{u}$ and $\vec{v}$. Find the scalars α and β such that vectors $\alpha\vec{a}+\beta\vec{b}$ and $\vec{u}-\vec{v}$ are equivalent.

21) Consider the vector $\vec{a}(t)=\langle \cos t, \sin t\rangle$ with components that depend on a real number t. As the number t varies, the components of $\vec{a}(t)$ change as well, depending on the functions that define them.

a. Write the vectors $\vec{a}(0)$ and $\vec{a}(\pi)$ in component form.

b. Show that the magnitude $\| \vec{a}(t) \|$ of vector $\vec{a}(t)$ remains constant for any real number t.

c. As t varies, show that the terminal point of vector $\vec{a}(t)$ describes a circle centered at the origin of radius 1.

Answer

a. $\vec{\mathbf{a}}(0) = \langle 1, 0\rangle, \quad \vec{\mathbf{a}}(\pi) = \langle -1, 0\rangle$

b. Answers may vary

c. Answers may vary

22) Consider vector $\vec{\mathbf{a}}(x) = \langle x, \sqrt{1-x^2}\rangle$ with components that depend on a real number $x \in [-1, 1]$. As the number x varies, the components of $\vec{\mathbf{a}}(x)$ change as well, depending on the functions that define them.

a. Write the vectors $\vec{\mathbf{a}}(0)$ and $\vec{\mathbf{a}}(1)$ in component form.

b. Show that the magnitude $\| \vec{\mathbf{a}}(x) \|$ of vector $\vec{\mathbf{a}}(x)$ remains constant for any real number x.

c. As x varies, show that the terminal point of vector $\vec{\mathbf{a}}(x)$ describes a circle centered at the origin of radius 1.

23) Show that vectors $\vec{\mathbf{a}}(t) = \langle \cos t, \sin t\rangle$ and $\vec{\mathbf{a}}(x) = \langle x, \sqrt{1-x^2}\rangle$ are equivalent for $x = 1$ and $t = 2k\pi$, where k is an integer.

Answer Answers may vary

24) Show that vectors $\vec{\mathbf{a}}(t) = \langle \cos t, \sin t\rangle$ and $\vec{\mathbf{a}}(x) = \langle x, \sqrt{1-x^2}\rangle$ are opposite for $x = 1$ and $t = \pi + 2k\pi$, where k is an integer.

For exercises 25-28, find a vector $\vec{\mathbf{v}}$ with the given magnitude and in the same direction as the vector $\vec{\mathbf{u}}$.

25) $\|\vec{\mathbf{v}}\| = 7, \quad \vec{\mathbf{u}} = \langle 3, 4\rangle$

Answer

$\vec{\mathbf{v}} = \langle \frac{21}{5}, \frac{28}{5}\rangle$

26) $\|\vec{\mathbf{v}}\| = 3, \quad \vec{\mathbf{u}} = \langle -2, 5\rangle$

27) $\|\vec{\mathbf{v}}\| = 7, \quad \vec{\mathbf{u}} = \langle 3, -5\rangle$

Answer

$\vec{\mathbf{v}} = \langle \frac{21\sqrt{34}}{34}, -\frac{35\sqrt{34}}{34}\rangle$

28) $\|\vec{\mathbf{v}}\| = 10, \quad \vec{\mathbf{u}} = \langle 2, -1\rangle$

For exercises 29-34, find the component form of vector $\vec{\mathbf{u}}$, given its magnitude and the angle the vector makes with the positive x-axis. Give exact answers when possible.

29) $\|\vec{\mathbf{u}}\| = 2, \theta = 30°$

Answer

$\vec{\mathbf{u}} = \langle \sqrt{3}, 1\rangle$

30) $\|\vec{\mathbf{u}}\| = 6, \theta = 60°$

31) $\|\vec{\mathbf{u}}\| = 5, \theta = \frac{\pi}{2}$

Answer

$\vec{\mathbf{u}} = \langle 0, 5\rangle$

32) $\|\vec{\mathbf{u}}\| = 8, \theta = \pi$

33) $\|\vec{\mathbf{u}}\| = 10, \theta = \frac{5\pi}{6}$

Answer

$\vec{\mathbf{u}} = \langle -5\sqrt{3}, 5\rangle$

34) $\|\vec{\mathbf{u}}\| = 50, \theta = \frac{3\pi}{4}$

For exercises 35 and 36, vector $\vec{\mathbf{u}}$ is given. Find the angle $\theta \in [0, 2\pi)$ that vector $\vec{\mathbf{u}}$ makes with the positive direction of the x-axis, in a counter-clockwise direction.

35) $\vec{\mathbf{u}} = 5\sqrt{2}\hat{\mathbf{i}} - 5\sqrt{2}\hat{\mathbf{j}}$

Answer

$\theta = \frac{7\pi}{4}$

36) $\vec{\mathbf{u}} = -\sqrt{3}\hat{\mathbf{i}} - \hat{\mathbf{j}}$

37) Let $\vec{\mathbf{a}} = \langle a_1, a_2 \rangle$, $\vec{\mathbf{b}} = \langle b_1, b_2 \rangle$, and $\vec{\mathbf{c}} = \langle c_1, c_2 \rangle$ be three nonzero vectors. If $a_1 b_2 - a_2 b_1 \neq 0$, then show there are two scalars, α and β, such that $\vec{\mathbf{c}} = \alpha\vec{\mathbf{a}} + \beta\vec{\mathbf{b}}$.

Answer Answers may vary

38) Consider vectors $\vec{\mathbf{a}} = \langle 2, -4 \rangle$, $\vec{\mathbf{b}} = \langle -1, 2 \rangle$, and $\vec{\mathbf{c}} = \vec{\mathbf{0}}$ Determine the scalars α and β such that $\vec{\mathbf{c}} = \alpha\vec{\mathbf{a}} + \beta\vec{\mathbf{b}}$.

39) Let $P(x_0, f(x_0))$ be a fixed point on the graph of the differentiable function f with a domain that is the set of real numbers.

a. Determine the real number z_0 such that point $Q(x_0 + 1, z_0)$ is situated on the line tangent to the graph of f at point P.

b. Determine the unit vector $\vec{\mathbf{u}}$ with initial point P and terminal point Q.

Answer

$a.\quad z_0 = f(x_0) + f'(x_0); \quad b.\quad \vec{\mathbf{u}} = \frac{1}{\sqrt{1+[f'(x_0)]^2}}\langle 1, f'(x_0) \rangle$

40) Consider the function $f(x) = x^4$, where $x \in R$.

a. Determine the real number z_0 such that point $Q(2, z_0)$ s situated on the line tangent to the graph of f at point $P(1, 1)$.

b. Determine the unit vector $\vec{\mathbf{u}}$ with initial point P and terminal point Q.

41) Consider f and g two functions defined on the same set of real numbers D. Let $\vec{\mathbf{a}} = \langle x, f(x) \rangle$ and $\vec{\mathbf{b}} = \langle x, g(x) \rangle$ be two vectors that describe the graphs of the functions, where $x \in D$. Show that if the graphs of the functions f and g do not intersect, then the vectors $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ are not equivalent.

42) Find $x \in R$ such that vectors $\vec{\mathbf{a}} = \langle x, \sin x \rangle$ and $\vec{\mathbf{b}} = \langle x, \cos x \rangle$ are equivalent.

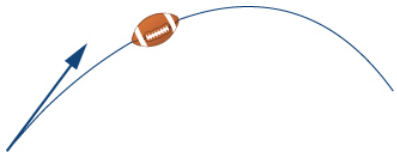
43) Calculate the coordinates of point D such that $ABCD$ is a parallelogram, with $A(1, 1)$, $B(2, 4)$, and $C(7, 4)$.

Answer

$D(6, 1)$

44) Consider the points $A(2, 1), B(10, 6), C(13, 4)$, and $D(16, -2)$. Determine the component form of vector $\overrightarrow{AD}$.

45) The **speed** of an object is the magnitude of its related velocity vector. A football thrown by a quarterback has an initial speed of 70 mph and an angle of elevation of $30°$. Determine the velocity vector in mph and express it in component form. (Round to two decimal places.)

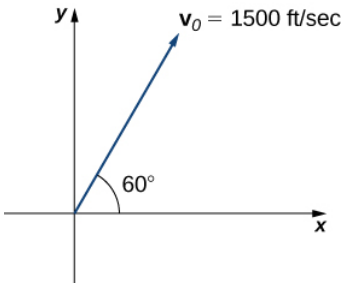


Answer

$\langle 60.62, 35 \rangle$

46) A baseball player throws a baseball at an angle of $30°$ with the horizontal. If the initial speed of the ball is 100 mph, find the horizontal and vertical components of the initial velocity vector of the baseball. (Round to two decimal places.)

47) A bullet is fired with an initial velocity of 1500 ft/sec at an angle of $60°$ with the horizontal. Find the horizontal and vertical components of the velocity vector of the bullet. (Round to two decimal places.)

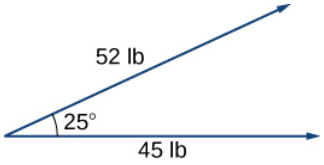


Answer

The horizontal and vertical components are 750 ft/sec and 1299.04 ft/sec, respectively.

48) [T] A 65-kg sprinter exerts a force of 798 N at a $19°$ angle with respect to the ground on the starting block at the instant a race begins. Find the horizontal component of the force. (Round to two decimal places.)

49) [T] Two forces, a horizontal force of 45 lb and another of 52 lb, act on the same object. The angle between these forces is $25°$. Find the magnitude and direction angle from the positive x-axis of the resultant force that acts on the object. (Round to two decimal places.)



Answer

The magnitude of resultant force is 94.71 lb; the direction angle is $13.42°$.

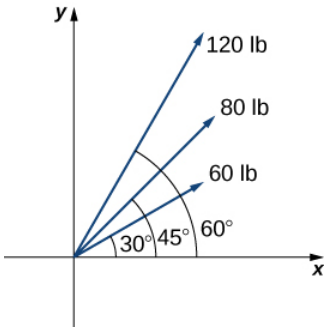
50) [T] Two forces, a vertical force of 26 lb and another of 45 lb, act on the same object. The angle between these forces is $55°$. Find the magnitude and direction angle from the positive x-axis of the resultant force that acts on the object. (Round to two decimal places.)

51) [T] Three forces act on object. Two of the forces have the magnitudes 58 N and 27 N, and make angles $53°$ and $152°$, respectively, with the positive x-axis. Find the magnitude and the direction angle from the positive x-axis of the third force such that the resultant force acting on the object is zero. (Round to two decimal places.)

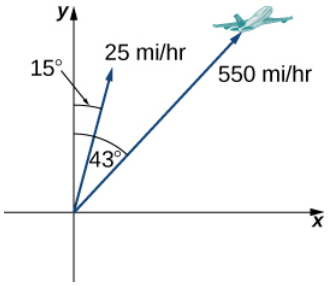
Answer

The magnitude of the third vector is 60.03 N; the direction angle is $259.38°$.

52) Three forces with magnitudes **80** lb, **120** lb, and **60** lb act on an object at angles of $45°, 60°$ and $30°$, respectively, with the positive x-axis. Find the magnitude and direction angle from the positive x-axis of the resultant force. (Round to two decimal places.)



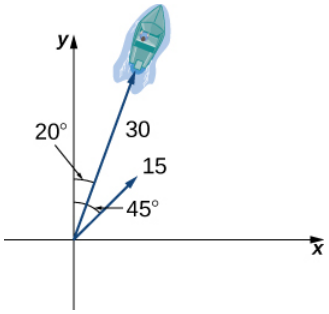
53) [T] An airplane is flying in the direction of $43°$ east of north (also abbreviated as $N43E$ at a speed of 550 mph. A wind with speed 25 mph comes from the southwest at a bearing of $N15E$. What are the ground speed and new direction of the airplane?



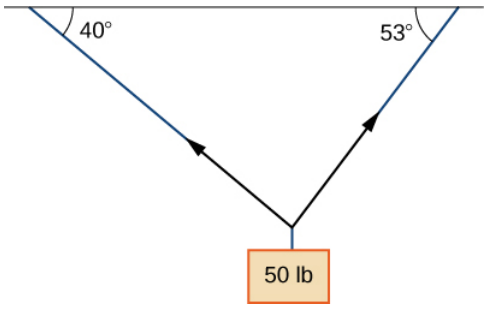
Answer

The new ground speed of the airplane is 572.19 mph; the new direction is $N41.82E$.

54) [T] A boat is traveling in the water at 30 mph in a direction of $N20E$ (that is, $20°$ east of north). A strong current is moving at 15 mph in a direction of $N45E$. What are the new speed and direction of the boat?



55) [T] A 50-lb weight is hung by a cable so that the two portions of the cable make angles of $40°$ and $53°$, respectively, with the horizontal. Find the magnitudes of the forces of tension $\vec{\mathbf{T}}_1$ and $\vec{\mathbf{T}}_2$ in the cables if the resultant force acting on the object is zero. (Round to two decimal places.)

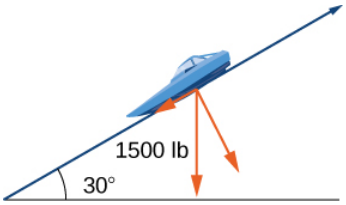


Answer

$\|\vec{\mathbf{T}}_1\| = 30.13\,lb, \quad \|\vec{\mathbf{T}}_2\| = 38.35\,lb$

56) [T] A 62-lb weight hangs from a rope that makes the angles of $29°$ and $61°$, respectively, with the horizontal. Find the magnitudes of the forces of tension $\vec{\mathbf{T}}_1$ and $\vec{\mathbf{T}}_2$ in the cables if the resultant force acting on the object is zero. (Round to two decimal places.)

57) [T] A 1500-lb boat is parked on a ramp that makes an angle of $30°$ with the horizontal. The boat's weight vector points downward and is a sum of two vectors: a horizontal vector $\vec{\mathbf{v}}_1$ that is parallel to the ramp and a vertical vector $\vec{\mathbf{v}}_2$ that is perpendicular to the inclined surface. The magnitudes of vectors $\vec{\mathbf{v}}_1$ and $\vec{\mathbf{v}}_2$ are the horizontal and vertical component, respectively, of the boat's weight vector. Find the magnitudes of $\vec{\mathbf{v}}_1$ and $\vec{\mathbf{v}}_2$. (Round to the nearest integer.)

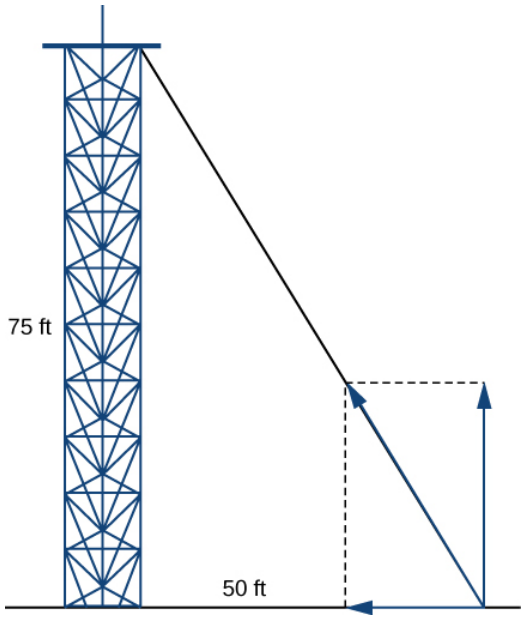


Answer

$\|\vec{v}_1\| = 750\,lb, \quad \|\vec{v}_2\| = 1299\,lb$

58) [T] An 85-lb box is at rest on a $26°$ incline. Determine the magnitude of the force parallel to the incline necessary to keep the box from sliding. (Round to the nearest integer.)

59) A guy-wire supports a pole that is 75 ft high. One end of the wire is attached to the top of the pole and the other end is anchored to the ground 50 ft from the base of the pole. Determine the horizontal and vertical components of the force of tension in the wire if its magnitude is 50 lb. (Round to the nearest integer.)



Answer

The two horizontal and vertical components of the force of tension are 28 lb and 42 lb, respectively.

60) A telephone pole guy-wire has an angle of elevation of $35°$ with respect to the ground. The force of tension in the guy-wire is 120 lb. Find the horizontal and vertical components of the force of tension. (Round to the nearest integer.)

Contributors

Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.

This page titled 12.1E: Exercises for Section 12.1 is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

12.2: Vectors in Three Dimensions

Learning Objectives

- Describe three-dimensional space mathematically.
- Locate points in space using coordinates.
- Write the distance formula in three dimensions.
- Write the equations for simple planes and spheres.
- Perform vector operations in $\mathbb{R}^3$.

Vectors are useful tools for solving two-dimensional problems. Life, however, happens in three dimensions. To expand the use of vectors to more realistic applications, it is necessary to create a framework for describing three-dimensional space. For example, although a two-dimensional map is a useful tool for navigating from one place to another, in some cases the topography of the land is important. Does your planned route go through the mountains? Do you have to cross a river? To appreciate fully the impact of these geographic features, you must use three dimensions. This section presents a natural extension of the two-dimensional Cartesian coordinate plane into three dimensions.

Three-Dimensional Coordinate Systems

As we have learned, the two-dimensional rectangular coordinate system contains two perpendicular axes: the horizontal x-axis and the vertical y-axis. We can add a third dimension, the z-axis, which is perpendicular to both the x-axis and the y-axis. We call this system the three-dimensional rectangular coordinate system. It represents the three dimensions we encounter in real life.

Definition: Three-dimensional Rectangular Coordinate System

The three-dimensional rectangular coordinate system consists of three perpendicular axes: the x-axis, the y-axis, and the z-axis. Because each axis is a number line representing all real numbers in $\mathbb{R}$, the three-dimensional system is often denoted by $\mathbb{R}^3$.

In Figure 12.2.1a, the positive z-axis is shown above the plane containing the x- and y-axes. The positive x-axis appears to the left and the positive y-axis is to the right. A natural question to ask is: How was this arrangement determined? The system displayed follows the **right-hand rule**. If we take our right hand and align the fingers with the positive x-axis, then curl the fingers so they point in the direction of the positive y-axis, our thumb points in the direction of the positive z-axis (Figure 12.2.1b). In this text, we always work with coordinate systems set up in accordance with the right-hand rule. Some systems do follow a left-hand rule, but the right-hand rule is considered the standard representation.

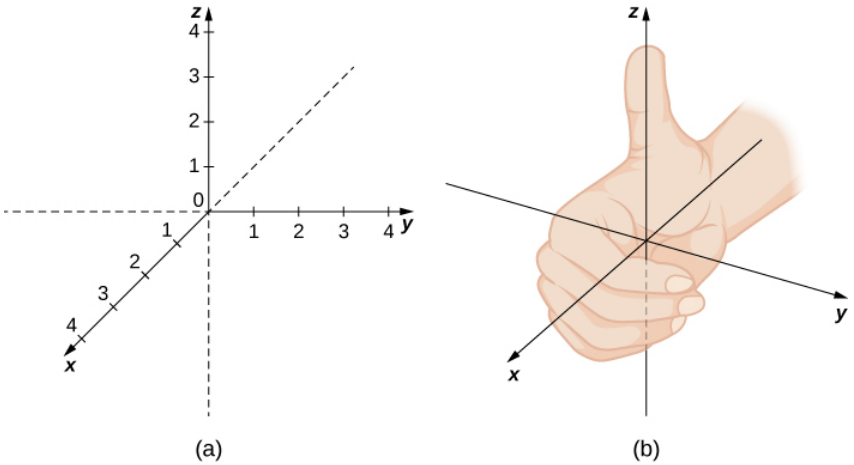


Figure 12.2.1: (a) We can extend the two-dimensional rectangular coordinate system by adding a third axis, the z-axis, that is perpendicular to both the x-axis and the y-axis. (b) The right-hand rule is used to determine the placement of the coordinate axes in the standard Cartesian plane.

In two dimensions, we describe a point in the plane with the coordinates (x, y). Each coordinate describes how the point aligns with the corresponding axis. In three dimensions, a new coordinate, z, is appended to indicate alignment with the z-axis: (x, y, z). A point in space is identified by all three coordinates (Figure 12.2.2). To plot the point (x, y, z), go x units along the x-axis, then y units in the direction of the y-axis, then z units in the direction of the z-axis.

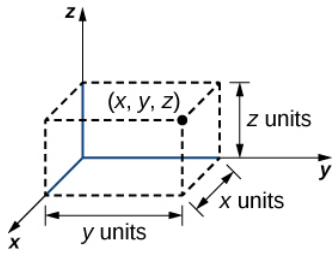


Figure 12.2.2: To plot the point (x, y, z) go x units along the x-axis, then y units in the direction of the y-axis, then z units in the direction of the z-axis.

Example 12.2.1: Locating Points in Space

Sketch the point $(1, -2, 3)$ in three-dimensional space.

Solution

To sketch a point, start by sketching three sides of a rectangular prism along the coordinate axes: one unit in the positive x direction, 2 units in the negative y direction, and 3 units in the positive z direction. Complete the prism to plot the point (Figure 12.2.3).

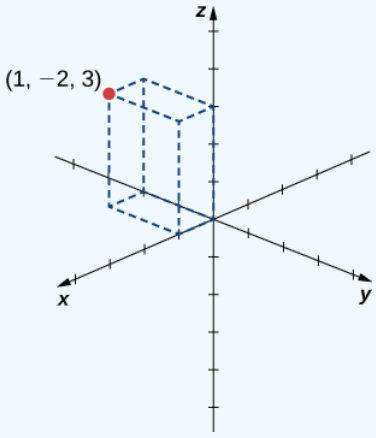


Figure 12.2.3: Sketching the point $(1, -2, 3)$.

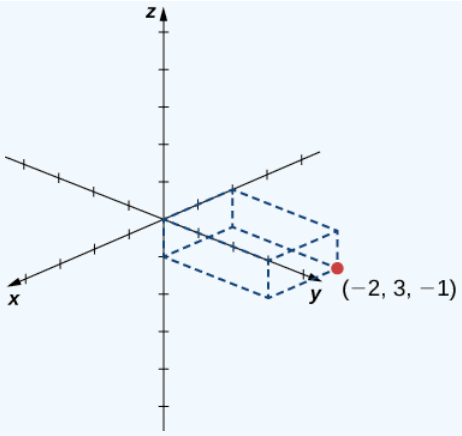
Exercise 12.2.1

Sketch the point $(-2, 3, -1)$ in three-dimensional space.

Hint

Start by sketching the coordinate axes. e.g., Figure 12.2.3. Then sketch a rectangular prism to help find the point in space.

Answer



In two-dimensional space, the coordinate plane is defined by a pair of perpendicular axes. These axes allow us to name any location within the plane. In three dimensions, we define **coordinate planes** by the coordinate axes, just as in two dimensions. There are three axes now, so there are three intersecting pairs of axes. Each pair of axes forms a coordinate plane: the xy-plane, the xz-plane, and the yz-plane (Figure 12.2.4). We define the xy-plane formally as the following set: $\{(x, y, 0) : x, y \in \mathbb{R}\}$. Similarly, the xz-plane and the yz-plane are defined as $\{(x, 0, z) : x, z \in \mathbb{R}\}$ and $\{(0, y, z) : y, z \in \mathbb{R}\}$, respectively.

To visualize this, imagine you're building a house and are standing in a room with only two of the four walls finished. (Assume the two finished walls are adjacent to each other.) If you stand with your back to the corner where the two finished walls meet, facing out into the room, the floor is the xy-plane, the wall to your right is the xz-plane, and the wall to your left is the yz-plane.

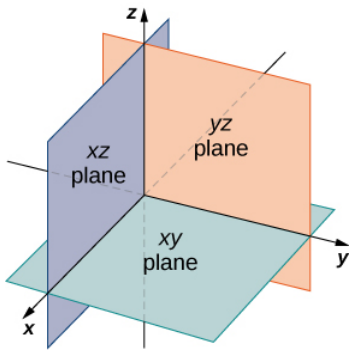


Figure 12.2.4: The plane containing the x- *and* y-axes is called the xy-plane. The plane containing the x- and z-axes is called the xz-plane, and the y- and z-axes define the yz-plane.

In two dimensions, the coordinate axes partition the plane into four quadrants. Similarly, the coordinate planes divide space between them into eight regions about the origin, called **octants.** The octants fill $\mathbb{R}^3$ in the same way that quadrants fill $\mathbb{R}^2$, as shown in Figure 12.2.5.

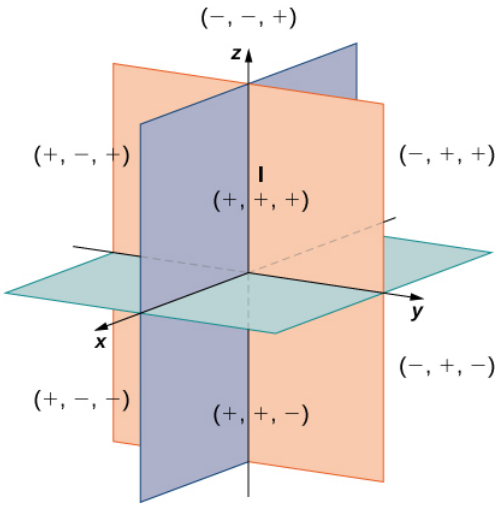


Figure 12.2.5: Points that lie in octants have three nonzero coordinates.

Most work in three-dimensional space is a comfortable extension of the corresponding concepts in two dimensions. In this section, we use our knowledge of circles to describe spheres, then we expand our understanding of vectors to three dimensions. To accomplish these goals, we begin by adapting the distance formula to three-dimensional space.

If two points lie in the same coordinate plane, then it is straightforward to calculate the distance between them. We know that the distance d between two points (x_1, y_1) and (x_2, y_2) in the xy-coordinate plane is given by the formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

The formula for the distance between two points in space is a natural extension of this formula.

The Distance between Two Points in Space

The distance d between points (x_1, y_1, z_1) and (x_2, y_2, z_2) is given by the formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}. \tag{12.2.1}$$

The proof of this theorem is left as an exercise. (Hint: First find the distance d_1 between the points (x_1, y_1, z_1) and (x_2, y_2, z_1) as shown in Figure 12.2.6.)

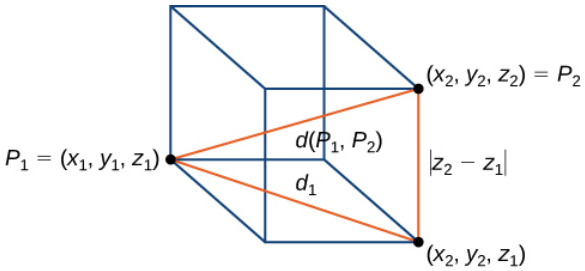


Figure 12.2.6: The distance between P_1 and P_2 is the length of the diagonal of the rectangular prism having P_1 and P_2 as opposite corners.

Example 12.2.2: Distance in Space

Find the distance between points $P_1 = (3, -1, 5)$ and $P_2 = (2, 1, -1)$.

https://math.libretexts.org/@go/page/2587

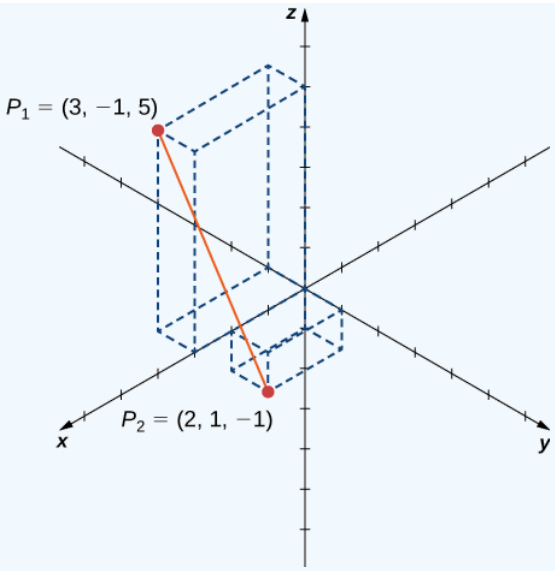


Figure 12.2.7: Find the distance between the two points.

Solution

Substitute values directly into the distance formula (Equation 12.2.1):

$$\begin{aligned} d(P_1, P_2) &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} \\ &= \sqrt{(2-3)^2 + (1-(-1))^2 + (-1-5)^2} \\ &= \sqrt{(-1)^2 + 2^2 + (-6)^2} \\ &= \sqrt{41}. \end{aligned}$$

? Exercise 12.2.2

Find the distance between points $P_1 = (1, -5, 4)$ and $P_2 = (4, -1, -1)$.

Hint

$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$

Answer

$$5\sqrt{2}$$

Before moving on to the next section, let's get a feel for how $\mathbb{R}^3$ differs from $\mathbb{R}^2$. For example, in $\mathbb{R}^2$, lines that are not parallel must always intersect. This is not the case in $\mathbb{R}^3$. For example, consider the lines shown in Figure 12.2.8. These two lines are not parallel, nor do they intersect.

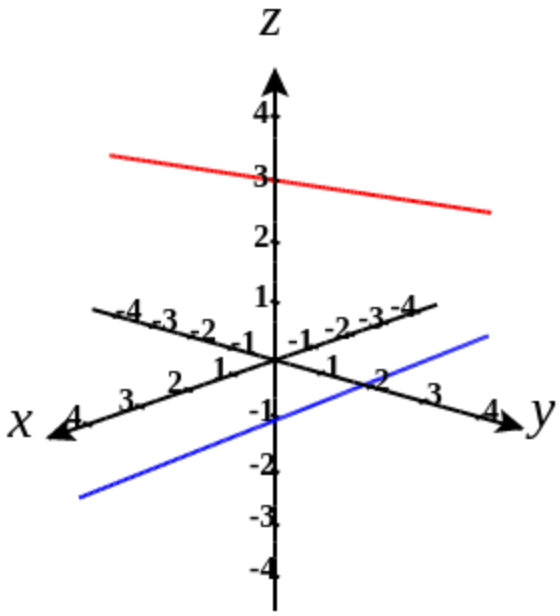


Figure 12.2.8: These two lines are not parallel, but still do not intersect.

You can also have circles that are interconnected but have no points in common, as in Figure 12.2.9.

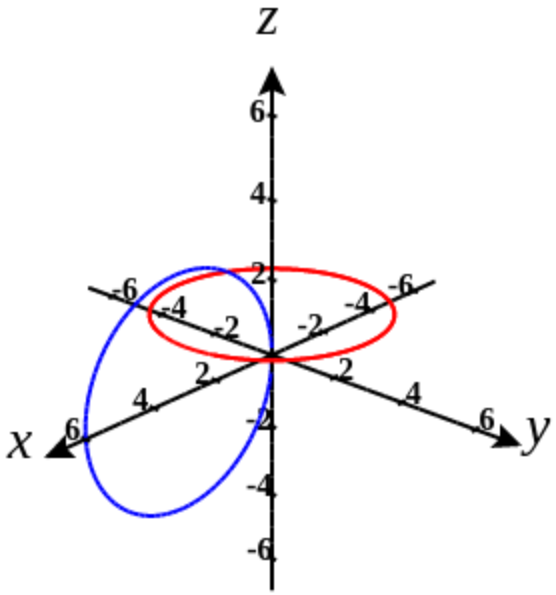


Figure 12.2.9: These circles are interconnected, but have no points in common.

We have a lot more flexibility working in three dimensions than we do if we stuck with only two dimensions.

Writing Equations in $\mathbb{R}^3$

Now that we can represent points in space and find the distance between them, we can learn how to write equations of geometric objects such as lines, planes, and curved surfaces in $\mathbb{R}^3$. First, we start with a simple equation. Compare the graphs of the equation $x = 0$ in $\mathbb{R}$, $\mathbb{R}^2$,and $\mathbb{R}^3$ (Figure 12.2.10). From these graphs, we can see the same equation can describe a point, a line, or a plane.

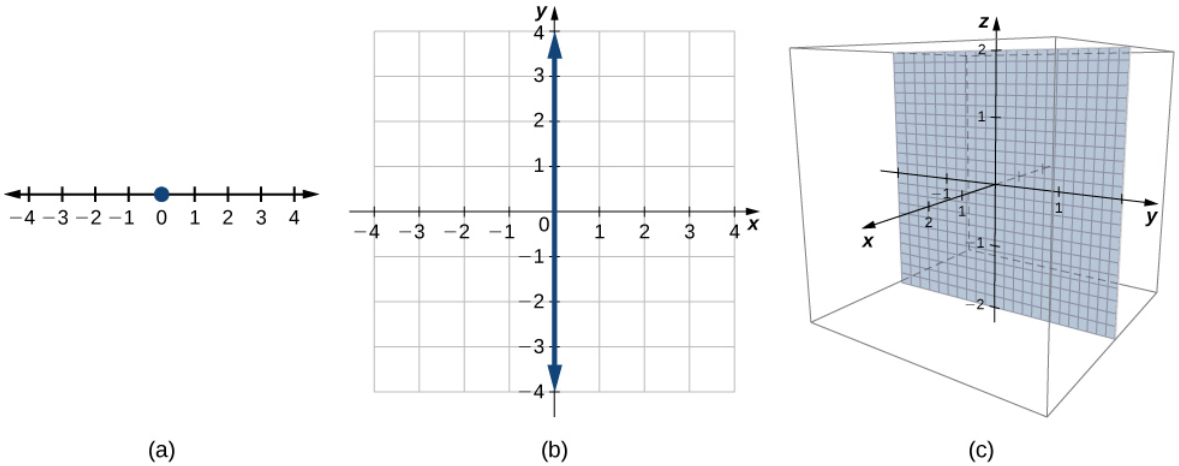


Figure 12.2.10: (a) In $\mathbb{R}$, the equation $x = 0$ describes a single point. (b) In $\mathbb{R}^2$, the equation $x = 0$ describes a line, the y-axis. (c) In $\mathbb{R}^3$, the equation $x = 0$ describes a plane, the yz-plane.

In space, the equation $x = 0$ describes all points $(0, y, z)$. This equation defines the yz-plane. Similarly, the xy-plane contains all points of the form $(x, y, 0)$. The equation $z = 0$ defines the xy-plane and the equation $y = 0$ describes the xz-plane (Figure 12.2.11).

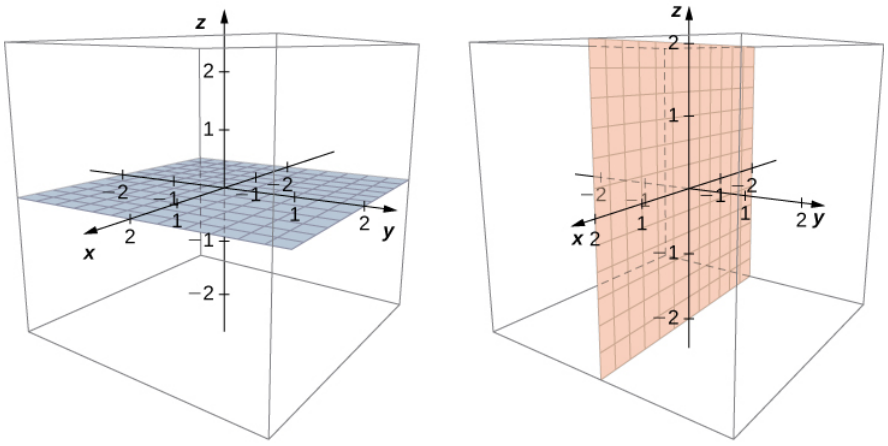


Figure 12.2.11: (a) In space, the equation $z = 0$ describes the xy-plane. (b) All points in the xz-plane satisfy the equation $y = 0$.

Understanding the equations of the coordinate planes allows us to write an equation for any plane that is parallel to one of the coordinate planes. When a plane is parallel to the xy-plane, for example, the z-coordinate of each point in the plane has the same constant value. Only the x- and y-coordinates of points in that plane vary from point to point.

Equations of Planes Parallel to Coordinate Planes

1. The plane in space that is parallel to the xy-plane and contains point (a, b, c) can be represented by the equation $z = c$.
2. The plane in space that is parallel to the xz-plane and contains point (a, b, c) can be represented by the equation $y = b$.
3. The plane in space that is parallel to the yz-plane and contains point (a, b, c) can be represented by the equation $x = a$.

Example 12.2.3: Writing Equations of Planes Parallel to Coordinate Planes

a. Write an equation of the plane passing through point $(3, 11, 7)$ that is parallel to the yz-plane.
b. Find an equation of the plane passing through points $(6, -2, 9), (0, -2, 4)$, and $(1, -2, -3)$.

Solution

a. When a plane is parallel to the yz-plane, only the y- and z-coordinates may vary. The x-coordinate has the same constant value for all points in this plane, so this plane can be represented by the equation $x = 3$.
b. Each of the points $(6, -2, 9), (0, -2, 4)$, and $(1, -2, -3)$ has the same y-coordinate. This plane can be represented by the equation $y = -2$.

Exercise 12.2.3

Write an equation of the plane passing through point $(1, -6, -4)$ that is parallel to the xy-plane.

Hint

If a plane is parallel to the xy-plane, the z-coordinates of the points in that plane do not vary.

Answer

$$z = -4$$

As we have seen, in $\mathbb{R}^2$ the equation $x = 5$ describes the vertical line passing through point $(5, 0)$. This line is parallel to the y-axis. In a natural extension, the equation $x = 5$ in $\mathbb{R}^3$ describes the plane passing through point $(5, 0, 0)$, which is parallel to the yz-plane. Another natural extension of a familiar equation is found in the equation of a sphere.

Definition: Sphere

A sphere is the set of all points in space equidistant from a fixed point, the center of the sphere (Figure 12.2.12), just as the set of all points in a plane that are equidistant from the center represents a circle. In a sphere, as in a circle, the distance from the center to a point on the sphere is called the *radius*.

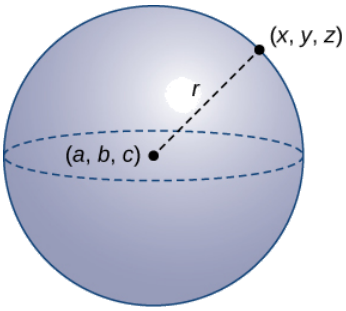


Figure 12.2.12: Each point (x, y, z) on the surface of a sphere is r units away from the center (a, b, c).

The equation of a circle is derived using the distance formula in two dimensions. In the same way, the equation of a sphere is based on the three-dimensional formula for distance.

Standard Equation of a Sphere

The sphere with center (a, b, c) and radius r can be represented by the equation

$$(x-a)^2+(y-b)^2+(z-c)^2=r^2.$$

This equation is known as the *standard equation of a sphere*.

Example 12.2.4: Finding an Equation of a Sphere

Find the standard equation of the sphere with center $(10, 7, 4)$ and point $(-1, 3, -2)$, as shown in Figure 12.2.13.

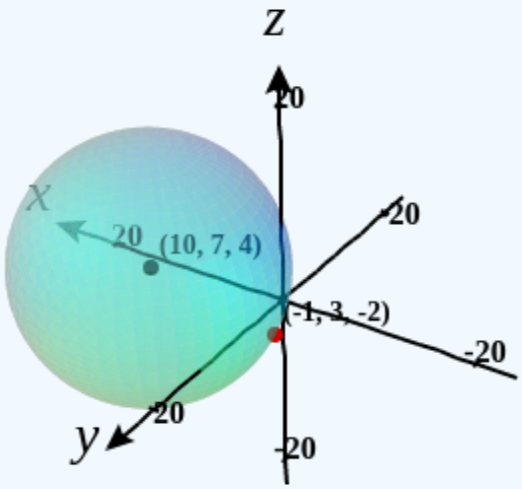


Figure 12.2.13: The sphere centered at $(10, 7, 4)$ containing point $(-1, 3, -2)$.

Solution

Use the distance formula to find the radius r of the sphere:

$$\begin{aligned} r &= \sqrt{(-1-10)^2 + (3-7)^2 + (-2-4)^2} \\ &= \sqrt{(-11)^2 + (-4)^2 + (-6)^2} \\ &= \sqrt{173} \end{aligned}$$

The standard equation of the sphere is

$$(x-10)^2 + (y-7)^2 + (z-4)^2 = 173.$$

? Exercise 12.2.4

Find the standard equation of the sphere with center $(-2, 4, -5)$ containing point $(4, 4, -1)$.

Hint

First use the distance formula to find the radius of the sphere.

Answer

$$(x+2)^2 + (y-4)^2 + (z+5)^2 = 52$$

Example 12.2.5: Finding the Equation of a Sphere

Let $P = (-5, 2, 3)$ and $Q = (3, 4, -1)$, and suppose line segment $\overline{PQ}$ forms the diameter of a sphere (Figure 12.2.14). Find the equation of the sphere.

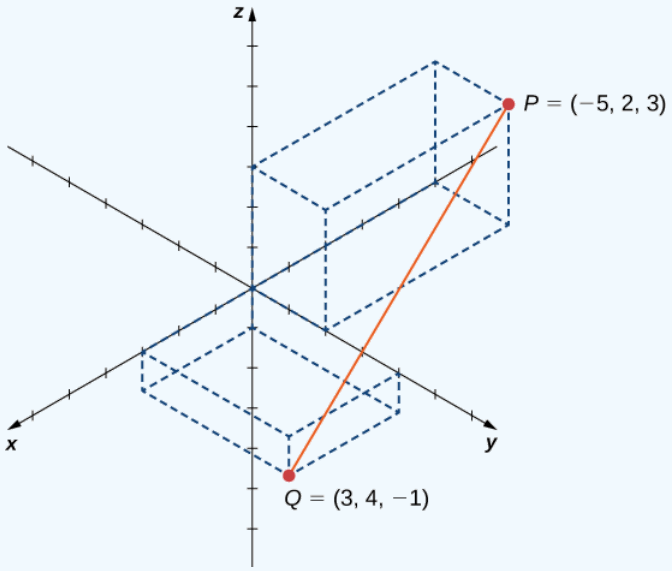


Figure 12.2.14: Line segment $\overline{PQ}$.

Solution:

Since $\overline{PQ}$ is a diameter of the sphere, we know the center of the sphere is the midpoint of $\overline{PQ}$.Then,

$$C = \left(\frac{-5+3}{2}, \frac{2+4}{2}, \frac{3+(-1)}{2}\right) = (-1, 3, 1).$$

Furthermore, we know the radius of the sphere is half the length of the diameter. This gives

$$\begin{aligned} r &= \frac{1}{2}\sqrt{(-5-3)^2 + (2-4)^2 + (3-(-1))^2} \\ &= \frac{1}{2}\sqrt{64+4+16} \\ &= \sqrt{21} \end{aligned}$$

Then, the equation of the sphere is $(x+1)^2 + (y-3)^2 + (z-1)^2 = 21$.

Exercise 12.2.5

Find the equation of the sphere with diameter $\overline{PQ}$, where $P = (2, -1, -3)$ and $Q = (-2, 5, -1)$.

Hint

Find the midpoint of the diameter first.

Answer

$$x^2 + (y-2)^2 + (z+2)^2 = 14$$

Example 12.2.6: Graphing Other Equations in Three Dimensions

Describe the set of points that satisfies $(x-4)(z-2) = 0$, and graph the set.

Solution

We must have either $x - 4 = 0$ or $z - 2 = 0$, so the set of points forms the two planes $x = 4$ and $z = 2$ (Figure 12.2.15).

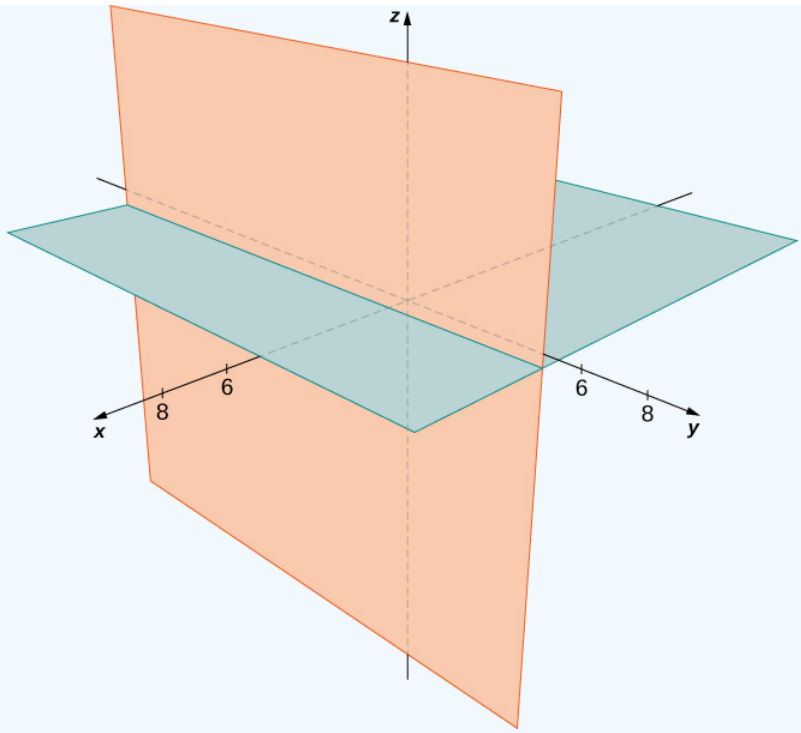


Figure 12.2.15: The set of points satisfying $(x-4)(z-2)=0$ forms the two planes $x=4$ and $z=2$.

Exercise 12.2.6

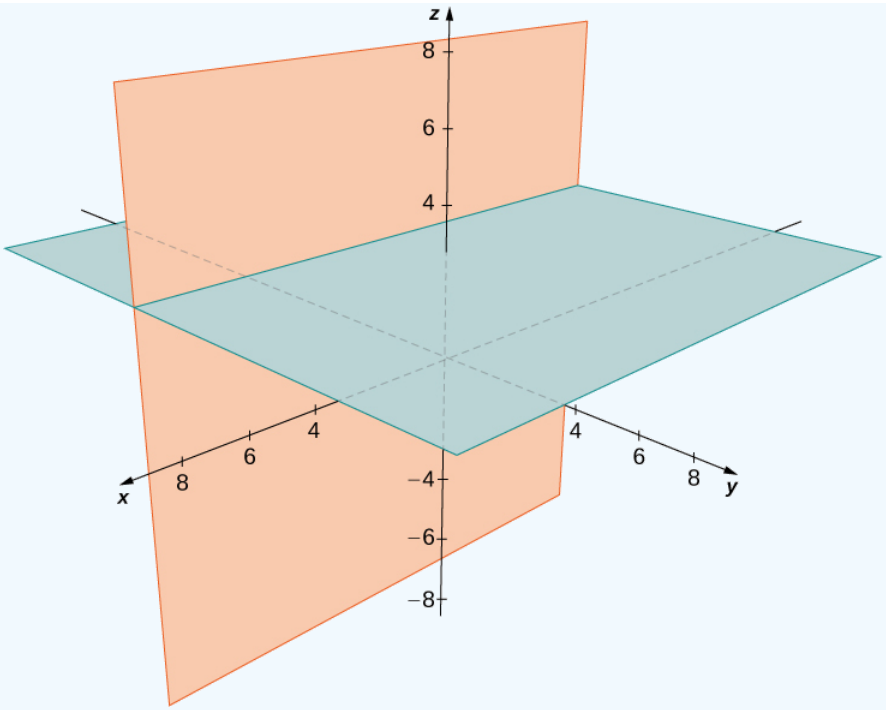
Describe the set of points that satisfies $(y+2)(z-3)=0$, and graph the set.

Hint

One of the factors must be zero.

Answer

The set of points forms the two planes $y=-2$ and $z=3$.



Example 12.2.7: Graphing Other Equations in Three Dimensions

Describe the set of points in three-dimensional space that satisfies $(x-2)^2+(y-1)^2=4$, and graph the set.

Solution

The x- and y-coordinates form a circle in the xy-plane of radius 2, centered at $(2,1)$. Since there is no restriction on the z-coordinate, the three-dimensional result is a circular cylinder of radius 2 centered on the line with $x=2$ and $y=1$. The cylinder extends indefinitely in the z-direction (Figure 12.2.16).

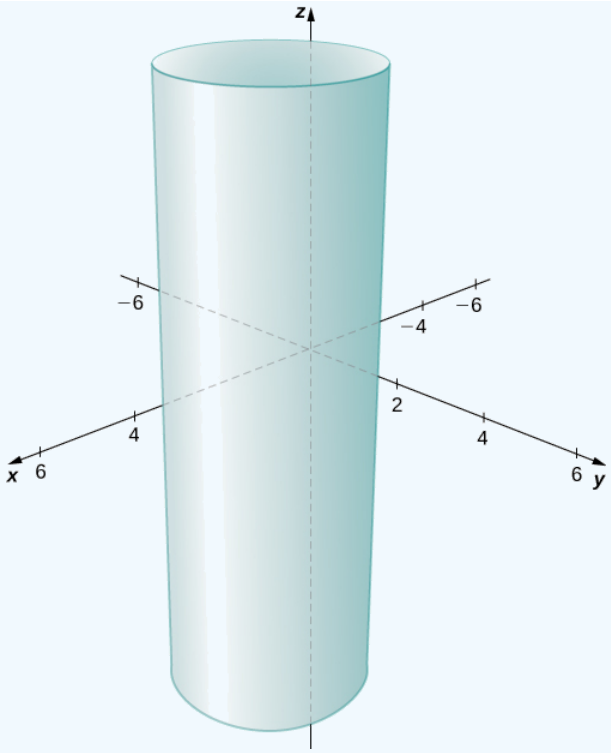


Figure 12.2.16: The set of points satisfying $(x-2)^2+(y-1)^2=4$. This is a cylinder of radius 2 centered on the line with $x=2$ and $y=1$.

? Exercise 12.2.7

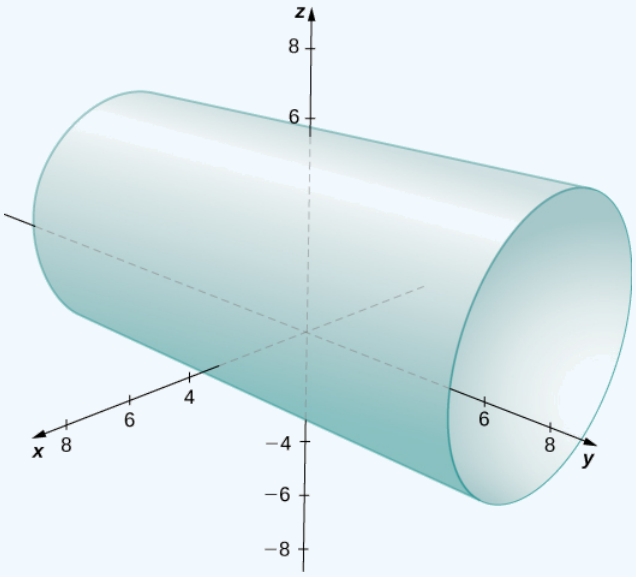
Describe the set of points in three dimensional space that satisfies $x^2+(z-2)^2=16$, and graph the surface.

Hint

Think about what happens if you plot this equation in two dimensions in the xz-plane.

Answer

A cylinder of radius 4 centered on the line with $x=0$ and $z=2$.



https://math.libretexts.org/@go/page/2587

Working with Vectors in $\mathbb{R}^3$

Just like two-dimensional vectors, three-dimensional vectors are quantities with both magnitude and direction, and they are represented by directed line segments (arrows). With a three-dimensional vector, we use a three-dimensional arrow.

Three-dimensional vectors can also be represented in component form. The notation $\vec{\mathbf{v}} = \langle x, y, z \rangle$ is a natural extension of the two-dimensional case, representing a vector with the initial point at the origin, $(0, 0, 0)$, and terminal point (x, y, z). The zero vector is $\vec{\mathbf{0}} = \langle 0, 0, 0 \rangle$. So, for example, the three dimensional vector $\vec{\mathbf{v}} = \langle 2, 4, 1 \rangle$ is represented by a directed line segment from point $(0, 0, 0)$ to point $(2, 4, 1)$ (Figure 12.2.17).

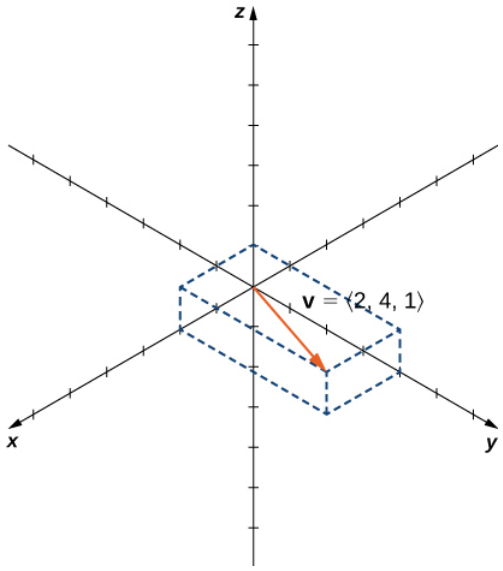


Figure 12.2.17: Vector $\vec{\mathbf{v}} = \langle 2, 4, 1 \rangle$ is represented by a directed line segment from point $(0, 0, 0)$ to point $(2, 4, 1)$.

Vector addition and scalar multiplication are defined analogously to the two-dimensional case. If $\vec{\mathbf{v}} = \langle x_1, y_1, z_1 \rangle$ and $\vec{\mathbf{w}} = \langle x_2, y_2, z_2 \rangle$ are vectors, and k is a scalar, then

$$\vec{\mathbf{v}} + \vec{\mathbf{w}} = \langle x_1 + x_2, y_1 + y_2, z_1 + z_2 \rangle$$

and

$$k\vec{\mathbf{v}} = \langle kx_1, ky_1, kz_1 \rangle.$$

If $k = -1$, then $k\vec{\mathbf{v}} = (-1)\vec{\mathbf{v}}$ is written as $-\vec{\mathbf{v}}$, and vector subtraction is defined by $\vec{\mathbf{v}} - \vec{\mathbf{w}} = \vec{\mathbf{v}} + (-\vec{\mathbf{w}}) = \vec{\mathbf{v}} + (-1)\vec{\mathbf{w}}$.

The standard unit vectors extend easily into three dimensions as well, $\hat{\mathbf{i}} = \langle 1, 0, 0 \rangle$, $\hat{\mathbf{j}} = \langle 0, 1, 0 \rangle$, and $\hat{\mathbf{k}} = \langle 0, 0, 1 \rangle$, and we use them in the same way we used the standard unit vectors in two dimensions. Thus, we can represent a vector in $\mathbb{R}^3$ in the following ways:

$$\vec{\mathbf{v}} = \langle x, y, z \rangle = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$$

.

Example 12.2.8: Vector Representations

Let $\overrightarrow{PQ}$ be the vector with initial point $P = (3, 12, 6)$ and terminal point $Q = (-4, -3, 2)$ as shown in Figure 12.2.18. Express $\overrightarrow{PQ}$ in both component form and using standard unit vectors.

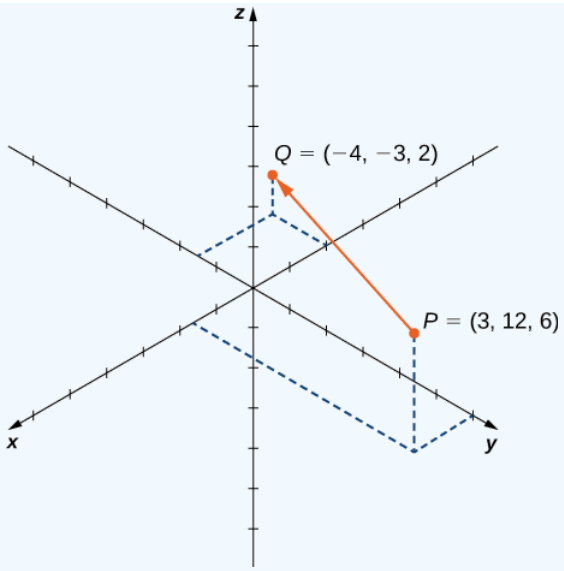


Figure 12.2.18: The vector with initial point $P = (3, 12, 6)$ and terminal point $Q = (-4, -3, 2)$.

Solution

In component form,

$$\begin{aligned}\overrightarrow{PQ} &= \langle x_2 - x_1, y_2 - y_1, z_2 - z_1\rangle \\ &= \langle -4-3, -3-12, 2-6\rangle \\ &= \langle -7, -15, -4\rangle.\end{aligned}$$

In standard unit form,

$$\overrightarrow{PQ} = -7\hat{\mathbf{i}} - 15\hat{\mathbf{j}} - 4\hat{\mathbf{k}}.$$

? Exercise 12.2.8

Let $S = (3, 8, 2)$ and $T = (2, -1, 3)$. Express $\vec{ST}$ in component form and in standard unit form.

Hint

Write $\overrightarrow{ST}$ in component form first. T is the terminal point of $\overrightarrow{ST}$.

Answer

$\overrightarrow{ST} = \langle -1, -9, 1\rangle = -\hat{\mathbf{i}} - 9\hat{\mathbf{j}} + \hat{\mathbf{k}}$

As described earlier, vectors in three dimensions behave in the same way as vectors in a plane. The geometric interpretation of vector addition, for example, is the same in both two- and three-dimensional space (Figure 12.2.19).

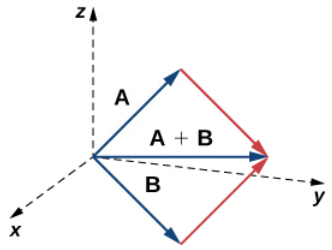


Figure 12.2.19: To add vectors in three dimensions, we follow the same procedures we learned for two dimensions.

We have already seen how some of the algebraic properties of vectors, such as vector addition and scalar multiplication, can be extended to three dimensions. Other properties can be extended in similar fashion. They are summarized here for our reference.

Properties of Vectors in Space

Let $\vec{\mathbf{v}} = \langle x_1, y_1, z_1 \rangle$ and $\vec{\mathbf{w}} = \langle x_2, y_2, z_2 \rangle$ be vectors, and let k be a scalar.

- **Scalar multiplication**:

$$k\vec{\mathbf{v}} = \langle kx_1, ky_1, kz_1 \rangle$$

- **Vector addition**:

$$\vec{\mathbf{v}} + \vec{\mathbf{w}} = \langle x_1, y_1, z_1 \rangle + \langle x_2, y_2, z_2 \rangle = \langle x_1 + x_2, y_1 + y_2, z_1 + z_2 \rangle$$

- **Vector subtraction**:

$$\vec{\mathbf{v}} - \vec{\mathbf{w}} = \langle x_1, y_1, z_1 \rangle - \langle x_2, y_2, z_2 \rangle = \langle x_1 - x_2, y_1 - y_2, z_1 - z_2 \rangle$$

- **Vector magnitude**:

$$\|\vec{\mathbf{v}}\| = \sqrt{x_1^2 + y_1^2 + z_1^2}$$

- **Unit vector in the direction of $\vec{\mathbf{v}}$**:

$$\frac{1}{\|\vec{\mathbf{v}}\|}\vec{\mathbf{v}} = \frac{1}{\|\vec{\mathbf{v}}\|}\langle x_1, y_1, z_1 \rangle = \langle \frac{x_1}{\|\vec{\mathbf{v}}\|}, \frac{y_1}{\|\vec{\mathbf{v}}\|}, \frac{z_1}{\|\vec{\mathbf{v}}\|} \rangle, \quad \text{if } \vec{\mathbf{v}} \neq \vec{\mathbf{0}}$$

We have seen that vector addition in two dimensions satisfies the commutative, associative, and additive inverse properties. These properties of vector operations are valid for three-dimensional vectors as well. Scalar multiplication of vectors satisfies the distributive property, and the zero vector acts as an additive identity. The proofs to verify these properties in three dimensions are straightforward extensions of the proofs in two dimensions.

Example 12.2.9: Vector Operations in Three Dimensions

Let $\vec{\mathbf{v}} = \langle -2, 9, 5 \rangle$ and $\vec{\mathbf{w}} = \langle 1, -1, 0 \rangle$ (Figure 12.2.20). Find the following vectors.

a. $3\vec{\mathbf{v}} - 2\vec{\mathbf{w}}$
b. $5\|\vec{\mathbf{w}}\|$
c. $\|5\vec{\mathbf{w}}\|$
d. A unit vector in the direction of $\vec{\mathbf{v}}$

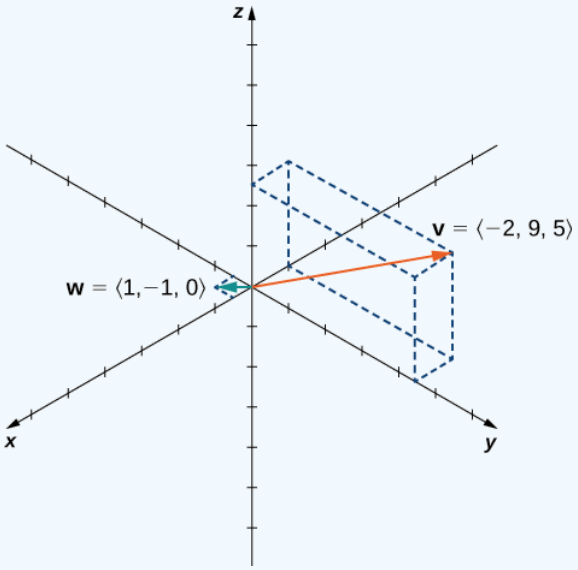


Figure 12.2.20: The vectors $\vec{\mathbf{v}} = \langle -2, 9, 5 \rangle$ and $\vec{\mathbf{w}} = \langle 1, -1, 0 \rangle$.

Solution

a. First, use scalar multiplication of each vector, then subtract:

$$\begin{aligned} 3\vec{\mathbf{v}} - 2\vec{\mathbf{w}} &= 3\langle -2, 9, 5\rangle - 2\langle 1, -1, 0\rangle \\ &= \langle -6, 27, 15\rangle - \langle 2, -2, 0\rangle \\ &= \langle -6-2, 27-(-2), 15-0\rangle \\ &= \langle -8, 29, 15\rangle. \end{aligned}$$

b. Write the equation for the magnitude of the vector, then use scalar multiplication:

$$5\|\vec{\mathbf{w}}\| = 5\sqrt{1^2 + (-1)^2 + 0^2} = 5\sqrt{2}.$$

c. First, use scalar multiplication, then find the magnitude of the new vector. Note that the result is the same as for part b.:

$$\|5\vec{\mathbf{w}}\| = \| \langle 5, -5, 0\rangle \| = \sqrt{5^2 + (-5)^2 + 0^2} = \sqrt{50} = 5\sqrt{2}$$

d. Recall that to find a unit vector in two dimensions, we divide a vector by its magnitude. The procedure is the same in three dimensions:

$$\begin{aligned} \frac{\vec{\mathbf{v}}}{\|\vec{\mathbf{v}}\|} &= \frac{1}{\|\vec{\mathbf{v}}\|}\langle -2, 9, 5\rangle \\ &= \frac{1}{\sqrt{(-2)^2 + 9^2 + 5^2}}\langle -2, 9, 5\rangle \\ &= \frac{1}{\sqrt{110}}\langle -2, 9, 5\rangle \\ &= \left\langle \frac{-2}{\sqrt{110}}, \frac{9}{\sqrt{110}}, \frac{5}{\sqrt{110}}\right\rangle. \end{aligned}$$

? Exercise 12.2.9:

Let $\vec{\mathbf{v}} = \langle -1, -1, 1\rangle$ and $\vec{\mathbf{w}} = \langle 2, 0, 1\rangle$. Find a unit vector in the direction of $5\vec{\mathbf{v}} + 3\vec{\mathbf{w}}$.

Hint

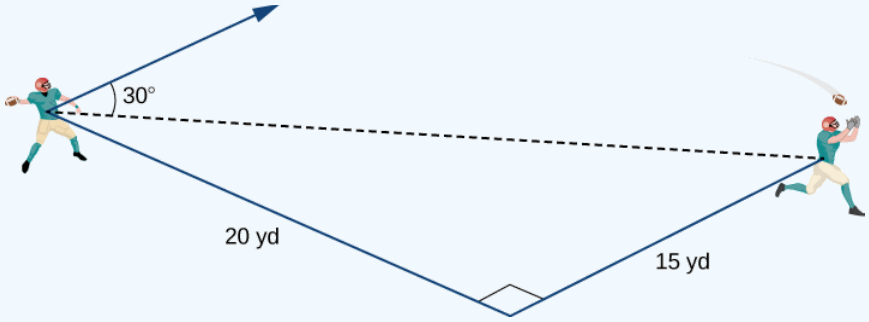
Start by writing $5\vec{\mathbf{v}} + 3\vec{\mathbf{w}}$ in component form.

Answer

$$\left\langle \frac{1}{3\sqrt{10}}, -\frac{5}{3\sqrt{10}}, \frac{8}{3\sqrt{10}}\right\rangle$$

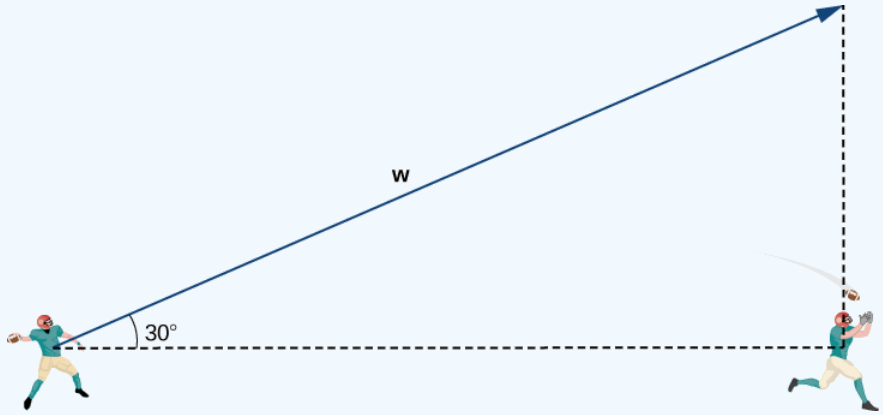
✓ Example 12.2.10: Throwing a Forward Pass

A quarterback is standing on the football field preparing to throw a pass. His receiver is standing 20 yd down the field and 15 yd to the quarterback's left. The quarterback throws the ball at a velocity of 60 mph toward the receiver at an upward angle of $30°$ (see the following figure). Write the initial velocity vector of the ball, $\vec{\mathbf{v}}$, in component form.



Solution

The first thing we want to do is find a vector in the same direction as the velocity vector of the ball. We then scale the vector appropriately so that it has the right magnitude. Consider the vector $\vec{w}$ extending from the quarterback's arm to a point directly above the receiver's head at an angle of $30°$ (see the following figure). This vector would have the same direction as $\vec{v}$, but it may not have the right magnitude.



The receiver is 20 yd down the field and 15 yd to the quarterback's left. Therefore, the straight-line distance from the quarterback to the receiver is

$$\textbf{Dist from QB to receiver} = \sqrt{15^2+20^2} = \sqrt{225+400} = \sqrt{625} = 25 \text{ yd}.$$

We have $\dfrac{25}{\|\vec{w}\|} = \cos 30°$. Then the magnitude of $\vec{w}$ is given by

$$\|\vec{w}\| = \frac{25}{\cos 30°} = \frac{25 \cdot 2}{\sqrt{3}} = \frac{50}{\sqrt{3}} \text{ yd}$$

and the vertical distance from the receiver to the terminal point of $\vec{w}$ is

$$\textbf{Vert dist from receiver to terminal point of } \vec{w} = \|\vec{w}\| \sin 30° = \frac{50}{\sqrt{3}} \cdot \frac{1}{2} = \frac{25}{\sqrt{3}} \text{ yd}.$$

Then $\vec{w} = \langle 20, 15, \dfrac{25}{\sqrt{3}} \rangle$, and has the same direction as $\vec{v}$.

Recall, though, that we calculated the magnitude of $\vec{w}$ to be $\|\vec{w}\| = \dfrac{50}{\sqrt{3}}$ yd, and $\vec{v}$ has magnitude 60 mph. So, we need to multiply vector $\vec{w}$ by an appropriate constant, k. We want to find a value of k so that $\| k\vec{w} \| = 60$ mph*. We have

$$\|k\vec{w}\| = k\|\vec{w}\| = k\frac{50}{\sqrt{3}} \text{ yd},$$

so we want

$$k\left(\frac{50}{\sqrt{3}} \text{ yd}\right) = 60 \text{ mph}$$

$$k = \frac{60\sqrt{3}}{50} \text{ mph / yd}$$

$$k = \frac{6\sqrt{3}}{5} \text{ mph / yd}.$$

Then

$$\vec{v} = k\vec{w} = k\langle 20, 15, \frac{25}{\sqrt{3}} \rangle = \frac{6\sqrt{3}}{5} \langle 20, 15, \frac{25}{\sqrt{3}} \rangle = \langle 24\sqrt{3}, 18\sqrt{3}, 30 \rangle.$$

Let's double-check that $\|\vec{v}\| = 60$ mph. We have

$$\|\vec{v}\| = \sqrt{(24\sqrt{3})^2 + (18\sqrt{3})^2 + (30)^2} = \sqrt{1728 + 972 + 900} = \sqrt{3600} = 60 \text{ mph}.$$

So, we have found the correct components for $\vec{v}$.

Note *

Readers who have been watching the units of measurement may be wondering what exactly is going on at this point: haven't we just mixed yards and miles per hour? We haven't, but the reason is subtle. One way to understand it is to realize that there are really two parallel coordinate systems in this problem: one gives positions down the field, across the field, and up into the air in units of yards; the other gives speeds down the field, across the field, and up into the air in units of miles per hour. The vector $\vec{\mathbf{w}}$ is calculated in the position coordinate system; vector $\vec{\mathbf{v}}$ will be in the speed system. Because corresponding axes in each system are parallel, directions in the two systems are also parallel, so the claim that $\vec{\mathbf{w}}$ and $\vec{\mathbf{v}}$ point in the same direction is correct. The constant k that we're looking for is a conversion factor between the magnitudes of these two vectors, converting from the position system to the speed one in the process. And as seen above, our calculation of k produces the right units for such a conversion, namely miles per hour per yard.

? Exercise 12.2.10

Assume the quarterback and the receiver are in the same place as in the previous example. This time, however, the quarterback throws the ball at velocity of 40 mph and an angle of $45°$. Write the initial velocity vector of the ball, $\vec{\mathbf{v}}$, in component form.

Hint

Follow the process used in the previous example.

Answer

$v = \langle 16\sqrt{2}, 12\sqrt{2}, 20\sqrt{2} \rangle$

Key Concepts

- The three-dimensional coordinate system is built around a set of three axes that intersect at right angles at a single point, the origin. Ordered triples (x, y, z) are used to describe the location of a point in space.
- The distance d between points (x_1, y_1, z_1) and (x_2, y_2, z_2) is given by the formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$

- In three dimensions, the equations $x = a,\ y = b$, and $z = c$ describe planes that are parallel to the coordinate planes.
- The standard equation of a sphere with center (a, b, c) and radius r is

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = r^2.$$

- In three dimensions, as in two, vectors are commonly expressed in component form, $\vec{\mathbf{v}} = \langle x, y, z \rangle$, or in terms of the standard unit vectors, $\vec{\mathbf{v}} = x\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}} + z\,\hat{\mathbf{k}}$.
- Properties of vectors in space are a natural extension of the properties for vectors in a plane. Let $\vec{\mathbf{v}} = \langle x_1, y_1, z_1 \rangle$ and $\vec{\mathbf{w}} = \langle x_2, y_2, z_2 \rangle$ be vectors, and let k be a scalar.

  Scalar multiplication:

$$k\vec{\mathbf{v}} = \langle kx_1, ky_1, kz_1 \rangle$$

  Vector addition:

$$\vec{\mathbf{v}} + \vec{\mathbf{w}} = \langle x_1, y_1, z_1 \rangle + \langle x_2, y_2, z_2 \rangle = \langle x_1 + x_2, y_1 + y_2, z_1 + z_2 \rangle$$

  Vector subtraction:

$$\vec{\mathbf{v}} - \vec{\mathbf{w}} = \langle x_1, y_1, z_1 \rangle - \langle x_2, y_2, z_2 \rangle = \langle x_1 - x_2, y_1 - y_2, z_1 - z_2 \rangle$$

  Vector magnitude:

$$\|\vec{\mathbf{v}}\| = \sqrt{x_1^2 + y_1^2 + z_1^2}$$

Unit vector in the direction of $\vec{v}$:

$$\frac{\vec{\mathbf{v}}}{\|\vec{\mathbf{v}}\|} = \frac{1}{\|\vec{\mathbf{v}}\|}\langle x_1, y_1, z_1\rangle = \langle \frac{x_1}{\|\vec{\mathbf{v}}\|}, \frac{y_1}{\|\vec{\mathbf{v}}\|}, \frac{z_1}{\|\vec{\mathbf{v}}\|}\rangle, \ \vec{\mathbf{v}} \neq \vec{\mathbf{0}}$$

Key Equations

Distance between two points in space:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Sphere with center (a, b, c) and radius r:

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = r^2$$

Glossary

coordinate plane
a plane containing two of the three coordinate axes in the three-dimensional coordinate system, named by the axes it contains: the xy-plane, xz-plane, or the yz-plane

right-hand rule
a common way to define the orientation of the three-dimensional coordinate system; when the right hand is curved around the z-axis in such a way that the fingers curl from the positive x-axis to the positive y-axis, the thumb points in the direction of the positive z-axis

octants
the eight regions of space created by the coordinate planes

sphere
the set of all points equidistant from a given point known as the *center*

standard equation of a sphere
$(x - a)^2 + (y - b)^2 + (z - c)^2 = r^2$ describes a sphere with center (a, b, c) and radius r

three-dimensional rectangular coordinate system
a coordinate system defined by three lines that intersect at right angles; every point in space is described by an ordered triple (x, y, z) that plots its location relative to the defining axes

Contributors

Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.

Example 12.2.10 has been modified by Doug Baldwin and Paul Seeburger to clarify the units of measurement that it uses and how it uses them.

Paul Seeburger also created dynamic versions of Figures 12.2.8, 12.2.9 and 12.2.13 using CalcPlot3D.

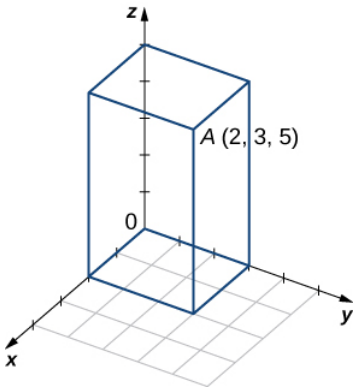
This page titled 12.2: Vectors in Three Dimensions is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

12.2E: Exercises for Section 12.2

1) Consider a rectangular box with one of the vertices at the origin, as shown in the following figure. If point $A(2,3,5)$ is the opposite vertex to the origin, then find

a. the coordinates of the other six vertices of the box and

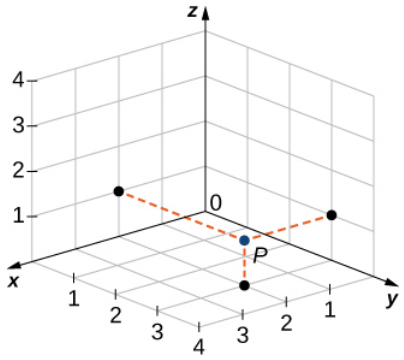
b. the length of the diagonal of the box determined by the vertices O and A.



Answer

a. $(2,0,5),(2,0,0),(2,3,0),(0,3,0),(0,3,5),(0,0,5)$ b. $\sqrt{38}$

2) Find the coordinates of point P and determine its distance to the origin.

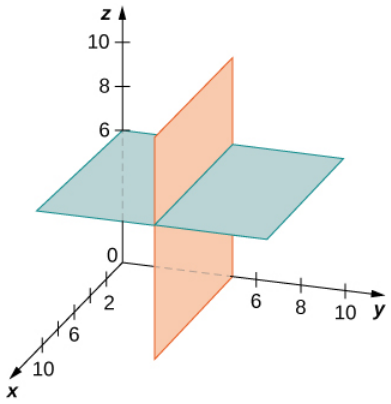


For exercises 3-6, describe and graph the set of points that satisfies the given equation.

3) $(y-5)(z-6)=0$

Answer

A union of two planes: $y=5$ (a plane parallel to the xz-plane) and $z=6$ (a plane parallel to the xy-plane)

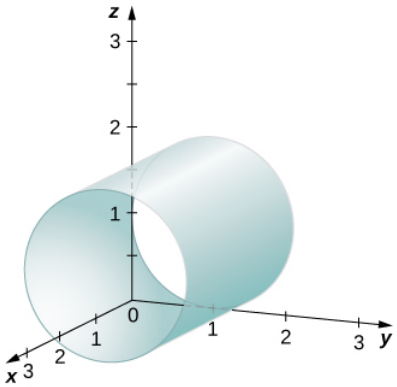


4) $(z-2)(z-5)=0$

5) $(y-1)^2+(z-1)^2=1$

Answer

A cylinder of radius 1 centered on the line $y=1, z=1$



6) $(x-2)^2+(z-5)^2=4$

7) Write the equation of the plane passing through point $(1,1,1)$ that is parallel to the xy-plane.

Answer

$z=1$

8) Write the equation of the plane passing through point $(1,-3,2)$ that is parallel to the xz-plane.

9) Find an equation of the plane passing through points $(1,-3,-2), (0,3,-2)$, and $(1,0,-2)$.

Answer

$z=-2$

10) Find an equation of the plane passing through points $(1,9,2), (1,3,6)$, and $(1,-7,8)$.

For exercises 11-14, find the equation of the sphere in standard form that satisfies the given conditions.

11) Center $C(-1,7,4)$ and radius 4

Answer

$(x+1)^2+(y-7)^2+(z-4)^2=16$

12) Center $C(-4,7,2)$ and radius 6

13) Diameter PQ, where $P(-1,5,7)$ and $Q(-5,2,9)$

Answer

$x+3)^2+(y-3.5)^2+(z-8)^2=\dfrac{29}{4}$

14) Diameter PQ, where $P(-16,-3,9)$ and $Q(-2,3,5)$

For exercises 15 and 16, find the center and radius of the sphere with an equation in general form that is given.

15) $x^2+y^2+z^2-4z+3=0$

Answer

Center $C(0,0,2)$ and radius 1

16) $x^2+y^2+z^2-6x+8y-10z+25=0$

For exercises 17-20, express vector $\overrightarrow{PQ}$ with the initial point at P and the terminal point at Q

a. in component form and

b. by using standard unit vectors.

17) $P(3,0,2)$ and $Q(-1,-1,4)$

Answer

$a.\overrightarrow{PQ}=\langle -4,-1,2\rangle$
$b.\overrightarrow{PQ}=-4\hat{\mathbf{i}}-\hat{\mathbf{j}}+2\hat{\mathbf{k}}$

18) $P(0,10,5)$ and $Q(1,1,-3)$

19) $P(-2,5,-8)$ and $M(1,-7,4)$, where M is the midpoint of the line segment $\overline{PQ}$

Answer

$a.\overrightarrow{PQ}=\langle 6,-24,24\rangle$
$b.\overrightarrow{PQ}=6\hat{\mathbf{i}}-24\hat{\mathbf{j}}+24\hat{\mathbf{k}}$

20) $Q(0,7,-6)$ and $M(-1,3,2)$, where M is the midpoint of the line segment $\overline{PQ}$

21) Find terminal point Q of vector $\overrightarrow{PQ}=\langle 7,-1,3\rangle$ with the initial point at $P(-2,3,5)$.

Answer

$Q(5,2,8)$

22) Find initial point P of vector $\overrightarrow{PQ}=\langle -9,1,2\rangle$ with the terminal point at $Q(10,0,-1)$.

For exercises 23-26, use the given vectors $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ to find and express the vectors $\vec{\mathbf{a}}+\vec{\mathbf{b}}$, $4\vec{\mathbf{a}}$, and $-5\vec{\mathbf{a}}+3\vec{\mathbf{b}}$ in component form.

23) $\vec{\mathbf{a}}=\langle -1,-2,4\rangle, \quad \vec{\mathbf{b}}=\langle -5,6,-7\rangle$

Answer

$\vec{\mathbf{a}}+\vec{\mathbf{b}}=\langle -6,4,-3\rangle, 4\vec{\mathbf{a}}=\langle -4,-8,16\rangle, -5\vec{\mathbf{a}}+3\vec{\mathbf{b}}=\langle -10,28,-41\rangle$

24) $\vec{\mathbf{a}}=\langle 3,-2,4\rangle, \quad \vec{\mathbf{b}}=\langle -5,6,-9\rangle$

25) $\vec{\mathbf{a}}=-\hat{\mathbf{k}}, \quad \vec{\mathbf{b}}=-\hat{\mathbf{i}}$

Answer

$\vec{\mathbf{a}}+\vec{\mathbf{b}}=\langle -1,0,-1\rangle, 4\vec{\mathbf{a}}=\langle 0,0,-4\rangle, -5\vec{\mathbf{a}}+3\vec{\mathbf{b}}=\langle -3,0,5\rangle$

26) $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \quad \vec{b} = 2\hat{i} - 3\hat{j} + 2\hat{k}$

For exercises 27-30, vectors $\vec{u}$ and $\vec{v}$ are given. Find the magnitudes of vectors $\vec{u} - \vec{v}$ and $-2\vec{u}$.

27) $\vec{u} = 2\hat{i} + 3\hat{j} + 4\hat{k}, \quad \vec{v} = -\hat{i} + 5\hat{j} - \hat{k}$

Answer

$\|\vec{u} - \vec{v}\| = \sqrt{38}, \quad \|-2\vec{u}\| = 2\sqrt{29}$

28) $\vec{u} = \hat{i} + \hat{j}, \quad \vec{v} = \hat{j} - \hat{k}$

29) $\vec{u} = \langle 2\cos t, -2\sin t, 3\rangle, \quad \vec{v} = \langle 0, 0, 3\rangle, \quad$ where t is a real number.

Answer

$\|\vec{u} - \vec{v}\| = 2, \quad \|-2\vec{u}\| = 2\sqrt{13}$

30) $\vec{u} = \langle 0, 1, \sinh t\rangle, \quad \vec{v} = \langle 1, 1, 0\rangle, \quad$ where t is a real number.

For exercises 31-36, find the unit vector in the direction of the given vector $\vec{a}$ and express it using standard unit vectors.

31) $\vec{a} = 3\hat{i} - 4\hat{j}$

Answer

$\frac{3}{5}\hat{i} - \frac{4}{5}\hat{j}$

32) $\vec{a} = \langle 4, -3, 6\rangle$

33) $\vec{a} = \overrightarrow{PQ}$, where $P(-2, 3, 1)$ and $Q(0, -4, 4)$

Answer

$\frac{\sqrt{62}}{31}\hat{i} - \frac{7\sqrt{62}}{62}\hat{j} + \frac{3\sqrt{62}}{62}\hat{k}$

34) $\vec{a} = \overrightarrow{OP}$, where $P(-1, -1, 1)$

35) $\vec{a} = \vec{u} - \vec{v} + \vec{w}$, where $\vec{u} = \hat{i} - \hat{j} - \hat{k}, \quad \vec{v} = 2\hat{i} - \hat{j} + \hat{k}, \quad$ and $\vec{w} = -\hat{i} + \hat{j} + 3\hat{k}$

Answer

$-\frac{\sqrt{6}}{3}\hat{i} + \frac{\sqrt{6}}{6}\hat{j} + \frac{\sqrt{6}}{6}\hat{k}$

36) $\vec{a} = 2\vec{u} + \vec{v} - \vec{w}, \quad$ where $\vec{u} = \hat{i} - \hat{k}, \quad \vec{v} = 2\hat{j} \quad$, and $\vec{w} = \hat{i} - \hat{j}$

37) Determine whether $\overrightarrow{AB}$ and $\overrightarrow{PQ}$ are equivalent vectors, where $A(1, 1, 1)$, $B(3, 3, 3)$, $P(1, 4, 5)$, and $Q(3, 6, 7)$.

Answer

Equivalent vectors

38) Determine whether the vectors $\overrightarrow{AB}$ and $\overrightarrow{PQ}$ are equivalent, where $A(1, 4, 1)$, $B(-2, 2, 0)$, $P(2, 5, 7)$, and $Q(-3, 2, 1)$.

For exercises 39-42, find vector $\vec{u}$ with a magnitude that is given and satisfies the given conditions.

39) $\vec{v} = \langle 7, -1, 3\rangle$, $\|\vec{u}\| = 10$, and $\vec{u}$ and $\vec{v}$ have the same direction

Answer

$\vec{u} = \langle \frac{70\sqrt{59}}{59}, -\frac{10\sqrt{59}}{59}, \frac{30\sqrt{59}}{59}\rangle$

40) $\vec{v} = \langle 2, 4, 1\rangle$, $\|\vec{u}\| = 15$, and $\vec{u}$ and $\vec{v}$ have the same direction

41) $\vec{v} = \langle 2\sin t,\ 2\cos t, 1\rangle$, $\|\vec{u}\| = 2$, $\vec{u}$ and $\vec{v}$ have opposite directions for any t, where t is a real number

Answer

$\vec{\mathbf{u}} = \langle -\frac{4\sqrt{5}}{5}\sin t, -\frac{4\sqrt{5}}{5}\cos t, -\frac{2\sqrt{5}}{5}\rangle$

42) $\vec{\mathbf{v}} = \langle 3\sinh t, 0, 3\rangle$, $\|\vec{\mathbf{u}}\| = 5$, and $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ have opposite directions for any t, where t is a real number

43) Determine a vector of magnitude 5 in the direction of vector $\overrightarrow{AB}$, where $A(2, 1, 5)$ and $B(3, 4, -7)$.

Answer

$\langle \frac{5\sqrt{154}}{154}, \frac{15\sqrt{154}}{154}, -\frac{30\sqrt{154}}{77}\rangle$

44) Find a vector of magnitude 2 that points in the opposite direction than vector $\overrightarrow{AB}$, where $A(-1, -1, 1)$ and $B(0, 1, 1)$. Express the answer in component form.

45) Consider the points $A(2, \alpha, 0)$, $B(0, 1, \beta)$, and $C(1, 1, \beta)$, where α and β are negative real numbers. Find α and β such that $\|\overrightarrow{OA} - \overrightarrow{OB} + \overrightarrow{OC}\| = \|\overrightarrow{OB}\| = 4$.

Answer

$\alpha = -\sqrt{7}$, $\beta = -\sqrt{15}$

46) Consider points $A(\alpha, 0, 0)$, $B(0, \beta, 0)$, and $C(\alpha, \beta, \beta)$, where α and β are positive real numbers. Find α and β such that $\|\overrightarrow{OA} + \overrightarrow{OB}\| = \sqrt{2}$ and $\|\overrightarrow{OC}\| = \sqrt{3}$.

47) Let $P(x, y, z)$ be a point situated at an equal distance from points $A(1, -1, 0)$ and $B(-1, 2, 1)$. Show that point P lies on the plane of equation $-2x + 3y + z = 2$.

48) Let $P(x, y, z)$ be a point situated at an equal distance from the origin and point $A(4, 1, 2)$. Show that the coordinates of point P satisfy the equation $8x + 2y + 4z = 21$.

49) The points A, B, and C are collinear (in this order) if the relation $\|\overrightarrow{AB}\| + \|\overrightarrow{BC}\| = \|\overrightarrow{AC}\|$ is satisfied. Show that $A(5, 3, -1)$, $B(-5, -3, 1)$, and $C(-15, -9, 3)$ are collinear points.

50) Show that points $A(1, 0, 1)$, $B(0, 1, 1)$, and $C(1, 1, 1)$ are not collinear.

51) [T] A force $\vec{\mathbf{F}}$ of $50\ N$ acts on a particle in the direction of the vector $\overrightarrow{OP}$, where $P(3, 4, 0)$.

a. Express the force as a vector in component form.

b. Find the angle between force $\vec{\mathbf{F}}$ and the positive direction of the x-axis. Express the answer in degrees rounded to the nearest integer.

Answer

$a. \vec{\mathbf{F}} = \langle 30, 40, 0\rangle$; $\quad b. 53°$

52) [T] A force $\vec{\mathbf{F}}$ of $40\ N$ acts on a box in the direction of the vector $\overrightarrow{OP}$, where $P(1, 0, 2)$.

a. Express the force as a vector by using standard unit vectors.

b. Find the angle between force $\vec{\mathbf{F}}$ and the positive direction of the x-axis.

53) If $\vec{\mathbf{F}}$ is a force that moves an object from point $P_1(x_1, y_1, z_1)$ to another point $P_2(x_2, y_2, z_2)$, then the displacement vector is defined as $\vec{\mathbf{D}} = (x_2 - x_1)\hat{\mathbf{i}} + (y_2 - y_1)\hat{\mathbf{j}} + (z_2 - z_1)\hat{\mathbf{k}}$. A metal container is lifted 10 m vertically by a constant force $\vec{\mathbf{F}}$. Express the displacement vector $\vec{\mathbf{D}}$ by using standard unit vectors.

Answer

$\vec{\mathbf{D}} = 10\hat{\mathbf{k}}$

54) A box is pulled 4 yd horizontally in the x-direction by a constant force $\vec{\mathbf{F}}$. Find the displacement vector in component form.

55) The sum of the forces acting on an object is called the resultant or net force. An object is said to be in static equilibrium if the resultant force of the forces that act on it is zero. Let $\vec{\mathbf{F}}_1 = \langle 10, 6, 3\rangle$, $\vec{\mathbf{F}}_2 = \langle 0, 4, 9\rangle$, and $\vec{\mathbf{F}}_3 = \langle 10, -3, -9\rangle$ be three forces acting on a box. Find the force $\vec{\mathbf{F}}_4$ acting on the box such that the box is in static equilibrium. Express the answer in component form.

Answer

$\vec{\mathbf{F}}_4 = \langle -20, -7, -3 \rangle$

56) [T] Let $\vec{\mathbf{F}}_k = \langle 1, k, k^2 \rangle, k = 1, ..., n$ be n forces acting on a particle, with $n \geq 2$.

a. Find the net force $\vec{\mathbf{F}} = \sum_{k=1}^{n} \vec{\mathbf{F}}_k$. Express the answer using standard unit vectors.

b. Use a computer algebra system (CAS) to find n such that $\|\vec{\mathbf{F}}\| < 100$.

57) The force of gravity $\vec{\mathbf{F}}$ acting on an object is given by $\vec{\mathbf{F}} = m\vec{\mathbf{g}}$, where m is the mass of the object (expressed in kilograms) and $\vec{\mathbf{g}}$ is acceleration resulting from gravity, with $\|\vec{\mathbf{g}}\| = 9.8\, N/kg$. A 2-kg disco ball hangs by a chain from the ceiling of a room.

a. Find the force of gravity $\vec{\mathbf{F}}$ acting on the disco ball and find its magnitude.

b. Find the force of tension $\vec{\mathbf{T}}$ in the chain and its magnitude.

Express the answers using standard unit vectors.



Figure 18: (credit: modification of work by Kenneth Lu, Flickr)

Answer

$a. \vec{\mathbf{F}} = -19.6\hat{\mathbf{k}}, \quad \|\vec{\mathbf{F}}\| = 19.6\ N$

$b. \vec{\mathbf{T}} = 19.6\hat{\mathbf{k}}, \quad \|\vec{\mathbf{T}}\| = 19.6\ N$

58) A 5-kg pendant chandelier is designed such that the alabaster bowl is held by four chains of equal length, as shown in the following figure.

a. Find the magnitude of the force of gravity acting on the chandelier.

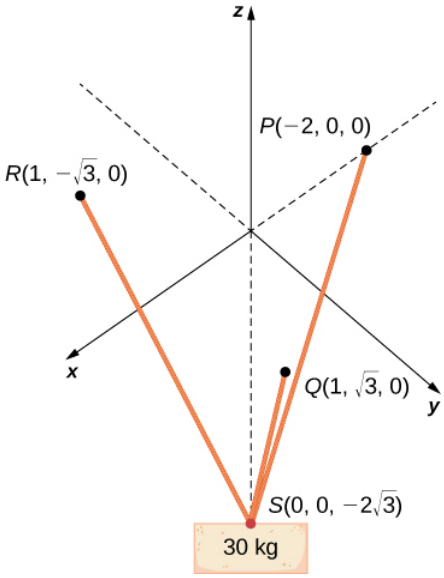
b. Find the magnitudes of the forces of tension for each of the four chains (assume chains are essentially vertical).



59) [T] A 30-kg block of cement is suspended by three cables of equal length that are anchored at points $P(-2,0,0), Q(1,\sqrt{3},0)$, and $R(1,-\sqrt{3},0)$. The load is located at $S(0,0,-2\sqrt{3})$, as shown in the following figure. Let $\vec{\mathbf{F}}_1, \vec{\mathbf{F}}_2$, and $\vec{\mathbf{F}}_3$ be the forces of tension resulting from the load in cables RS, QS, and PS, respectively.

a. Find the gravitational force $\vec{\mathbf{F}}$ acting on the block of cement that counterbalances the sum $\vec{\mathbf{F}}_1 + \vec{\mathbf{F}}_2 + \vec{\mathbf{F}}_3$ of the forces of tension in the cables.

b. Find forces $\vec{\mathbf{F}}_1, \vec{\mathbf{F}}_2$, and $\vec{\mathbf{F}}_3$. Express the answer in component form.

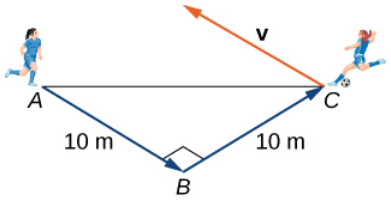


Answer

a. $\vec{\mathbf{F}} = -294\hat{\mathbf{k}}$ N;

b. $\vec{\mathbf{F}}_1 = \langle -\frac{49\sqrt{3}}{3}, 49, -98\rangle, \vec{\mathbf{F}}_2 = \langle -\frac{49\sqrt{3}}{3}, -49, -98\rangle$, and $\vec{\mathbf{F}}_3 = \langle \frac{98\sqrt{3}}{3}, 0, -98\rangle$ (each component is expressed in newtons)

60) Two soccer players are practicing for an upcoming game. One of them runs 10 m from point A to point B. She then turns left at $90°$ and runs 10 m until she reaches point C. Then she kicks the ball with a speed of 10 m/sec at an upward angle of $45°$ to her teammate, who is located at point A. Write the velocity of the ball in component form.



61) Let $\vec{\mathbf{r}}(t) = \langle x(t),\ y(t),\ z(t)\rangle$ be the position vector of a particle at the time $t \in [0, T]$, where x, y, and z are smooth functions on $[0, T]$. The instantaneous velocity of the particle at time t is defined by vector $\vec{\mathbf{v}}(t) = \langle x'(t),\ y'(t),\ z'(t)\rangle$, with components that are the derivatives with respect to t, of the functions x, y, and z, respectively. The magnitude $\| \vec{\mathbf{v}}(t) \|$ of the instantaneous velocity vector is called the speed of the particle at time t. Vector $\vec{\mathbf{a}}(t) = \langle x''(t),\ y''(t),\ z''(t)\rangle$, with components that are the second derivatives with respect to t, of the functions x, y, and z, respectively, gives the acceleration of the particle at time t. Consider $\vec{\mathbf{r}}(t) = \langle \cos t,\ \sin t,\ 2t\rangle$ the position vector of a particle at time $t \in [0, 30]$, where the components of $\vec{\mathbf{r}}$ are expressed in centimeters and time is expressed in seconds.

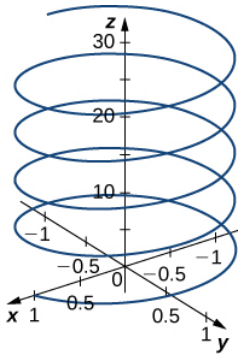
a. Find the instantaneous velocity, speed, and acceleration of the particle after the first second. Round your answer to two decimal places.

b. Use a CAS to visualize the path of the particle—that is, the set of all points of coordinates $(\cos t, \sin t, 2t)$, where $t \in [0, 30]$.

Answer

a. $\vec{\mathbf{v}}(1) = \langle -0.84, 0.54, 2\rangle$ (each component is expressed in centimeters per second); $\| \vec{\mathbf{v}}(1) \| = 2.24$ (expressed in centimeters per second); $\vec{\mathbf{a}}(1) = \langle -0.54, -0.84, 0\rangle$ (each component expressed in centimeters per second squared);

b.



62) [T] Let $\vec{\mathbf{r}}(t) = \langle t, 2t^2, 4t^2\rangle$ be the position vector of a particle at time t (in seconds), where $t \in [0, 10]$ (here the components of $\vec{\mathbf{r}}$ are expressed in centimeters).

a. Find the instantaneous velocity, speed, and acceleration of the particle after the first two seconds. Round your answer to two decimal places.

b. Use a CAS to visualize the path of the particle defined by the points $(t,\ 2t^2,\ 4t^2)$, where $t \in [0,\ 60]$.

Contributors

Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.

This page titled 12.2E: Exercises for Section 12.2 is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

12.3: The Dot Product

Learning Objectives

- Calculate the dot product of two given vectors.
- Determine whether two given vectors are perpendicular.
- Find the direction cosines of a given vector.
- Explain what is meant by the vector projection of one vector onto another vector, and describe how to compute it.
- Calculate the work done by a given force.

If we apply a force to an object so that the object moves, we say that work is done by the force. Previously, we looked at a constant force and we assumed the force was applied in the direction of motion of the object. Under those conditions, work can be expressed as the product of the force acting on an object and the distance the object moves. In this chapter, however, we have seen that both force and the motion of an object can be represented by vectors.

In this section, we develop an operation called the dot product, which allows us to calculate work in the case when the force vector and the motion vector have different directions. The dot product essentially tells us how much of the force vector is applied in the direction of the motion vector. The dot product can also help us measure the angle formed by a pair of vectors and the position of a vector relative to the coordinate axes. It even provides a simple test to determine whether two vectors meet at a right angle.

The Dot Product and Its Properties

We have already learned how to add and subtract vectors. In this chapter, we investigate two types of vector multiplication. The first type of vector multiplication is called the dot product, based on the notation we use for it, and it is defined as follows:

Definition: dot product

The *dot product* of vectors $\vec{\mathbf{u}} = \langle u_1, u_2, u_3 \rangle$ and $\vec{\mathbf{v}} = \langle v_1, v_2, v_3 \rangle$ is given by the sum of the products of the components

$$\vec{\mathbf{u}} \cdot \vec{\mathbf{v}} = u_1 v_1 + u_2 v_2 + u_3 v_3.$$

Note that if u and v are two-dimensional vectors, we calculate the dot product in a similar fashion. Thus, if $\vec{\mathbf{u}} = \langle u_1, u_2 \rangle$ and $\vec{\mathbf{v}} = \langle v_1, v_2 \rangle$, then

$$\vec{\mathbf{u}} \cdot \vec{\mathbf{v}} = u_1 v_1 + u_2 v_2.$$

When two vectors are combined under addition or subtraction, the result is a vector. When two vectors are combined using the dot product, the result is a scalar. For this reason, the dot product is often called the *scalar product*. It may also be called the *inner product*.

Example 12.3.1: Calculating Dot Products

a. Find the dot product of $\vec{\mathbf{u}} = \langle 3, 5, 2 \rangle$ and $\vec{\mathbf{v}} = \langle -1, 3, 0 \rangle$.
b. Find the scalar product of $\vec{\mathbf{p}} = 10\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + 7\hat{\mathbf{k}}$ and $\vec{\mathbf{q}} = -2\hat{\mathbf{i}} + \hat{\mathbf{j}} + 6\hat{\mathbf{k}}$.

Solution:

a. Substitute the vector components into the formula for the dot product:

$$\begin{aligned} \vec{\mathbf{u}} \cdot \vec{\mathbf{v}} &= u_1 v_1 + u_2 v_2 + u_3 v_3 \\ &= 3(-1) + 5(3) + 2(0) \\ &= -3 + 15 + 0 \\ &= 12. \end{aligned}$$

b. The calculation is the same if the vectors are written using standard unit vectors. We still have three components for each vector to substitute into the formula for the dot product:

$$\begin{aligned}\vec{\mathbf{p}} \cdot \vec{\mathbf{q}} &= p_1q_1 + p_2q_2 + p_3q_3 \\ &= 10(-2) + (-4)(1) + (7)(6) \\ &= -20 - 4 + 42 \\ &= 18.\end{aligned}$$

Exercise 12.3.1

Find $\vec{\mathbf{u}} \cdot \vec{\mathbf{v}}$, where $\vec{\mathbf{u}} = \langle 2, 9, -1\rangle$ and $\vec{\mathbf{v}} = \langle -3, 1, -4\rangle$.

Hint

Multiply corresponding components and then add their products.

Answer

7

Like vector addition and subtraction, the dot product has several algebraic properties. We prove three of these properties and leave the rest as exercises.

Properties of the Dot Product

Let $\vec{\mathbf{u}}$, $\vec{\mathbf{v}}$, and $\vec{\mathbf{w}}$ be vectors, and let c be a scalar.

i. *Commutative property*

$$\vec{\mathbf{u}} \cdot \vec{\mathbf{v}} = \vec{\mathbf{v}} \cdot \vec{\mathbf{u}}$$

ii. *Distributive property*

$$\vec{\mathbf{u}} \cdot (\vec{\mathbf{v}} + \vec{\mathbf{w}}) = \vec{\mathbf{u}} \cdot \vec{\mathbf{v}} + \vec{\mathbf{u}} \cdot \vec{\mathbf{w}}$$

iii. *Associative property*

$$c(\vec{\mathbf{u}} \cdot \vec{\mathbf{v}}) = (c\vec{\mathbf{u}}) \cdot \vec{\mathbf{v}} = \vec{\mathbf{u}} \cdot (c\vec{\mathbf{v}})$$

iv. *Property of magnitude*

$$\vec{\mathbf{v}} \cdot \vec{\mathbf{v}} = \|\vec{\mathbf{v}}\|^2$$

Proof

Let $\vec{\mathbf{u}} = \langle u_1, u_2, u_3\rangle$ and $\vec{\mathbf{v}} = \langle v_1, v_2, v_3\rangle$. Then

$$\begin{aligned}\vec{\mathbf{u}} \cdot \vec{\mathbf{v}} &= \langle u_1, u_2, u_3\rangle \cdot \langle v_1, v_2, v_3\rangle \\ &= u_1v_1 + u_2v_2 + u_3v_3 \\ &= v_1u_1 + v_2u_2 + v_3u_3 \\ &= \langle v_1, v_2, v_3\rangle \cdot \langle u_1, u_2, u_3\rangle \\ &= \vec{\mathbf{v}} \cdot \vec{\mathbf{u}}.\end{aligned}$$

The associative property looks like the associative property for real-number multiplication, but pay close attention to the difference between scalar and vector objects:

$$\begin{aligned} c(\vec{\mathbf{u}} \cdot \vec{\mathbf{v}}) &= c(u_1v_1 + u_2v_2 + u_3v_3) \\ &= c(u_1v_1) + c(u_2v_2) + c(u_3v_3) \\ &= (cu_1)v_1 + (cu_2)v_2 + (cu_3)v_3 \\ &= \langle cu_1, cu_2, cu_3 \rangle \cdot \langle v_1, v_2, v_3 \rangle \\ &= c\langle u_1, u_2, u_3 \rangle \cdot \langle v_1, v_2, v_3 \rangle \\ &= (c\vec{\mathbf{u}}) \cdot \vec{\mathbf{v}}. \end{aligned}$$

The proof that $c(\vec{\mathbf{u}} \cdot \vec{\mathbf{v}}) = \vec{\mathbf{u}} \cdot (c\vec{\mathbf{v}})$ is similar.

The fourth property shows the relationship between the magnitude of a vector and its dot product with itself:

$$\begin{aligned} \vec{\mathbf{v}} \cdot \vec{\mathbf{v}} &= \langle v_1, v_2, v_3 \rangle \cdot \langle v_1, v_2, v_3 \rangle \\ &= (v_1)^2 + (v_2)^2 + (v_3)^2 \\ &= \left[\sqrt{(v_1)^2 + (v_2)^2 + (v_3)^2}\right]^2 \\ &= \|\vec{\mathbf{v}}\|^2. \end{aligned}$$

□

Note that the definition of the dot product yields $\vec{\mathbf{0}} \cdot \vec{\mathbf{v}} = 0$. By property iv. if $\vec{\mathbf{v}} \cdot \vec{\mathbf{v}} = 0$, then $\vec{\mathbf{v}} = \vec{\mathbf{0}}$.

Example 12.3.2: Using Properties of the Dot Product

Let $\vec{\mathbf{a}} = \langle 1, 2, -3 \rangle$, $\vec{\mathbf{b}} = \langle 0, 2, 4 \rangle$, and $\vec{\mathbf{c}} = \langle 5, -1, 3 \rangle$.

Find each of the following products.

a. $(\vec{\mathbf{a}} \cdot \vec{\mathbf{b}})\vec{\mathbf{c}}$
b. $\vec{\mathbf{a}} \cdot (2\vec{\mathbf{c}})$
c. $\|\vec{\mathbf{b}}\|^2$

Solution

a. Note that this expression asks for the scalar multiple of $\vec{\mathbf{c}}$ by $\vec{\mathbf{a}} \cdot \vec{\mathbf{b}}$:

$$\begin{aligned} (\vec{\mathbf{a}} \cdot \vec{\mathbf{b}})\vec{\mathbf{c}} &= (\langle 1, 2, -3 \rangle \cdot \langle 0, 2, 4 \rangle)\langle 5, -1, 3 \rangle \\ &= (1(0) + 2(2) + (-3)(4))\langle 5, -1, 3 \rangle \\ &= -8\langle 5, -1, 3 \rangle \\ &= \langle -40, 8, -24 \rangle. \end{aligned}$$

b. This expression is a dot product of vector $\vec{\mathbf{a}}$ and scalar multiple $2\vec{\mathbf{c}}$:

$$\begin{aligned} \vec{\mathbf{a}} \cdot (2\vec{\mathbf{c}}) &= 2(\vec{\mathbf{a}} \cdot \vec{\mathbf{c}}) \\ &= 2(\langle 1, 2, -3 \rangle \cdot \langle 5, -1, 3 \rangle) \\ &= 2(1(5) + 2(-1) + (-3)(3)) \\ &= 2(-6) = -12. \end{aligned}$$

c. Simplifying this expression is a straightforward application of the dot product:

$$\begin{aligned} \|\vec{\mathbf{b}}\|^2 &= \vec{\mathbf{b}} \cdot \vec{\mathbf{b}} \\ &= \langle 0, 2, 4 \rangle \cdot \langle 0, 2, 4 \rangle \\ &= 0^2 + 2^2 + 4^2 \\ &= 0 + 4 + 16 \\ &= 20. \end{aligned}$$

Exercise 12.3.2

Find the following products for $\vec{\mathbf{p}} = \langle 7, 0, 2\rangle$, $\vec{\mathbf{q}} = \langle -2, 2, -2\rangle$, and $\vec{\mathbf{r}} = \langle 0, 2, -3\rangle$.

a. $(\vec{\mathbf{r}} \cdot \vec{\mathbf{p}})\vec{\mathbf{q}}$
b. $\|\vec{\mathbf{p}}\|^2$

Hint

$\vec{\mathbf{r}} \cdot \vec{\mathbf{p}}$ is a scalar.

Answer

a. $(\vec{\mathbf{r}} \cdot \vec{\mathbf{p}})\vec{\mathbf{q}} = \langle 12, -12, 12\rangle$; *b.* $\|\vec{\mathbf{p}}\|^2 = 53$

Using the Dot Product to Find the Angle between Two Vectors

When two nonzero vectors are placed in standard position, whether in two dimensions or three dimensions, they form an angle between them (Figure 12.3.1). The dot product provides a way to find the measure of this angle. This property is a result of the fact that we can express the dot product in terms of the cosine of the angle formed by two vectors.

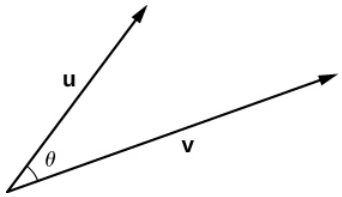


Figure 12.3.1: Let θ be the angle between two nonzero vectors $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ such that $0 \le \theta \le \pi$.

Evaluating a Dot Product

The dot product of two vectors is the product of the magnitude of each vector and the cosine of the angle between them:

$$\vec{\mathbf{u}} \cdot \vec{\mathbf{v}} = \|\vec{\mathbf{u}}\|\|\vec{\mathbf{v}}\| \cos\theta. \tag{12.3.1}$$

Proof

Place vectors $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ in standard position and consider the vector $\vec{\mathbf{v}} - \vec{\mathbf{u}}$ (Figure 12.3.2). These three vectors form a triangle with side lengths $\|\vec{\mathbf{u}}\|$, $\|\vec{\mathbf{v}}\|$, and $\|\vec{\mathbf{v}} - \vec{\mathbf{u}}\|$.

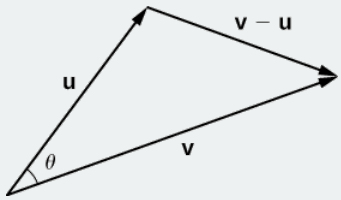


Figure 12.3.2: The lengths of the sides of the triangle are given by the magnitudes of the vectors that form the triangle.

Recall from trigonometry that the law of cosines describes the relationship among the side lengths of the triangle and the angle θ. Applying the law of cosines here gives

$$\|\vec{\mathbf{v}} - \vec{\mathbf{u}}\|^2 = \|\vec{\mathbf{u}}\|^2 + \|\vec{\mathbf{v}}\|^2 - 2\|\vec{\mathbf{u}}\|\|\vec{\mathbf{v}}\| \cos\theta. \tag{12.3.2}$$

The dot product provides a way to rewrite the left side of Equation 12.3.2:

$$\begin{aligned}
\|\vec{\mathbf{v}} - \vec{\mathbf{u}}\|^2 &= (\vec{\mathbf{v}} - \vec{\mathbf{u}}) \cdot (\vec{\mathbf{v}} - \vec{\mathbf{u}}) \\
&= (\vec{\mathbf{v}} - \vec{\mathbf{u}}) \cdot \vec{\mathbf{v}} - (\vec{\mathbf{v}} - \vec{\mathbf{u}}) \cdot \vec{\mathbf{u}} \\
&= \vec{\mathbf{v}} \cdot \vec{\mathbf{v}} - \vec{\mathbf{u}} \cdot \vec{\mathbf{v}} - \vec{\mathbf{v}} \cdot \vec{\mathbf{u}} + \vec{\mathbf{u}} \cdot \vec{\mathbf{u}} \\
&= \vec{\mathbf{v}} \cdot \vec{\mathbf{v}} - \vec{\mathbf{u}} \cdot \vec{\mathbf{v}} - \vec{\mathbf{u}} \cdot \vec{\mathbf{v}} + \vec{\mathbf{u}} \cdot \vec{\mathbf{u}} \\
&= \|\vec{\mathbf{v}}\|^2 - 2\vec{\mathbf{u}} \cdot \vec{\mathbf{v}} + \|\vec{\mathbf{u}}\|^2.
\end{aligned}$$

Substituting into the law of cosines yields

$$\begin{aligned}\|\vec{\mathbf{v}}-\vec{\mathbf{u}}\|^2 &= \|\vec{\mathbf{u}}\|^2+\|\vec{\mathbf{v}}\|^2-2\|\vec{\mathbf{u}}\|\|\vec{\mathbf{v}}\|\cos\theta\\ \|\vec{\mathbf{v}}\|^2-2\vec{\mathbf{u}}\cdot\vec{\mathbf{v}}+\|\vec{\mathbf{u}}\|^2 &= \|\vec{\mathbf{u}}\|^2+\|\vec{\mathbf{v}}\|^2-2\|\vec{\mathbf{u}}\|\|\vec{\mathbf{v}}\|\cos\theta\\ -2\vec{\mathbf{u}}\cdot\vec{\mathbf{v}} &= -2\|\vec{\mathbf{u}}\|\|\vec{\mathbf{v}}\|\cos\theta\\ \vec{\mathbf{u}}\cdot\vec{\mathbf{v}} &= \|\vec{\mathbf{u}}\|\|\vec{\mathbf{v}}\|\cos\theta.\end{aligned}$$

□

We can use the form of the dot product in Equation 12.3.1 to find the measure of the angle between two nonzero vectors by rearranging Equation 12.3.1 to solve for the cosine of the angle:

$$\cos\theta=\frac{\vec{\mathbf{u}}\cdot\vec{\mathbf{v}}}{\|\vec{\mathbf{u}}\|\|\vec{\mathbf{v}}\|}. \tag{12.3.3}$$

Using this equation, we can find the cosine of the angle between two nonzero vectors. Since we are considering the smallest angle between the vectors, we assume $0°\leq\theta\leq 180°$ (or $0\leq\theta\leq\pi$ if we are working in radians). The inverse cosine is unique over this range, so we are then able to determine the measure of the angle θ.

Example 12.3.3: Finding the Angle between Two Vectors

Find the measure of the angle between each pair of vectors.

a. $\hat{\mathbf{i}}+\hat{\mathbf{j}}+\hat{\mathbf{k}}$ and $2\hat{\mathbf{i}}-\hat{\mathbf{j}}-3\hat{\mathbf{k}}$

b. $\langle 2,5,6\rangle$ and $\langle -2,-4,4\rangle$

Solution

a. To find the cosine of the angle formed by the two vectors, substitute the components of the vectors into Equation 12.3.3:

$$\begin{aligned}\cos\theta &= \frac{(\hat{\mathbf{i}}+\hat{\mathbf{j}}+\hat{\mathbf{k}})\cdot(2\hat{\mathbf{i}}-\hat{\mathbf{j}}-3\hat{\mathbf{k}})}{\|\,\hat{\mathbf{i}}+\hat{\mathbf{j}}+\hat{\mathbf{k}}\,\|\cdot\|\,2\hat{\mathbf{i}}-\hat{\mathbf{j}}-3\hat{\mathbf{k}}\,\|}\\ &= \frac{1(2)+(1)(-1)+(1)(-3)}{\sqrt{1^2+1^2+1^2}\sqrt{2^2+(-1)^2+(-3)^2}}\\ &= \frac{-2}{\sqrt{3}\sqrt{14}}=\frac{-2}{\sqrt{42}}.\end{aligned}$$

Therefore, $\theta=\arccos\dfrac{-2}{\sqrt{42}}$ rad.

b. Start by finding the value of the cosine of the angle between the vectors:

$$\begin{aligned}\cos\theta &= \frac{\langle 2,5,6\rangle\cdot\langle -2,-4,4\rangle}{\|\,\langle 2,5,6\rangle\,\|\cdot\|\,\langle -2,-4,4\rangle\,\|}\\ &= \frac{2(-2)+(5)(-4)+(6)(4)}{\sqrt{2^2+5^2+6^2}\sqrt{(-2)^2+(-4)^2+4^2}}\\ &= \frac{0}{\sqrt{65}\sqrt{36}}=0.\end{aligned}$$

Now, $\cos\theta=0$ and $0\leq\theta\leq\pi$, so $\theta=\pi/2$.

Exercise 12.3.3

Find the measure of the angle, in radians, formed by vectors $\vec{\mathbf{a}}=\langle 1,2,0\rangle$ and $\vec{\mathbf{b}}=\langle 2,4,1\rangle$. Round to the nearest hundredth.

Hint

Use the Equation 12.3.3.

Answer

$\theta \approx 0.22$ rad

The angle between two vectors can be acute $(0 < \cos\theta < 1)$, obtuse $(-1 < \cos\theta < 0)$, or straight $(\cos\theta = -1)$. If $\cos\theta = 1$, then both vectors have the same direction. If $\cos\theta = 0$, then the vectors, when placed in standard position, form a right angle (Figure 12.3.3). We can formalize this result into a theorem regarding orthogonal (perpendicular) vectors.

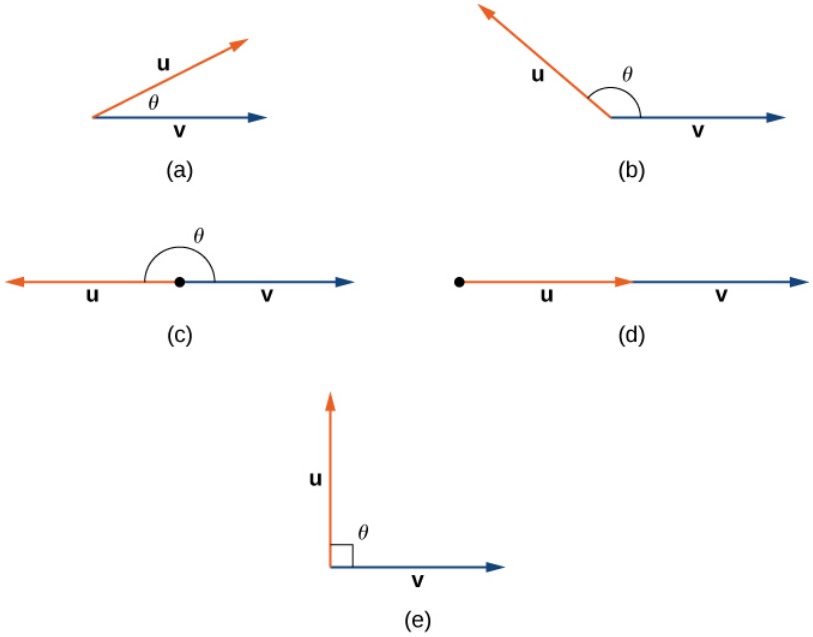


Figure 12.3.3: (a) An acute angle has $0 < \cos\theta < 1$. (b) An obtuse angle has $-1 < \cos\theta < 0$. (c) A straight line has $\cos\theta = -1$. (d) If the vectors have the same direction, $\cos\theta = 1$. (e) If the vectors are orthogonal (perpendicular), $\cos\theta = 0$.

Orthogonal Vectors

The nonzero vectors $\vec{u}$ and $\vec{v}$ are **orthogonal vectors** if and only if $\vec{u} \cdot \vec{v} = 0$.

Proof

Let $\vec{u}$ and $\vec{v}$ be nonzero vectors, and let θ denote the angle between them. First, assume $\vec{u} \cdot \vec{v} = 0$.Then

$$\|\vec{u}\|\|\vec{v}\| \cos\theta = 0.$$

However, $\|\vec{u}\| \neq 0$ and $\|\vec{v}\| \neq 0$, so we must have $\cos\theta = 0$. Hence, $\theta = 90°$, and the vectors are orthogonal.

Now assume $\vec{u}$ and $\vec{v}$ are orthogonal. Then $\theta = 90°$ and we have

$$\begin{aligned} \vec{u} \cdot \vec{v} &= \|\vec{u}\|\|\vec{v}\| \cos\theta \\ &= \|\vec{u}\|\|\vec{v}\| \cos 90° \\ &= \|\vec{u}\|\|\vec{v}\|(0) \\ &= 0. \end{aligned}$$

□

The terms **orthogonal, perpendicular,** and **normal** each indicate that mathematical objects are intersecting at right angles. The use of each term is determined mainly by its context. We say that vectors are orthogonal and lines are perpendicular. The term *normal* is used most often when measuring the angle made with a plane or other surface.

Example 12.3.4: Identifying Orthogonal Vectors

Determine whether $\vec{p} = \langle 1, 0, 5 \rangle$ and $\vec{q} = \langle 10, 3, -2 \rangle$ are orthogonal vectors.

Solution

Using the definition, we need only check the dot product of the vectors:

$$\vec{p} \cdot \vec{q} = 1(10) + (0)(3) + (5)(-2) = 10 + 0 - 10 = 0.$$

Because $\vec{p} \cdot \vec{q} = 0$, the vectors are orthogonal (Figure 12.3.4).

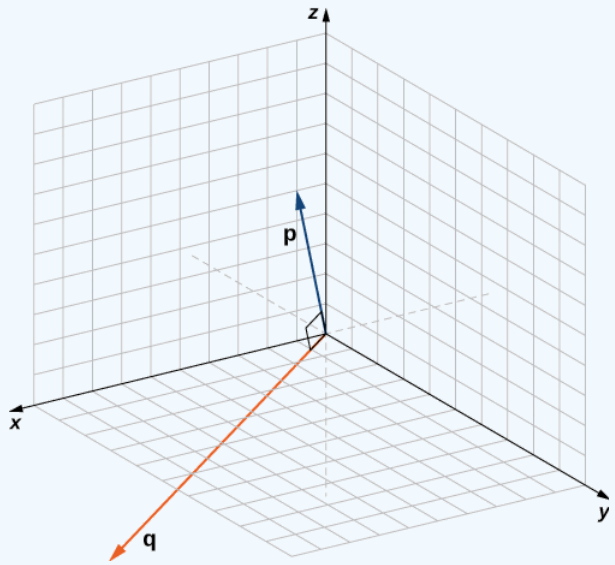


Figure 12.3.4: Vectors $\vec{p}$ and $\vec{q}$ form a right angle when their initial points are aligned.

Exercise 12.3.4

For which value of x is $\vec{p} = \langle 2, 8, -1 \rangle$ orthogonal to $\vec{q} = \langle x, -1, 2 \rangle$?

Hint

Vectors $\vec{p}$ and $\vec{q}$ are orthogonal if and only if $\vec{p} \cdot \vec{q} = 0$.

Answer

$x = 5$

Example 12.3.5: Measuring the Angle Formed by Two Vectors

Let $\vec{v} = \langle 2, 3, 3 \rangle$. Find the measures of the angles formed by the following vectors.

a. $\vec{v}$ and $\hat{i}$
b. $\vec{v}$ and $\hat{j}$
c. $\vec{v}$ and $\hat{k}$

Solution

a.Let α be the angle formed by $\vec{v}$ and $\hat{i}$:

$$\cos\alpha = \frac{\vec{v} \cdot \hat{i}}{\|\vec{v}\| \cdot \|\hat{i}\|} = \frac{\langle 2, 3, 3 \rangle \cdot \langle 1, 0, 0 \rangle}{\sqrt{2^2 + 3^2 + 3^2}\sqrt{1}} = \frac{2}{\sqrt{22}}$$

$$\alpha = \arccos \frac{2}{\sqrt{22}} \approx 1.130 \, \text{rad}.$$

b. Let β represent the angle formed by $\vec{v}$ and $\hat{j}$:

$$\cos\beta = \frac{\vec{v}\cdot\hat{j}}{\|\vec{v}\|\cdot\|\hat{j}\|} = \frac{\langle 2,3,3\rangle\cdot\langle 0,1,0\rangle}{\sqrt{2^2+3^2+3^2}\sqrt{1}} = \frac{3}{\sqrt{22}}$$

$$\beta = \arccos\frac{3}{\sqrt{22}} \approx 0.877\,\text{rad}.$$

c. Let γ represent the angle formed by $\vec{v}$ and $\hat{k}$:

$$\cos\gamma = \frac{\vec{v}\cdot\hat{k}}{\|\vec{v}\|\cdot\|\hat{k}\|} = \frac{\langle 2,3,3\rangle\cdot\langle 0,0,1\rangle}{\sqrt{2^2+3^2+3^2}\sqrt{1}} = \frac{3}{\sqrt{22}}$$

$$\gamma = \arccos\frac{3}{\sqrt{22}} \approx 0.877\,\text{rad}.$$

Exercise 12.3.5

Let $\vec{v} = \langle 3,-5,1\rangle$. Find the measure of the angles formed by each pair of vectors.

a. $\vec{v}$ and $\hat{i}$
b. $\vec{v}$ and $\hat{j}$
c. $\vec{v}$ and $\hat{k}$

Hint

$\hat{i} = \langle 1,0,0\rangle, \hat{j} = \langle 0,1,0\rangle$, and $\hat{k} = \langle 0,0,1\rangle$

Answer

$a.\alpha \approx 1.04$ rad; b. $\beta \approx 2.58$ rad; c. $\gamma \approx 1.40$ rad

The angle a vector makes with each of the coordinate axes, called a direction angle, is very important in practical computations, especially in a field such as engineering. For example, in astronautical engineering, the angle at which a rocket is launched must be determined very precisely. A very small error in the angle can lead to the rocket going hundreds of miles off course. Direction angles are often calculated by using the dot product and the cosines of the angles, called the direction cosines. Therefore, we define both these angles and their cosines.

Definition: direction angles

The angles formed by a nonzero vector and the coordinate axes are called the **direction angles** for the vector (Figure 12.3.5). The cosines for these angles are called the **direction cosines.**

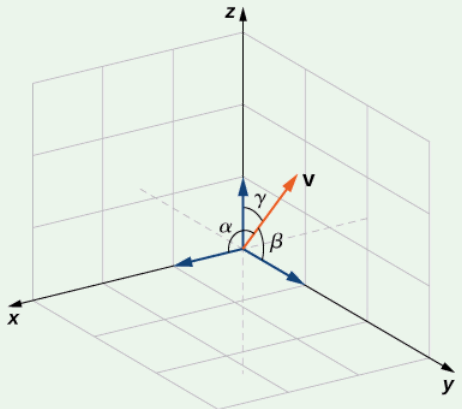


Figure 12.3.5: Angle α is formed by vector $\vec{v}$ and unit vector $\hat{i}$. Angle β is formed by vector $\vec{v}$ and unit vector $\hat{j}$. Angle γ is formed by vector $\vec{v}$ and unit vector $\hat{k}$.

In Example 12.3.5, the direction cosines of $\vec{\mathbf{v}} = \langle 2, 3, 3 \rangle$ are $\cos\alpha = \dfrac{2}{\sqrt{22}}, \cos\beta = \dfrac{3}{\sqrt{22}}$, and $\cos\gamma = \dfrac{3}{\sqrt{22}}$. The direction angles of $\vec{\mathbf{v}}$ are $\alpha = 1.130$ rad, $\beta = 0.877$ rad, and $\gamma = 0.877$ rad.

So far, we have focused mainly on vectors related to force, movement, and position in three-dimensional physical space. However, vectors are often used in more abstract ways. For example, suppose a fruit vendor sells apples, bananas, and oranges. On a given day, he sells 30 apples, 12 bananas, and 18 oranges. He might use a quantity vector, $\vec{\mathbf{q}} = \langle 30, 12, 18 \rangle$, to represent the quantity of fruit he sold that day. Similarly, he might want to use a price vector, $\vec{\mathbf{p}} = \langle 0.50, 0.25, 1 \rangle$, to indicate that he sells his apples for 50¢ each, bananas for 25¢ each, and oranges for $1 apiece. In this example, although we could still graph these vectors, we do not interpret them as literal representations of position in the physical world. We are simply using vectors to keep track of particular pieces of information about apples, bananas, and oranges.

This idea might seem a little strange, but if we simply regard vectors as a way to order and store data, we find they can be quite a powerful tool. Going back to the fruit vendor, let's think about the dot product, $\vec{\mathbf{q}} \cdot \vec{\mathbf{p}}$. We compute it by multiplying the number of apples sold (30) by the price per apple (50¢), the number of bananas sold by the price per banana, and the number of oranges sold by the price per orange. We then add all these values together. So, in this example, the dot product tells us how much money the fruit vendor had in sales on that particular day.

When we use vectors in this more general way, there is no reason to limit the number of components to three. What if the fruit vendor decides to start selling grapefruit? In that case, he would want to use four-dimensional quantity and price vectors to represent the number of apples, bananas, oranges, and grapefruit sold, and their unit prices. As you might expect, to calculate the dot product of four-dimensional vectors, we simply add the products of the components as before, but the sum has four terms instead of three.

Example 12.3.6: Using Vectors in an Economic Context

AAA Party Supply Store sells invitations, party favors, decorations, and food service items such as paper plates and napkins. When AAA buys its inventory, it pays 25¢ per package for invitations and party favors. Decorations cost AAA 50¢ each, and food service items cost 20¢ per package. AAA sells invitations for $2.50 per package and party favors for $1.50 per package. Decorations sell for $4.50 each and food service items for $1.25 per package.

During the month of May, AAA Party Supply Store sells 1258 invitations, 342 party favors, 2426 decorations, and 1354 food service items. Use vectors and dot products to calculate how much money AAA made in sales during the month of May. How much did the store make in profit?

Solution

The cost, price, and quantity vectors are

$$\begin{aligned}\vec{\mathbf{c}} &= \langle 0.25, 0.25, 0.50, 0.20 \rangle \\ \vec{\mathbf{p}} &= \langle 2.50, 1.50, 4.50, 1.25 \rangle \\ \vec{\mathbf{q}} &= \langle 1258, 342, 2426, 1354 \rangle.\end{aligned}$$

AAA sales for the month of May can be calculated using the dot product $\vec{\mathbf{p}} \cdot \vec{\mathbf{q}}$. We have

$$\begin{aligned}\vec{\mathbf{p}} \cdot \vec{\mathbf{q}} &= \langle 2.50, 1.50, 4.50, 1.25 \rangle \cdot \langle 1258, 342, 2426, 1354 \rangle \\ &= 3145 + 513 + 10917 + 1692.5 \\ &= 16267.5.\end{aligned}$$

So, AAA took in $16,267.50 during the month of May. To calculate the profit, we must first calculate how much AAA paid for the items sold. We use the dot product $\vec{\mathbf{c}} \cdot \vec{\mathbf{q}}$ to get

$$\begin{aligned}\vec{\mathbf{c}} \cdot \vec{\mathbf{q}} &= \langle 0.25, 0.25, 0.50, 0.20 \rangle \cdot \langle 1258, 342, 2426, 1354 \rangle \\ &= 314.5 + 85.5 + 1213 + 270.8 \\ &= 1883.8.\end{aligned}$$

So, AAA paid $1,883.80 for the items they sold. Their profit, then, is given by

$$\vec{\mathbf{p}} \cdot \vec{\mathbf{q}} - \vec{\mathbf{c}} \cdot \vec{\mathbf{q}} = 16267.5 - 1883.8 = 14383.7.$$

Therefore, AAA Party Supply Store made $14,383.70 in May.

Exercise 12.3.6

On June 1, AAA Party Supply Store decided to increase the price they charge for party favors to $2 per package. They also changed suppliers for their invitations, and are now able to purchase invitations for only 10¢ per package. All their other costs and prices remain the same. If AAA sells 1408 invitations, 147 party favors, 2112 decorations, and 1894 food service items in the month of June, use vectors and dot products to calculate their total sales and profit for June.

Hint

Use four-dimensional vectors for cost, price, and quantity sold.

Answer

Sales = $15,685.50; profit = $14,073.15

Projections

As we have seen, addition combines two vectors to create a resultant vector. But what if we are given a vector and we need to find its component parts? We use vector projections to perform the opposite process; they can break down a vector into its components. The magnitude of a vector projection is a scalar projection. For example, if a child is pulling the handle of a wagon at a 55° angle, we can use projections to determine how much of the force on the handle is actually moving the wagon forward (12.3.6). We return to this example and learn how to solve it after we see how to calculate projections.

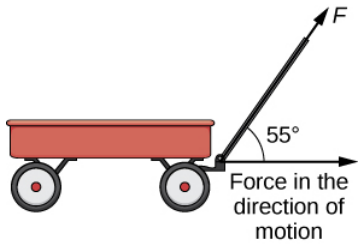


Figure 12.3.6: When a child pulls a wagon, only the horizontal component of the force propels the wagon forward.

Definition: Vector and Projection

The **vector projection** of $\vec{\mathbf{v}}$ onto $\vec{\mathbf{u}}$ is the vector labeled $\text{proj}_{\vec{\mathbf{u}}}\vec{\mathbf{v}}$ in Figure 12.3.7. It has the same initial point as $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ and the same direction as $\vec{\mathbf{u}}$, and represents the component of $\vec{\mathbf{v}}$ that acts in the direction of $\vec{\mathbf{u}}$. If θ represents the angle between $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$, then, by properties of triangles, we know the length of $\text{proj}_{\vec{\mathbf{u}}}\vec{\mathbf{v}}$ is $\|\text{proj}_{\vec{\mathbf{u}}}\vec{\mathbf{v}}\| = \|\vec{\mathbf{v}}\|\cos\theta$. When expressing $\cos\theta$ in terms of the dot product, this becomes

$$\|\text{proj}_{\vec{\mathbf{u}}}\vec{\mathbf{v}}\| = \|\vec{\mathbf{v}}\|\cos\theta = \|\vec{\mathbf{v}}\|\left(\frac{\vec{\mathbf{u}}\cdot\vec{\mathbf{v}}}{\|\vec{\mathbf{u}}\|\|\vec{\mathbf{v}}\|}\right) = \frac{\vec{\mathbf{u}}\cdot\vec{\mathbf{v}}}{\|\vec{\mathbf{u}}\|}.$$

We now multiply by a unit vector in the direction of $\vec{\mathbf{u}}$ to get $\text{proj}_{\vec{\mathbf{u}}}\vec{\mathbf{v}}$:

$$\text{proj}_{\vec{\mathbf{u}}}\vec{\mathbf{v}} = \frac{\vec{\mathbf{u}}\cdot\vec{\mathbf{v}}}{\|\vec{\mathbf{u}}\|}\left(\frac{1}{\|\vec{\mathbf{u}}\|}\vec{\mathbf{u}}\right) = \frac{\vec{\mathbf{u}}\cdot\vec{\mathbf{v}}}{\|\vec{\mathbf{u}}\|^2}\vec{\mathbf{u}}.$$

The length of this vector is also known as the **scalar projection** of $\vec{\mathbf{v}}$ onto $\vec{\mathbf{u}}$ and is denoted by

$$\|\text{proj}_{\vec{\mathbf{u}}}\vec{\mathbf{v}}\| = \text{comp}_{\vec{\mathbf{u}}}\vec{\mathbf{v}} = \frac{\vec{\mathbf{u}}\cdot\vec{\mathbf{v}}}{\|\vec{\mathbf{u}}\|}.$$

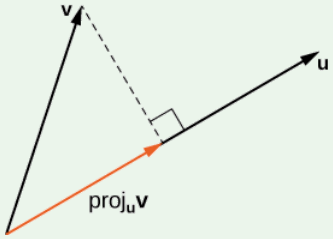


Figure 12.3.7: The projection of $\vec{v}$ onto $\vec{u}$ shows the component of vector $\vec{v}$ in the direction of $\vec{u}$.

Example 12.3.7: Finding Projections

Find the projection of $\vec{v}$ onto $\vec{u}$.

a. $\vec{v} = \langle 3, 5, 1\rangle$ and $\vec{u} = \langle -1, 4, 3\rangle$
b. $\vec{v} = 3\hat{i} - 2\hat{j}$ and $\vec{u} = \hat{i} + 6\hat{j}$

Solution

a. Substitute the components of $\vec{v}$ and $\vec{u}$ into the formula for the projection:

$$\begin{aligned}\text{proj}_{\vec{u}}\vec{v} &= \frac{\vec{u}\cdot\vec{v}}{\|\vec{u}\|^2}\vec{u} \\ &= \frac{\langle -1,4,3\rangle\cdot\langle 3,5,1\rangle}{\|\langle -1,4,3\rangle\|^2}\langle -1,4,3\rangle \\ &= \frac{-3+20+3}{(-1)^2+4^2+3^2}\langle -1,4,3\rangle \\ &= \frac{20}{26}\langle -1,4,3\rangle \\ &= \langle -\frac{10}{13}, \frac{40}{13}, \frac{30}{13}\rangle.\end{aligned}$$

b. To find the two-dimensional projection, simply adapt the formula to the two-dimensional case:

$$\begin{aligned}\text{proj}_{\vec{u}}\vec{v} &= \frac{\vec{u}\cdot\vec{v}}{\|\vec{u}\|^2}\vec{u} \\ &= \frac{(\hat{i}+6\hat{j})\cdot(3\hat{i}-2\hat{j})}{\|\hat{i}+6\hat{j}\|^2}(\hat{i}+6\hat{j}) \\ &= \frac{1(3)+6(-2)}{1^2+6^2}(\hat{i}+6\hat{j}) \\ &= -\frac{9}{37}(\hat{i}+6\hat{j}) \\ &= -\frac{9}{37}\hat{i} - \frac{54}{37}\hat{j}.\end{aligned}$$

Sometimes it is useful to decompose vectors—that is, to break a vector apart into a sum. This process is called the **resolution of a vector into components.** Projections allow us to identify two orthogonal vectors having a desired sum. For example, let $\vec{v} = \langle 6, -4\rangle$ and let $\vec{u} = \langle 3, 1\rangle$. We want to decompose the vector $\vec{v}$ into orthogonal components such that one of the component vectors has the same direction as $\vec{u}$.

We first find the component that has the same direction as $\vec{u}$ by projecting $\vec{v}$ onto $\vec{u}$. Let $\vec{p} = \text{proj}_{\vec{u}}\vec{v}$. Then, we have

$$\begin{aligned}\vec{p} &= \frac{\vec{u}\cdot\vec{v}}{\|\vec{u}\|^2}\vec{u} \\ &= \frac{18-4}{9+1}\vec{u} \\ = \frac{7}{5}\vec{u} = \frac{7}{5}\langle 3,1\rangle &= \langle \frac{21}{5}, \frac{7}{5}\rangle.\end{aligned}$$

Now consider the vector $\vec{\mathbf{q}} = \vec{\mathbf{v}} - \vec{\mathbf{p}}$. We have

$$\begin{aligned}\vec{\mathbf{q}} &= \vec{\mathbf{v}} - \vec{\mathbf{p}} \\ &= \langle 6, -4\rangle - \langle \frac{21}{5}, \frac{7}{5}\rangle \\ &= \langle \frac{9}{5}, -\frac{27}{5}\rangle.\end{aligned}$$

Clearly, by the way we defined $\vec{\mathbf{q}}$, we have $\vec{\mathbf{v}} = \vec{\mathbf{q}} + \vec{\mathbf{p}}$, and

$$\begin{aligned}\vec{\mathbf{q}} \cdot \vec{\mathbf{p}} &= \langle \frac{9}{5}, -\frac{27}{5}\rangle \cdot \langle \frac{21}{5}, \frac{7}{5}\rangle \\ &= \frac{9(21)}{25} + -\frac{27(7)}{25} \\ &= \frac{189}{25} - \frac{189}{25} = 0.\end{aligned}$$

Therefore, $\vec{\mathbf{q}}$ and $\vec{\mathbf{p}}$ are orthogonal.

Example 12.3.8: Resolving Vectors into Components

Express $\vec{\mathbf{v}} = \langle 8, -3, -3\rangle$ as a sum of orthogonal vectors such that one of the vectors has the same direction as $\vec{\mathbf{u}} = \langle 2, 3, 2\rangle$.

Solution

Let $\vec{\mathbf{p}}$ represent the projection of $\vec{\mathbf{v}}$ onto $\vec{\mathbf{u}}$:

$$\begin{aligned}\vec{\mathbf{p}} &= \text{proj}_{\vec{\mathbf{u}}}\vec{\mathbf{v}} \\ &= \frac{\vec{\mathbf{u}} \cdot \vec{\mathbf{v}}}{\|\vec{\mathbf{u}}\|^2}\vec{\mathbf{u}} \\ &= \frac{\langle 2, 3, 2\rangle \cdot \langle 8, -3, -3\rangle}{\| \langle 2, 3, 2\rangle \|^2}\langle 2, 3, 2\rangle \\ &= \frac{16 - 9 - 6}{2^2 + 3^2 + 2^2}\langle 2, 3, 2\rangle \\ &= \frac{1}{17}\langle 2, 3, 2\rangle \\ &= \langle \frac{2}{17}, \frac{3}{17}, \frac{2}{17}\rangle.\end{aligned}$$

Then,

$$\begin{aligned}\vec{\mathbf{q}} = \vec{\mathbf{v}} - \vec{\mathbf{p}} &= \langle 8, -3, -3\rangle - \langle \frac{2}{17}, \frac{3}{17}, \frac{2}{17}\rangle \\ &= \langle \frac{134}{17}, -\frac{54}{17}, -\frac{53}{17}\rangle.\end{aligned}$$

To check our work, we can use the dot product to verify that $\vec{\mathbf{p}}$ and $\vec{\mathbf{q}}$ are orthogonal vectors:

$$\begin{aligned}\vec{\mathbf{p}} \cdot \vec{\mathbf{q}} &= \langle \frac{2}{17}, \frac{3}{17}, \frac{2}{17}\rangle \cdot \langle \frac{134}{17}, -\frac{54}{17}, -\frac{53}{17}\rangle \\ &= \frac{268}{289} - \frac{162}{289} - \frac{106}{289} = 0.\end{aligned}$$

Then,

$$\vec{\mathbf{v}} = \vec{\mathbf{p}} + \vec{\mathbf{q}} = \langle \frac{2}{17}, \frac{3}{17}, \frac{2}{17}\rangle + \langle \frac{134}{17}, -\frac{54}{17}, -\frac{53}{17}\rangle.$$

Exercise 12.3.7

Express $\vec{v} = 5\hat{i} - \hat{j}$ as a sum of orthogonal vectors such that one of the vectors has the same direction as $\vec{u} = 4\hat{i} + 2\hat{j}$.

Hint

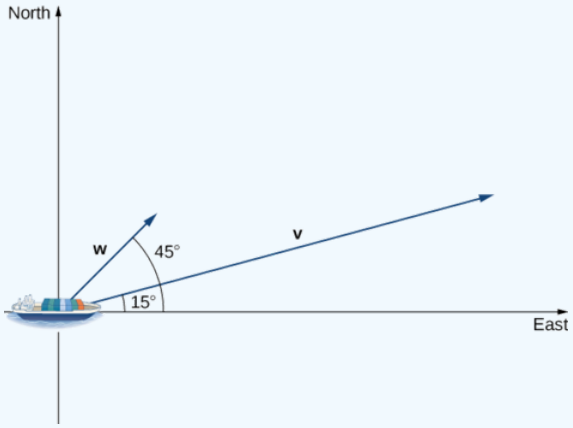
Start by finding the projection of $\vec{v}$ onto $\vec{u}$.

Answer

$\vec{v} = \vec{p} + \vec{q}$, where $\vec{p} = \frac{18}{5}\hat{i} + \frac{9}{5}\hat{j}$ and $\vec{q} = \frac{7}{5}\hat{i} - \frac{14}{5}\hat{j}$

Example 12.3.9: Scalar Projection of Velocity

A container ship leaves port traveling $15°$ north of east. Its engine generates a speed of 20 knots along that path (see the following figure). In addition, the ocean current moves the ship northeast at a speed of 2 knots. Considering both the engine and the current, how fast is the ship moving in the direction $15°$ north of east? Round the answer to two decimal places.



Solution

Let $\vec{v}$ be the velocity vector generated by the engine, and let $\vec{w}$ be the velocity vector of the current. We already know $\|\vec{v}\| = 20$ along the desired route. We just need to add in the scalar projection of $\vec{w}$ onto $\vec{v}$. We get

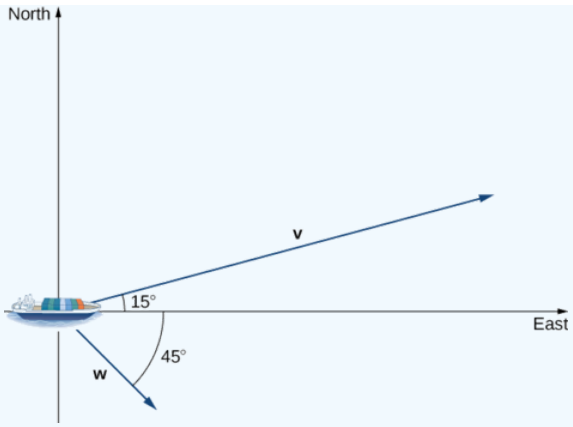
$$\text{comp}_{\vec{v}}\vec{w} = \frac{\vec{v} \cdot \vec{w}}{\|\vec{v}\|}$$

$$= \frac{\|\vec{v}\|\|\vec{w}\|\cos(30°)}{\|\vec{v}\|} = \|\vec{w}\|\cos(30°) = 2\frac{\sqrt{3}}{2} = \sqrt{3} \approx 1.73 \text{ knots}.$$

The ship is moving at 21.73 knots in the direction $15°$ north of east.

Exercise 12.3.8

Repeat the previous example, but assume the ocean current is moving southeast instead of northeast, as shown in the following figure.



Hint

Compute the scalar projection of $\vec{w}$ onto $\vec{v}$.

Answer

21 knots

Work

Now that we understand dot products, we can see how to apply them to real-life situations. The most common application of the dot product of two vectors is in the calculation of work.

From physics, we know that work is done when an object is moved by a force. When the force is constant and applied in the same direction the object moves, then we define the work done as the product of the force and the distance the object travels: $W = Fd$. We saw several examples of this type in earlier chapters. Now imagine the direction of the force is different from the direction of motion, as with the example of a child pulling a wagon. To find the work done, we need to multiply the component of the force that acts in the direction of the motion by the magnitude of the displacement. The dot product allows us to do just that. If we represent an applied force by a vector $\vec{\mathbf{F}}$ and the displacement of an object by a vector $\vec{s}$, then the **work done by the force** is the dot product of $\vec{\mathbf{F}}$ and $\vec{s}$.

Definition: Constant Force

When a constant force is applied to an object so the object moves in a straight line from point P to point Q, the work W done by the force $\vec{\mathbf{F}}$, acting at an angle θ from the line of motion, is given by

$$W = \vec{\mathbf{F}} \cdot \overrightarrow{PQ} = \| \vec{\mathbf{F}} \| \| \overrightarrow{PQ} \| \cos\theta.$$

Let's revisit the problem of the child's wagon introduced earlier. Suppose a child is pulling a wagon with a force having a magnitude of 8 lb on the handle at an angle of 55°. If the child pulls the wagon 50 ft, find the work done by the force (Figure 12.3.8).

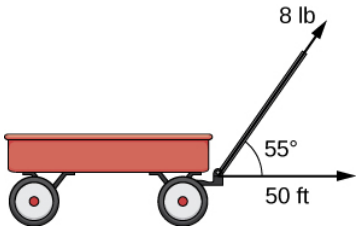


Figure 12.3.8: The horizontal component of the force is the projection of $\vec{\mathbf{F}}$ onto the positive x-axis.

We have

$$W = \| \vec{\mathbf{F}} \| \| \overrightarrow{PQ} \| \cos\theta = 8(50)(\cos(55°)) \approx 229\,\text{ft·lb}.$$

In U.S. standard units, we measure the magnitude of force $\| \vec{\mathbf{F}} \|$ in pounds. The magnitude of the displacement vector $\| \overrightarrow{PQ} \|$ tells us how far the object moved, and it is measured in feet. The customary unit of measure for work, then, is the foot-pound. One foot-pound is the amount of work required to move an object weighing 1 lb a distance of 1 ft straight up. In the metric system, the unit of measure for force is the newton (N), and the unit of measure of magnitude for work is a newton-meter (N·m), or a joule (J).

Example 12.3.10: Calculating Work

A conveyor belt generates a force $\vec{\mathbf{F}} = 5\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + \hat{\mathbf{k}}$ that moves a suitcase from point $(1, 1, 1)$ to point $(9, 4, 7)$ along a straight line. Find the work done by the conveyor belt. The distance is measured in meters and the force is measured in newtons.

Solution

The displacement vector $\overrightarrow{PQ}$ has initial point $(1, 1, 1)$ and terminal point $(9, 4, 7)$:

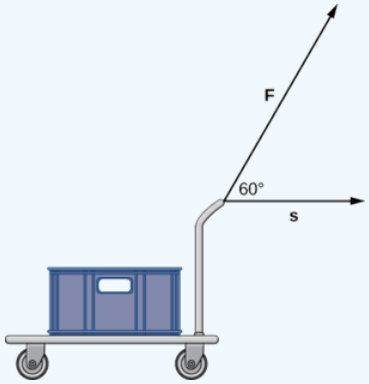
$$\overrightarrow{PQ} = \langle 9 - 1, 4 - 1, 7 - 1 \rangle = \langle 8, 3, 6 \rangle = 8\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 6\hat{\mathbf{k}}.$$

Work is the dot product of force and displacement:

$$\begin{aligned} W &= \vec{\mathbf{F}} \cdot \overrightarrow{PQ} \\ &= (5\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + \hat{\mathbf{k}}) \cdot (8\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 6\hat{\mathbf{k}}) \\ &= 5(8) + (-3)(3) + 1(6) \\ &= 37\,\text{N·m} \\ &= 37\,\text{J} \end{aligned}$$

Exercise 12.3.9

A constant force of 30 lb is applied at an angle of 60° to pull a handcart 10 ft across the ground. What is the work done by this force?



Hint

Use the definition of work as the dot product of force and distance.

Answer

150 ft-lb

Key Concepts

- The dot product, or scalar product, of two vectors $\vec{\mathbf{u}} = \langle u_1, u_2, u_3 \rangle$ and $\vec{\mathbf{v}} = \langle v_1, v_2, v_3 \rangle$ is $\vec{\mathbf{u}} \cdot \vec{\mathbf{v}} = u_1 v_1 + u_2 v_2 + u_3 v_3$.
- The dot product satisfies the following properties:
  - $\vec{\mathbf{u}} \cdot \vec{\mathbf{v}} = \vec{\mathbf{v}} \cdot \vec{\mathbf{u}}$
  - $\vec{\mathbf{u}} \cdot (\vec{\mathbf{v}} + \vec{\mathbf{w}}) = \vec{\mathbf{u}} \cdot \vec{\mathbf{v}} + \vec{\mathbf{u}} \cdot \vec{\mathbf{w}}$
  - $c(\vec{\mathbf{u}} \cdot \vec{\mathbf{v}}) = (c\vec{\mathbf{u}}) \cdot \vec{\mathbf{v}} = \vec{\mathbf{u}} \cdot (c\vec{\mathbf{v}})$

- $\vec{\mathbf{v}} \cdot \vec{\mathbf{v}} = \|\vec{\mathbf{v}}\|^2$

- The dot product of two vectors can be expressed, alternatively, as $\vec{\mathbf{u}} \cdot \vec{\mathbf{v}} = \|\vec{\mathbf{u}}\|\|\vec{\mathbf{v}}\| \cos\theta$. This form of the dot product is useful for finding the measure of the angle formed by two vectors.
- Vectors $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ are orthogonal if $\vec{\mathbf{u}} \cdot \vec{\mathbf{v}} = 0$.
- The angles formed by a nonzero vector and the coordinate axes are called the *direction angles* for the vector. The cosines of these angles are known as the *direction cosines.*
- The vector projection of $\vec{\mathbf{v}}$ onto $\vec{\mathbf{u}}$ is the vector $\text{proj}_{\vec{\mathbf{u}}}\vec{\mathbf{v}} = \dfrac{\vec{\mathbf{u}} \cdot \vec{\mathbf{v}}}{\|\vec{\mathbf{u}}\|^2}\vec{\mathbf{u}}$. The magnitude of this vector is known as the *scalar projection* of $\vec{\mathbf{v}}$ onto $\vec{\mathbf{u}}$, given by $\text{comp}_{\vec{\mathbf{u}}}\vec{\mathbf{v}} = \dfrac{\vec{\mathbf{u}} \cdot \vec{\mathbf{v}}}{\|\vec{\mathbf{u}}\|}$.
- Work is done when a force is applied to an object, causing displacement. When the force is represented by the vector $\vec{\mathbf{F}}$ and the displacement is represented by the vector $\vec{\mathbf{s}}$, then the work done W is given by the formula $W = \vec{\mathbf{F}} \cdot \vec{\mathbf{s}} = \| \vec{\mathbf{F}} \| \, \|\vec{\mathbf{s}}\| \cos\theta$.

Key Equations

- **Dot product of $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$**

  $$\vec{\mathbf{u}} \cdot \vec{\mathbf{v}} = u_1v_1 + u_2v_2 + u_3v_3 = \|\vec{\mathbf{u}}\|\|\vec{\mathbf{v}}\| \cos\theta$$

- **Cosine of the angle formed by $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$**

  $$\cos\theta = \frac{\vec{\mathbf{u}} \cdot \vec{\mathbf{v}}}{\|\vec{\mathbf{u}}\|\|\vec{\mathbf{v}}\|}$$

- **Vector projection of $\vec{\mathbf{v}}$ onto $\vec{\mathbf{u}}$**

  $$\text{proj}_{\vec{\mathbf{u}}}\vec{\mathbf{v}} = \frac{\vec{\mathbf{u}} \cdot \vec{\mathbf{v}}}{\|\vec{\mathbf{u}}\|^2}\vec{\mathbf{u}}$$

- **Scalar projection of $\vec{\mathbf{v}}$ onto $\vec{\mathbf{u}}$**

  $$\text{comp}_{\vec{\mathbf{u}}}\vec{\mathbf{v}} = \frac{\vec{\mathbf{u}} \cdot \vec{\mathbf{v}}}{\|\vec{\mathbf{u}}\|}$$

- **Work done by a force $\vec{\mathbf{F}}$ to move an object through displacement vector $\overrightarrow{PQ}$**

  $$W = \vec{\mathbf{F}} \cdot \overrightarrow{PQ} = \| \vec{\mathbf{F}} \| \, \| \overrightarrow{PQ} \| \cos\theta$$

Glossary

direction angles

the angles formed by a nonzero vector and the coordinate axes

direction cosines

the cosines of the angles formed by a nonzero vector and the coordinate axes

dot product or scalar product

$\vec{\mathbf{u}} \cdot \vec{\mathbf{v}} = u_1v_1 + u_2v_2 + u_3v_3$ where $\vec{\mathbf{u}} = \langle u_1, u_2, u_3 \rangle$ and $\vec{\mathbf{v}} = \langle v_1, v_2, v_3 \rangle$

scalar projection

the magnitude of the vector projection of a vector

orthogonal vectors

vectors that form a right angle when placed in standard position

vector projection

the component of a vector that follows a given direction

work done by a force

work is generally thought of as the amount of energy it takes to move an object; if we represent an applied force by a vector $\vec{\mathbf{F}}$ and the displacement of an object by a vector $\vec{\mathbf{s}}$, then the work done by the force is the dot product of $\vec{\mathbf{F}}$ and $\vec{\mathbf{s}}$.

- Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.
- edited for vector notation by Paul Seeburger

This page titled 12.3: The Dot Product is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

12.3E: Exercises for Section 12.3

For exercises 1-4, the vectors $\vec{u}$ and $\vec{v}$ are given. Calculate the dot product $\vec{u} \cdot \vec{v}$.

1) $\vec{u} = \langle 3, 0 \rangle, \quad \vec{v} = \langle 2, 2 \rangle$

Answer

$\vec{u} \cdot \vec{v} = 6$

2) $\vec{u} = \langle 3, -4 \rangle, \quad \vec{v} = \langle 4, 3 \rangle$

3) $\vec{u} = \langle 2, 2, -1 \rangle, \quad \vec{v} = \langle -1, 2, 2 \rangle$

Answer

$\vec{u} \cdot \vec{v} = 0$

4) $\vec{u} = \langle 4, 5, -6 \rangle, \quad \vec{v} = \langle 0, -2, -3 \rangle$

For exercises 5-8, the vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$ are given. Determine the vectors $(\vec{a} \cdot \vec{b})\vec{c}$ and $(\vec{a} \cdot \vec{c})\vec{b}$. Express the vectors in component form.

5) $\vec{a} = \langle 2, 0, -3 \rangle, \quad \vec{b} = \langle -4, -7, 1 \rangle, \quad \vec{c} = \langle 1, 1, -1 \rangle$

Answer

$(\vec{a} \cdot \vec{b})\vec{c} = \langle -11, -11, 11 \rangle; (\vec{a} \cdot \vec{c})\vec{b} = \langle -20, -35, 5 \rangle$

6) $\vec{a} = \langle 0, 1, 2 \rangle, \quad \vec{b} = \langle -1, 0, 1 \rangle, \quad \vec{c} = \langle 1, 0, -1 \rangle$

7) $\vec{a} = \hat{i} + \hat{j}, \quad \vec{b} = \hat{i} - \hat{k}, \quad \vec{c} = \hat{i} - 2\hat{k}$

Answer

$(\vec{a} \cdot \vec{b})\vec{c} = \langle 1, 0, -2 \rangle; (\vec{a} \cdot \vec{c})\vec{b} = \langle 1, 0, -1 \rangle$

8) $\vec{a} = \hat{i} - \hat{j} + \hat{k}, \quad \vec{b} = \hat{j} + 3\hat{k}, \quad \vec{c} = -\hat{i} + 2\hat{j} - 4\hat{k}$

For exercises 9-12, two vectors are given.

a. Find the measure of the angle θ between these two vectors. Express the answer in radians rounded to two decimal places, if it is not possible to express it exactly.

b. Is θ an acute angle?

9) [T] $\vec{a} = \langle 3, -1 \rangle, \quad \vec{b} = \langle -4, 0 \rangle$

Answer

$a. \theta = 2.82$ rad; $b. \theta$ is not acute.

10) [T] $\vec{a} = \langle 2, 1 \rangle, \quad \vec{b} = \langle -1, 3 \rangle$

11) $\vec{u} = 3\hat{i}, \quad \vec{v} = 4\hat{i} + 4\hat{j}$

Answer

$a. \theta = \frac{\pi}{4}$ rad; $b. \theta$ is acute.

12) $\vec{u} = 5\hat{i}, \quad \vec{v} = -6\hat{i} + 6\hat{j}$

For exercises 13-18, find the measure of the angle between the three-dimensional vectors $\vec{a}$ and $\vec{b}$. Express the answer in radians rounded to two decimal places, if it is not possible to express it exactly.

13) $\vec{a} = \langle 3, -1, 2 \rangle, \quad \vec{b} = \langle 1, -1, -2 \rangle$

Answer

$\theta = \frac{\pi}{2}$

14) $\vec{\mathbf{a}} = \langle 0, -1, -3 \rangle, \quad \vec{\mathbf{b}} = \langle 2, 3, -1 \rangle$

15) $\vec{\mathbf{a}} = \hat{\mathbf{i}} + \hat{\mathbf{j}}, \quad \vec{\mathbf{b}} = \hat{\mathbf{j}} - \hat{\mathbf{k}}$

Answer

$\theta = \frac{\pi}{3}$

16) $\vec{\mathbf{a}} = \hat{\mathbf{i}} - 2\hat{\mathbf{j}} + \hat{\mathbf{k}}, \quad \vec{\mathbf{b}} = \hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}}$

17) [T] $\vec{\mathbf{a}} = 3\hat{\mathbf{i}} - \hat{\mathbf{j}} - 2\hat{\mathbf{k}}, \quad \vec{\mathbf{b}} = \vec{\mathbf{v}} + \vec{\mathbf{w}}$, where $\vec{\mathbf{v}} = -2\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$ and $\vec{\mathbf{w}} = \hat{\mathbf{i}} + 2\hat{\mathbf{k}}$

Answer

$\theta = 2$ rad

18) [T] $\vec{\mathbf{a}} = 3\hat{\mathbf{i}} - \hat{\mathbf{j}} + 2\hat{\mathbf{k}}, \quad \vec{\mathbf{b}} = \vec{\mathbf{v}} - \vec{\mathbf{w}}$, where $\vec{\mathbf{v}} = 2\hat{\mathbf{i}} + \hat{\mathbf{j}} + 4\hat{\mathbf{k}}$ and $\vec{\mathbf{w}} = 6\hat{\mathbf{i}} + \hat{\mathbf{j}} + 2\hat{\mathbf{k}}$

For exercises 19-22, determine whether the given vectors are orthogonal.

19) $\vec{\mathbf{a}} = \langle x, y \rangle, \quad \vec{\mathbf{b}} = \langle -y, x \rangle$, where x and y are nonzero real numbers

Answer

Orthogonal

20) $\vec{\mathbf{a}} = \langle x, x \rangle, \quad \vec{\mathbf{b}} = \langle -y, y \rangle$, where x and y are nonzero real numbers

21) $\vec{\mathbf{a}} = 3\hat{\mathbf{i}} - \hat{\mathbf{j}} - 2\hat{\mathbf{k}}, \quad \vec{\mathbf{b}} = -2\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + \hat{\mathbf{k}}$

Answer

Not orthogonal

22) $\vec{\mathbf{a}} = \hat{\mathbf{i}} - \hat{\mathbf{j}}, \quad \vec{\mathbf{b}} = 7\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - \hat{\mathbf{k}}$

23) Find all two-dimensional vectors $\vec{\mathbf{a}}$ orthogonal to vector $\vec{\mathbf{b}} = \langle 3, 4 \rangle$. Express the answer in component form.

Answer

$\vec{\mathbf{a}} = \langle -\frac{4\alpha}{3}, \alpha \rangle$, where $\alpha \neq 0$ is a real number

24) Find all two-dimensional vectors $\vec{\mathbf{a}}$ orthogonal to vector $\vec{\mathbf{b}} = \langle 5, -6 \rangle$. Express the answer by using standard unit vectors.

25) Determine all three-dimensional vectors $\vec{\mathbf{u}}$ orthogonal to vector $\vec{\mathbf{v}} = \langle 1, 1, 0 \rangle$. Express the answer by using standard unit vectors.

Answer

$\vec{\mathbf{u}} = -\alpha\hat{\mathbf{i}} + \alpha\hat{\mathbf{j}} + \beta\hat{\mathbf{k}}$, where α and β are real numbers such that $\alpha^2 + \beta^2 \neq 0$

26) Determine all three-dimensional vectors $\vec{\mathbf{u}}$ orthogonal to vector $\vec{\mathbf{v}} = \hat{\mathbf{i}} - \hat{\mathbf{j}} - \hat{\mathbf{k}}$. Express the answer in component form.

27) Determine the real number α such that vectors $\vec{\mathbf{a}} = 2\hat{\mathbf{i}} + 3\hat{\mathbf{j}}$ and $\vec{\mathbf{b}} = 9\hat{\mathbf{i}} + \alpha\hat{\mathbf{j}}$ are orthogonal.

Answer

$\alpha = -6$

28) Determine the real number α such that vectors $\vec{\mathbf{a}} = -3\hat{\mathbf{i}} + 2\hat{\mathbf{j}}$ and $\vec{\mathbf{b}} = 2\hat{\mathbf{i}} + \alpha\hat{\mathbf{j}}$ are orthogonal.

29) [T] Consider the points $P(4, 5)$ and $Q(5, -7)$, and note that O represents the origin.

a. Determine vectors $\overrightarrow{OP}$ and $\overrightarrow{OQ}$. Express the answer by using standard unit vectors.

b. Determine the measure of angle O in triangle OPQ. Express the answer in degrees rounded to two decimal places.

Answer

$a. \overrightarrow{OP} = 4\hat{\mathbf{i}} + 5\hat{\mathbf{j}}, \quad \overrightarrow{OQ} = 5\hat{\mathbf{i}} - 7\hat{\mathbf{j}}; \quad b. 105.8°$

30) [T] Consider points $A(1,1), B(2,-7)$, and $C(6,3)$.

a. Determine vectors $\overrightarrow{BA}$ and $\overrightarrow{BC}$. Express the answer in component form.

b. Determine the measure of angle B in triangle ABC. Express the answer in degrees rounded to two decimal places.

31) Determine the measure of angle A in triangle ABC, where $A(1,1,8), B(4,-3,-4)$, and $C(-3,1,5)$. Express your answer in degrees rounded to two decimal places.

Answer

$68.33°$

32) Consider points $P(3,7,-2)$ and $Q(1,1,-3)$. Determine the angle between vectors $\overrightarrow{OP}$ and $\overrightarrow{OQ}$. [Remember that O represents the origin.] Express the answer in degrees rounded to two decimal places.

For exercises 33-34, determine which (if any) pairs of the following vectors are orthogonal.

33) $\vec{\mathbf{u}} = \langle 3,7,-2\rangle, \quad \vec{\mathbf{v}} = \langle 5,-3,-3\rangle, \quad \vec{\mathbf{w}} = \langle 0,1,-1\rangle$

Answer

$\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ are orthogonal; $\vec{\mathbf{v}}$ and $\vec{\mathbf{w}}$ are orthogonal, but $\vec{\mathbf{u}}$ and $\vec{\mathbf{w}}$ are **not** orthogonal.

34) $\vec{\mathbf{u}} = \hat{\mathbf{i}} - \hat{\mathbf{k}}, \quad \vec{\mathbf{v}} = 5\hat{\mathbf{j}} - 5\hat{\mathbf{k}}, \quad \vec{\mathbf{w}} = 10\hat{\mathbf{j}}$

35) Use vectors to show that a parallelogram with equal diagonals is a **rectangle**.

36) Use vectors to show that the diagonals of a **rhombus** are perpendicular.

37) Show that $\vec{\mathbf{u}} \cdot (\vec{\mathbf{v}} + \vec{\mathbf{w}}) = \vec{\mathbf{u}} \cdot \vec{\mathbf{v}} + \vec{\mathbf{u}} \cdot \vec{\mathbf{w}}$ is true for any vectors $\vec{\mathbf{u}}, \vec{\mathbf{v}}$, and $\vec{\mathbf{w}}$.

38) Verify the identity $\vec{\mathbf{u}} \cdot (\vec{\mathbf{v}} + \vec{\mathbf{w}}) = \vec{\mathbf{u}} \cdot \vec{\mathbf{v}} + \vec{\mathbf{u}} \cdot \vec{\mathbf{w}}$ for vectors $\vec{\mathbf{u}} = \langle 1,0,4\rangle, \vec{\mathbf{v}} = \langle -2,3,5\rangle$, and $\vec{\mathbf{w}} = \langle 4,-2,6\rangle$.

For exercises 39-41, determine $\vec{\mathbf{u}} \cdot \vec{\mathbf{v}}$ using the given information.

39) $\|\vec{\mathbf{u}}\| = 5, \quad \|\vec{\mathbf{v}}\| = 3$, and the angle between $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ is $\pi/6$ rad.

Answer

$\vec{\mathbf{u}} \cdot \vec{\mathbf{v}} = (5)(3)\cos\frac{\pi}{6} = 15 \cdot \frac{\sqrt{3}}{2} = \frac{15\sqrt{3}}{2}$

40) $\|\vec{\mathbf{u}}\| = 20, \quad \|\vec{\mathbf{v}}\| = 15$, and the angle between $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ is $5\pi/4$ rad.

41) $\|\vec{\mathbf{u}}\| = 7, \quad \|\vec{\mathbf{v}}\| = 12$, and the angle between $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ is $\pi/2$ rad.

Answer

$\vec{\mathbf{u}} \cdot \vec{\mathbf{v}} = (7)(12)\cos\frac{\pi}{2} = 84 \cdot 0 = 0$

42) Considering the definition of the dot product, $\vec{\mathbf{u}} \cdot \vec{\mathbf{v}} = \|\vec{\mathbf{u}}\|\|\vec{\mathbf{v}}\|\cos\theta$, where θ is the angle between $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$, what can you say about the angle θ between two nonzero vectors $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ if:

a. $\vec{\mathbf{u}} \cdot \vec{\mathbf{v}} > 0$?

b. $\vec{\mathbf{u}} \cdot \vec{\mathbf{v}} < 0$?

c. $\vec{\mathbf{u}} \cdot \vec{\mathbf{v}} = 0$?

In exercises 43-45, you are given the vertices of a triangle. Use dot products to determine whether each triangle is *acute*, *obtuse*, or *right*.

43) $(2,3,0),\ (3,1,-2),\ (-1,4,5)$

Answer

Label the points P, Q and R. Then determine the vectors $\overrightarrow{PQ}$ and $\overrightarrow{PR}$.

Here we see that, if $P=(2,3,0),\ Q=(3,1,-2)$ and $R=(-1,4,5)$, so we have $\overrightarrow{PQ}=<1,-2,-2>$ and $\overrightarrow{PR}=<-3,1,5>$.

Then $\overrightarrow{PQ}\cdot\overrightarrow{PR}=-15<0$, so angle QPR is obtuse. Therefore this triangle must be an ***obtuse triangle***.

44) $(5,1,0),\ (7,1,1),\ (6,3,2)$

45) $(6,1,4),\ (3,2,-1),\ (2,3,1)$

Solution:

Label the points P, Q and R. Then determine the vectors $\overrightarrow{PQ}$ and $\overrightarrow{PR}$.

Here we see that, if $P=(6,1,4),\ Q=(3,2,-1)$ and $R=(2,3,1)$, so we have $\overrightarrow{PQ}=<-3,1,-5>$ and $\overrightarrow{PR}=<-4,2,-3>$.

Then $\overrightarrow{PQ}\cdot\overrightarrow{PR}=29>0$, so angle QPR is **acute**. This triangle could still be any of the three types, so we need to check another angle.

Considering angle PQR we find $\overrightarrow{QP}=<3,-1,5>$ and $\overrightarrow{QR}=<-1,1,2>$, so $\overrightarrow{QP}\cdot\overrightarrow{QR}=6>0$, showing angle PQR is also **acute**.

We then must check the third angle to determine whether this is a right triangle or an acute triangle.

To check angle PRQ we find $\overrightarrow{RP}=<4,-2,3>$ and $\overrightarrow{RQ}=<1,-1,-2>$, giving us $\overrightarrow{RP}\cdot\overrightarrow{RQ}=0$. Angle PRQ is thus a **right** angle.

Therefore this triangle is a ***right triangle***.

46) Which of the following operations are allowed for nonzero vectors $\vec{u}$, $\vec{v}$, and $\vec{w}$, and which are not. Explain your reasoning.

a. $\vec{u}+(\vec{v}\cdot\vec{w})$ b. $(\vec{u}\cdot\vec{v})\cdot\vec{w}$ c. $(\vec{u}\cdot\vec{v})\,\vec{w}$
d. $(\vec{u}+\vec{v})\cdot\vec{w}$ e. $(\vec{u}+\vec{v})\cdot\|\vec{w}\|$ f. $\|\vec{u}+\vec{v}\|\|\vec{w}\|$

Projections

For the exercises 47-50, given the vectors $\vec{u}$ and $\vec{v}$:

a. Find the vector projection $\text{Proj}_{\vec{u}}\vec{v}$ of vector $\vec{v}$ onto vector $\vec{u}$ and the component of $\vec{v}$ that is orthogonal to $\vec{u}$, i.e., $\vec{v}_{\text{perp}}$. Express your answers in component form.

b. Find the scalar projection $\text{comp}_{\vec{u}}\vec{v}=\|\text{Proj}_{\vec{u}}\vec{v}\|$ of vector $\vec{v}$ onto vector $\vec{u}$.

c. Find the vector projection $\text{Proj}_{\vec{v}}\vec{u}$ of vector $\vec{u}$ onto vector $\vec{v}$ and the component of $\vec{u}$ that is orthogonal to $\vec{v}$, i.e., $\vec{u}_{\text{perp}}$. Express your answers in unit vector form.

d. Find the scalar projection $\text{comp}_{\vec{v}}\vec{u}$ of vector $\vec{u}$ onto vector $\vec{v}$.

47) $\vec{u}=5\hat{i}+2\hat{j},\quad \vec{v}=2\hat{i}+3\hat{j}$

Answer

a. $\text{Proj}_{\vec{u}}\vec{v}=\langle\frac{80}{29},\frac{32}{29}\rangle$ and $\vec{v}_{\text{perp}}=<-\frac{22}{29},\frac{55}{29}>$; b. $\text{comp}_{\vec{u}}\vec{v}=\frac{16}{\sqrt{29}}=\frac{16\sqrt{29}}{29}$;
c. $\text{Proj}_{\vec{v}}\vec{u}=\frac{32}{13}\hat{i}+\frac{48}{13}\hat{j}$ and $\vec{u}_{\text{perp}}=\frac{33}{13}\hat{i}-\frac{22}{13}\hat{j}$; d. $\text{comp}_{\vec{v}}\vec{u}=\frac{16}{\sqrt{13}}=\frac{16\sqrt{13}}{13}$

48) $\vec{u}=\langle-4,7\rangle,\quad \vec{v}=\langle3,5\rangle$

49) $\vec{u}=3\hat{i}+2\hat{k},\quad \vec{v}=2\hat{j}+4\hat{k}$

Answer

a. $\text{Proj}_{\vec{u}}\vec{v}=\langle\frac{24}{13},0,\frac{16}{13}\rangle$ and $\vec{v}_{\text{perp}}=<-\frac{24}{13},2,\frac{36}{13}>$; b. $\text{comp}_{\vec{u}}\vec{v}=\frac{8}{\sqrt{13}}=\frac{8\sqrt{13}}{13}$
c. $\text{Proj}_{\vec{v}}\vec{u}=\frac{4}{5}\hat{j}+\frac{8}{5}\hat{k}$ and $\vec{u}_{\text{perp}}=3\hat{i}-\frac{4}{5}\hat{j}+\frac{2}{5}\hat{k}$; d. $\text{comp}_{\vec{v}}\vec{u}=\frac{\sqrt{80}}{5}=\frac{4\sqrt{5}}{5}$;

50) $\vec{u}=\langle4,4,0\rangle,\quad \vec{v}=\langle0,4,1\rangle$

51) Consider the vectors $\vec{u} = 4\hat{i} - 3\hat{j}$ and $\vec{v} = 3\hat{i} + 2\hat{j}$.

a. Find the component form of vector $\text{Proj}_{\vec{u}}\vec{v}$ that represents the projection of $\vec{v}$ onto $\vec{u}$.

b. Write the decomposition $\vec{v} = \vec{w} + \vec{q}$ of vector $\vec{v}$ into the orthogonal components $\vec{w}$ and $\vec{q}$, where $\vec{w}$ is the projection of $\vec{v}$ onto $\vec{u}$ and $\vec{q}$ is the vector component of $\vec{v}$ orthogonal to the direction of $\vec{u}$. That is, $\vec{q} = \vec{v}_{\text{perp}}$.

Answer

a. $\text{Proj}_{\vec{u}}\vec{v} = \langle \frac{24}{25}, -\frac{18}{25} \rangle$; b. $\vec{q} = \vec{v}_{\text{perp}} = \langle \frac{51}{25}, \frac{68}{25} \rangle$,
$\vec{v} = \vec{w} + \vec{q} = \text{Proj}_{\vec{u}}\vec{v} + \vec{v}_{\text{perp}} = \langle \frac{24}{25}, -\frac{18}{25} \rangle + \langle \frac{51}{25}, \frac{68}{25} \rangle$
So we have that, $\vec{v} = \langle \frac{24}{25}, -\frac{18}{25} \rangle + \langle \frac{51}{25}, \frac{68}{25} \rangle$.

52) Consider vectors $\vec{u} = 2\hat{i} + 4\hat{j}$ and $\vec{v} = 4\hat{j} + 2\hat{k}$.

a. Find the component form of vector $\vec{w} = \text{Proj}_{\vec{u}}\vec{v}$ that represents the projection of $\vec{v}$ onto $\vec{u}$.

b. Write the decomposition $\vec{v} = \vec{w} + \vec{q}$ of vector $\vec{v}$ into the orthogonal components $\vec{w}$ and $\vec{q}$, where $\vec{w}$ is the projection of $\vec{v}$ onto $\vec{u}$ and $\vec{q}$ is a vector orthogonal to the direction of $\vec{u}$.

53) A $50,000$-pound truck is parked on a hill with a $5°$ slope above the horizontal (in the positive x-direction). Considering only the force due to gravity, find

a. the component of the weight of the truck that is pulling the truck down the hill (aligned with the road)

b. the component of the weight of the truck that is perpendicular to the road.

c. the magnitude of the force needed to keep the truck from rolling down the hill

d. the magnitude of the force that is perpendicular to the road.

Solution:

Let $\vec{F}$ be the force due to gravity on the truck (its weight) and let $\vec{r}$ be the unit vector pointing up the hill to the right.

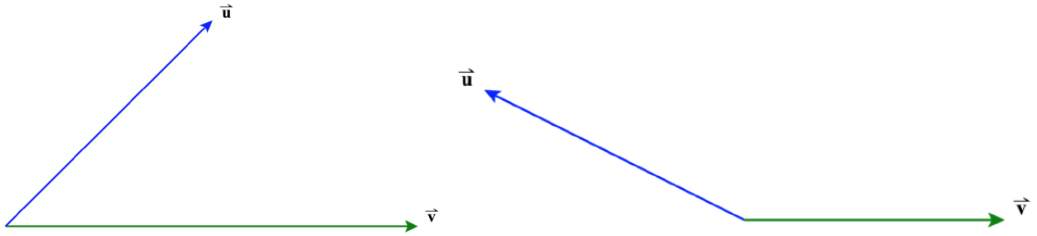
Then
a. the component of the weight of the truck that is pulling the truck down the hill is $\text{Proj}_{\vec{r}}\vec{F}$.
Now $\vec{F} = -50,000\hat{j}$ and $\vec{r} = \cos 5°\,\hat{i} + \sin 5°\,\hat{j}$ and $\vec{F} \cdot \vec{r} = -50,000 \sin 5° \approx -4357.78714$
Note that $\|\vec{r}\| = 1$.
Then: $\text{Proj}_{\vec{r}}\vec{F} = \left(\frac{\vec{F}\cdot\vec{r}}{\|\vec{r}\|^2}\right)\vec{r} = (-50,000 \sin 5°)\,\vec{r} = -50000 \sin 5° \cos 5°\,\hat{i} - 50000 \sin^2 5°\,\hat{j}$.
Using the identity, $\sin 2\theta = 2\cos\theta \sin\theta$, we can simplify this as
$\text{Proj}_{\vec{r}}\vec{F} = -25000 \sin 10°\,\hat{i} - 50000 \sin^2 5°\,\hat{j} \approx -4341.2\,\hat{i} - 379.8\,\hat{j}$.

b. the component of the weight of the truck that is perpendicular to the road is $\vec{F}_{\text{perp}}$.
Since we know that $\vec{F} = \text{Proj}_{\vec{r}}\vec{F} + \vec{F}_{\text{perp}}$, we have that $\vec{F}_{\text{perp}} = \vec{F} - \text{Proj}_{\vec{r}}\vec{F}$.
So $\vec{F}_{\text{perp}} = -50000\hat{j} - \left(-25000 \sin 10°\,\hat{i} - 50000 \sin^2 5°\,\hat{j}\right)$
$= 25000 \sin 10°\,\hat{i} + (50000 \sin^2 5° - 50000)\,\hat{j} = 25000 \sin 10°\,\hat{i} - 50000 \cos^2 5°\,\hat{j} \approx 4341.2\,\hat{i} - 49620.2\,\hat{j}$.

c. the magnitude of the force needed to keep the truck from rolling down the hill is $\|\text{Proj}_{\vec{r}}\vec{F}\| = 50,000 \sin 5°$ lbs ≈ 4357.8 lbs. Note that this force is actually the negative of the projection vector, but its magnitude is the same.

d. the magnitude of the truck's weight that is perpendicular to the road is $\|\vec{F}_{\text{perp}}\| \approx 49809.7$ lbs.

54) Given the vectors $\vec{u}$ and $\vec{v}$ shown in each diagram below, draw in $\text{Proj}_{\vec{v}}\vec{u}$ and $\vec{u}_{\text{perp}}$.



Work

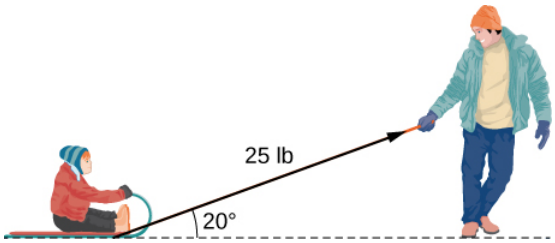
55) Find the work done by force $\vec{\mathbf{F}} = \langle 5, 6, -2 \rangle$ (measured in Newtons) that moves a particle from point $P(3, -1, 0)$ to point $Q(2, 3, 1)$ along a straight line (the distance is measured in meters).

Answer

$17\ \text{N}\cdot\text{m}$

56) [T] A sled is pulled by exerting a force of 100 N on a rope that makes an angle of $25°$ with the horizontal. Find the work done in pulling the sled 40 m. (Round the answer to one decimal place.)

57) [T] A father is pulling his son on a sled at an angle of $20°$ with the horizontal with a force of 25 lb (see the following image). He pulls the sled in a straight path of 50 ft. How much work was done by the man pulling the sled? (Round the answer to the nearest integer.)



Answer

1175 ft·lb

58) [T] A car is towed using a force of 1600 N. The rope used to pull the car makes an angle of 25° with the horizontal. Find the work done in towing the car 2 km. Express the answer in joules ($1\ \text{J} = 1\ \text{N}\cdot\text{m}$) rounded to the nearest integer.

59) [T] A boat sails north aided by a wind blowing in a direction of $N30°E$ with a magnitude of 500 lb. How much work is performed by the wind as the boat moves 100 ft? (Round the answer to two decimal places.)

Answer

$25000\sqrt{3}$ ft-lbs $\approx 43,301.27$ ft-lbs

Solution:

Vector representing the wind: $\vec{\mathbf{w}} = 500\cos 60°\hat{\mathbf{i}} + 500\sin 60°\hat{\mathbf{j}}$

Vector representing the displacement to the north: $\vec{\mathbf{d}} = 100\hat{\mathbf{j}}$

Work done by the wind: $W = \vec{\mathbf{w}} \cdot \vec{\mathbf{d}} = 25000\sqrt{3}$ ft-lbs $\approx 43,301.27$ ft-lbs

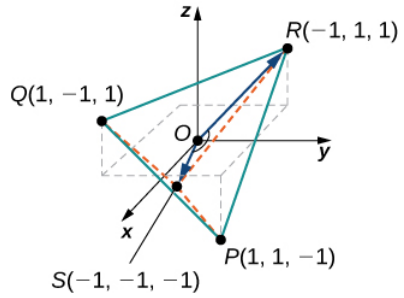
Other Applications of the Dot Product

60) [T] Find the vectors that join the center of a clock to the hours 1:00, 2:00, and 3:00. Assume the clock is circular with a radius of 1 unit.

61) A methane molecule has a carbon atom situated at the origin and four hydrogen atoms located at points $P(1, 1, -1), Q(1, -1, 1), R(-1, 1, 1)$, and $S(-1, -1, -1)$ (see figure).

a. Find the distance between the hydrogen atoms located at $P$and R.

b. Find the angle between vectors $\overrightarrow{OS}$ and $\overrightarrow{OR}$ that connect the carbon atom with the hydrogen atoms located at S and R, which is also called the *bond angle*. Express the answer in degrees rounded to two decimal places.



Answer

$a. 2\sqrt{2}; \quad b. 109.47°$

62) Vector $\vec{\mathbf{p}} = \langle 150, 225, 375 \rangle$ represents the price of certain models of bicycles sold by a bicycle shop. Vector $\vec{\mathbf{n}} = \langle 10, 7, 9 \rangle$ represents the number of bicycles sold of each model, respectively. Compute the dot product $\vec{\mathbf{p}} \cdot \vec{\mathbf{n}}$ and state its meaning.

63) [T] Two forces $\vec{\mathbf{F}}_1$ and $\vec{\mathbf{F}}_2$ are represented by vectors with initial points that are at the origin. The first force has a magnitude of 20 lb and passes through the point $P(1, 1, 0)$. The second force has a magnitude of 40 lb and passes through the point $Q(0, 1, 1)$. Let $\vec{\mathbf{F}}$ be the resultant force of forces $\vec{\mathbf{F}}_1$ and $\vec{\mathbf{F}}_2$.

a. Find the magnitude of $\vec{\mathbf{F}}$. (Round the answer to one decimal place.)

b. Find the direction angles of $\vec{\mathbf{F}}$. (Express the answer in degrees rounded to one decimal place.)

Answer

$a. \parallel \vec{\mathbf{F}}_1 + \vec{\mathbf{F}}_2 \parallel = 52.9$ lb; b. The direction angles are $\alpha = 74.5°$, $\beta = 36.7°$, and $\gamma = 57.7°$.

64) [T] Consider $\vec{\mathbf{r}}(t) = \langle \cos t, \sin t, 2t \rangle$ the position vector of a particle at time $t \in [0, 30]$, where the components of $\vec{\mathbf{r}}$ are expressed in centimeters and time in seconds. Let $\overrightarrow{OP}$ be the position vector of the particle after 1 sec.

a. Show that all vectors $\overrightarrow{PQ}$, where $Q(x, y, z)$ is an arbitrary point, orthogonal to the instantaneous velocity vector $\vec{\mathbf{v}}(1)$ of the particle after 1 sec, can be expressed as $\overrightarrow{PQ} = \langle x - \cos 1, y - \sin 1, z - 2 \rangle$, where $x \sin 1 - y \cos 1 - 2z + 4 = 0$. The set of point Q describes a plane called the *normal plane* to the path of the particle at point P.

b. Use a CAS to visualize the instantaneous velocity vector and the normal plane at point P along with the path of the particle.

Contributors

Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.

Problems 39-46 and 53 & 54 were created by Paul Seeburger (Monroe Community College).
Solution to Problem 59 also was added by Paul Seeburger.

This page titled 12.3E: Exercises for Section 12.3 is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

12.4: The Cross Product

Learning Objectives

- Calculate the cross product of two given vectors.
- Use determinants to calculate a cross product.
- Find a vector orthogonal to two given vectors.
- Determine areas and volumes by using the cross product.
- Calculate the torque of a given force and position vector.

Imagine a mechanic turning a wrench to tighten a bolt. The mechanic applies a force at the end of the wrench. This creates rotation, or torque, which tightens the bolt. We can use vectors to represent the force applied by the mechanic, and the distance (radius) from the bolt to the end of the wrench. Then, we can represent torque by a vector oriented along the axis of rotation. Note that the torque vector is orthogonal to both the force vector and the radius vector.

In this section, we develop an operation called the **cross product**, which allows us to find a vector orthogonal to two given vectors. Calculating torque is an important application of cross products, and we examine torque in more detail later in the section.

The Cross Product and Its Properties

The dot product is a multiplication of two vectors that results in a scalar. In this section, we introduce a product of two vectors that generates a third vector orthogonal to the first two. Consider how we might find such a vector. Let $\vec{\mathbf{u}} = \langle u_1, u_2, u_3 \rangle$ and $\vec{\mathbf{v}} = \langle v_1, v_2, v_3 \rangle$ be nonzero vectors. We want to find a vector $\vec{\mathbf{w}} = \langle w_1, w_2, w_3 \rangle$ orthogonal to both $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$—that is, we want to find $\vec{\mathbf{w}}$ such that $\vec{\mathbf{u}} \cdot \vec{\mathbf{w}} = 0$ and $\vec{\mathbf{v}} \cdot \vec{\mathbf{w}} = 0$. Therefore, w_1, w_2, and w_3 must satisfy

$$u_1 w_1 + u_2 w_2 + u_3 w_3 = 0 \tag{12.4.1}$$

$$v_1 w_1 + v_2 w_2 + v_3 w_3 = 0. \tag{12.4.2}$$

If we multiply the top equation by v_3 and the bottom equation by u_3 and subtract, we can eliminate the variable w_3, which gives

$$(u_1 v_3 - v_1 u_3) w_1 + (u_2 v_3 - v_2 u_3) w_2 = 0.$$

If we select

$$w_1 = u_2 v_3 - u_3 v_2$$
$$w_2 = -(u_1 v_3 - u_3 v_1),$$

we get a possible solution vector. Substituting these values back into the original equations (Equations 12.4.1 and 12.4.2) gives

$$w_3 = u_1 v_2 - u_2 v_1.$$

That is, vector

$$\vec{\mathbf{w}} = \langle u_2 v_3 - u_3 v_2, -(u_1 v_3 - u_3 v_1), u_1 v_2 - u_2 v_1 \rangle$$

is orthogonal to both $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$, which leads us to define the following operation, called the *cross product*.

Definition: Cross Product

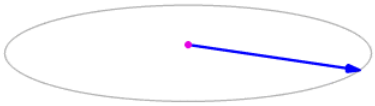
Let $\vec{\mathbf{u}} = \langle u_1, u_2, u_3 \rangle$ and $\vec{\mathbf{v}} = \langle v_1, v_2, v_3 \rangle$. Then, the *cross product* $\vec{\mathbf{u}} \times \vec{\mathbf{v}}$ is vector

$$\begin{aligned} \vec{\mathbf{u}} \times \vec{\mathbf{v}} &= (u_2 v_3 - u_3 v_2)\hat{\mathbf{i}} - (u_1 v_3 - u_3 v_1)\hat{\mathbf{j}} + (u_1 v_2 - u_2 v_1)\hat{\mathbf{k}} \\ &= \langle u_2 v_3 - u_3 v_2, -(u_1 v_3 - u_3 v_1), u_1 v_2 - u_2 v_1 \rangle. \end{aligned} \tag{12.4.3}$$

From the way we have developed $\vec{\mathbf{u}} \times \vec{\mathbf{v}}$, it should be clear that the cross product is orthogonal to both $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$. However, it never hurts to check. To show that $\vec{\mathbf{u}} \times \vec{\mathbf{v}}$ is orthogonal to $\vec{\mathbf{u}}$, we calculate the dot product of $\vec{\mathbf{u}}$ and $\vec{\mathbf{u}} \times \vec{\mathbf{v}}$.

$$\begin{aligned}\vec{\mathbf{u}} \cdot (\vec{\mathbf{u}} \times \vec{\mathbf{v}}) &= \langle u_1, u_2, u_3 \rangle \cdot \langle u_2v_3 - u_3v_2, -u_1v_3 + u_3v_1, u_1v_2 - u_2v_1 \rangle \\ &= u_1(u_2v_3 - u_3v_2) + u_2(-u_1v_3 + u_3v_1) + u_3(u_1v_2 - u_2v_1) \\ &= u_1u_2v_3 - u_1u_3v_2 - u_1u_2v_3 + u_2u_3v_1 + u_1u_3v_2 - u_2u_3v_1 \\ &= (u_1u_2v_3 - u_1u_2v_3) + (-u_1u_3v_2 + u_1u_3v_2) + (u_2u_3v_1 - u_2u_3v_1) \\ &= 0\end{aligned}$$

In a similar manner, we can show that the cross product is also orthogonal to $\vec{\mathbf{v}}$.



The cross product $\vec{\mathbf{a}} \times \vec{\mathbf{b}}$ (vertical, in pink) changes as the angle between the vectors $\vec{\mathbf{a}}$ (blue) and $\vec{\mathbf{b}}$ (red) changes. The cross product (purple) is always perpendicular to both vectors, and has magnitude zero when the vectors are parallel and maximum magnitude $\|\vec{\mathbf{a}}\|\|\vec{\mathbf{b}}\|$ when they are perpendicular. (Public Domain; LucasVB).

Example 12.4.1: Finding a Cross Product

Let $\vec{\mathbf{p}} = \langle -1, 2, 5 \rangle$ and $\vec{\mathbf{q}} = \langle 4, 0, -3 \rangle$ (Figure 12.4.1). Find $\vec{\mathbf{p}} \times \vec{\mathbf{q}}$.

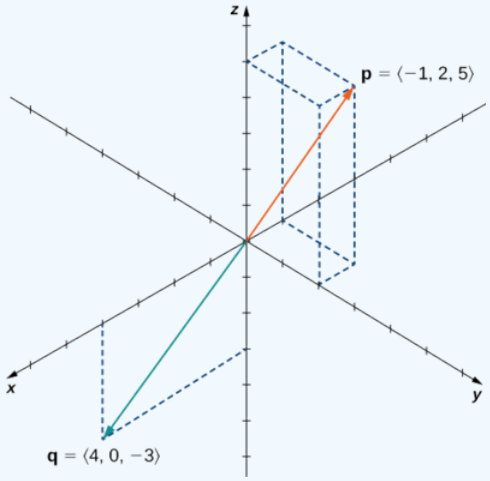


Figure 12.4.1: Finding a cross product to two given vectors.

Solution

Substitute the components of the vectors into Equation 12.4.3:

$$\begin{aligned}\vec{\mathbf{p}} \times \vec{\mathbf{q}} &= \langle -1, 2, 5 \rangle \times \langle 4, 0, -3 \rangle \\ &= \langle p_2q_3 - p_3q_2, -(p_1q_3 - p_3q_1), p_1q_2 - p_2q_1 \rangle \\ &= \langle 2(-3) - 5(0), -(-1)(-3) + 5(4), (-1)(0) - 2(4) \rangle \\ &= \langle -6, 17, -8 \rangle.\end{aligned}$$

Exercise 12.4.1

Find $\vec{\mathbf{p}} \times \vec{\mathbf{q}}$ for $\vec{\mathbf{p}} = \langle 5, 1, 2 \rangle$ and $\vec{\mathbf{q}} = \langle -2, 0, 1 \rangle$. Express the answer using standard unit vectors.

Hint

Use the formula $\vec{\mathbf{u}} \times \vec{\mathbf{v}} = (u_2 v_3 - u_3 v_2)\hat{\mathbf{i}} - (u_1 v_3 - u_3 v_1)\hat{\mathbf{j}} + (u_1 v_2 - u_2 v_1)\hat{\mathbf{k}}$.

Answer

$\vec{\mathbf{p}} \times \vec{\mathbf{q}} = \hat{\mathbf{i}} - 9\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$

Although it may not be obvious from Equation 12.4.3, the direction of $\vec{\mathbf{u}} \times \vec{\mathbf{v}}$ is given by the right-hand rule. If we hold the right hand out with the fingers pointing in the direction of $\vec{\mathbf{u}}$, then curl the fingers toward vector $\vec{\mathbf{v}}$, the thumb points in the direction of the cross product, as shown in Figure 12.4.2.

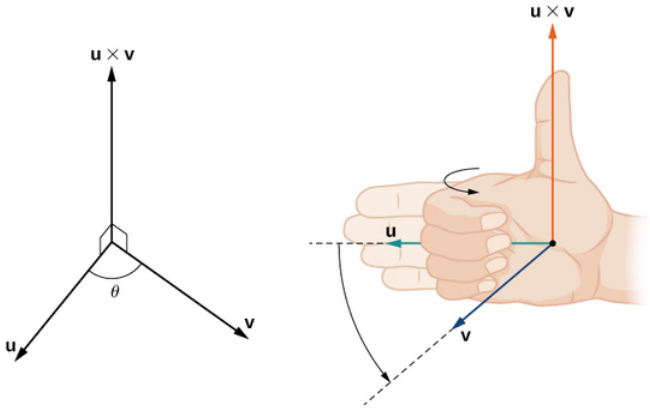


Figure 12.4.2: The direction of $\vec{\mathbf{u}} \times \vec{\mathbf{v}}$ is determined by the right-hand rule.

Notice what this means for the direction of $\vec{\mathbf{v}} \times \vec{\mathbf{u}}$. If we apply the right-hand rule to $\vec{\mathbf{v}} \times \vec{\mathbf{u}}$, we start with our fingers pointed in the direction of $\vec{\mathbf{v}}$, then curl our fingers toward the vector $\vec{\mathbf{u}}$. In this case, the thumb points in the opposite direction of $\vec{\mathbf{u}} \times \vec{\mathbf{v}}$. (Try it!)

Example 12.4.2: Anticommutativity of the Cross Product

Let $\vec{\mathbf{u}} = \langle 0, 2, 1 \rangle$ and $\vec{\mathbf{v}} = \langle 3, -1, 0 \rangle$. Calculate $\vec{\mathbf{u}} \times \vec{\mathbf{v}}$ and $\vec{\mathbf{v}} \times \vec{\mathbf{u}}$ and graph them.

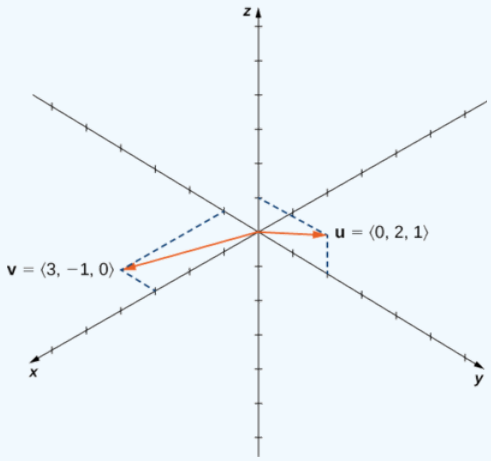


Figure 12.4.3: Are the cross products $\vec{\mathbf{u}} \times \vec{\mathbf{v}}$ and $\vec{\mathbf{v}} \times \vec{\mathbf{u}}$ in the same direction?

Solution

We have

$$\vec{\mathbf{u}} \times \vec{\mathbf{v}} = \langle (0 + 1), -(0 - 3), (0 - 6) \rangle = \langle 1, 3, -6 \rangle$$

$$\vec{\mathbf{v}} \times \vec{\mathbf{u}} = \langle (-1 - 0), -(3 - 0), (6 - 0) \rangle = \langle -1, -3, 6 \rangle.$$

We see that, in this case, $\vec{u} \times \vec{v} = -(\vec{v} \times \vec{u})$ (Figure 12.4.4). We prove this in general later in this section.

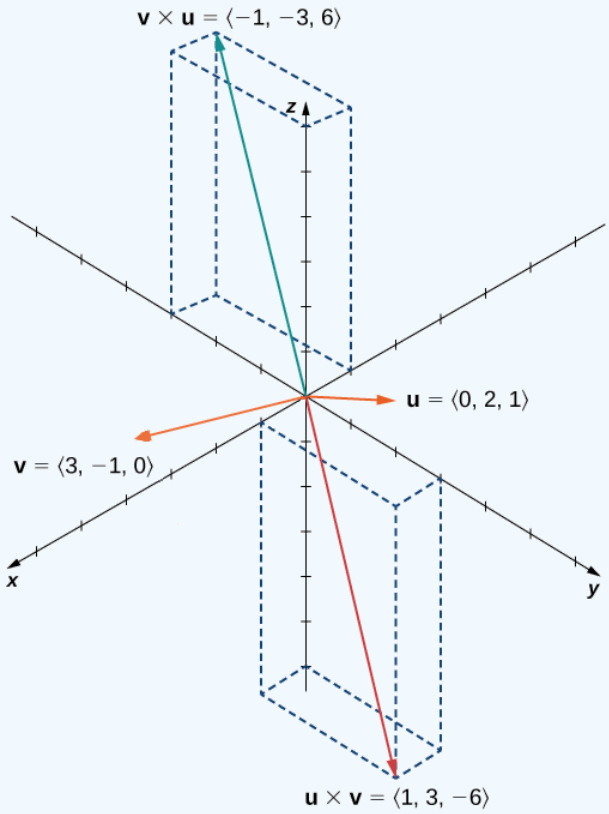


Figure 12.4.4: The cross products $\vec{u} \times \vec{v}$ and $\vec{v} \times \vec{u}$ are both orthogonal to $\vec{u}$ and $\vec{v}$, but in opposite directions.

Exercise 12.4.2

Suppose vectors $\vec{u}$ and $\vec{v}$ lie in the xy-plane (the z-component of each vector is zero). Now suppose the x- and y-components of $\vec{u}$ and the y-component of $\vec{v}$ are all positive, whereas the x-component of $\vec{v}$ is negative. Assuming the coordinate axes are oriented in the usual positions, in which direction does $\vec{u} \times \vec{v}$ point?

Hint

Remember the right-hand rule (Figure 12.4.2).

Answer

Up (the positive z-direction)

The cross products of the standard unit vectors $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{k}}$ can be useful for simplifying some calculations, so let's consider these cross products. A straightforward application of the definition shows that

$$\hat{\mathbf{i}} \times \hat{\mathbf{i}} = \hat{\mathbf{j}} \times \hat{\mathbf{j}} = \hat{\mathbf{k}} \times \hat{\mathbf{k}} = \vec{\mathbf{0}}.$$

(The cross product of two vectors is a vector, so each of these products results in the zero vector, not the scalar 0.) It's up to you to verify the calculations on your own.

Furthermore, because the cross product of two vectors is orthogonal to each of these vectors, we know that the cross product of $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ is parallel to $\hat{\mathbf{k}}$. Similarly, the vector product of $\hat{\mathbf{i}}$ and $\hat{\mathbf{k}}$ is parallel to $\hat{\mathbf{j}}$, and the vector product of $\hat{\mathbf{j}}$ and $\hat{\mathbf{k}}$ is parallel to $\hat{\mathbf{i}}$.

We can use the right-hand rule to determine the direction of each product. Then we have

$$\hat{\mathbf{i}} \times \hat{\mathbf{j}} = \hat{\mathbf{k}}$$
$$\hat{\mathbf{j}} \times \hat{\mathbf{i}} = -\hat{\mathbf{k}}$$
$$\hat{\mathbf{j}} \times \hat{\mathbf{k}} = \hat{\mathbf{i}}$$
$$\hat{\mathbf{k}} \times \hat{\mathbf{j}} = -\hat{\mathbf{i}}$$
$$\hat{\mathbf{k}} \times \hat{\mathbf{i}} = \hat{\mathbf{j}}$$
$$\hat{\mathbf{i}} \times \hat{\mathbf{k}} = -\hat{\mathbf{j}}.$$

These formulas come in handy later.

Example 12.4.3: Cross Product of Standard Unit Vectors

Find $\hat{\mathbf{i}} \times (\hat{\mathbf{j}} \times \hat{\mathbf{k}})$.

Solution

We know that $\hat{\mathbf{j}} \times \hat{\mathbf{k}} = \hat{\mathbf{i}}$. Therefore, $\hat{\mathbf{i}} \times (\hat{\mathbf{j}} \times \hat{\mathbf{k}}) = \hat{\mathbf{i}} \times \hat{\mathbf{i}} = \vec{\mathbf{0}}$.

Exercise 12.4.3

Find $(\hat{\mathbf{i}} \times \hat{\mathbf{j}}) \times (\hat{\mathbf{k}} \times \hat{\mathbf{i}})$.

Hint

Remember the right-hand rule (Figure 12.4.2).

Answer

$-\hat{\mathbf{i}}$

As we have seen, the dot product is often called the *scalar product* because it results in a scalar. The cross product results in a vector, so it is sometimes called the **vector product.** These operations are both versions of vector multiplication, but they have very different properties and applications. Let's explore some properties of the cross product. We prove only a few of them. Proofs of the other properties are left as exercises.

Properties of the Cross Product

Let $\vec{\mathbf{u}}, \vec{\mathbf{v}}$, and $\vec{\mathbf{w}}$ be vectors in space, and let c be a scalar.

i. Anticommutative property:

$$\vec{\mathbf{u}} \times \vec{\mathbf{v}} = -(\vec{\mathbf{v}} \times \vec{\mathbf{u}})$$

ii. Distributive property:

$$\vec{\mathbf{u}} \times (\vec{\mathbf{v}} + \vec{\mathbf{w}}) = \vec{\mathbf{u}} \times \vec{\mathbf{v}} + \vec{\mathbf{u}} \times \vec{\mathbf{w}}$$

iii. Multiplication by a constant:

$$c(\vec{\mathbf{u}} \times \vec{\mathbf{v}}) = (c\vec{\mathbf{u}}) \times \vec{\mathbf{v}} = \vec{\mathbf{u}} \times (c\vec{\mathbf{v}})$$

iv. Cross product of the zero vector:

$$\vec{\mathbf{u}} \times \vec{\mathbf{0}} = \vec{\mathbf{0}} \times \vec{\mathbf{u}} = \vec{\mathbf{0}}$$

v. Cross product of a vector with itself:

$$\vec{\mathbf{v}} \times \vec{\mathbf{v}} = \vec{\mathbf{0}}$$

vi. Triple scalar product:

$$\vec{\mathbf{u}} \cdot (\vec{\mathbf{v}} \times \vec{\mathbf{w}}) = (\vec{\mathbf{u}} \times \vec{\mathbf{v}}) \cdot \vec{\mathbf{w}}$$

vii. Triple cross product:

$$\vec{\mathbf{u}} \times (\vec{\mathbf{v}} \times \vec{\mathbf{w}}) = (\vec{\mathbf{u}} \cdot \vec{\mathbf{w}})\, \vec{\mathbf{v}} - (\vec{\mathbf{u}} \cdot \vec{\mathbf{v}})\, \vec{\mathbf{w}}$$

Proof

For property i, we want to show $\vec{\mathbf{u}} \times \vec{\mathbf{v}} = -(\vec{\mathbf{v}} \times \vec{\mathbf{u}})$. We have

$$\begin{aligned} \vec{\mathbf{u}} \times \vec{\mathbf{v}} &= \langle u_1, u_2, u_3 \rangle \times \langle v_1, v_2, v_3 \rangle \\ &= \langle u_2 v_3 - u_3 v_2, -u_1 v_3 + u_3 v_1, u_1 v_2 - u_2 v_1 \rangle \\ &= -\langle u_3 v_2 - u_2 v_3, -u_3 v_1 + u_1 v_3, u_2 v_1 - u_1 v_2 \rangle \\ &= -\langle v_1, v_2, v_3 \rangle \times \langle u_1, u_2, u_3 \rangle \\ &= -(\vec{\mathbf{v}} \times \vec{\mathbf{u}}). \end{aligned}$$

Unlike most operations we've seen, the cross product is not commutative. This makes sense if we think about the right-hand rule.

For property iv., this follows directly from the definition of the cross product. We have

$$\vec{\mathbf{u}} \times \vec{\mathbf{0}} = \langle u_2(0) - u_3(0), -(u_1(0) - u_3(0)), u_1(0) - u_2(0) \rangle = \langle 0, 0, 0 \rangle = \vec{\mathbf{0}}.$$

Then, by property i., $\vec{\mathbf{0}} \times \vec{\mathbf{u}} = \vec{\mathbf{0}}$ as well. Remember that the dot product of a vector and the zero vector is the *scalar* 0, whereas the cross product of a vector with the zero vector is the *vector* $\vec{\mathbf{0}}$.

Property vi. looks like the associative property, but note the change in operations:

$$\begin{aligned} \vec{\mathbf{u}} \cdot (\vec{\mathbf{v}} \times \vec{\mathbf{w}}) &= u \cdot \langle v_2 w_3 - v_3 w_2, -v_1 w_3 + v_3 w_1, v_1 w_2 - v_2 w_1 \rangle \\ &= u_1(v_2 w_3 - v_3 w_2) + u_2(-v_1 w_3 + v_3 w_1) + u_3(v_1 w_2 - v_2 w_1) \\ &= u_1 v_2 w_3 - u_1 v_3 w_2 - u_2 v_1 w_3 + u_2 v_3 w_1 + u_3 v_1 w_2 - u_3 v_2 w_1 \\ &= (u_2 v_3 - u_3 v_2) w_1 + (u_3 v_1 - u_1 v_3) w_2 + (u_1 v_2 - u_2 v_1) w_3 \\ &= \langle u_2 v_3 - u_3 v_2, u_3 v_1 - u_1 v_3, u_1 v_2 - u_2 v_1 \rangle \cdot \langle w_1, w_2, w_3 \rangle = (\vec{\mathbf{u}} \times \vec{\mathbf{v}}) \cdot \vec{\mathbf{w}}. \end{aligned}$$

□

Example 12.4.4: Using the Properties of the Cross Product

Use the cross product properties to calculate $(2\hat{\mathbf{i}} \times 3\hat{\mathbf{j}}) \times \hat{\mathbf{j}}$.

Solution

$$\begin{aligned} (2\hat{\mathbf{i}} \times 3\hat{\mathbf{j}}) \times \hat{\mathbf{j}} &= 2(\hat{\mathbf{i}} \times 3\hat{\mathbf{j}}) \times \hat{\mathbf{j}} \\ &= 2(3)(\hat{\mathbf{i}} \times \hat{\mathbf{j}}) \times \hat{\mathbf{j}} \\ &= (6\hat{\mathbf{k}}) \times \hat{\mathbf{j}} \\ &= 6(\hat{\mathbf{k}} \times \hat{\mathbf{j}}) \\ &= 6(-\hat{\mathbf{i}}) = -6\hat{\mathbf{i}}. \end{aligned}$$

Exercise 12.4.4

Use the properties of the cross product to calculate $(\hat{\mathbf{i}} \times \hat{\mathbf{k}}) \times (\hat{\mathbf{k}} \times \hat{\mathbf{j}})$.

Hint

$\vec{\mathbf{u}} \times \vec{\mathbf{v}} = -(\vec{\mathbf{v}} \times \vec{\mathbf{u}})$

Answer

$-\hat{\mathbf{k}}$

So far in this section, we have been concerned with the direction of the vector $\vec{\mathbf{u}} \times \vec{\mathbf{v}}$, but we have not discussed its magnitude. It turns out there is a simple expression for the magnitude of $\vec{\mathbf{u}} \times \vec{\mathbf{v}}$ involving the magnitudes of $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$, and the sine of the angle between them.

Magnitude of the Cross Product

Let $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ be vectors, and let θ be the angle between them. Then, $\|\vec{\mathbf{u}} \times \vec{\mathbf{v}}\| = \|\vec{\mathbf{u}}\| \cdot \|\vec{\mathbf{v}}\| \cdot \sin\theta$.

Proof

Let $\vec{\mathbf{u}} = \langle u_1, u_2, u_3 \rangle$ and $\vec{\mathbf{v}} = \langle v_1, v_2, v_3 \rangle$ be vectors, and let θ denote the angle between them. Then

$$\begin{aligned}
\|\vec{\mathbf{u}} \times \vec{\mathbf{v}}\|^2 &= (u_2v_3 - u_3v_2)^2 + (u_3v_1 - u_1v_3)^2 + (u_1v_2 - u_2v_1)^2 \\
&= u_2^2v_3^2 - 2u_2u_3v_2v_3 + u_3^2v_2^2 + u_3^2v_1^2 - 2u_1u_3v_1v_3 + u_1^2v_3^2 + u_1^2v_2^2 - 2u_1u_2v_1v_2 + u_2^2v_1^2 \\
&= u_1^2v_1^2 + u_1^2v_2^2 + u_1^2v_3^2 + u_2^2v_1^2 + u_2^2v_2^2 + u_2^2v_3^2 + u_3^2v_1^2 + u_3^2v_2^2 + u_3^2v_3^2 \\
&\quad - (u_1^2v_1^2 + u_2^2v_2^2 + u_3^2v_3^2 + 2u_1u_2v_1v_2 + 2u_1u_3v_1v_3 + 2u_2u_3v_2v_3) \\
&= (u_1^2 + u_2^2 + u_3^2)(v_1^2 + v_2^2 + v_3^2) - (u_1v_1 + u_2v_2 + u_3v_3)^2 \\
&= \|\vec{\mathbf{u}}\|^2\|\vec{\mathbf{v}}\|^2 - (\vec{\mathbf{u}} \cdot \vec{\mathbf{v}})^2 \\
&= \|\vec{\mathbf{u}}\|^2\|\vec{\mathbf{v}}\|^2 - \|\vec{\mathbf{u}}\|^2\|\vec{\mathbf{v}}\|^2\cos^2\theta \\
&= \|\vec{\mathbf{u}}\|^2\|\vec{\mathbf{v}}\|^2(1 - \cos^2\theta) \\
&= \|\vec{\mathbf{u}}\|^2\|\vec{\mathbf{v}}\|^2(\sin^2\theta).
\end{aligned}$$

Taking square roots and noting that $\sqrt{\sin^2\theta} = \sin\theta$ for $0 \le \theta \le 180°$, we have the desired result:

$$\|\vec{\mathbf{u}} \times \vec{\mathbf{v}}\| = \|\vec{\mathbf{u}}\|\|\vec{\mathbf{v}}\|\sin\theta.$$

□

This definition of the cross product allows us to visualize or interpret the product geometrically. It is clear, for example, that the cross product is defined only for vectors in three dimensions, not for vectors in two dimensions. In two dimensions, it is impossible to generate a vector simultaneously orthogonal to two nonparallel vectors.

Example 12.4.5: Calculating the Cross Product

Use "Magnitude of the Cross Product" to find the magnitude of the cross product of $\vec{\mathbf{u}} = \langle 0, 4, 0 \rangle$ and $\vec{\mathbf{v}} = \langle 0, 0, -3 \rangle$.

Solution

We have

$$\begin{aligned}
\|\vec{\mathbf{u}} \times \vec{\mathbf{v}}\| &= \|\vec{\mathbf{u}}\| \cdot \|\vec{\mathbf{v}}\| \cdot \sin\theta \\
&= \sqrt{0^2 + 4^2 + 0^2} \cdot \sqrt{0^2 + 0^2 + (-3)^2} \cdot \sin\frac{\pi}{2} \\
&= 4(3)(1) = 12
\end{aligned}$$

Exercise 12.4.5

Use "Magnitude of the Cross Product" to find the magnitude of $\vec{\mathbf{u}} \times \vec{\mathbf{v}}$, where $\vec{\mathbf{u}} = \langle -8, 0, 0 \rangle$ and $\vec{\mathbf{v}} = \langle 0, 2, 0 \rangle$.

Hint

Vectors $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ are orthogonal.

Answer

Determinants and the Cross Product

Using Equation 12.4.3 to find the cross product of two vectors is straightforward, and it presents the cross product in the useful component form. The formula, however, is complicated and difficult to remember. Fortunately, we have an alternative. We can calculate the cross product of two vectors using determinant notation.

A 2×2 determinant is defined by

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = a_1 b_2 - b_1 a_2.$$

For example,

$$\begin{vmatrix} 3 & -2 \\ 5 & 1 \end{vmatrix} = 3(1) - 5(-2) = 3 + 10 = 13.$$

A 3×3 determinant is defined in terms of 2×2 determinants as follows:

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}. \tag{12.4.4}$$

Equation 12.4.4 is referred to as the expansion of the determinant along the first row. Notice that the multipliers of each of the 2×2 determinants on the right side of this expression are the entries in the first row of the 3×3 determinant. Furthermore, each of the 2×2 determinants contains the entries from the 3×3 determinant that would remain if you crossed out the row and column containing the multiplier. Thus, for the first term on the right, a_1 is the multiplier, and the 2×2 determinant contains the entries that remain if you cross out the first row and first column of the 3×3 determinant. Similarly, for the second term, the multiplier is a_2, and the 2×2 determinant contains the entries that remain if you cross out the first row and second column of the 3×3 determinant. Notice, however, that the coefficient of the second term is negative. The third term can be calculated in similar fashion.

Example 12.4.6: Using Expansion Along the First Row to Compute a 3×3 Determinant

Evaluate the determinant $\begin{vmatrix} 2 & 5 & -1 \\ -1 & 1 & 3 \\ -2 & 3 & 4 \end{vmatrix}$.

Solution

We have

$$\begin{aligned} \begin{vmatrix} 2 & 5 & -1 \\ -1 & 1 & 3 \\ -2 & 3 & 4 \end{vmatrix} &= 2 \begin{vmatrix} 1 & 3 \\ 3 & 4 \end{vmatrix} - 5 \begin{vmatrix} -1 & 3 \\ -2 & 4 \end{vmatrix} - 1 \begin{vmatrix} -1 & 1 \\ -2 & 3 \end{vmatrix} \\ &= 2(4-9) - 5(-4+6) - 1(-3+2) \\ &= 2(-5) - 5(2) - 1(-1) = -10 - 10 + 1 \\ &= -19 \end{aligned}$$

Exercise 12.4.6

Evaluate the determinant $\begin{vmatrix} 1 & -2 & -1 \\ 3 & 2 & -3 \\ 1 & 5 & 4 \end{vmatrix}$.

Hint

Expand along the first row. Don't forget the second term is negative!

Answer

Technically, determinants are defined only in terms of arrays of real numbers. However, the determinant notation provides a useful mnemonic device for the cross product formula.

Rule: Cross Product Calculated by a Determinant

Let $\vec{\mathbf{u}} = \langle u_1, u_2, u_3 \rangle$ and $\vec{\mathbf{v}} = \langle v_1, v_2, v_3 \rangle$ be vectors. Then the cross product $\vec{\mathbf{u}} \times \vec{\mathbf{v}}$ is given by

$$\vec{\mathbf{u}} \times \vec{\mathbf{v}} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} \hat{\mathbf{i}} - \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} \hat{\mathbf{j}} + \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \hat{\mathbf{k}}.$$

Example 12.4.7: Using Determinant Notation to find $\vec{\mathbf{p}} \times \vec{\mathbf{q}}$

Let $\vec{\mathbf{p}} = \langle -1, 2, 5 \rangle$ and $\vec{\mathbf{q}} = \langle 4, 0, -3 \rangle$. Find $\vec{\mathbf{p}} \times \vec{\mathbf{q}}$.

Solution

We set up our determinant by putting the standard unit vectors across the first row, the components of $\vec{\mathbf{u}}$ in the second row, and the components of $\vec{\mathbf{v}}$ in the third row. Then, we have

$$\begin{aligned} \vec{\mathbf{p}} \times \vec{\mathbf{q}} &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ -1 & 2 & 5 \\ 4 & 0 & -3 \end{vmatrix} = \begin{vmatrix} 2 & 5 \\ 0 & -3 \end{vmatrix} \hat{\mathbf{i}} - \begin{vmatrix} -1 & 5 \\ 4 & -3 \end{vmatrix} \hat{\mathbf{j}} + \begin{vmatrix} -1 & 2 \\ 4 & 0 \end{vmatrix} \hat{\mathbf{k}} \\ &= (-6 - 0)\hat{\mathbf{i}} - (3 - 20)\hat{\mathbf{j}} + (0 - 8)\hat{\mathbf{k}} \\ &= -6\hat{\mathbf{i}} + 17\hat{\mathbf{j}} - 8\hat{\mathbf{k}}. \end{aligned}$$

Notice that this answer confirms the calculation of the cross product in Example 12.4.1.

Exercise 12.4.7

Use determinant notation to find $\vec{\mathbf{a}} \times \vec{\mathbf{b}}$, where $\vec{\mathbf{a}} = \langle 8, 2, 3 \rangle$ and $\vec{\mathbf{b}} = \langle -1, 0, 4 \rangle$.

Hint

Calculate the determinant $\begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 8 & 2 & 3 \\ -1 & 0 & 4 \end{vmatrix}$.

Answer

$\vec{\mathbf{a}} \times \vec{\mathbf{b}} = 8\hat{\mathbf{i}} - 35\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$

Using the Cross Product

The cross product is very useful for several types of calculations, including finding a vector orthogonal to two given vectors, computing areas of triangles and parallelograms, and even determining the volume of the three-dimensional geometric shape made of parallelograms known as a *parallelepiped*. The following examples illustrate these calculations.

Example 12.4.8: Finding a Unit Vector Orthogonal to Two Given Vectors

Let $\vec{\mathbf{a}} = \langle 5, 2, -1 \rangle$ and $\vec{\mathbf{b}} = \langle 0, -1, 4 \rangle$. Find a unit vector orthogonal to both $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$.

Solution

The cross product $\vec{\mathbf{a}} \times \vec{\mathbf{b}}$ is orthogonal to both vectors $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$. We can calculate it with a determinant:

$$\vec{\mathbf{a}} \times \vec{\mathbf{b}} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 5 & 2 & -1 \\ 0 & -1 & 4 \end{vmatrix} = \begin{vmatrix} 2 & -1 \\ -1 & 4 \end{vmatrix} \hat{\mathbf{i}} - \begin{vmatrix} 5 & -1 \\ 0 & 4 \end{vmatrix} \hat{\mathbf{j}} + \begin{vmatrix} 5 & 2 \\ 0 & -1 \end{vmatrix} \hat{\mathbf{k}}$$
$$= (8-1)\hat{\mathbf{i}} - (20-0)\hat{\mathbf{j}} + (-5-0)\hat{\mathbf{k}}$$
$$= 7\hat{\mathbf{i}} - 20\hat{\mathbf{j}} - 5\hat{\mathbf{k}}.$$

Normalize this vector to find a unit vector in the same direction:

$$\|\vec{\mathbf{a}} \times \vec{\mathbf{b}}\| = \sqrt{(7)^2 + (-20)^2 + (-5)^2} = \sqrt{474}.$$

Thus, $\left\langle \frac{7}{\sqrt{474}}, \frac{-20}{\sqrt{474}}, \frac{-5}{\sqrt{474}} \right\rangle$ is a unit vector orthogonal to $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$.

Simplified, this vector becomes $\left\langle \frac{7\sqrt{474}}{474}, \frac{-10\sqrt{474}}{237}, \frac{-5\sqrt{474}}{474} \right\rangle$.

Exercise 12.4.8

Find a unit vector orthogonal to both $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$, where $\vec{\mathbf{a}} = \langle 4, 0, 3 \rangle$ and $\vec{\mathbf{b}} = \langle 1, 1, 4 \rangle$.

Hint

Normalize the cross product.

Answer

$\left\langle \frac{-3}{\sqrt{194}}, \frac{-13}{\sqrt{194}}, \frac{4}{\sqrt{194}} \right\rangle$ or, simplified as $\left\langle \frac{-3\sqrt{194}}{194}, \frac{-13\sqrt{194}}{194}, \frac{2\sqrt{194}}{97} \right\rangle$

To use the cross product for calculating areas, we state and prove the following theorem.

Area of a Parallelogram

If we locate vectors $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ such that they form adjacent sides of a parallelogram, then the area of the parallelogram is given by $\|\vec{\mathbf{u}} \times \vec{\mathbf{v}}\|$ (Figure 12.4.5).

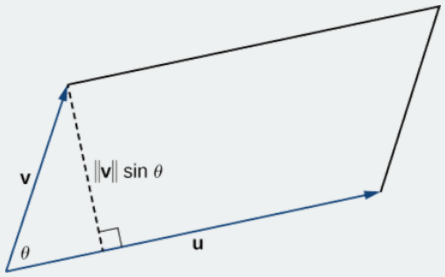


Figure 12.4.5: The parallelogram with adjacent sides $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ has base $\|\vec{\mathbf{u}}\|$ and height $\|\vec{\mathbf{v}}\| \sin\theta$.

Proof

We show that the magnitude of the cross product is equal to the base times height of the parallelogram.

$$\begin{aligned} \text{Area of a parallelogram} &= \text{base} \times \text{height} \\ &= \|\vec{\mathbf{u}}\|(\|\vec{\mathbf{v}}\| \sin\theta) \\ &= \|\vec{\mathbf{u}} \times \vec{\mathbf{v}}\| \end{aligned}$$

□

Example 12.4.9: Finding the Area of a Triangle

Let $P = (1, 0, 0), Q = (0, 1, 0)$, and $R = (0, 0, 1)$ be the vertices of a triangle (Figure 12.4.6). Find its area.

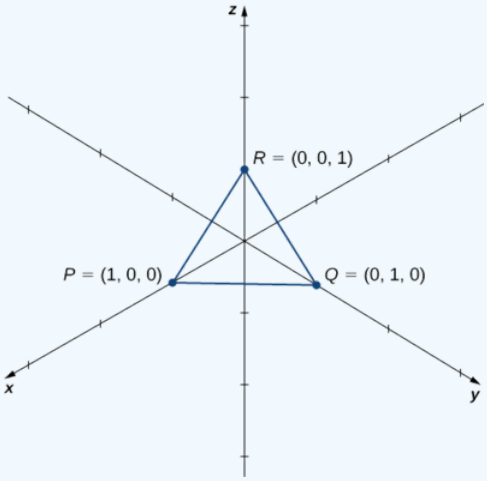


Figure 12.4.6: Finding the area of a triangle by using the cross product.

Solution

We have $\overrightarrow{PQ} = \langle 0-1, 1-0, 0-0\rangle = \langle -1, 1, 0\rangle$ and $\overrightarrow{PR} = \langle 0-1, 0-0, 1-0\rangle = \langle -1, 0, 1\rangle$. The area of the parallelogram with adjacent sides $\overrightarrow{PQ}$ and $\overrightarrow{PR}$ is given by $\| \overrightarrow{PQ} \times \overrightarrow{PR} \|$:

$$\begin{aligned} \overrightarrow{PQ} \times \overrightarrow{PR} &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix} \\ &= (1-0)\hat{\mathbf{i}} - (-1-0)\hat{\mathbf{j}} + (0-(-1))\hat{\mathbf{k}} \\ &= \hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}} \end{aligned}$$

$$\begin{aligned} \| \overrightarrow{PQ} \times \overrightarrow{PR} \| &= \| \langle 1, 1, 1\rangle \| \\ &= \sqrt{1^2 + 1^2 + 1^2} \\ &= \sqrt{3}. \end{aligned}$$

The area of ΔPQR is half the area of the parallelogram or $\sqrt{3}/2 \text{ units}^2$.

Exercise 12.4.9

Find the area of the parallelogram $PQRS$ with vertices $P(1, 1, 0)$, $Q(7, 1, 0)$, $R(9, 4, 2)$, and $S(3, 4, 2)$.

Hint

Sketch the parallelogram and identify two vectors that form adjacent sides of the parallelogram.

Answer

$6\sqrt{13} \text{ units}^2$

The Triple Scalar Product

Because the cross product of two vectors is a vector, it is possible to combine the dot product and the cross product. The dot product of a vector with the cross product of two other vectors is called the **triple scalar product** because the result is a scalar.

Definition: Triple Scalar Product

The triple scalar product of vectors $\vec{\mathbf{u}}$, $\vec{\mathbf{v}}$, and $\vec{\mathbf{w}}$ is

$$\vec{\mathbf{u}} \cdot (\vec{\mathbf{v}} \times \vec{\mathbf{w}}).$$

Calculating a Triple Scalar Product

The triple scalar product of vectors

$$\vec{\mathbf{u}} = u_1\hat{\mathbf{i}} + u_2\hat{\mathbf{j}} + u_3\hat{\mathbf{k}}$$

$$\vec{\mathbf{v}} = v_1\hat{\mathbf{i}} + v_2\hat{\mathbf{j}} + v_3\hat{\mathbf{k}}$$

and

$$\vec{\mathbf{w}} = w_1\hat{\mathbf{i}} + w_2\hat{\mathbf{j}} + w_3\hat{\mathbf{k}}$$

is the determinant of the 3×3 matrix formed by the components of the vectors:

$$\vec{\mathbf{u}} \cdot (\vec{\mathbf{v}} \times \vec{\mathbf{w}}) = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}. \tag{12.4.5}$$

Proof

The calculation is straightforward.

$$\begin{aligned} \vec{\mathbf{u}} \cdot (\vec{\mathbf{v}} \times \vec{\mathbf{w}}) &= \langle u_1, u_2, u_3 \rangle \cdot \langle v_2w_3 - v_3w_2, -v_1w_3 + v_3w_1, v_1w_2 - v_2w_1 \rangle \\ &= u_1(v_2w_3 - v_3w_2) + u_2(-v_1w_3 + v_3w_1) + u_3(v_1w_2 - v_2w_1) \\ &= u_1(v_2w_3 - v_3w_2) - u_2(v_1w_3 - v_3w_1) + u_3(v_1w_2 - v_2w_1) \\ &= \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}. \end{aligned}$$

□

Example 12.4.10: Calculating the Triple Scalar Product

Let $\vec{\mathbf{u}} = \langle 1, 3, 5 \rangle$, $\vec{\mathbf{v}} = \langle 2, -1, 0 \rangle$ and $\vec{\mathbf{w}} = \langle -3, 0, -1 \rangle$. Calculate the triple scalar product $\vec{\mathbf{u}} \cdot (\vec{\mathbf{v}} \times \vec{\mathbf{w}})$.

Solution

Apply Equation 12.4.5 directly:

$$\begin{aligned} \vec{\mathbf{u}} \cdot (\vec{\mathbf{v}} \times \vec{\mathbf{w}}) &= \begin{vmatrix} 1 & 3 & 5 \\ 2 & -1 & 0 \\ -3 & 0 & -1 \end{vmatrix} \\ &= 1\begin{vmatrix} -1 & 0 \\ 0 & -1 \end{vmatrix} - 3\begin{vmatrix} 2 & 0 \\ -3 & -1 \end{vmatrix} + 5\begin{vmatrix} 2 & -1 \\ -3 & 0 \end{vmatrix} \\ &= (1 - 0) - 3(-2 - 0) + 5(0 - 3) \\ &= 1 + 6 - 15 = -8. \end{aligned}$$

Exercise 12.4.10

Calculate the triple scalar product $\vec{\mathbf{a}} \cdot (\vec{\mathbf{b}} \times \vec{\mathbf{c}})$, where $\vec{\mathbf{a}} = \langle 2, -4, 1 \rangle$, $\vec{\mathbf{b}} = \langle 0, 3, -1 \rangle$, and $\vec{\mathbf{c}} = \langle 5, -3, 3 \rangle$.

Hint

Place the vectors as the rows of a 3×3 matrix, then calculate the determinant.

Answer

17

When we create a matrix from three vectors, we must be careful about the order in which we list the vectors. If we list them in a matrix in one order and then rearrange the rows, the absolute value of the determinant remains unchanged. However, each time two rows switch places, the determinant changes sign:

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = d \qquad \begin{vmatrix} b_1 & b_2 & b_3 \\ a_1 & a_2 & a_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = -d \qquad \begin{vmatrix} b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \\ a_1 & a_2 & a_3 \end{vmatrix} = d \qquad \begin{vmatrix} c_1 & c_2 & c_3 \\ b_1 & b_2 & b_3 \\ a_1 & a_2 & a_3 \end{vmatrix} = -d$$

Verifying this fact is straightforward, but rather messy. Let's take a look at this with an example:

$$\begin{aligned} \begin{vmatrix} 1 & 2 & 1 \\ -2 & 0 & 3 \\ 4 & 1 & -1 \end{vmatrix} &= \begin{vmatrix} 0 & 3 \\ 1 & -1 \end{vmatrix} - 2 \begin{vmatrix} -2 & 3 \\ 4 & -1 \end{vmatrix} + \begin{vmatrix} -2 & 0 \\ 4 & 1 \end{vmatrix} \\ &= (0-3) - 2(2-12) + (-2-0) \\ &= -3 + 20 - 2 = 15. \end{aligned}$$

Switching the top two rows we have

$$\begin{aligned} \begin{vmatrix} -2 & 0 & 3 \\ 1 & 2 & 1 \\ 4 & 1 & -1 \end{vmatrix} &= -2 \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} + 3 \begin{vmatrix} 1 & 2 \\ 4 & 1 \end{vmatrix} \\ &= -2(-2-1) + 3(1-8) \\ &= 6 - 21 = -15. \end{aligned}$$

Rearranging vectors in the triple products is equivalent to reordering the rows in the matrix of the determinant. Let $\vec{\mathbf{u}} = u_1\hat{\mathbf{i}} + u_2\hat{\mathbf{j}} + u_3\hat{\mathbf{k}}$, $\vec{\mathbf{v}} = v_1\hat{\mathbf{i}} + v_2\hat{\mathbf{j}} + v_3\hat{\mathbf{k}}$, and $\vec{\mathbf{w}} = w_1\hat{\mathbf{i}} + w_2\hat{\mathbf{j}} + w_3\hat{\mathbf{k}}$. Applying Calculating a Triple Scalar Product, we have

$$\vec{\mathbf{u}} \cdot (\vec{\mathbf{v}} \times \vec{\mathbf{w}}) = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}$$

and

$$\vec{\mathbf{u}} \cdot (\vec{\mathbf{w}} \times \vec{\mathbf{v}}) = \begin{vmatrix} u_1 & u_2 & u_3 \\ w_1 & w_2 & w_3 \\ v_1 & v_2 & v_3 \end{vmatrix}.$$

We can obtain the determinant for calculating $\vec{\mathbf{u}} \cdot (\vec{\mathbf{w}} \times \vec{\mathbf{v}})$ by switching the bottom two rows of $\vec{\mathbf{u}} \cdot (\vec{\mathbf{v}} \times \vec{\mathbf{w}})$. Therefore, $\vec{\mathbf{u}} \cdot (\vec{\mathbf{v}} \times \vec{\mathbf{w}}) = -\vec{\mathbf{u}} \cdot (\vec{\mathbf{w}} \times \vec{\mathbf{v}})$.

Following this reasoning and exploring the different ways we can interchange variables in the triple scalar product lead to the following identities:

$$\vec{\mathbf{u}} \cdot (\vec{\mathbf{v}} \times \vec{\mathbf{w}}) = -\vec{\mathbf{u}} \cdot (\vec{\mathbf{w}} \times \vec{\mathbf{v}}) \tag{12.4.6}$$

$$\vec{\mathbf{u}} \cdot (\vec{\mathbf{v}} \times \vec{\mathbf{w}}) = \vec{\mathbf{v}} \cdot (\vec{\mathbf{w}} \times \vec{\mathbf{u}}) = \vec{\mathbf{w}} \cdot (\vec{\mathbf{u}} \times \vec{\mathbf{v}}). \tag{12.4.7}$$

Let $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ be two vectors in standard position. If $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ are not scalar multiples of each other, then these vectors form adjacent sides of a parallelogram. We saw in Area of a Parallelogram that the area of this parallelogram is $\|\vec{\mathbf{u}} \times \vec{\mathbf{v}}\|$. Now suppose we add a third vector $\vec{\mathbf{w}}$ that does not lie in the same plane as $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ but still shares the same initial point. Then these vectors form three edges of a **parallelepiped**, a three-dimensional prism with six faces that are each parallelograms, as shown in Figure 12.4.7. The volume of this prism is the product of the figure's height and the area of its base. The triple scalar product of $\vec{\mathbf{u}}, \vec{\mathbf{v}}$, and $\vec{\mathbf{w}}$ provides a simple method for calculating the volume of the parallelepiped defined by these vectors.

Volume of a Parallelepiped

The volume of a parallelepiped with adjacent edges given by the vectors $\vec{u}, \vec{v}$, and $\vec{w}$ is the absolute value of the triple scalar product (Figure 12.4.7):

$$V = |\vec{u} \cdot (\vec{v} \times \vec{w})|.$$

Note that, as the name indicates, the triple scalar product produces a scalar. The volume formula just presented uses the absolute value of a scalar quantity.

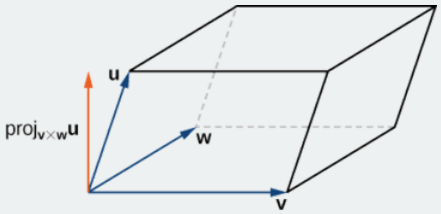


Figure 12.4.7: The height of the parallelepiped is given by $\|\text{proj}_{\vec{v}\times\vec{w}}\vec{u}\|$.

Proof

The area of the base of the parallelepiped is given by $\|\vec{v} \times \vec{w}\|$. The height of the figure is given by $\|\text{proj}_{\vec{v}\times\vec{w}}\vec{u}\|$. The volume of the parallelepiped is the product of the height and the area of the base, so we have

$$\begin{aligned} V &= \| \text{ proj}_{\vec{v}\times\vec{w}}\vec{u} \| \, \|\vec{v} \times \vec{w}\| \\ &= \left| \frac{\vec{u} \cdot (\vec{v} \times \vec{w})}{\|\vec{v} \times \vec{w}\|} \right| \|\vec{v} \times \vec{w}\| \\ &= |\vec{u} \cdot (\vec{v} \times \vec{w})|. \end{aligned}$$

□

Example 12.4.11: Calculating the Volume of a Parallelepiped

Let $\vec{u} = \langle -1, -2, 1\rangle$, $\vec{v} = \langle 4, 3, 2\rangle$, and $\vec{w} = \langle 0, -5, -2\rangle$. Find the volume of the parallelepiped with adjacent edges $\vec{u}, \vec{v}$, and $\vec{w}$ (Figure 12.4.8).

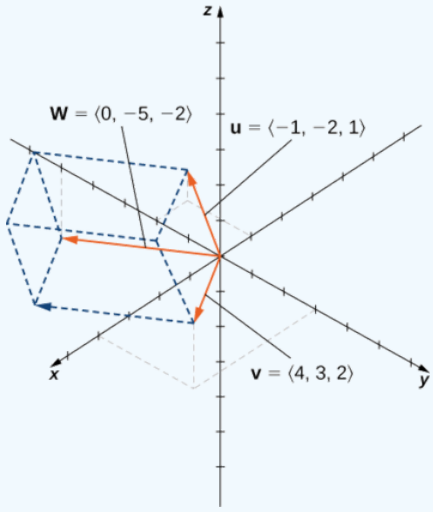


Figure 12.4.8

Solution

We have

$$\vec{\mathbf{u}} \cdot (\vec{\mathbf{v}} \times \vec{\mathbf{w}}) = \begin{vmatrix} -1 & -2 & 1 \\ 4 & 3 & 2 \\ 0 & -5 & -2 \end{vmatrix}$$
$$= (-1)\begin{vmatrix} 3 & 2 \\ -5 & -2 \end{vmatrix} + 2\begin{vmatrix} 4 & 2 \\ 0 & -2 \end{vmatrix} + \begin{vmatrix} 4 & 3 \\ 0 & -5 \end{vmatrix}$$
$$= (-1)(-6+10) + 2(-8-0) + (-20-0)$$
$$= -4 - 16 - 20$$
$$= -40.$$

Thus, the volume of the parallelepiped is $|-40| = 40$ units^3

? Exercise 12.4.11

Find the volume of the parallelepiped formed by the vectors $\vec{\mathbf{a}} = 3\hat{\mathbf{i}} + 4\hat{\mathbf{j}} - \hat{\mathbf{k}}$, $\vec{\mathbf{b}} = 2\hat{\mathbf{i}} - \hat{\mathbf{j}} - \hat{\mathbf{k}}$, and $\vec{\mathbf{c}} = 3\hat{\mathbf{j}} + \hat{\mathbf{k}}$.

Hint

Calculate the triple scalar product by finding a determinant.

Answer

8 units^3

Applications of the Cross Product

The cross product appears in many practical applications in mathematics, physics, and engineering. Let's examine some of these applications here, including the idea of torque, with which we began this section. Other applications show up in later chapters, particularly in our study of vector fields such as gravitational and electromagnetic fields (Introduction to Vector Calculus).

✓ Example 12.4.12: Using the Triple Scalar Product

Use the triple scalar product to show that vectors $\vec{\mathbf{u}} = \langle 2, 0, 5\rangle$, $\vec{\mathbf{v}} = \langle 2, 2, 4\rangle$, and $\vec{\mathbf{w}} = \langle 1, -1, 3\rangle$ are coplanar—that is, show that these vectors lie in the same plane.

Solution

Start by calculating the triple scalar product to find the volume of the parallelepiped defined by $\vec{\mathbf{u}}$, $\vec{\mathbf{v}}$, and $\vec{\mathbf{w}}$:

$$\vec{\mathbf{u}} \cdot (\vec{\mathbf{v}} \times \vec{\mathbf{w}}) = \begin{vmatrix} 2 & 0 & 5 \\ 2 & 2 & 4 \\ 1 & -1 & 3 \end{vmatrix}$$
$$= [2(2)(3) + (0)(4)(1) + 5(2)(-1)] - [5(2)(1) + (2)(4)(-1) + (0)(2)(3)]$$
$$= 2 - 2 = 0.$$

The volume of the parallelepiped is 0 units^3, so one of the dimensions must be zero. Therefore, the three vectors all lie in the same plane.

? Exercise 12.4.12

Are the vectors $\vec{\mathbf{a}} = \hat{\mathbf{i}} + \hat{\mathbf{j}} - \hat{\mathbf{k}}$, $\vec{\mathbf{b}} = \hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}$, and $\vec{\mathbf{c}} = \hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$ coplanar?

Hint

Calculate the triple scalar product.

Answer

No, the triple scalar product is $-4 \neq 0$, so the three vectors form the adjacent edges of a parallelepiped. They are not coplanar.

Example 12.4.13: Finding an Orthogonal Vector

Only a single plane can pass through any set of three noncolinear points. Find a vector orthogonal to the plane containing points $P = (9, -3, -2), Q = (1, 3, 0)$, and $R = (-2, 5, 0)$.

Solution

The plane must contain vectors $\overrightarrow{PQ}$ and $\overrightarrow{QR}$:

$$\overrightarrow{PQ} = \langle 1 - 9, 3 - (-3), 0 - (-2) \rangle = \langle -8, 6, 2 \rangle$$

$$\overrightarrow{QR} = \langle -2 - 1, 5 - 3, 0 - 0 \rangle = \langle -3, 2, 0 \rangle.$$

The cross product $\overrightarrow{PQ} \times \overrightarrow{QR}$ produces a vector orthogonal to both $\overrightarrow{PQ}$ and $\overrightarrow{QR}$. Therefore, the cross product is orthogonal to the plane that contains these two vectors:

$$\begin{aligned} \overrightarrow{PQ} \times \overrightarrow{QR} &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ -8 & 6 & 2 \\ -3 & 2 & 0 \end{vmatrix} \\ &= 0\hat{\mathbf{i}} - 6\hat{\mathbf{j}} - 16\hat{\mathbf{k}} - (-18\hat{\mathbf{k}} + 4\hat{\mathbf{i}} + 0\hat{\mathbf{j}}) \\ &= -4\hat{\mathbf{i}} - 6\hat{\mathbf{j}} + 2\hat{\mathbf{k}}. \end{aligned}$$

We have seen how to use the triple scalar product and how to find a vector orthogonal to a plane. Now we apply the cross product to real-world situations.

Sometimes a force causes an object to rotate. For example, turning a screwdriver or a wrench creates this kind of rotational effect, called torque.

Definition: Torque

Torque, $\vec{\tau}$ (the Greek letter tau), measures the tendency of a force to produce rotation about an axis of rotation. Let $\vec{\mathbf{r}}$ be a vector with an initial point located on the axis of rotation and with a terminal point located at the point where the force is applied, and let vector $\vec{\mathbf{F}}$ represent the force. Then torque is equal to the cross product of $\vec{\mathbf{r}}$ and $\vec{\mathbf{F}}$:

$$\vec{\tau} = \vec{\mathbf{r}} \times \vec{\mathbf{F}}.$$

See Figure 12.4.9.

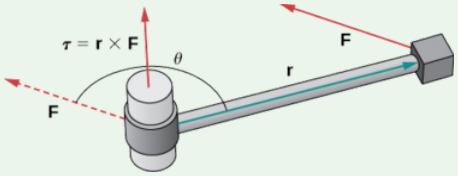


Figure 12.4.9: Torque measures how a force causes an object to rotate.

Think about using a wrench to tighten a bolt. The torque τ applied to the bolt depends on how hard we push the wrench (force) and how far up the handle we apply the force (distance). The torque increases with a greater force on the wrench at a greater distance from the bolt. Common units of torque are the newton-meter or foot-pound. Although torque is dimensionally equivalent to work (it has the same units), the two concepts are distinct. Torque is used specifically in the context of rotation, whereas work typically involves motion along a line.

Example 12.4.14: Evaluating Torque

A bolt is tightened by applying a force of 6 N to a 0.15-m wrench (Figure 12.4.10). The angle between the wrench and the force vector is 40°. Find the magnitude of the torque about the center of the bolt. Round the answer to two decimal places.

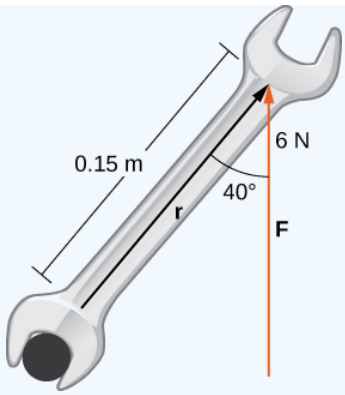


Figure 12.4.10: Torque describes the twisting action of the wrench.

Solution:

Substitute the given information into the equation defining torque:

$$\begin{aligned} \|\vec{\tau}\| &= \|\vec{r} \times \vec{F}\| \\ &= \|\vec{r}\| \, \| \, \vec{F} \, \| \sin\theta \\ &= (0.15\,\text{m})(6\,\text{N}) \sin 40° \\ &\approx 0.58\,\text{N·m}. \end{aligned}$$

Exercise 12.4.14

Calculate the force required to produce 15 N·m torque at an angle of 30º from a 150-cm rod.

Hint

$\|\vec{\tau}\| = 15$ N·m and $\|\vec{r}\| = 1.5$ m

Answer

20 N

Key Concepts

- The cross product $\vec{u} \times \vec{v}$ of two vectors $\vec{u} = \langle u_1, u_2, u_3 \rangle$ and $\vec{v} = \langle v_1, v_2, v_3 \rangle$ is a vector orthogonal to both $\vec{u}$ and $\vec{v}$. Its length is given by $\|\vec{u} \times \vec{v}\| = \|\vec{u}\| \cdot \|\vec{v}\| \cdot \sin\theta$, where θ is the angle between $\vec{u}$ and $\vec{v}$. Its direction is given by the right-hand rule.
- The algebraic formula for calculating the cross product of two vectors,

  $\vec{u} = \langle u_1, u_2, u_3 \rangle$ and $\vec{v} = \langle v_1, v_2, v_3 \rangle$, is

  $\vec{u} \times \vec{v} = (u_2 v_3 - u_3 v_2)\hat{i} - (u_1 v_3 - u_3 v_1)\hat{j} + (u_1 v_2 - u_2 v_1)\hat{k}$.
- The cross product satisfies the following properties for vectors $\vec{u}, \vec{v}$, and $\vec{w}$, and scalar c:

  $$\vec{u} \times \vec{v} = -(\vec{v} \times \vec{u})$$

  $$\vec{u} \times (\vec{v} + \vec{w}) = \vec{u} \times \vec{v} + \vec{u} \times \vec{w}$$

  $$c(\vec{u} \times \vec{v}) = (c\vec{u}) \times \vec{v} = \vec{u} \times (c\vec{v})$$

  $$\vec{u} \times \vec{0} = \vec{0} \times \vec{u} = \vec{0}$$

  $$\vec{v} \times \vec{v} = \vec{0}$$

  $$\vec{u} \cdot (\vec{v} \times \vec{w}) = (\vec{u} \times \vec{v}) \cdot \vec{w}$$
- The cross product of vectors $\vec{u} = \langle u_1, u_2, u_3 \rangle$ and $\vec{v} = \langle v_1, v_2, v_3 \rangle$ is the determinant $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$

- If vectors $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ form adjacent sides of a parallelogram, then the area of the parallelogram is given by $\|\vec{\mathbf{u}} \times \vec{\mathbf{v}}\|$.
- The triple scalar product of vectors $\vec{\mathbf{u}}, \vec{\mathbf{v}}$, and $\vec{\mathbf{w}}$ is $\vec{\mathbf{u}} \cdot (\vec{\mathbf{v}} \times \vec{\mathbf{w}})$.
- The volume of a parallelepiped with adjacent edges given by vectors $\vec{\mathbf{u}}, \vec{\mathbf{v}}$, and $\vec{\mathbf{w}}$ is $V = |\vec{\mathbf{u}} \cdot (\vec{\mathbf{v}} \times \vec{\mathbf{w}})|$.
- If the triple scalar product of vectors $\vec{\mathbf{u}}, \vec{\mathbf{v}}$, and $\vec{\mathbf{w}}$ is zero, then the vectors are coplanar. The converse is also true: If the vectors are coplanar, then their triple scalar product is zero.
- The cross product can be used to identify a vector orthogonal to two given vectors or to a plane.
- Torque $\vec{\boldsymbol{\tau}}$ measures the tendency of a force to produce rotation about an axis of rotation. If force $\vec{\mathbf{F}}$ is acting at a distance (displacement) $\vec{\mathbf{r}}$ from the axis, then torque is equal to the cross product of $\vec{\mathbf{r}}$ and $\vec{\mathbf{F}}$: $\vec{\boldsymbol{\tau}} = \vec{\mathbf{r}} \times \vec{\mathbf{F}}$.

Key Equations

- **The cross product of two vectors in terms of the unit vectors**

$$\vec{\mathbf{u}} \times \vec{\mathbf{v}} = (u_2 v_3 - u_3 v_2)\hat{\mathbf{i}} - (u_1 v_3 - u_3 v_1)\hat{\mathbf{j}} + (u_1 v_2 - u_2 v_1)\hat{\mathbf{k}}$$

Glossary

cross product

$\vec{\mathbf{u}} \times \vec{\mathbf{v}} = (u_2 v_3 - u_3 v_2)\hat{\mathbf{i}} - (u_1 v_3 - u_3 v_1)\hat{\mathbf{j}} + (u_1 v_2 - u_2 v_1)\hat{\mathbf{k}}$, where $\vec{\mathbf{u}} = \langle u_1, u_2, u_3 \rangle$ and $\vec{\mathbf{v}} = \langle v_1, v_2, v_3 \rangle$

determinant

a real number associated with a square matrix

parallelepiped

a three-dimensional prism with six faces that are parallelograms

torque

the effect of a force that causes an object to rotate

triple scalar product

the dot product of a vector with the cross product of two other vectors: $\vec{\mathbf{u}} \cdot (\vec{\mathbf{v}} \times \vec{\mathbf{w}})$

vector product

the cross product of two vectors

This page titled 12.4: The Cross Product is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

12.4E: Exercises for Section 12.4

For exercises 1-4, the vectors $\vec{u}$ and $\vec{v}$ are given.

a. Find the cross product $\vec{u} \times \vec{v}$ of the vectors $\vec{u}$ and $\vec{v}$. Express the answer in component form.

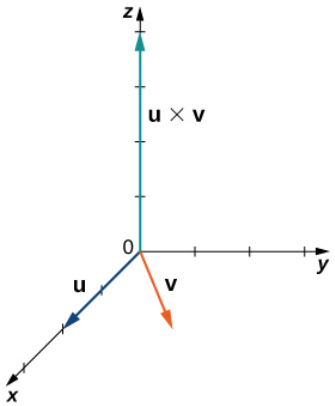
b. Sketch the vectors $\vec{u}$, $\vec{v}$, and $\vec{u} \times \vec{v}$.

1) $\vec{u} = \langle 2, 0, 0 \rangle, \quad \vec{v} = \langle 2, 2, 0 \rangle$

Answer

$a. \vec{u} \times \vec{v} = \langle 0, 0, 4 \rangle;$

$b.$



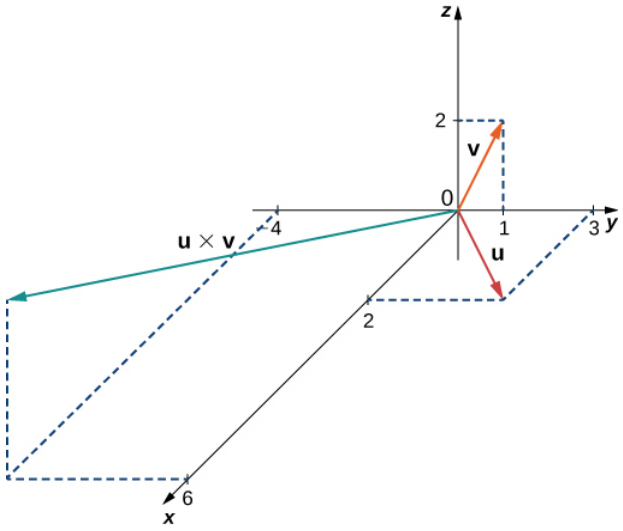
2) $\vec{u} = \langle 3, 2, -1 \rangle, \quad \vec{v} = \langle 1, 1, 0 \rangle$

3) $\vec{u} = 2\hat{i} + 3\hat{j}, \quad \vec{v} = \hat{j} + 2\hat{k}$

Answer

$a. \vec{u} \times \vec{v} = \langle 6, -4, 2 \rangle;$

$b.$



4) $\vec{u} = 2\hat{j} + 3\hat{k}, \quad \vec{v} = 3\hat{i} + \hat{k}$

5) Simplify $(\hat{i} \times \hat{i} - 2\hat{i} \times \hat{j} - 4\hat{i} \times \hat{k} + 3\hat{j} \times \hat{k}) \times \hat{i}$.

.

Answer

$-2\hat{\mathbf{j}} - 4\hat{\mathbf{k}}$

6) Simplify $\hat{\mathbf{j}} \times (\hat{\mathbf{k}} \times \hat{\mathbf{j}} + 2\hat{\mathbf{j}} \times \hat{\mathbf{i}} - 3\hat{\mathbf{j}} \times \hat{\mathbf{j}} + 5\hat{\mathbf{i}} \times \hat{\mathbf{k}})$.

In exercises 7-10, vectors $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ are given. Find unit vector $\vec{\mathbf{w}}$ in the direction of the cross product vector $\vec{\mathbf{u}} \times \vec{\mathbf{v}}$. Express your answer using standard unit vectors.

7) $\vec{\mathbf{u}} = \langle 3, -1, 2 \rangle, \quad \vec{\mathbf{v}} = \langle -2, 0, 1 \rangle$

Answer

$\vec{\mathbf{w}} = -\frac{\sqrt{6}}{18}\hat{\mathbf{i}} - \frac{7\sqrt{6}}{18}\hat{\mathbf{j}} - \frac{\sqrt{6}}{9}\hat{\mathbf{k}}$

8) $\vec{\mathbf{u}} = \langle 2, 6, 1 \rangle, \quad \vec{\mathbf{v}} = \langle 3, 0, 1 \rangle$

9) $\vec{\mathbf{u}} = \overrightarrow{AB}, \quad \vec{\mathbf{v}} = \overrightarrow{AC}$, where $A(1, 0, 1)$, $B(1, -1, 3)$, and $C(0, 0, 5)$

Answer

$\vec{\mathbf{w}} = -\frac{4\sqrt{21}}{21}\hat{\mathbf{i}} - \frac{2\sqrt{21}}{21}\hat{\mathbf{j}} - \frac{\sqrt{21}}{21}\hat{\mathbf{k}}$

10) $\vec{\mathbf{u}} = \overrightarrow{OP}, \quad \vec{\mathbf{v}} = \overrightarrow{PQ}$, where $P(-1, 1, 0)$ and $Q(0, 2, 1)$

11) Determine the real number α such that $\vec{\mathbf{u}} \times \vec{\mathbf{v}}$ and $\hat{\mathbf{i}}$ are orthogonal, where $\vec{\mathbf{u}} = 3\hat{\mathbf{i}} + \hat{\mathbf{j}} - 5\hat{\mathbf{k}}$ and $\vec{\mathbf{v}} = 4\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + \alpha\hat{\mathbf{k}}$.

Answer

$\alpha = 10$

12) Show that $\vec{\mathbf{u}} \times \vec{\mathbf{v}}$ and $2\hat{\mathbf{i}} - 14\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$ cannot be orthogonal for any α real number, where $\vec{\mathbf{u}} = \hat{\mathbf{i}} + 7\hat{\mathbf{j}} - \hat{\mathbf{k}}$ and $\vec{\mathbf{v}} = \alpha\hat{\mathbf{i}} + 5\hat{\mathbf{j}} + \hat{\mathbf{k}}$.

13) Show that $\vec{\mathbf{u}} \times \vec{\mathbf{v}}$ is orthogonal to $\vec{\mathbf{u}} + \vec{\mathbf{v}}$ and $\vec{\mathbf{u}} - \vec{\mathbf{v}}$, where $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ are nonzero vectors.

14) Show that $\vec{\mathbf{v}} \times \vec{\mathbf{u}}$ is orthogonal to $(\vec{\mathbf{u}} \cdot \vec{\mathbf{v}})(\vec{\mathbf{u}} + \vec{\mathbf{v}}) + \vec{\mathbf{u}}$, where $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ are nonzero vectors.

15) Calculate the determinant $\begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 1 & -1 & 7 \\ 2 & 0 & 3 \end{vmatrix}$.

Answer

$-3\hat{\mathbf{i}} + 11\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$

16) Calculate the determinant $\begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 0 & 3 & -4 \\ 1 & 6 & -1 \end{vmatrix}$.

For exercises 17-18, the vectors $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ are given. Use determinant notation to find vector $\vec{\mathbf{w}}$ orthogonal to vectors $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$.

17) $\vec{\mathbf{u}} = \langle -1, 0, e^t \rangle, \quad \vec{\mathbf{v}} = \langle 1, e^{-t}, 0 \rangle$, where t is a real number

Answer

$\vec{\mathbf{w}} = \langle -1, e^t, -e^{-t} \rangle$

18) $\vec{\mathbf{u}} = \langle 1, 0, x \rangle, \quad \vec{\mathbf{v}} = \langle \frac{2}{x}, 1, 0 \rangle$, where x is a nonzero real number

19) Find vector $(\vec{\mathbf{a}} - 2\vec{\mathbf{b}}) \times \vec{\mathbf{c}}$, where $\vec{\mathbf{a}} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 2 & -1 & 5 \\ 0 & 1 & 8 \end{vmatrix}$, $\vec{\mathbf{b}} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 0 & 1 & 1 \\ 2 & -1 & -2 \end{vmatrix}$, and $\vec{\mathbf{c}} = \hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$.

Answer

$-26\hat{\mathbf{i}} + 17\hat{\mathbf{j}} + 9\hat{\mathbf{k}}$

20) Find vector $\vec{c} \times (\vec{a} + 3\vec{b})$, where $\vec{a} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 5 & 0 & 9 \\ 0 & 1 & 0 \end{vmatrix}$, $\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & -1 & 1 \\ 7 & 1 & -1 \end{vmatrix}$, and $\vec{c} = \hat{i} - \hat{k}$.

21) [T] Use the cross product $\vec{u} \times \vec{v}$ to find the acute angle between vectors $\vec{u}$ and $\vec{v}$, where $\vec{u} = \hat{i} + 2\hat{j}$ and $\vec{v} = \hat{i} + \hat{k}$. Express the answer in degrees rounded to the nearest integer.

Answer

$72°$

22) [T] Use the cross product $\vec{u} \times \vec{v}$ to find the obtuse angle between vectors $\vec{u}$ and $\vec{v}$, where $\vec{u} = -\hat{i} + 3\hat{j} + \hat{k}$ and $\vec{v} = \hat{i} - 2\hat{j}$. Express the answer in degrees rounded to the nearest integer.

23) Use the sine and cosine of the angle between two nonzero vectors $\vec{u}$ and $\vec{v}$ to prove Lagrange's identity: $\|\vec{u} \times \vec{v}\|^2 = \|\vec{u}\|^2\|\vec{v}\|^2 - (\vec{u} \cdot \vec{v})^2$.

24) Verify Lagrange's identity $\|\vec{u} \times \vec{v}\|^2 = \|\vec{u}\|^2\|\vec{v}\|^2 - (\vec{u} \cdot \vec{v})^2$ for vectors $\vec{u} = -\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{v} = 2\hat{i} - \hat{j}$.

25) Nonzero vectors $\vec{u}$ and $\vec{v}$ are called collinear if there exists a nonzero scalar α such that $\vec{v} = \alpha\vec{u}$. Show that $\vec{u}$ and $\vec{v}$ are collinear if and only if $\vec{u} \times \vec{v} = 0$.

26) Nonzero vectors $\vec{u}$ and $\vec{v}$ are called *collinear* if there exists a nonzero scalar α such that $\vec{v} = \alpha\vec{u}$. Show that vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are collinear, where $A(4, 1, 0)$, $B(6, 5, -2)$, and $C(5, 3, -1)$.

27) Find the area of the parallelogram with adjacent sides $\vec{u} = \langle 3, 2, 0 \rangle$ and $\vec{v} = \langle 0, 2, 1 \rangle$.

Answer

7

28) Find the area of the parallelogram with adjacent sides $\vec{u} = \hat{i} + \hat{j}$ and $\vec{v} = \hat{i} + \hat{k}$.

29) Consider points $A(3, -1, 2)$, $B(2, 1, 5)$, and $C(1, -2, -2)$.

a. Find the area of parallelogram $ABCD$ with adjacent sides $\overrightarrow{AB}$ and $\overrightarrow{AC}$.

b. Find the area of triangle ABC.

c. Find the distance from point A to line BC.

Answer

a. $5\sqrt{6}$; b. $\frac{5\sqrt{6}}{2}$; c. $\frac{5\sqrt{6}}{\sqrt{59}} = \frac{5\sqrt{354}}{59}$

30) Consider points $A(2, -3, 4)$, $B(0, 1, 2)$, and $C(-1, 2, 0)$.

a. Find the area of parallelogram $ABCD$ with adjacent sides $\overrightarrow{AB}$ and $\overrightarrow{AC}$.

b. Find the area of triangle ABC.

c. Find the distance from point B to line AC.

In exercises 31-32, vectors $\vec{u}$, $\vec{v}$, and $\vec{w}$ are given.

a. Find the triple scalar product $\vec{u} \cdot (\vec{v} \times \vec{w})$.

b. Find the volume of the parallelepiped with the adjacent edges $\vec{u}$, $\vec{v}$, and $\vec{w}$.

31) $\vec{u} = \hat{i} + \hat{j}$, $\vec{v} = \hat{j} + \hat{k}$, and $\vec{w} = \hat{i} + \hat{k}$

Answer

$a. 2$; $b. 2$ units^3

32) $\vec{u} = \langle -3, 5, -1 \rangle$, $\vec{v} = \langle 0, 2, -2 \rangle$, and $\vec{w} = \langle 3, 1, 1 \rangle$

33) Calculate the triple scalar products $\vec{v} \cdot (\vec{u} \times \vec{w})$ and $\vec{w} \cdot (\vec{u} \times \vec{v})$, where $\vec{u} = \langle 1, 1, 1 \rangle, \vec{v} = \langle 7, 6, 9 \rangle$, and $\vec{w} = \langle 4, 2, 7 \rangle$.

Answer

$\vec{v} \cdot (\vec{u} \times \vec{w}) = -1, \quad \vec{w} \cdot (\vec{u} \times \vec{v}) = 1$

34) Calculate the triple scalar products $\vec{w} \cdot (\vec{v} \times \vec{u})$ and $\vec{u} \cdot (\vec{w} \times \vec{v})$, where $\vec{u} = \langle 4, 2, -1 \rangle, \vec{v} = \langle 2, 5, -3 \rangle$, and $\vec{w} = \langle 9, 5, -10 \rangle$.

35) Find vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$ with a triple scalar product given by the determinant $\begin{vmatrix} 1 & 2 & 3 \\ 0 & 2 & 5 \\ 8 & 9 & 2 \end{vmatrix}$. Determine their triple scalar product.

Answer

$\vec{a} = \langle 1, 2, 3 \rangle, \quad \vec{b} = \langle 0, 2, 5 \rangle, \quad \vec{c} = \langle 8, 9, 2 \rangle; \quad \vec{a} \cdot (\vec{b} \times \vec{c}) = -9$

36) The triple scalar product of vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$ is given by the determinant $\begin{vmatrix} 0 & -2 & 1 \\ 0 & 1 & 4 \\ 1 & -3 & 7 \end{vmatrix}$. Find vector $\vec{a} - \vec{b} + \vec{c}$.

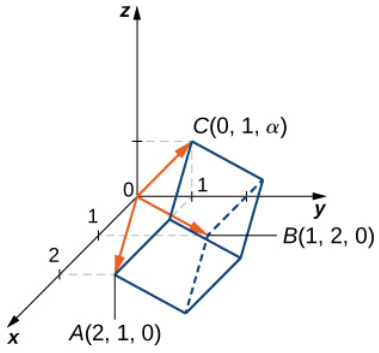
37) Consider the parallelepiped with edges OA, OB, and OC, where $A(2, 1, 0), B(1, 2, 0)$, and $C(0, 1, \alpha)$.

a. Find the real number $\alpha > 0$ such that the volume of the parallelepiped is 3 units^3.

b. For $\alpha = 1$, find the height h from vertex C of the parallelepiped. Sketch the parallelepiped.

Answer

$a.\ \alpha = 1; \quad b.\ h = 1$ unit,



38) Consider points $A(\alpha, 0, 0), B(0, \beta, 0)$, and $C(0, 0, \gamma)$, with α, β, and γ positive real numbers.

a. Determine the volume of the parallelepiped with adjacent sides $\overrightarrow{OA}, \overrightarrow{OB}$, and $\overrightarrow{OC}$.

b. Find the volume of the tetrahedron with vertices O, A, B, and C. (Hint: The volume of the tetrahedron is $1/6$ of the volume of the parallelepiped.)

c. Find the distance from the origin to the plane determined by A, B, and C. Sketch the parallelepiped and tetrahedron.

39) Let u, v, and w be three-dimensional vectors and c be a real number. Prove the following properties of the cross product.

a. $\vec{u} \times \vec{u} = \vec{0}$

b. $\vec{u} \times (\vec{v} + \vec{w}) = (\vec{u} \times \vec{v}) + (\vec{u} \times \vec{w})$

c. $c(\vec{u} \times \vec{v}) = (c\vec{u}) \times \vec{v} = \vec{u} \times (c\vec{v})$

d. $\vec{u} \cdot (\vec{u} \times \vec{v}) = \vec{0}$

40) Show that vectors $\vec{u} = \langle 1, 0, -8 \rangle, \vec{v} = \langle 0, 1, 6 \rangle$, and $\vec{w} = \langle -1, 9, 3 \rangle$ satisfy the following properties of the cross product.

a. $\vec{u} \times \vec{u} = \vec{0}$

b. $\vec{u} \times (\vec{v} + \vec{w}) = (\vec{u} \times \vec{v}) + (\vec{u} \times \vec{w})$

c. $c(\vec{\mathbf{u}} \times \vec{\mathbf{v}}) = (c\vec{\mathbf{u}}) \times \vec{\mathbf{v}} = \vec{\mathbf{u}} \times (c\vec{\mathbf{v}})$

d. $\vec{\mathbf{u}} \cdot (\vec{\mathbf{u}} \times \vec{\mathbf{v}}) = \vec{\mathbf{0}}$

41) Nonzero vectors $\vec{\mathbf{u}}$, $\vec{\mathbf{v}}$, and $\vec{\mathbf{w}}$ are said to be linearly dependent if one of the vectors is a linear combination of the other two. For instance, there exist two nonzero real numbers α and β such that $\vec{\mathbf{w}} = \alpha\vec{\mathbf{u}} + \beta\vec{\mathbf{v}}$. Otherwise, the vectors are called linearly independent. Show that $\vec{\mathbf{u}}, \vec{\mathbf{v}}$, and $\vec{\mathbf{w}}$ could be placed on the same plane if and only if they are linear dependent.

42) Consider vectors $\vec{\mathbf{u}} = \langle 1, 4, -7\rangle$, $\vec{\mathbf{v}} = \langle 2, -1, 4\rangle$, $\vec{\mathbf{w}} = \langle 0, -9, 18\rangle$, and $\vec{\mathbf{p}} = \langle 0, -9, 17\rangle$.

a. Show that $\vec{\mathbf{u}}$, $\vec{\mathbf{v}}$, and $\vec{\mathbf{w}}$ can be placed on the same plane by using their triple scalar product

b. Show that $\vec{\mathbf{u}}$, $\vec{\mathbf{v}}$, and $\vec{\mathbf{w}}$ can be placed on the same plane by using the definition that there exist two nonzero real numbers α and β such that $w = \alpha u + \beta v$.

c. Show that $\vec{\mathbf{u}}$, $\vec{\mathbf{v}}$, and $\vec{\mathbf{p}}$ are linearly independent—that is, none of the vectors is a linear combination of the other two.

43) Consider points $A(0, 0, 2), B(1, 0, 2), C(1, 1, 2)$, and $D(0, 1, 2)$. Are vectors $\overrightarrow{AB}, \overrightarrow{AC}$, and $\overrightarrow{AD}$ linearly dependent (that is, one of the vectors is a linear combination of the other two)?

Answer

Yes, $\overrightarrow{AD} = \alpha\overrightarrow{AB} + \beta\overrightarrow{AC}$, where $\alpha = -1$ and $\beta = 1$.

44) Show that vectors $\hat{\mathbf{i}} + \hat{\mathbf{j}}, \hat{\mathbf{i}} - \hat{\mathbf{j}}$, and $\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$ are linearly independent—that is, there exist two nonzero real numbers α and β such that $\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}} = \alpha(\hat{\mathbf{i}} + \hat{\mathbf{j}}) + \beta(\hat{\mathbf{i}} - \hat{\mathbf{j}})$.

45) Let $\vec{\mathbf{u}} = \langle u_1, u_2\rangle$ and $\vec{\mathbf{v}} = \langle v_1, v_2\rangle$ be two-dimensional vectors. The cross product of vectors $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ is not defined. However, if the vectors are regarded as the three-dimensional vectors $\tilde{\vec{\mathbf{u}}} = \langle u_1, u_2, 0\rangle$ and $\tilde{\vec{\mathbf{v}}} = \langle v_1, v_2, 0\rangle$, respectively, then, in this case, we can define the cross product of $\tilde{\vec{\mathbf{u}}}$ and $\tilde{\vec{\mathbf{v}}}$. In particular, in determinant notation, the cross product of $\tilde{\vec{\mathbf{u}}}$ and $\tilde{\vec{\mathbf{v}}}$ is given by

$$\tilde{\vec{\mathbf{u}}} \times \tilde{\vec{\mathbf{v}}} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ u_1 & u_2 & 0 \\ v_1 & v_2 & 0 \end{vmatrix}.$$

Use this result to compute $(\cos\theta\,\hat{\mathbf{i}} + \sin\theta\,\hat{\mathbf{j}}) \times (\sin\theta\,\hat{\mathbf{i}} - \cos\theta\,\hat{\mathbf{j}})$, where θ is a real number.

Answer

$-\hat{\mathbf{k}}$

46) Consider points $P(2, 1), Q(4, 2)$, and $R(1, 2)$.

a. Find the area of triangle PQR.

b. Determine the distance from point R to the line passing through P and Q.

47) Determine a vector of magnitude 10 perpendicular to the plane passing through the x-axis and point $P(1, 2, 4)$.

Answer

$\langle 0, \pm 4\sqrt{5}, 2\sqrt{5}\rangle$

48) Determine a unit vector perpendicular to the plane passing through the z-axis and point $A(3, 1, -2)$.

49) Consider $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ two three-dimensional vectors. If the magnitude of the cross product vector $\vec{\mathbf{u}} \times \vec{\mathbf{v}}$ is k times larger than the magnitude of vector $\vec{\mathbf{u}}$, show that the magnitude of $\vec{\mathbf{v}}$ is greater than or equal to k, where k is a natural number.

50) [T] Assume that the magnitudes of two nonzero vectors $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ are known. The function $f(\theta) = \|\vec{\mathbf{u}}\|\|\vec{\mathbf{v}}\| \sin\theta$ defines the magnitude of the cross product vector $\vec{\mathbf{u}} \times \vec{\mathbf{v}}$, where $\theta \in [0, \pi]$ is the angle between $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$.

a. Graph the function f.

b. Find the absolute minimum and maximum of function f. Interpret the results.

c. If $\|\vec{\mathbf{u}}\| = 5$ and $\|\vec{\mathbf{v}}\| = 2$, find the angle between $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ if the magnitude of their cross product vector is equal to 9.

51) Find all vectors $\vec{\mathbf{w}} = \langle w_1, w_2, w_3 \rangle$ that satisfy the equation $\langle 1, 1, 1 \rangle \times \vec{\mathbf{w}} = \langle -1, -1, 2 \rangle$. Hint: You should be able to write all components of $\vec{\mathbf{w}}$ in terms of one of the constants w_1, w_2, or w_3.

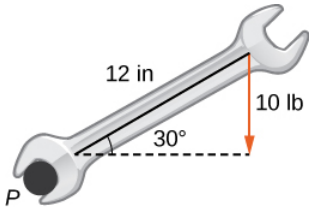
Answer

Writing all components in terms of the constant w_3, one way to represent these vectors is: $\vec{\mathbf{w}} = \langle w_3 - 1, w_3 + 1, w_3 \rangle$, where w_3 is any real number.

Note that we could use any parameter we wish here. We could set $w_3 = a$. Then $\vec{\mathbf{w}} = \langle a - 1, a + 1, a \rangle$ would also represent these vectors.

52) Solve the equation $\vec{\mathbf{w}} \times \langle 1, 0, -1 \rangle = \langle 3, 0, 3 \rangle$, where $\vec{\mathbf{w}} = \langle w_1, w_2, w_3 \rangle$ is a nonzero vector with a magnitude of 3.

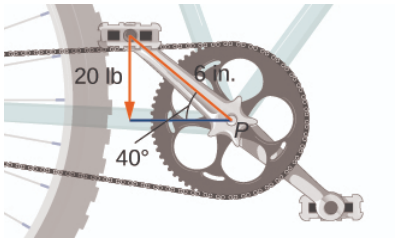
53) [T] A mechanic uses a 12-in. wrench to turn a bolt. The wrench makes a $30°$ angle with the horizontal. If the mechanic applies a vertical force of 10 lb on the wrench handle, what is the magnitude of the torque at point P (see the following figure)? Express the answer in foot-pounds rounded to two decimal places.



Answer

8.66 ft-lb

54) [T] A boy applies the brakes on a bicycle by applying a downward force of 20 lb on the pedal when the 6-in. crank makes a $40°$ angle with the horizontal (see the following figure). Find the torque at point P. Express your answer in foot-pounds rounded to two decimal places.



55) [T] Find the magnitude of the force that needs to be applied to the end of a 20-cm wrench located on the positive direction of the y-axis if the force is applied in the direction $\langle 0, 1, -2 \rangle$ and it produces a 100 N·m torque to the bolt located at the origin.

Answer

$250\sqrt{5}$ N ≈ 559 N

56) [T] What is the magnitude of the force required to be applied to the end of a 1-ft wrench at an angle of $35°$ to produce a torque of 20 N·m?

57) [T] The force vector $\vec{\mathbf{F}}$ acting on a proton with an electric charge of $1.6 \times 10^{-19}\ C$ (in coulombs) moving in a magnetic field $\vec{\mathbf{B}}$ where the velocity vector $\vec{\mathbf{v}}$ is given by $\vec{\mathbf{F}} = 1.6 \times 10^{-19}(\vec{\mathbf{v}} \times \vec{\mathbf{B}})$ (here, $\vec{\mathbf{v}}$ is expressed in meters per second, $\vec{\mathbf{B}}$ is in tesla [T], and $\vec{\mathbf{F}}$ is in newtons [N]). Find the force that acts on a proton that moves in the xy-plane at velocity $\vec{\mathbf{v}} = 10^5\hat{\mathbf{i}} + 10^5\hat{\mathbf{j}}$ (in meters per second) in a magnetic field given by $\vec{\mathbf{B}} = 0.3\hat{\mathbf{j}}$.

Answer

$\vec{\mathbf{F}} = 4.8 \times 10^{-15}\ kN$

58) [T] The force vector $\vec{\mathbf{F}}$ acting on a proton with an electric charge of $1.6 \times 10^{-19}\ C$ moving in a magnetic field $\vec{\mathbf{B}}$ where the velocity vector $\vec{\mathbf{v}}$ is given by $\vec{\mathbf{F}} = 1.6 \times 10^{-19}(\vec{\mathbf{v}} \times \vec{\mathbf{B}})$ (here, $\vec{\mathbf{v}}$ is expressed in meters per second, $\vec{\mathbf{B}}$ in T, and $\vec{\mathbf{F}}$ in N). If the magnitude of force $\vec{\mathbf{F}}$ acting on a proton is $5.9 \times 10^{-17}\ N$ and the proton is moving at the speed of 300 m/sec in magnetic field $\vec{\mathbf{B}}$ of magnitude 2.4 T, find the angle between velocity vector $\vec{\mathbf{v}}$ of the proton and magnetic field $\vec{\mathbf{B}}$. Express the answer in degrees rounded to the nearest integer.

59) [T] Consider $\vec{\mathbf{r}}(t) = \langle \cos t,\ \sin t,\ 2t \rangle$ the position vector of a particle at time $t \in [0, 30]$, where the components of $\vec{\mathbf{r}}$ are expressed in centimeters and time in seconds. Let $\overrightarrow{OP}$ be the position vector of the particle after 1 sec.

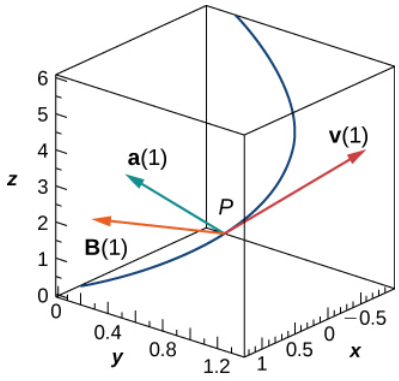
a. Determine unit vector $\vec{\mathbf{B}}(t)$ (called the binormal unit vector) that has the direction of cross product vector $\vec{\mathbf{v}}(t) \times \vec{\mathbf{a}}(t)$, where $\vec{\mathbf{v}}(t)$ and $\vec{\mathbf{a}}(t)$ are the instantaneous velocity vector and, respectively, the acceleration vector of the particle after t seconds.

b. Use a CAS to visualize vectors $\vec{\mathbf{v}}(1)$, $\vec{\mathbf{a}}(1)$, and $\vec{\mathbf{B}}(1)$ as vectors starting at point P along with the path of the particle.

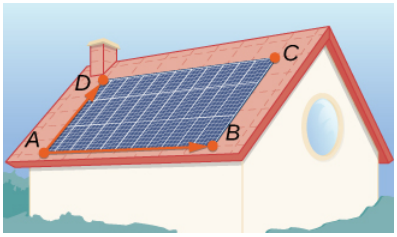
Answer

a. $\vec{\mathbf{B}}(t) = \langle \frac{2\sqrt{5}\sin t}{5}, -\frac{2\sqrt{5}\cos t}{5}, \frac{\sqrt{5}}{5} \rangle$;

b.



60) A solar panel is mounted on the roof of a house. The panel may be regarded as positioned at the points of coordinates (in meters) $A(8, 0, 0), B(8, 18, 0), C(0, 18, 8)$, and $D(0, 0, 8)$ (see the following figure).



a. Find vector $\vec{\mathbf{n}} = \overrightarrow{AB} \times \overrightarrow{AD}$ perpendicular to the surface of the solar panels. Express the answer using standard unit vectors. Note that the magnitude of this vector should give us the area of rectangle $ABCD$.

b. Assume unit vector $\vec{\mathbf{s}} = \frac{1}{\sqrt{3}}\hat{\mathbf{i}} + \frac{1}{\sqrt{3}}\hat{\mathbf{j}} + \frac{1}{\sqrt{3}}\hat{\mathbf{k}}$ points toward the Sun at a particular time of the day and the flow of solar energy is $\vec{\mathbf{F}} = 900\vec{\mathbf{s}}$ (in watts per square meter $[W/m^2]$). Find the predicted amount of electrical power the panel can produce, which is given by the dot product of vectors $\vec{\mathbf{F}}$ and $\vec{\mathbf{n}}$ (expressed in watts).

c. Determine the angle of elevation of the Sun above the solar panel. Express the answer in degrees rounded to the nearest whole number. (Hint: The angle between vectors $\vec{\mathbf{n}}$ and $\vec{\mathbf{s}}$ and the angle of elevation are complementary.)

Contributors

Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.

This page titled 12.4E: Exercises for Section 12.4 is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

12.5: Equations of Lines and Planes in Space

Learning Objectives

- Write the vector, parametric, and symmetric equations of a line through a given point in a given direction, and a line through two given points.
- Find the distance from a point to a given line.
- Write the vector and scalar equations of a plane through a given point with a given normal.
- Find the distance from a point to a given plane.
- Find the angle between two planes.

By now, we are familiar with writing equations that describe a line in two dimensions. To write an equation for a line, we must know two points on the line, or we must know the direction of the line and at least one point through which the line passes. In two dimensions, we use the concept of slope to describe the orientation, or direction, of a line. In three dimensions, we describe the direction of a line using a vector parallel to the line. In this section, we examine how to use equations to describe lines and planes in space.

Equations for a Line in Space

Let's first explore what it means for two vectors to be parallel. Recall that parallel vectors must have the same or opposite directions. If two nonzero vectors, $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$, are parallel, we claim there must be a scalar, k, such that $\vec{\mathbf{u}} = k\vec{\mathbf{v}}$. If $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ have the same direction, simply choose

$$k = \frac{\|\vec{\mathbf{u}}\|}{\|\vec{\mathbf{v}}\|}.$$

If $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ have opposite directions, choose

$$k = -\frac{\|\vec{\mathbf{u}}\|}{\|\vec{\mathbf{v}}\|}.$$

Note that the converse holds as well. If $\vec{\mathbf{u}} = k\vec{\mathbf{v}}$ for some scalar k, then either $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ have the same direction $(k > 0)$ or opposite directions $(k < 0)$, so $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ are parallel. Therefore, two nonzero vectors $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ are parallel if and only if $\vec{\mathbf{u}} = k\vec{\mathbf{v}}$ for some scalar k. By convention, the zero vector $\vec{\mathbf{0}}$ is considered to be parallel to all vectors.

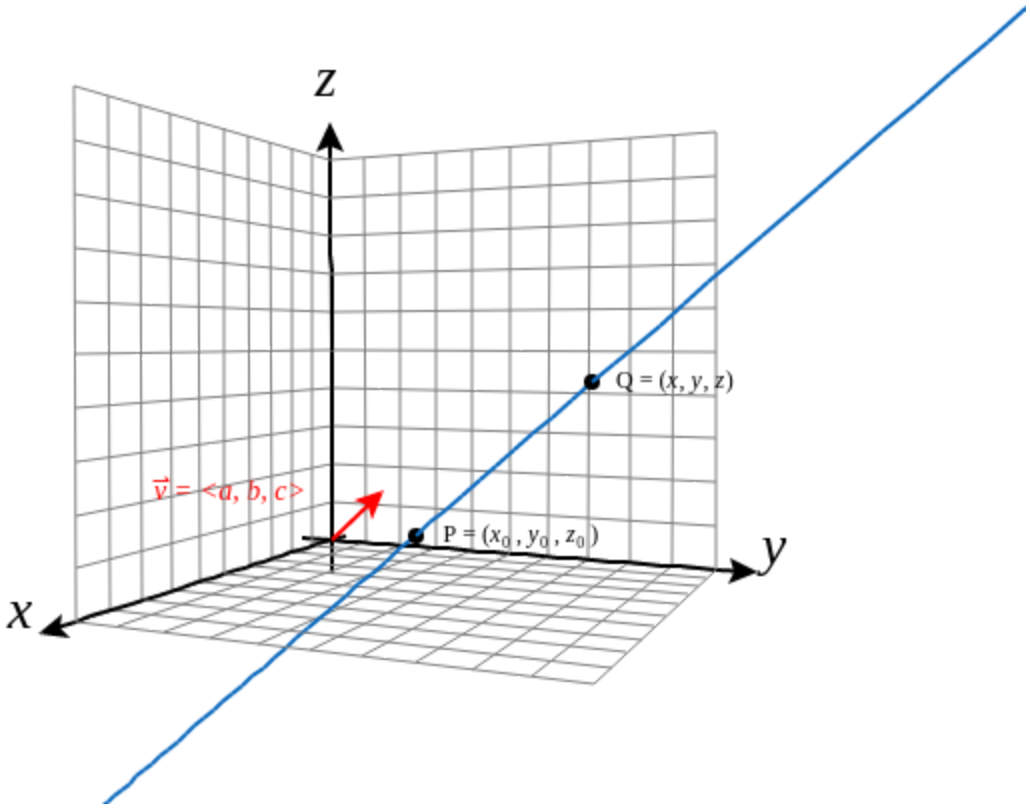


Figure 12.5.1: Vector $\vec{\mathbf{v}}$ is the direction vector for $\overrightarrow{PQ}$.

As in two dimensions, we can describe a line in space using a point on the line and the direction of the line, or a parallel vector, which we call the **direction vector** (Figure 12.5.1). Let L be a line in space passing through point $P(x_0, y_0, z_0)$. Let $\vec{\mathbf{v}} = \langle a, b, c\rangle$ be a vector parallel to L. Then, for any point on line $Q(x, y, z)$, we know that $\overrightarrow{PQ}$ is parallel to $\vec{\mathbf{v}}$. Thus, as we just discussed, there is a scalar, t, such that $\overrightarrow{PQ} = t\vec{\mathbf{v}}$, which gives

$$\begin{aligned} \overrightarrow{PQ} &= t\vec{\mathbf{v}} \\ \langle x - x_0, y - y_0, z - z_0\rangle &= t\langle a, b, c\rangle \\ \langle x - x_0, y - y_0, z - z_0\rangle &= \langle ta, tb, tc\rangle. \end{aligned} \tag{12.5.1}$$

Using vector operations, we can rewrite Equation 12.5.1

$$\begin{aligned} \langle x - x_0, y - y_0, z - z_0\rangle &= \langle ta, tb, tc\rangle \\ \langle x, y, z\rangle - \langle x_0, y_0, z_0\rangle &= t\langle a, b, c\rangle \\ \underbrace{\langle x, y, z\rangle}_{\vec{\mathbf{r}}} &= \underbrace{\langle x_0, y_0, z_0\rangle}_{\vec{\mathbf{r}}_o} + t\underbrace{\langle a, b, c\rangle}_{\vec{\mathbf{v}}}. \end{aligned}$$

Setting $\vec{\mathbf{r}} = \langle x, y, z\rangle$ and $\vec{\mathbf{r}}_0 = \langle x_0, y_0, z_0\rangle$, we now have the **vector equation of a line:**

$$\vec{\mathbf{r}} = \vec{\mathbf{r}}_0 + t\vec{\mathbf{v}}. \tag{12.5.2}$$

Equating components, Equation 12.5.2 shows that the following equations are simultaneously true: $x - x_0 = ta, y - y_0 = tb$, and $z - z_0 = tc$. If we solve each of these equations for the component variables x, y, and z, we get a set of equations in which each variable is defined in terms of the parameter t and that, together, describe the line. This set of three equations forms a set of **parametric equations of a line:**

$$x = x_0 + ta$$

$$y = y_0 + tb$$

$$z = z_0 + tc.$$

If we solve each of the equations for t assuming $a, b,$ and c are nonzero, we get a different description of the same line:

$$\frac{x - x_0}{a} = t$$
$$\frac{y - y_0}{b} = t$$
$$\frac{z - z_0}{c} = t.$$

Because each expression equals t, they all have the same value. We can set them equal to each other to create **symmetric equations of a line:**

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

We summarize the results in the following theorem.

Theorem: Parametric and Symmetric Equations of a Line

A line L parallel to vector $\vec{\mathbf{v}} = \langle a, b, c \rangle$ and passing through point $P(x_0, y_0, z_0)$ can be described by the following parametric equations:

$$x = x_0 + ta, y = y_0 + tb,$$

and

$$z = z_0 + tc.$$

If the constants $a, b,$ and c are all nonzero, then L can be described by the symmetric equation of the line:

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

The parametric equations of a line are not unique. Using a different parallel vector or a different point on the line leads to a different, equivalent representation. Each set of parametric equations leads to a related set of symmetric equations, so it follows that a symmetric equation of a line is not unique either.

Example 12.5.1: Equations of a Line in Space

Find parametric and symmetric equations of the line passing through points $(1, 4, -2)$ and $(-3, 5, 0)$.

Solution

First, identify a vector parallel to the line:

$$\vec{\mathbf{v}} = \langle -3 - 1, 5 - 4, 0 - (-2) \rangle = \langle -4, 1, 2 \rangle.$$

Use either of the given points on the line to complete the parametric equations:

$$x = 1 - 4t$$
$$y = 4 + t,$$

and

$$z = -2 + 2t.$$

Solve each equation for t to create the symmetric equation of the line:

$$\frac{x-1}{-4} = y-4 = \frac{z+2}{2}.$$

Exercise 12.5.1

Find parametric and symmetric equations of the line passing through points $(1,-3,2)$ and $(5,-2,8)$.

Hint:

Start by finding a vector parallel to the line.

Answer

Possible set of parametric equations: $x = 1+4t,\ y = -3+t,\ z = 2+6t$; related set of symmetric equations:

$$\frac{x-1}{4} = y+3 = \frac{z-2}{6}$$

Sometimes we don't want the equation of a whole line, just a line segment. In this case, we limit the values of our parameter t. For example, let $P(x_0,y_0,z_0)$ and $Q(x_1,y_1,z_1)$ be points on a line, and let $\vec{\mathbf{p}} = \langle x_0,y_0,z_0\rangle$ and $\vec{\mathbf{q}} = \langle x_1,y_1,z_1\rangle$ be the associated position vectors. In addition, let $\vec{\mathbf{r}} = \langle x,y,z\rangle$. We want to find a vector equation for the line segment between P and Q. Using P as our known point on the line, and $\overrightarrow{PQ} = \langle x_1-x_0, y_1-y_0, z_1-z_0\rangle$ as the direction vector equation, Equation 12.5.2 gives

$$\vec{\mathbf{r}} = \vec{\mathbf{p}} + t(\overrightarrow{PQ}). \tag{12.5.3}$$

Equation 12.5.3 can be expanded using properties of vectors:

$$\begin{aligned}
\vec{\mathbf{r}} &= \vec{\mathbf{p}} + t(\overrightarrow{PQ}) \\
&= \langle x_0,y_0,z_0\rangle + t\langle x_1-x_0, y_1-y_0, z_1-z_0\rangle \\
&= \langle x_0,y_0,z_0\rangle + t(\langle x_1,y_1,z_1\rangle - \langle x_0,y_0,z_0\rangle) \\
&= \langle x_0,y_0,z_0\rangle + t\langle x_1,y_1,z_1\rangle - t\langle x_0,y_0,z_0\rangle \\
&= (1-t)\langle x_0,y_0,z_0\rangle + t\langle x_1,y_1,z_1\rangle \\
&= (1-t)\vec{\mathbf{p}} + t\vec{\mathbf{q}}.
\end{aligned}$$

Thus, the vector equation of the line passing through P and Q is

$$\vec{\mathbf{r}} = (1-t)\vec{\mathbf{p}} + t\vec{\mathbf{q}}.$$

Remember that we did not want the equation of the whole line, just the line segment between P and Q. Notice that when $t=0$, we have $\vec{\mathbf{r}} = \vec{\mathbf{p}}$, and when $t=1$, we have $\vec{\mathbf{r}} = \vec{\mathbf{q}}$. Therefore, the vector equation of the line segment between P and Q is

$$\vec{\mathbf{r}} = (1-t)\vec{\mathbf{p}} + t\vec{\mathbf{q}}, 0 \le t \le 1.$$

Going back to Equation 12.5.2, we can also find parametric equations for this line segment. We have

$$\begin{aligned}
\vec{\mathbf{r}} &= \vec{\mathbf{p}} + t(\overrightarrow{PQ}) \\
\langle x,y,z\rangle &= \langle x_0,y_0,z_0\rangle + t\langle x_1-x_0, y_1-y_0, z_1-z_0\rangle \\
&= \langle x_0+t(x_1-x_0), y_0+t(y_1-y_0), z_0+t(z_1-z_0)\rangle.
\end{aligned}$$

Then, the parametric equations are

$$\begin{aligned}
x &= x_0 + t(x_1-x_0) \\
y &= y_0 + t(y_1-y_0) \\
z &= z_0 + t(z_1-z_0), \quad 0 \le t \le 1.
\end{aligned} \tag{12.5.4}$$

Example 12.5.2: Parametric Equations of a Line Segment

Find parametric equations of the line segment between the points $P(2,1,4)$ and $Q(3,-1,3)$.

Solution

Start with the parametric equations for a line (Equations 12.5.4) and work with each component separately:

$$\begin{aligned} x &= x_0 + t(x_1 - x_0) \\ &= 2 + t(3-2) \\ &= 2 + t, \end{aligned}$$

$$\begin{aligned} y &= y_0 + t(y_1 - y_0) \\ &= 1 + t(-1-1) \\ &= 1 - 2t, \end{aligned}$$

and

$$\begin{aligned} z &= z_0 + t(z_1 - z_0) \\ &= 4 + t(3-4) \\ &= 4 - t. \end{aligned}$$

Therefore, the parametric equations for the line segment are

$$\begin{aligned} x &= 2 + t \\ y &= 1 - 2t \\ z &= 4 - t, \quad 0 \le t \le 1. \end{aligned}$$

Exercise 12.5.2

Find parametric equations of the line segment between points $P(-1,3,6)$ and $Q(-8,2,4)$.

Answer

$x = -1 - 7t,\ y = 3 - t,\ z = 6 - 2t, \quad 0 \le t \le 1$

Distance between a Point and a Line

We already know how to calculate the distance between two points in space. We now expand this definition to describe the distance between a point and a line in space. Several real-world contexts exist when it is important to be able to calculate these distances. When building a home, for example, builders must consider "setback" requirements, when structures or fixtures have to be a certain distance from the property line. Air travel offers another example. Airlines are concerned about the distances between populated areas and proposed flight paths.

Let L be a line in the plane and let M be any point not on the line. Then, we define distance d from M to L as the length of line segment $\overline{MP}$, where P is a point on L such that $\overline{MP}$ is perpendicular to L (Figure 12.5.2).

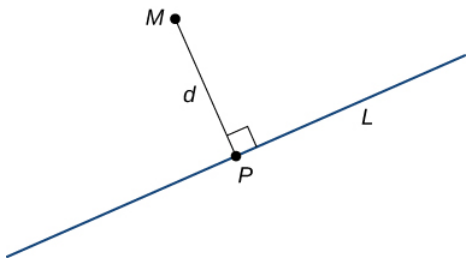


Figure 12.5.2: The distance from point M to line L is the length of $\overline{MP}$.

When we're looking for the distance between a line and a point in space, Figure 12.5.2 still applies. We still define the distance as the length of the perpendicular line segment connecting the point to the line. In space, however, there is no clear way to know

which point on the line creates such a perpendicular line segment, so we select an arbitrary point on the line and use properties of vectors to calculate the distance. Therefore, let P be an arbitrary point on line L and let $\vec{\mathbf{v}}$ be a direction vector for L (Figure 12.5.3).

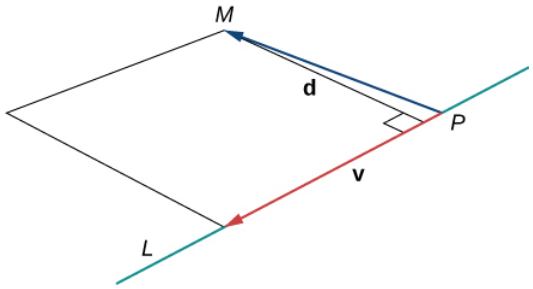


Figure 12.5.3: Vectors $\overrightarrow{PM}$ and $\vec{\mathbf{v}}$ form two sides of a parallelogram with base $\|\vec{\mathbf{v}}\|$ and height d, which is the distance between a line and a point in space.

Vectors $\overrightarrow{PM}$ and $\vec{\mathbf{v}}$ form two sides of a parallelogram with area $\|\overrightarrow{PM} \times \vec{\mathbf{v}}\|$. Using a formula from geometry, the area of this parallelogram can also be calculated as the product of its base and height:

$$\|\overrightarrow{PM} \times \vec{\mathbf{v}}\| = \|\vec{\mathbf{v}}\| d.$$

We can use this formula to find a general formula for the distance between a line in space and any point not on the line.

Distance from a Point to a Line

Let L be a line in space passing through point P with direction vector $\vec{\mathbf{v}}$. If M is any point not on L, then the distance from M to L is

$$d = \frac{\|\overrightarrow{PM} \times \vec{\mathbf{v}}\|}{\|\vec{\mathbf{v}}\|}.$$

Example 12.5.3: Calculating the Distance from a Point to a Line

Find the distance between the point $M = (1, 1, 3)$ and line $\dfrac{x-3}{4} = \dfrac{y+1}{2} = z - 3$.

Solution:

From the symmetric equations of the line, we know that vector $\vec{\mathbf{v}} = \langle 4, 2, 1 \rangle$ is a direction vector for the line. Setting the symmetric equations of the line equal to zero, we see that point $P(3, -1, 3)$ lies on the line. Then,

$$\begin{aligned} \overrightarrow{PM} &= \langle 1-3, 1-(-1), 3-3 \rangle \\ &= \langle -2, 2, 0 \rangle. \end{aligned}$$

To calculate the distance, we need to find $\overrightarrow{PM} \times \vec{\mathbf{v}}$:

$$\begin{aligned} \overrightarrow{PM} \times \vec{\mathbf{v}} &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ -2 & 2 & 0 \\ 4 & 2 & 1 \end{vmatrix} \\ &= (2-0)\hat{\mathbf{i}} - (-2-0)\hat{\mathbf{j}} + (-4-8)\hat{\mathbf{k}} \\ &= 2\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 12\hat{\mathbf{k}}. \end{aligned}$$

Therefore, the distance between the point and the line is (Figure 12.5.4)

$$d = \frac{\|\overrightarrow{PM} \times \vec{\mathbf{v}}\|}{\|\vec{\mathbf{v}}\|}$$
$$= \frac{\sqrt{2^2 + 2^2 + 12^2}}{\sqrt{4^2 + 2^2 + 1^2}}$$
$$= \frac{2\sqrt{38}}{\sqrt{21}}$$
$$= \frac{2\sqrt{798}}{21} \text{ units}$$

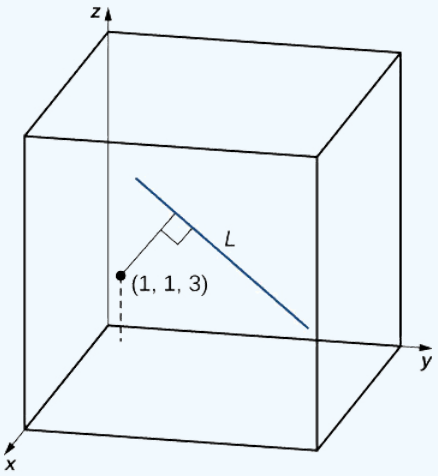


Figure 12.5.4: Point $(1, 1, 3)$ is approximately 2.7 units from the line with symmetric equations $\frac{x-3}{4} = \frac{y+1}{2} = z - 3$.

? Exercise 12.5.3

Find the distance between point $(0, 3, 6)$ and the line with parametric equations $x = 1 - t,\ y = 1 + 2t,\ z = 5 + 3t$.

Hint

Find a vector with initial point $(0, 3, 6)$ and a terminal point on the line, and then find a direction vector for the line.

Answer

$\sqrt{\frac{10}{7}} = \frac{\sqrt{70}}{7}$ units

Relationships between Lines

Given two lines in the two-dimensional plane, the lines are equal, they are parallel but not equal, or they intersect in a single point. In three dimensions, a fourth case is possible. If two lines in space are not parallel, but do not intersect, then the lines are said to be **skew lines** (Figure 12.5.5).

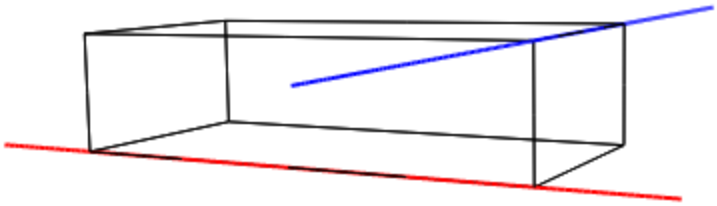


Figure 12.5.5: In three dimensions, it is possible that two lines do not cross, even when they have different directions.

To classify lines as parallel but not equal, equal, intersecting, or skew, we need to know two things: whether the direction vectors are parallel and whether the lines share a point (Figure 12.5.6).

Direction Vectors Are Parallel? \ Lines Share A Common Point?	Yes	No
Yes	Equal	Parallel but not equal
No	Intersecting	Skew

Figure 12.5.6: Determine the relationship between two lines based on whether their direction vectors are parallel and whether they share a point.

Example 12.5.4: Classifying Lines in Space

For each pair of lines, determine whether the lines are equal, parallel but not equal, skew, or intersecting.

a.